Core Java Interview Questions

17 essential questions with explanations and code examples

1. What Is the Difference Between JDK, JRE, and JVM?

Imagine JVM as a virtual engine that runs Java programs, JRE as a ready-to-use car with the engine installed, and JDK as a complete workshop with tools, parts, and manuals for building and maintaining cars.

JVM (Java Virtual Machine)

Definition: JVM is an abstract computing machine that enables Java bytecode to be executed on different hardware platforms.

Functionality: It interprets Java bytecode into machine code or executes it using Just-In-Time (JIT) compilation, providing platform independence for Java applications.

Use Case: Suppose you have a Java application compiled into bytecode. The JVM executes this bytecode, translating it into machine-specific instructions for the underlying hardware.

JRE (Java Runtime Environment)

Definition: JRE is a runtime environment that contains JVM, libraries, and other components required to run Java applications but does not include development tools.

Functionality: It provides the runtime environment necessary for executing Java applications, including the JVM, core libraries, and additional runtime resources.

Use Case: When you want to run a pre-built Java application on your system, you need JRE installed. It provides the necessary runtime environment for the application to execute.

JDK (Java Development Kit)

Definition: JDK is a software development kit that includes JRE, development tools, libraries, and documentation necessary for developing Java applications.

Functionality: It provides everything included in JRE, along with additional tools such as compilers, debuggers, and development APIs for creating Java applications.

Use Case: If you're a Java developer, you'll use JDK to write, compile, and debug Java code. It includes everything you need for Java development, from writing code to deploying applications.

2. What Is the Difference Between Stack and Heap?

The stack and heap are two types of memory storage in the Java Virtual Machine (JVM).

The stack is used for storing method invocation and local variables. Each thread in a program has its own stack, and it operates in a last-in, first-out manner. When a method is called, a new stack frame is created, and when the method execution is completed, the stack frame is removed. Stack memory is typically faster to access than heap memory.

The heap is a shared memory pool used for dynamic memory allocation. Objects created in Java are stored in the heap. Unlike the stack, memory in the heap can be accessed randomly. Garbage collection is responsible for managing memory in the heap by reclaiming memory occupied by objects that are no longer needed.

In summary, the stack is used for storing method call information and local variables with a predictable, limited size per thread, while the heap is used for dynamic memory allocation of objects with a larger, shared memory space.

Always remember that the stack has much less memory space than the heap and when a stack is full, the JVM will throw a StackOverflowError.

3. Why is the main method in Java declared as public static void main(String[] args)?

The main method is declared this way for the following reasons:

- **public:** So the JVM can access it from outside the class.
- **static:** Allows the JVM to invoke it without creating an instance of the class.
- **void:** Indicates that the method does not return any value.
- **String[] args:** Used to accept command-line arguments when the program runs.

4. What Is the void Type and When is it preferable to Use It?

In Java, the void type is used to indicate that a method does not return any value. Using void appropriately makes the code more readable and helps communicate the intent of the method to other developers. It indicates that the method performs an action or has a side effect without returning any result.

```
// Example 1: Method with side effects
void writeToLogFile(String message) {
    // Write message to log file
}

// Example 2: Event handler method
void onButtonClick() {
    // Perform action when button is clicked
}

// Example 3: Setter method
void setName(String name) {
    this.name = name;
}

// Example 4: Utility method
void printMessage(String message) {
    System.out.println(message);
}

// Example 5: Lifecycle method
void init() {
    // Initialize resources
}
```

5. Which Access Modifiers Are Available in Java and their Purpose?

In Java, there are four access modifiers available, each serving a specific purpose:

Public (public)

Purpose: Allows unrestricted access to the member (class, method, or field) from any other class or package.

Example: A public method in a class can be accessed by any other class, even if it's in a different package.

Private (private)

Purpose: Restricts access to the member to only within the same class.

Example: A private field in a class can only be accessed and modified by methods within the same class.

Protected (protected)

Purpose: Allows access to the member within the same package and by subclasses (whether they are in the same package or different package).

Example: A protected method in a superclass can be accessed by its subclass, regardless of whether the subclass is in the same package or a different package.

Default (Package-Private)

Purpose: Restricts access to the member to only within the same package. If no access modifier is specified, it is considered default (package-private).

Example: A default (package-private) class member can be accessed by any class within the same package but not by classes outside the package.

These access modifiers help control the visibility and accessibility of classes, methods, and fields, allowing developers to enforce encapsulation, data hiding, and code organization in their Java programs.

6. Why Is String an Immutable Class?

String is immutable in Java, meaning its state cannot be changed after it's created. There are several reasons why String is designed to be immutable:

Thread Safety

Immutable objects are inherently thread-safe because their state cannot be modified once created. This eliminates the need for synchronization when working with Strings in a multi-threaded environment.

String Pool Optimization

Strings are stored in a string pool, which is a pool of unique String literals in memory. Immutable strings allow for string interning, where multiple references to the same string literal point to the same memory location. This reduces memory overhead and improves performance.

Security

Immutable strings are inherently more secure because their values cannot be changed once created. This prevents malicious code from modifying sensitive string data.

Caching and Optimization

Immutable strings can be cached and reused, leading to performance optimizations in applications. Operations like substring, concatenation, and conversion to upper/lower case can be optimized by reusing existing string instances.

Hashing and Hashcode Stability

Immutable strings have a stable hashcode, meaning their hashcode value remains constant throughout their lifetime. This property is essential for objects used in hash-based collections like HashMap and HashSet.

7. Explain Immutable nature of String with example?

```
public static void main(String[] args) {
    // Creating a string "Hello" and assigning it to str1 (stored in the string pool in the
    heap)
    String str1 = "Hello";

    // Creating another string "Hello" (refers to the same object in the string pool)
```

```
    String str3 = "Hello"; // Due to string interning, it refers to the same object in the
    string pool

    // Concatenating "World" to str1 and assigning it to str2
    String str2 = str1 + " World"; // New string created and stored in the heap

    // Converting str1 to uppercase and assigning it to str1
    str1 = str1.toUpperCase(); // New string created and stored in the heap

    // Outputting the values of str1, str2, and str3
    System.out.println("str1: " + str1); // Output: "HELLO"
    System.out.println("str2: " + str2); // Output: "Hello World"
    System.out.println("str3: " + str3); // Output: "Hello"
}
```

8. What is the concept of String Intern?

The concept of string internment in Java refers to the process of storing unique string literals in a special memory area known as the "string pool." When a string literal is encountered in Java code, the JVM checks if an equivalent string already exists in the string pool. If it does, the JVM returns a reference to the existing string object from the pool rather than creating a new object.

This concept ensures that only one copy of each distinct string literal is maintained in memory, reducing memory usage and improving performance by facilitating string sharing and reuse.

String interning can be explicitly invoked using the intern() method provided by the String class. When called on a string object, the intern() method checks if an equivalent string already exists in the string pool. If not, it adds the string to the pool and returns a reference to it. If an equivalent string already exists, it returns a reference to the existing string object from the pool.

```
public static void main(String[] args) {
    String str1 = "Hello";
    String str2 = "Hello";
    String str3 = new String("Hello").intern();

    System.out.println(str1 == str2); // Output: true (Same object reference)
    System.out.println(str1 == str3); // Output: true (Same object reference)
}
```

9. What happens when you create a String with a new keyword?

When the new keyword is used to create a string object, the JVM creates the object in the heap memory, regardless of whether an equivalent string already exists in the string pool. If the string is not found in the string pool, the JVM creates a new string object in the heap and returns a reference to it.

Here's a simplified explanation of the process:

When new keyword is used to create a string object, the JVM creates the object in the heap memory. If the string already exists in the string pool, a new object is still created in the heap, and the string is not added to the pool again.

If the string does not exist in the string pool, the JVM adds it to the pool and returns a reference to the newly created object in the heap.

This behavior ensures that each call to new String("...") creates a new string object in the heap, regardless of whether an equivalent string exists in the string pool.

This can be useful in scenarios where you explicitly want to avoid string interning and ensure that a new object is created in the heap. However, it's important to note that using new unnecessarily can lead to

unnecessary memory usage and should be done judiciously.

```
public static void main(String[] args) {
    // Creating a string object using the new keyword
    // This creates a new object in the heap memory
    String str1 = new String("Hello");

    // Creating another string object with the same value
    // This also creates a new object in the heap memory
    String str2 = new String("Hello");

    // Checking if str1 and str2 reference the same object
    // They are not the same object because each call to new creates a new object
    System.out.println("str1 == str2: " + (str1 == str2)); // Output: false

    // Creating a string object without using the new keyword
    // This checks the string pool and creates a new object only if the string is not already
    present
    String str3 = "Hello";

    // Checking if str1 and str3 reference the same object
    // They are not the same object because str3 is created in the string pool
    System.out.println("str1 == str3: " + (str1 == str3)); // Output: false
}
```

10. What is JAR vs WAR vs EAR?

JAR, WAR, and EAR are all archive file formats used in Java for packaging and deploying applications. Here's a brief overview of each:

JAR (Java ARchive)

Purpose: JAR files are used to package Java class files, resources, and metadata into a single archive file.

Contents: They typically contain compiled Java classes, libraries (JAR files), configuration files, and resources (images, properties files, etc.).

Usage: JAR files are commonly used for packaging libraries, standalone applications, and executable Java applications.

Extension: .jar

WAR (Web ARchive)

Purpose: WAR files are specifically used for packaging and deploying web applications in Java EE (Enterprise Edition) environments.

Contents: They include web components such as servlets, JSP files, HTML pages, JavaScript files, CSS stylesheets, and other resources needed for web applications.

Structure: WAR files have a defined directory structure (e.g., WEB-INF) to organize web application resources and configuration files.

Usage: WAR files are deployed to servlet containers (e.g., Apache Tomcat, Jetty) to run Java web applications.

Extension: .war

EAR (Enterprise ARchive)

Purpose: EAR files are used for packaging and deploying enterprise-level Java EE applications, which typically consist of multiple modules and components.

Contents: They can contain one or more WAR files, EJB (Enterprise JavaBeans) JAR files, application client JAR files, library JAR files, and other resources.

Structure: EAR files have a hierarchical structure, with modules organized into directories and subdirectories.

Usage: EAR files are deployed to Java EE application servers (e.g., JBoss, WebSphere, WebLogic) to run large-scale enterprise applications.

Extension: .ear

In summary, JAR files are generic archive files used for packaging Java applications and libraries. WAR files are specialized for web applications and contain web-specific resources. EAR files are used for large-scale enterprise applications containing multiple modules and components. Each type of archive serves a specific purpose in Java application development and deployment.

11. Java is strictly pass-by-value language?

Yes, Java is strictly pass-by-value language. When you pass arguments to a method in Java, the values of the arguments are copied and passed to the method parameters.

However, it's important to note that the behavior can sometimes be confusing, especially when dealing with objects. When you pass an object to a method, you are passing a reference to the object, not the object itself. This reference is also passed by value, meaning a copy of the reference is passed to the method. This can lead to the misconception that Java uses pass-by-reference, but in reality, it's still pass-by-value.

Here's a brief summary:

- For primitive data types (int, double, boolean, etc.), the value itself is passed.
- For objects, the reference to the object is passed by value. This means changes to the object's state made inside the method are reflected outside the method, but reassigning the reference inside the method does not affect the reference outside the method.

```
public class PassByValueExample {
    public static void main(String[] args) {
        int num = 10;
        System.out.println("Before calling modifyValue: num = " + num);

        // Passing a primitive type (int) by value
        modifyValue(num);

        System.out.println("After calling modifyValue: num = " + num);

        StringBuilder sb = new StringBuilder("Hello");
        System.out.println("Before calling modifyReference: sb = " + sb);

        // Passing an object (StringBuilder) by value (passing reference by value)
        modifyReference(sb);

        System.out.println("After calling modifyReference: sb = " + sb);
    }

    // Method to modify a primitive value
    public static void modifyValue(int num) {
        num += 5;
        System.out.println("Inside modifyValue: num = " + num);
    }

    // Method to modify an object reference
    public static void modifyReference(StringBuilder sb) {
        sb.append(" World");
        System.out.println("Inside modifyReference: sb = " + sb);
```

```
    }
}

// Output
// Before calling modifyValue: num = 10
// Inside modifyValue: num = 15
// After calling modifyValue: num = 10
// Before calling modifyReference: sb = Hello
// Inside modifyReference: sb = Hello World
// After calling modifyReference: sb = Hello World
```

12. Explain about static, abstract and final keywords in Java?

static

- The static keyword in Java is used to declare members (variables and methods) that belong to the class itself, rather than to instances of the class.
- Variables declared as static are called class variables or static variables. They are shared among all instances of the class and can be accessed using the class name.
- Methods declared as static are called class methods or static methods. They belong to the class itself and can be called without creating an instance of the class.
- static variables and methods can be accessed directly using the class name, without the need to create an object of the class.

abstract

- The abstract keyword in Java is used to declare abstract classes and methods.
- Abstract classes cannot be instantiated directly; they serve as templates for concrete subclasses to extend and implement.
- Abstract methods are declared without an implementation and must be implemented by concrete subclasses.
- Abstract classes may contain both abstract and concrete methods, but a class with at least one abstract method must be declared abstract.
- Abstract methods are intended to be overridden by subclasses to provide specific implementations.

final

- The final keyword in Java is used to restrict the modification of classes, methods, and variables.
- When applied to a class, the final keyword prevents the class from being subclassed (i.e., it becomes a final class).
- When applied to a method, the final keyword prevents the method from being overridden by subclasses.
- When applied to a variable, the final keyword indicates that the variable's value cannot be changed after initialization. For primitive types, it makes the value immutable, while for reference types, it makes the reference immutable (i.e., the reference cannot be changed, but the object's state can).
- final variables must be initialized either when declared or in a constructor, and they cannot be reassigned.

In summary, the static keyword is used to define members that belong to the class itself, the abstract keyword is used to declare abstract classes and methods, and the final keyword is used to restrict modification of classes, methods, and variables.

```
// StaticExample class with a static variable and method
class StaticExample {
    // Static variable
```

```
    static int staticVariable = 10;

    // Static method
    static void staticMethod() {
        System.out.println("Inside staticMethod: staticVariable = " + staticVariable);
    }
}

// AbstractExample abstract class with an abstract method
abstract class AbstractExample {
    // Abstract method (no implementation)
    abstract void abstractMethod();
}

// Concrete subclass of AbstractExample
class ConcreteExample extends AbstractExample {
    // Implementation of abstract method
    void abstractMethod() {
        System.out.println("Inside abstractMethod in ConcreteExample");
    }
}

// FinalExample class with final variable and method
class FinalExample {
    // Final variable
    final int finalVariable = 20;

    // Final method
    final void finalMethod() {
        System.out.println("Inside finalMethod: finalVariable = " + finalVariable);
    }
}

public class KeywordExample {
    public static void main(String[] args) {
        // Accessing static variable and method using class name
        System.out.println("Static variable: " + StaticExample.staticVariable);
        StaticExample.staticMethod();

        // Creating an instance of ConcreteExample and calling abstractMethod
        ConcreteExample concrete = new ConcreteExample();
        concrete.abstractMethod();

        // Creating an instance of FinalExample and accessing final variable and method
        FinalExample finalExample = new FinalExample();
        System.out.println("Final variable: " + finalExample.finalVariable);
        finalExample.finalMethod();
    }
}

/* Output
Static variable: 10
Inside staticMethod: staticVariable = 10
Inside abstractMethod in ConcreteExample
Final variable: 20
Inside finalMethod: finalVariable = 20
*/
```

13. Difference between String, StringBuffer, and StringBuilder?

The main difference between String, StringBuffer, and StringBuilder in Java lies in their mutability, thread safety, and performance characteristics:

String

- String objects are immutable, meaning their values cannot be changed after they are created.

- Once a String object is created, its value cannot be modified. Any operation that appears to modify a String actually creates a new String object with the modified value.
- String objects are thread-safe because they are immutable, making them safe for use in multi-threaded environments.
- Due to their immutability, String objects are safe for sharing across different threads without synchronization overhead.
- Use String when you need an immutable sequence of characters.

StringBuffer

- StringBuffer is a mutable sequence of characters, designed for use in multi-threaded environments.
- StringBuffer provides synchronized methods for appending, inserting, and modifying characters, making it thread-safe.
- Modifications to StringBuffer objects are performed in-place, avoiding unnecessary memory allocation and string copying.
- Use StringBuffer when you need a mutable sequence of characters in a multi-threaded environment.

StringBuilder

- StringBuilder is similar to StringBuffer in that it provides a mutable sequence of characters. However, it is not inherently thread-safe.
- Unlike StringBuffer, StringBuilder does not provide synchronized methods, making it more efficient in single-threaded scenarios.
- StringBuilder is recommended for use in single-threaded environments or when synchronization is not required.
- Use StringBuilder when you need a mutable sequence of characters in a single-threaded environment.

In summary, String is immutable and thread-safe, StringBuffer is mutable and thread-safe, and StringBuilder is mutable but not inherently thread-safe. The choice between them depends on the specific requirements of your application, such as mutability, thread safety, and performance considerations.

```
public static void main(String[] args) {
    // String Example
    String str = "Hello";
    System.out.println("Original String: " + str);

    // Concatenating a new string to the original string
    str = str + " World";
    System.out.println("Modified String (using String): " + str);

    // StringBuffer Example
    StringBuffer stringBuffer = new StringBuffer("Hello");
    System.out.println("Original StringBuffer: " + stringBuffer);

    // Appending a new string to the original StringBuffer
    stringBuffer.append(" World");
    System.out.println("Modified StringBuffer: " + stringBuffer);

    // StringBuilder Example
    StringBuilder stringBuilder = new StringBuilder("Hello");
    System.out.println("Original StringBuilder: " + stringBuilder);

    // Appending a new string to the original StringBuilder
    stringBuilder.append(" World");
    System.out.println("Modified StringBuilder: " + stringBuilder);
}
```

14. How Can We Reverse a String in Java?

The common approach to reverse a string in Java is by using either StringBuilder or StringBuffer along with their reverse() method as shown below:

```
String reversed_sBuilder = new StringBuilder("hungryCoders").reverse().toString();
System.out.println(reversed_sBuilder); // sredoCyrgnuh

String reversed_sBuffer = new StringBuffer("hungryCoders").reverse().toString();
System.out.println(reversed_sBuffer); // sredoCyrgnuh
```

15. What Is String.format() and How to Use It?

String.format() is a method in Java that allows you to create formatted strings by substituting placeholders with values.

```
String formattedString = String.format("Hello, %s! Welcome to %s.", "Friend", "HungryCoders.
    com");
System.out.println(formattedString); // Hello, Friend! Welcome to HungryCoders.com.
```

16. What Is an Enum and How to Use It?

An enum, short for "enumeration," is a special data type in Java used to define a collection of constants. It provides a way to represent a fixed set of predefined values.

```
// Defining an enum called DayOfWeek
enum DayOfWeek {
    SUNDAY, MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY
}

public class EnumExample {
    public static void main(String[] args) {
        // Using enum constants
        DayOfWeek today = DayOfWeek.MONDAY;
        System.out.println("Today is: " + today);

        // Switch statement with enum
        switch (today) {
            case MONDAY:
                System.out.println("It's the start of the week.");
                break;
            case FRIDAY:
                System.out.println("TGIF!");
                break;
            default:
                System.out.println("Just another day.");
        }

        // Iterating over enum constants
        System.out.println("Days of the week:");
        for (DayOfWeek day : DayOfWeek.values()) {
            System.out.println(day);
        }
    }
}

/* Output
Today is: MONDAY
It's the start of the week.
Days of the week:
```

```
SUNDAY
MONDAY
TUESDAY
WEDNESDAY
THURSDAY
FRIDAY
SATURDAY
*/
```

17. What Are Two Types of Casting in Java?

In Java, implicit casting (widening) is a form of upcasting, while explicit casting (narrowing) is a form of downcasting.

Upcasting (Implicit Casting)

- Upcasting involves converting an object of a subclass type to an object of its superclass type.
- It is performed automatically by the compiler when assigning a subclass object to a superclass reference.
- Upcasting is considered safe because it involves moving from a more specific type to a more general type.

Downcasting (Explicit Casting)

- Downcasting involves converting an object of a superclass type to an object of its subclass type.
- It requires an explicit cast operator and may result in a ClassCastException if the object being casted is not actually an instance of the target subclass.
- Downcasting is considered risky because it involves moving from a more general type to a more specific type, which may lead to runtime errors if not done properly.

In summary, implicit casting (upcasting) is converting from a subclass to a superclass type, while explicit casting (downcasting) is converting from a superclass to a subclass type. Upcasting is safe and implicit, while downcasting requires an explicit cast operator and may result in runtime errors if not done properly.

```
int num1 = 10;
double num2 = num1; // Implicit casting from int to double
System.out.println(num2); // 10.0

double num3 = 10.5;
int num4 = (int) num3; // Explicit casting from double to int
System.out.println(num4); // 10
```

18. Is Java statically and strongly typed? Explain in detail?

Java is both Statically and Strongly Typed.

Statically Typed: You cannot declare a variable without a type, and the type is checked at compile-time.

Strongly Typed: Type rules are strictly enforced, and incompatible types cannot mix without explicit conversions.

Statically Typed

- In Java, you must declare the type of every variable explicitly (or inferred, as with var introduced in Java 10)
- Type checking occurs during compilation. If there is a type mismatch, the compiler throws an error.
- Types are fixed after declaration and cannot change.

Being a statically typed language have lot of benefits such as:

- It prevents many runtime errors.
- It improves performance because type information is resolved during compilation.

```
int number = 10;  // Declared as int
number = "Hello"; // Compilation Error: Type mismatch
```

Strongly Typed

In Java, type conversions must follow strict rules, and incompatible types cannot be mixed without explicit conversion.

- Implicit type conversions (type coercion) are limited.
- Operations between incompatible types are not allowed.
- Java enforces strict type safety in collections (since Java 5 with generics)

Being a strongly typed language have lot of benefits such as:

- Avoids unpredictable behavior due to unintended type conversions.
- Ensures code reliability and readability.

```
int num = 10;
double result = num;  // Allowed: Implicit widening (int to double)
int value = result;   // Compilation Error: Explicit narrowing required

int x = 5;
String y = "10";
int sum = x + y; // Compilation Error: Incompatible types

List<String> list = new ArrayList<>();
list.add(42); // Compilation Error: Incompatible type
```

19. Does var keyword makes Java as a dynamically typed language?

Java introduced the var keyword in Java 10, which allows you to declare local variables without explicitly specifying their type. However, this does not mean Java has become a dynamically typed language. It is still statically and strongly typed, even with var.

Basically var is a reserved type name introduced to simplify local variable declarations. The compiler infers the type of the variable from the value assigned to it during initialization.

With var, the type is determined at compile-time, not at runtime. The compiler infers the type based on the assigned value and enforces type-checking.

This is still different from dynamically typed languages like Python or JavaScript, where variable types can change at runtime.

```
var number = 42;      // Compiler infers as int
number = "Hello";     // Compilation Error: Type mismatch
```

Even with var, Java still enforces strict rules about type usage and does not allow implicit type coercion.

```
var x = 10;         // Inferred as int
var y = "Hello";    // Inferred as String

int z = x + y;      // Compilation Error: Incompatible types
```

Key Restrictions of var:

- Cannot be used without initialization: The compiler needs the right-hand side value to infer the type.
- Type is fixed after inference: Once the type is inferred, it cannot change.
- Only for local variables: var cannot be used for Class fields, Method parameters and Return types
- Not allowed with null as an initializer

```
var num;  // Compilation Error: Cannot infer type for variable

var value = 10;  // Inferred as int
value = "Hello"; // Compilation Error: Type mismatch

var value = null; // Compilation Error: Cannot infer type for variable
```

20. What are the two main categories of data types in Java?

1. Primitive Data Types

These are the basic building blocks for data manipulation. They store simple values and include:

- boolean: true/false values.
- Numeric types like byte, short, int, long, float, and double.

- char: A single Unicode character.

2. Non-Primitive (Reference) Data Types

These store references (memory addresses) rather than the actual data. Examples:

- Strings: Represent sequences of characters.
- Arrays: Collections of elements of the same type.
- Classes and Objects: Represent complex structures with attributes and methods.

21. What are the 8 primitive data types in Java, and what does each represent?

Below are the 8 primitive data types in Java.

Boolean: Represents logical values (true or false).

```
boolean isJavaFun = true;
```

Byte: 8-bit signed integer. Used for Efficient memory storage, e.g., in large arrays.

```
byte b = 100;
```

Char: Represents a single 16-bit Unicode character. Used for Characters like a, 1, or symbols.

```
char grade = 'A';
```

Short: 16-bit signed integer. Used for Saving memory for small numbers.

```
short age = 200;
```

Int: 32-bit signed integer. Used for Default for whole numbers.

```
int salary = 50000;
```

Long: 64-bit signed integer. Used for Large numeric ranges.

```
long population = 8_000_000_000L;
```

Float: 32-bit single-precision floating-point. Used For decimals when precision is not critical.

```
float pi = 3.14f;
```

Double: 64-bit double-precision floating-point. Use for Default for decimals or scientific calculations.

```
double e = 2.718281828459045;
```

22. Why does char use 2 bytes in Java, and not 1 byte like in C/C++?

Java uses the Unicode system to support characters from all languages worldwide (e.g., Latin, Greek, Chinese).

But Unicode requires 2 bytes (16 bits) to represent all possible characters.

By contrast, ASCII, used in C/C++, is limited to 1 byte (8 bits), supporting only 256 characters.

23. What is the difference between float and double in Java? When would you use one over the other?

Double is more precise and better for computations requiring high accuracy, whereas float is more memory-efficient.

- Float is used when memory is limited (e.g., in embedded systems or large datasets).
- But Double is used as a default for decimal values, scientific and financial calculations.

```
float pi = 3.14159f; // Use 'f' to indicate float
double e = 2.718281828459045;
```

24. What is widening and narrowing type conversion? Provide examples of each.

Widening Conversion is process of Implicit conversion from a smaller to a larger type. Example: int to long or float to double

```
int num = 100;
double result = num; // Implicit widening
```

Narrowing Conversion is process of Explicit conversion from a larger to a smaller type (requires casting). Example: double to int

```
double value = 9.99;
int truncated = (int) value; // Explicit narrowing
```

25. What happens if you assign a larger data type to a smaller data type?

Java will throw a compile-time error, as narrowing requires explicit casting.

```
long bigNum = 100000L;
int smallerNum = bigNum; // Compilation Error
```

26. How does Java handle type safety in mixed-type arithmetic operations?

Java automatically promotes smaller types to the larger type in mixed operations.

```
int num = 5;
float value = 2.5f;
float result = num + value; // num promoted to float
```

27. When should you use byte and short instead of int?

Use byte and short when:

- Memory is constrained.
- You're working with fixed ranges (e.g., file streams or network protocols).

```
byte age = 127; // Fits within byte range
```

28. What are wrapper classes in Java, and how do they relate to primitive types?

Wrapper classes bridge the gap between primitive types and Java's object-oriented nature. Wrapper classes in Java are object representations of the primitive data types.

- They allow primitive values to be treated as objects.
- They enable primitives to interact with collections and provide utility methods.
- Autoboxing and auto-unboxing simplify conversions between primitives and objects.
- Use cases: Collections (e.g., List<Integer> instead of List<int>) and Utility methods (e.g., Integer.parseInt()).

```
int num = 10; // Primitive type
Integer obj = Integer.valueOf(num); // Primitive to Wrapper (Boxing)

int value = obj.intValue(); // Wrapper to Primitive (Unboxing)
```

- Boxing is the process of converting a primitive type to its corresponding wrapper class.
- Unboxing is the process of converting a wrapper class back to its primitive type.
- Wrapper classes allow primitives to be stored in collections.
- Wrapper classes provide methods to perform operations on primitive values.

• Wrapper classes can represent the absence of a value using null, while primitives cannot.

```
ArrayList<Integer> list = new ArrayList<>();
list.add(10); // Autoboxing of int to Integer

String str = "123";
int num = Integer.parseInt(str); // Converts String to int

Integer obj = null; // Allowed
int num = null;     // Compilation Error
```

```
public class WrapperExample {
    public static void main(String[] args) {
        // Primitive type
        int num = 5;

        // Wrapper class (Boxing)
        Integer obj = num; // Autoboxing

        // Using utility method
        String str = obj.toString(); // Converts Integer to String

        // Unboxing
        int value = obj; // Auto-unboxing
        System.out.println("Primitive: " + value);
    }
}
```

29. What are the three types of variables in Java, and how are they different?

Local Variable

A variable declared inside the body of a method. The variable can only be used within that specific method, and it is not accessible outside it.

• Local variables must be initialized before use as they do not have default values.

• They are stored in the stack memory.

• Cannot be declared as static since static implies shared across instances, while local variables are specific to a method's execution.

```
public void display() {
    int localVar = 10;  // Local variable
    System.out.println(localVar);
}
```

Instance Variable

A variable declared inside a class but outside any method or block, and not marked as static. These variables are associated with objects, meaning each object of the class has its own copy of the instance variable.

- Instance variables have default values if not explicitly initialized.
- Stored in the heap memory.
- Their values are instance-specific and not shared among different objects.

```
public class Example {
    int instanceVar;  // Instance variable

    public void show() {
        System.out.println(instanceVar); // Default value: 0
    }
}
```

Static Variable

A variable declared with the static keyword. It is not tied to any particular instance but belongs to the class itself.

- Shared across all objects of the class. Any change to a static variable reflects in all instances of the class.
- Static variables are initialized once when the class is loaded into memory.
- Stored in the method area of memory.
- Can be accessed using the class name directly, without creating an object.

```
public class Example {
    static int staticVar = 100;  // Static variable

    public static void show() {
        System.out.println(staticVar);
    }
}

// Accessing static variable
Example.staticVar = 200; // Accessed directly using the class name
```

30. What is the difference between final, static, and static final variables?

- final: Value cannot be reassigned after initialization.
- static: Shared across all instances; belongs to the class, not any instance.
- static final: A constant value shared across all instances; effectively a class-level constant.

Final Variable

A variable declared with the final keyword is a constant, meaning its value cannot be changed once it has been initialized.

- Must be initialized either at the time of declaration or inside the constructor (for instance variables).
- If the variable holds a reference to an object, the reference cannot be changed, but the object's state can be modified.
- Used to declare constants in a program.

```
public class FinalExample {
    final int MAX_AGE = 100;  // Final variable

    public void display() {
        // MAX_AGE = 101; // Compilation Error: Cannot reassign a final variable
        System.out.println(MAX_AGE);
    }
}
```

Static Variable

A variable declared with the static keyword belongs to the class rather than any specific instance of the class.

- Shared across all instances of the class.
- Memory is allocated for static variables only once, when the class is loaded into memory.
- Can be accessed using the class name directly (e.g., ClassName.variableName).
- Commonly used to store values that are shared among all instances, like configuration values or counters.

```
public class StaticExample {
    static int counter = 0;  // Static variable

    public StaticExample() {
        counter++;  // Increments for every new instance
    }

    public static void display() {
        System.out.println("Counter: " + counter);
    }
}
```

Static Final Variable

A variable that is both static and final is a constant shared across all instances of the class. It must be initialized only once and cannot be changed.

- Combines the properties of static (class-level variable) and final (constant).
- Typically used to define constants that are shared across all instances, like PI, mathematical constants, or configuration values
- The value must be initialized at the time of declaration or inside a static block.

```
public class StaticFinalExample {
    static final double PI = 3.14159;  // Static final variable
```

```
    public static void display() {
        System.out.println("Value of PI: " + PI);
    }
}
```

31. Can you declare a variable without initializing it? And what happens if you try to access an uninitialized local variable?

Yes, for instance and static variables, as they get default values. Local variables must be explicitly initialized before use.

The compiler throws an error if you try to access an uninitialized local variable because local variables do not get default values.

32. What is Object Oriented Programming (OOPs)?

Object-Oriented Programming (OOP) is a programming paradigm based on the concept of "objects," which can contain data in the form of fields (attributes or properties) and code in the form of procedures (methods or functions). OOP focuses on organizing software into objects that interact with each other to accomplish tasks.

OOP promotes modular, maintainable, and scalable software development by organizing code into reusable and understandable components. It emphasizes concepts such as data encapsulation, code reuse, and separation of concerns, which contribute to building robust and flexible software systems.

33. What are the 4 pillars of OOPs?

The four fundamental principles of Object-Oriented Programming (OOP) are commonly referred to as the "Four Pillars of OOP." They are:

Encapsulation

- Encapsulation refers to the bundling of data (attributes or properties) and methods (functions or procedures) that operate on the data into a single unit, called a class.
- It allows the internal state of an object to be hidden from outside interference and accessed only through well-defined interfaces (public methods).
- Encapsulation provides data protection, abstraction, and modularity, enhancing code maintainability and reusability.

Inheritance

- Inheritance is a mechanism that allows a class (subclass or derived class) to inherit properties and behavior (methods) from another class (superclass or base class).
- It promotes code reuse by enabling subclasses to inherit and extend the functionality defined in their superclass.
- Inheritance facilitates the creation of a hierarchy of classes with specialized behavior, promoting code organization and scalability.

Polymorphism

- Polymorphism enables objects to be treated as instances of their superclass, allowing them to take on multiple forms.
- It allows different objects to respond to the same message (method call) in different ways, based on their specific implementations.
- Polymorphism simplifies code by allowing methods to be written to work with objects of a superclass, without needing to know the exact subclass at compile time.
- Polymorphism is achieved through method overriding and method overloading.

Abstraction

- Abstraction involves simplifying complex systems by focusing on the essential properties while hiding unnecessary details.

- It allows developers to create abstract classes and interfaces to define common behavior without specifying implementation details.
- Abstraction helps in managing complexity, improving code maintainability, and promoting code reuse by providing a clear separation between interface and implementation.

These four principles form the foundation of Object-Oriented Programming and guide the design and implementation of software systems, promoting modular, maintainable, and scalable code.

```
class Person {
    // Private data members
    private String name;
    private int age;

    // Constructor
    public Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    // Public method to access private data
    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public int getAge() {
        return age;
    }

    public void setAge(int age) {
        if (age > 0) {
            this.age = age;
        }
    }
}

public class EncapsulationExample {
    public static void main(String[] args) {

        //Think of a capsule as a person.
        // The person's name and age are private information,
        // and they have methods to get or update this information safely.
        Person person = new Person("John", 30);
        System.out.println("Name: " + person.getName());
        System.out.println("Age: " + person.getAge());

        person.setAge(35); // updating age
```

```
        System.out.println("Updated Age: " + person.getAge());
    }
}
```

```
//Think of an animal family.
// Dogs and cats both inherit general
// animal characteristics but also
// have their own unique behaviors.
class Animal {
    public void eat() {
        System.out.println("This animal eats food.");
    }
}

class Dog extends Animal {
    public void bark() {
        System.out.println("The dog barks.");
    }
}

public class InheritanceExample {
    public static void main(String[] args) {
        Dog dog = new Dog();
        dog.eat(); // inherited method
        dog.bark(); // specific to Dog class
    }
}
```

```
//Runtime Polymorphism
class Animal {
    void sound() {
        System.out.println("Animal makes a sound");
    }
}

class Dog extends Animal {
    @Override
    void sound() {
        System.out.println("Dog barks");
    }
}

public class Main {
    public static void main(String[] args) {
        Animal animal = new Dog();  // Upcasting
        animal.sound();             // Calls Dog's overridden method
    }
```

```
}
```

```
//Think of an ATM machine.
// You interact with it to withdraw money,
// check your balance, etc.,
// without needing to know how it processes
// transactions internally.
abstract class ATM {
    abstract void withdrawMoney();
    abstract void checkBalance();
}

class MyATM extends ATM {
    @Override
    void withdrawMoney() {
        System.out.println("Money withdrawn.");
    }

    @Override
    void checkBalance() {
        System.out.println("Balance checked.");
    }
}

public class AbstractionExample {
    public static void main(String[] args) {
        ATM atm = new MyATM();
        atm.withdrawMoney(); // Outputs: Money withdrawn.
        atm.checkBalance(); // Outputs: Balance checked.
    }
}
```

34. What is Compile Time Polymorphism and Runtime Polymorphism?

Compile-time polymorphism and runtime polymorphism are two forms of polymorphism in object-oriented programming. They occur at different stages of program execution and are achieved through method overloading and method overriding, respectively.

Compile-Time Polymorphism (Static Binding)

- Compile-time polymorphism occurs when the decision about which method to call is made at compile time based on the method signature.
- It is achieved through method overloading, where multiple methods with the same name but different parameter lists (number or type of parameters) are defined within the same class.
- The compiler determines the appropriate method to invoke based on the number and types of arguments provided in the method call.

• Compile-time polymorphism is also known as static binding because the method call is resolved at compile time and cannot be changed at runtime.

```
class Calculator {
    // Method overloading
    int add(int a, int b) {
        return a + b;
    }

    double add(double a, double b) {
        return a + b;
    }
}

public class Main {
    public static void main(String[] args) {
        Calculator calc = new Calculator();
        int result1 = calc.add(2, 3);          // Calls int add(int a, int b)
        double result2 = calc.add(2.5, 3.5);   // Calls double add(double a, double b)
    }
}
```

Runtime Polymorphism (Dynamic Binding)

• Runtime polymorphism occurs when the decision about which method to call is made at runtime based on the actual type of the object.

• It is achieved through method overriding, where a subclass provides a specific implementation of a method that is already defined in its superclass.

• The JVM determines the appropriate method to invoke based on the actual object type (not the reference type) at runtime.

• Runtime polymorphism allows for flexibility and extensibility in object-oriented designs, as it enables subclass objects to exhibit behavior specific to their type.

```
class Animal {
    void sound() {
        System.out.println("Animal makes a sound");
    }
}

class Dog extends Animal {
    @Override
    void sound() {
        System.out.println("Dog barks");
    }
}

public class Main {
    public static void main(String[] args) {
```

```
        Animal animal = new Dog();  // Upcasting
        animal.sound();             // Calls Dog's overridden method
    }
}
```

In summary, compile-time polymorphism is achieved through method overloading and resolved at compile time, while runtime polymorphism is achieved through method overriding and resolved at runtime based on the actual object type. Each form of polymorphism provides flexibility and allows for code reuse in object-oriented programming.

35. What are the different types of inheritance?

Inheritance is a fundamental concept in object-oriented programming that allows a class (subclass or derived class) to inherit properties and behavior (methods) from another class (superclass or base class). There are several types of inheritance, based on the relationship between the subclass and superclass:

Single Inheritance

- Single inheritance occurs when a subclass inherits from only one superclass.
- It is the simplest form of inheritance and is widely supported in most object-oriented programming languages.

```
class Animal {
    // Animal properties and methods
}

class Dog extends Animal {
    // Dog inherits from Animal
}
```

Multilevel Inheritance

- Multilevel inheritance occurs when a subclass inherits from another subclass, forming a chain of inheritance.
- It allows for the creation of a hierarchy of classes with multiple levels of specialization.

```
class Animal {
    // Animal properties and methods
}

class Mammal extends Animal {
    // Mammal inherits from Animal
}

class Dog extends Mammal {
    // Dog inherits from Mammal
```

```
}
```

Hierarchical Inheritance

- Hierarchical inheritance occurs when multiple subclasses inherit from the same superclass.
- It allows for the creation of a hierarchy of classes with different levels of specialization, branching out from a common superclass.

```
class Animal {
    // Animal properties and methods
}

class Dog extends Animal {
    // Dog inherits from Animal
}

class Cat extends Animal {
    // Cat inherits from Animal
}
```

Multiple Inheritance via Interfaces in Java

- Multiple inheritance occurs when a subclass inherits from multiple superclasses.
- While Java does not support multiple inheritance of classes due to the diamond problem, it supports multiple inheritance through interfaces.
- Interfaces in Java allow a class to inherit abstract methods from multiple interfaces, enabling a form of multiple inheritance.

```
interface Swimmer {
    void swim();
}

interface Flyer {
    void fly();
}

class Bird implements Swimmer, Flyer {
    // Bird implements both Swimmer and Flyer interfaces
}
```

36. What is diamond problem in Java?

The diamond problem occurs when a class inherits from two or more classes, each of which inherits from the same superclass. This creates a diamond-shaped inheritance hierarchy, where a subclass inherits the same superclass through multiple paths.

The problem arises when the subclass attempts to inherit or override a method from the common superclass. In such cases, the compiler may not be able to determine which version of the method to use, leading to ambiguity and potential conflicts.

```
    A
   / \
  B   C
   \ /
    D
```

In this diamond-shaped inheritance hierarchy, both classes B and C inherit from class A, and class D inherits from both B and C. If class A has a method foo(), and classes B and C override foo() with different implementations, the compiler may not be able to determine which version of foo() to use in class D.

In Java, the diamond problem is avoided by allowing single inheritance of classes and supporting multiple inheritance only through interfaces. Interfaces do not contain any implementation, so there is no ambiguity in method resolution.

37. What is difference between abstract class and interface?

Abstract classes and interfaces are both used to achieve abstraction and define contracts for classes in Java, but they have some key differences:

Abstract Class

- An abstract class is a class that cannot be instantiated on its own and may contain abstract methods (methods without implementation) as well as concrete methods (methods with implementation).
- Abstract classes can have constructors, instance variables, and non-abstract methods, in addition to abstract methods.
- A subclass must provide implementations for all abstract methods defined in its abstract superclass, or it must itself be declared abstract.
- Abstract classes are used to define common behavior and characteristics shared among subclasses, serving as a blueprint for creating concrete subclasses.

```
abstract class Shape {
    // Abstract method
    abstract double area();

    // Concrete method
    void display() {
        System.out.println("Displaying shape");
    }
}
```

Interface

- An interface is a reference type similar to a class but contains only abstract methods, constants (static

final fields), and default/static methods (methods with default implementations).

• Interfaces cannot have constructors or instance variables, only method signatures.

• A class implements an interface by providing concrete implementations for all the methods declared in the interface.

• Interfaces are used to define contracts that classes can choose to implement, enabling multiple inheritance of type (a class can implement multiple interfaces).

```
interface Drawable {
    // Abstract method
    void draw();
}
```

Key Differences

• An abstract class can have both abstract and concrete methods, while an interface can have abstract methods, default methods, and static methods.

• A class can extend only one abstract class, but it can implement multiple interfaces.

• Abstract classes can have constructors, whereas interfaces cannot.

• Abstract classes are used to provide a common base for subclasses to extend, while interfaces define contracts that classes can implement to achieve polymorphism.

```
// Abstract class
abstract class Animal {
    private String name;

    // Constructor in abstract class
    public Animal(String name) {
        this.name = name;
    }

    // Abstract method (no implementation)
    public abstract void makeSound();

    // Concrete method
    public void sleep() {
        System.out.println(name + " is sleeping.");
    }
}

// Interface with abstract methods and default method
interface Pet {
    // Abstract method
    void play();

    // Default method (concrete method)
    default void beFriendly() {
        System.out.println("Being friendly.");
```

```
    }

    // Static method (concrete method)
    static void showAffection() {
        System.out.println("Showing affection.");
    }
}

// Class that extends abstract class and implements interface
class Dog extends Animal implements Pet {
    public Dog(String name) {
        super(name); // Call the constructor of the abstract class
    }

    @Override
    public void makeSound() {
        System.out.println("Dog barks.");
    }

    @Override
    public void play() {
        System.out.println("Dog is playing.");
    }
}


public class AbstractVSInterfaceExample {
    public static void main(String[] args) {
        Dog myDog = new Dog("Buddy");
        myDog.makeSound(); // Dog barks.
        myDog.sleep(); // Buddy is sleeping.
        myDog.play(); // Dog is playing.
        myDog.beFriendly(); // Being friendly.

        Pet.showAffection(); // Showing affection.
    }
}
```

38. When to use abstract class VS when to use interface?

Use Interface When

A is capable of [doing this] scenario fits in your requirement. For example: below Programmable interface is capable of doing Programming

```
interface Programmable {
        void programming();
}
```

• You want to define a contract for classes to implement: Interfaces specify a set of methods that implementing classes must provide, defining a clear contract for how classes interact with each other.

• You want to achieve multiple inheritance of type: Classes can implement multiple interfaces, allowing them to inherit behavior from multiple sources without the restrictions of single inheritance.

• You want to decouple implementation from interface: Interfaces provide a way to separate the specification of behavior from its implementation, promoting loose coupling and flexibility in design.

Use Abstract Class When

• A is B or C is B scenario fits in your requirement, for example: Dog is Animal, Cat is Animal..etc.

• You want to provide a common base implementation for subclasses: Abstract classes can contain both abstract and concrete methods, allowing you to define common behavior that subclasses can inherit.

• You want to define a class hierarchy: Abstract classes are useful for creating a hierarchical structure of classes where each subclass represents a more specialized version of the superclass.

• You want to share code among closely related classes: Abstract classes can contain state and behavior that is common to multiple subclasses, reducing code duplication.

39. What is Constructor and different types of Constructors?

A constructor in Java is a special type of method that is automatically called when an instance (object) of a class is created. It is used to initialize the newly created object and set its initial state. Constructors have the same name as the class and do not have a return type, not even void.

There are several types of constructors in Java:

Default Constructor

• A default constructor is a non-parameterized constructor automatically provided by the Java compiler if no other constructor is explicitly defined in a class.

• It initializes the object with default values (e.g., numeric fields are initialized to 0, references are initialized to null).

```
public class MyClass {
    // Default constructor
    public MyClass() {
        // Constructor body
    }
}
```

Parameterized Constructor

• A parameterized constructor is a constructor with parameters that allows you to initialize the object with specific values passed as arguments.

• It enables you to customize the initialization of objects based on the values provided at the time of object creation.

```
public class MyClass {
    // Parameterized constructor
    public MyClass(int x, int y) {
        // Constructor body
    }
}
```

Copy Constructor

- A copy constructor is a constructor that initializes an object by copying the state of another object of the same class.
- It allows you to create a new object with the same state as an existing object, providing a convenient way to clone objects.

```
public class MyClass {
    private int x;

    // Copy constructor
    public MyClass(MyClass obj) {
        this.x = obj.x;  // Copy state from obj
    }
}
```

40. Is there a destructor concept in Java?

No, Java does not have a destructor concept like some other programming languages such as C++. In Java, objects are automatically garbage collected by the JVM (Java Virtual Machine) when they are no longer referenced or needed.

There was finalize() method as a workaround for the java destructor but it is also deprecated since Java 9.

41. What are different Types of Loops in Java?

In Java, there are several types of loops that allow you to execute a block of code repeatedly until a certain condition is met. The main types of loops are:

for loop

- The for loop is used when you know in advance how many times you want to execute a block of code.
- It consists of three parts: initialization, condition, and increment/decrement.

```
for (int i = 0; i < 5; i++) {
    System.out.println("Iteration " + i);
}
/*Output
        Iteration 0
        Iteration 1
        Iteration 2
        Iteration 3
        Iteration 4*/
```

while loop

- The while loop is used when you want to execute a block of code repeatedly as long as a condition is true.
- It only consists of a condition, and the loop continues to execute as long as the condition evaluates to true.

```
int i = 0;
while (i < 5) {
    System.out.println("Iteration " + i);
    i++;
}
/*Output
        Iteration 0
        Iteration 1
        Iteration 2
        Iteration 3
        Iteration 4*/
```

do-while loop

- The do-while loop is similar to the while loop, but it guarantees that the block of code is executed at least once before checking the condition.
- It consists of a block of code followed by a while condition.

```
int i = 0;
do {
    System.out.println("Iteration " + i);
    i++;
```

```
} while (i < 5);
/*Output
        Iteration 0
        Iteration 1
        Iteration 2
        Iteration 3
        Iteration 4*/
```

enhanced for loop (for-each loop)

- The enhanced for loop (also known as the for-each loop) is used to iterate over elements of arrays or collections.
- It simplifies iterating through arrays or collections by hiding the details of the iterator.

```
int[] numbers = {1, 2, 3, 4, 5};
for (int num : numbers) {
    System.out.println(num);
}
/*Output
        1
        2
        3
        4
        5*/
```

42. How Can You Exit From a Loop?

Using the break statement

- The break statement is used to terminate the loop immediately and transfer control to the statement following the loop.
- It can be used with for, while, do-while, and switch statements.

```
for (int i = 0; i < 10; i++) {
    if (i == 5) {
        break; // Exit loop when i equals 5
    }
    System.out.println(i);
}
```

Using the return statement

- If the loop is contained within a method, you can use the return statement to exit the method and the loop simultaneously.
- This terminates the loop and returns control to the caller of the method.

```
public void someMethod() {
    for (int i = 0; i < 10; i++) {
        if (i == 5) {
            return; // Exit method and loop when i equals 5
        }
        System.out.println(i);
    }
}
```

Using the continue statement

- The continue statement is used to skip the rest of the current iteration and continue with the next iteration of the loop.
- It can be used with for, while, and do-while loops.

```
for (int i = 0; i < 10; i++) {
    if (i % 2 == 0) {
        continue; // Skip even numbers
    }
    System.out.println(i);
}
```

43. When to switch statement vs if-else statement?

Use a switch statement when:

- Multiple Conditions: You have multiple conditions to evaluate based on the value of a single expression or variable.
- Fixed Values: The conditions involve fixed, integral values (e.g., integers, characters, enums). switch statements are optimized for such scenarios.
- Readability: Using a switch statement can make the code more readable, especially when dealing with a large number of conditions.

```
switch (dayOfWeek) {
    case 1:
        System.out.println("Monday");
        break;
    case 2:
        System.out.println("Tuesday");
        break;
    // More cases...
    default:
        System.out.println("Invalid day");
}
```

Use a if-else statement when:

- Complex Conditions: You have complex conditions that cannot be easily expressed using a switch statement.
- Non-fixed Values: The conditions involve non-fixed values or expressions that cannot be directly compared to each other.
- Chained Conditions: You have a series of conditions where the evaluation of one condition depends on the result of the previous one.

```
if (age >= 18 && age < 65) {
    System.out.println("Adult");
} else if (age >= 65) {
    System.out.println("Senior Citizen");
} else {
    System.out.println("Minor");
}
```

44. How try/catch/finally works in Java?

In Java, the try, catch, and finally blocks are used for exception handling, allowing you to gracefully manage errors that might occur during the execution of your code.

Here's how they work:

- try: This block encloses the code that you want to monitor for exceptions. If an exception occurs within this block, it is thrown.
- catch: This block catches exceptions that are thrown within the corresponding try block. You can specify which type of exception you want to catch by specifying the exception type in the catch block. You can have multiple catch blocks to handle different types of exceptions.
- finally: This block, if present, is executed regardless of whether an exception is thrown or not. It's typically used to perform cleanup tasks, like closing file handles or releasing other resources, that should always occur, whether an exception is thrown or not.

```
try {
    // Code that might throw an exception
    int result = divide(10, 0); // This will throw an ArithmeticException
} catch (ArithmeticException e) {
    // Handle the specific exception
    System.out.println("An arithmetic exception occurred: " + e.getMessage());
} finally {
    // This block will always execute, regardless of whether an exception is caught or not
    System.out.println("Finally block executed.");
}
```

45. What are the scenarios in which finally block will not get executed?

In Java, the finally block is designed to execute regardless of whether an exception is thrown or not. However, there are a few scenarios in which the finally block may not get executed:

System.exit()

If the JVM terminates abruptly due to a call to System.exit(), the finally block will not get executed. This is because System.exit() immediately terminates the JVM and does not allow any further code execution.

Infinite loop

If the try or catch block contains an infinite loop or some other code that prevents control from reaching the finally block, then the finally block will not be executed until the loop terminates or control is transferred out of the block.

Fatal errors

If a fatal error occurs, such as a StackOverflowError or an OutOfMemoryError, the JVM may terminate abruptly without executing the finally block.

In all other scenarios, the finally block should execute as expected, even if an exception is thrown or caught within the try block.

46. How to use try-with-resources in Java?

Try-with-resources is a feature for automatic resource management. It's used to automatically close resources that implement the AutoCloseable interface, such as streams, database connections, or sockets, at the end of a try block.

In below code example, the FileReader is automatically closed at the end of the try block, even if an exception is thrown while reading from the file.

```
try (FileReader fileReader = new FileReader("example.txt")) {
    // Code that reads from the file
    int data = fileReader.read();
    while (data != -1) {
        System.out.print((char) data);
        data = fileReader.read();
    }
} catch (IOException e) {
    // Exception handling
    e.printStackTrace();
}
```

47. What Is the Difference Between a Checked and an Unchecked Exception?

In Java, exceptions are categorized into two main types: checked exceptions and unchecked exceptions. The main differences between them are related to how they are handled by the compiler and how they affect the

code:

Checked Exceptions

- Checked exceptions are subclasses of Exception (excluding RuntimeException and its subclasses).
- They must be either caught or declared to be thrown by the method using the throws keyword.
- Examples of checked exceptions include IOException, ClassNotFoundException, and SQLException.
- The compiler enforces handling of checked exceptions, which means that you must either catch them or declare that your method throws them. This helps to ensure that developers are aware of potential exceptional conditions and take appropriate action.

```
File file = new File("example.txt");
        try {
            FileReader fileReader = new FileReader(file);
            // Code that reads from the file
        } catch (IOException e) {
            // Handle the checked exception
            e.printStackTrace();
        }
```

Unchecked Exceptions (RuntimeExceptions)

- Unchecked exceptions are subclasses of RuntimeException.
- They do not need to be explicitly caught or declared by the method using the throws keyword.
- Examples of unchecked exceptions include NullPointerException, ArrayIndexOutOfBoundsException, and ArithmeticException.
- The compiler does not enforce handling of unchecked exceptions. This means that it's up to the developer to write code to handle or prevent them if necessary, but it's not mandatory to do so.
- Unchecked exceptions often represent programming errors or unexpected conditions that may occur at runtime, rather than conditions that can be reasonably anticipated and handled by the code.

```
String str = null;
        try {
            int length = str.length(); // This will throw a NullPointerException
        } catch (NullPointerException e) {
            // Handle the unchecked exception
            System.out.println("NullPointerException occurred: " + e.getMessage());
        }
```

In summary, checked exceptions are enforced by the compiler, must be handled explicitly, and typically represent anticipated exceptional conditions related to external factors (like file I/O or network operations). Unchecked exceptions, on the other hand, do not need to be handled explicitly and often represent programming errors or unexpected runtime conditions.

48. What Is the Difference Between an Unlabeled and a Labeled break Statement?

Unlabeled break Statement

- It only breaks out of the innermost loop or switch statement in which it appears.
- It is typically used in for, while, do-while, or switch statements.

Labeled break Statement

- It allows you to break out of a specific, outer loop or block of code.
- This is useful when dealing with nested loops or blocks, where you want to break out of a loop that is not the innermost one.

```
for (int i = 0; i < 5; i++) {
           for (int j = 0; j < 3; j++) {
               if (j == 2) {
                   System.out.println(j);
                   break; // Breaks out of the inner loop
               }
           }
           System.out.println(i);
       }

       System.out.println("Outer Loop");

       outerloop:
       for (int i = 0; i < 3; i++) {
           for (int j = 0; j < 5; j++) {
               System.out.println(i);
               System.out.println(j);
               if (i == 1 && j == 2) {
                   break outerloop; // Breaks out of the outer loop
               }
           }
       }
```

In summary, unlabeled breaks are used to exit the current loop or switch, while labeled breaks provide a way to exit from multiple nested loops or switches, though their use is generally discouraged in favor of clearer code structures.

49. What Is the Use of the Throw and Throws Keywords?

In Java, both throw and throws are related to exception handling, but they serve different purposes:

throw Keyword

- The throw keyword is used to explicitly throw an exception within a method or block of code.

- It's typically used when a particular condition occurs that warrants an exception to be thrown, either because of an error condition or because of some other exceptional circumstance.
- You can throw both checked and unchecked exceptions using the throw keyword.

Below is the code example:

```
public void validateAge(int age) {
    if (age < 0) {
        throw new IllegalArgumentException("Age cannot be negative");
    }
    // Other validation logic
}
```

throws Keyword

- The throws keyword is used in a method declaration to indicate that the method may throw certain types of exceptions.
- It's used to delegate the responsibility of handling exceptions to the caller of the method.
- When a method declares that it throws an exception using the throws keyword, it's informing the compiler that the method may throw that type of exception, and the calling code must either catch the exception or declare it to be thrown further up the call stack.
- Multiple exceptions can be declared in the throws clause, separated by commas.

```
public void readFile(String fileName) throws IOException, FileNotFoundException {
    // Code that reads from the file
}
```

In summary, throw is used to explicitly throw an exception within a method, while throws is used in the method signature to indicate that the method may throw certain types of exceptions, thereby shifting the responsibility of handling those exceptions to the caller of the method.

50. How to Catch Multiple Exceptions?

1. Using Multiple Catch Blocks

You can have multiple catch blocks, each handling a different type of exception:

```
try {
    // Code that may throw exceptions
} catch (IOException e) {
    // Handle IOException
} catch (SQLException e) {
    // Handle SQLException
} catch (Exception e) {
    // Handle other types of exceptions
}
```

2. Using Multi-Catch Blocks

You can use a single catch block to catch multiple exceptions if they have a common superclass or interface. This approach is useful when you want to handle exceptions in a similar way:

```
try {
    // Code that may throw exceptions
} catch (IOException | SQLException e) {
    // Handle IOException and SQLException
} catch (Exception e) {
    // Handle other types of exceptions
}
```

51. What Is the Difference Between an Error and Exception?

In Java, both errors and exceptions are subclasses of the Throwable class, but they serve different purposes and are meant to be handled differently:

Exception

- Exceptions represent exceptional conditions that occur during the execution of a program. They are typically caused by problems that are outside the control of the program, such as invalid input, network issues, or file I/O errors.
- Exceptions can be checked or unchecked. Checked exceptions must be either caught or declared in the method signature using the throws keyword, while unchecked exceptions do not need to be caught or declared.
- Examples of exceptions include IOException, NullPointerException, ArrayIndexOutOfBoundsException, etc.

Error

- Errors represent abnormal conditions that occur in the Java Virtual Machine (JVM) or in the runtime environment, which are typically not recoverable by the application itself.
- Errors are typically caused by serious problems that are beyond the control of the programmer, such as system failures, out-of-memory errors, or stack overflow errors.
- Errors are unchecked, which means they do not need to be caught or declared by the programmer. In fact, it's generally not recommended to catch errors because they often indicate serious problems that the application cannot recover from.
- Examples of errors include OutOfMemoryError, StackOverflowError, VirtualMachineError, etc.

In summary, exceptions are meant to be caught and handled within the program logic to gracefully recover from exceptional conditions, while errors indicate serious problems at the system level that may not be recoverable by the application itself. Handling errors is generally not recommended because they often indicate critical issues that may require intervention at a higher level.

52. How can you create a custom exception in Java?

To create a custom exception in Java, you need to create a new class that extends either Exception (for checked exceptions) or RuntimeException (for unchecked exceptions).

1. Define the Custom Exception Class

- Create a new class that extends Exception for a checked exception or RuntimeException for an unchecked exception.
- Provide constructors to initialize the exception.

2. Add Constructors

- Include at least one constructor that takes a String message to describe the error.
- Optionally, add constructors that take a Throwable cause and both a message and a cause.

```
//example of creating a custom checked exception
// Define the custom exception
public class CustomCheckedException extends Exception {
    // Constructor that accepts a message
    public CustomCheckedException(String message) {
        super(message);
    }

    // Constructor that accepts a message and a cause
    public CustomCheckedException(String message, Throwable cause) {
        super(message, cause);
    }

    // Constructor that accepts a cause
    public CustomCheckedException(Throwable cause) {
        super(cause);
    }
}
```

```
// example of creating a custom unchecked exception
//Define the custom exception
public class CustomUncheckedException extends RuntimeException {
    // Constructor that accepts a message
    public CustomUncheckedException(String message) {
        super(message);
    }

    // Constructor that accepts a message and a cause
    public CustomUncheckedException(String message, Throwable cause) {
        super(message, cause);
    }

    // Constructor that accepts a cause
```

```
    public CustomUncheckedException(Throwable cause) {
        super(cause);
    }
}
```

53. Can you have a try block without a catch block in Java?

Yes, you can have a try block without a catch block in Java. This is valid syntax and is often used when you want to handle exceptions at a higher level or simply when you want to execute some code that might throw an exception, without handling it immediately.

In such cases, you must follow the try block with either a catch block or a finally block, or both. Here are a few scenarios:

try-finally

You can have a try block followed by a finally block without a catch block. This ensures that the finally block is executed regardless of whether an exception is thrown or not.

```
try {
    // Code that may throw exceptions
} finally {
    // Code that always executes, whether exception occurs or not
}
```

try-with-resources

You can use try-with-resources statement to automatically close resources that implement the AutoCloseable interface. This also allows a try block without a catch block.

```
static String readFirstLineFromFile(String path) throws IOException {
    try (BufferedReader br = new BufferedReader(new FileReader(path))) {
        return br.readLine();
    }
}
```

54. Explain the difference between StackOverflowError and OutOfMemoryError in Java?

The main difference between StackOverflowError and OutOfMemoryError is their underlying causes and contexts:

- StackOverflowError occurs due to excessive stack usage, often caused by deep recursion or method call chains.
- OutOfMemoryError occurs due to insufficient heap memory, often caused by excessive object creation,

memory leaks, or large data processing tasks.

Both errors indicate serious issues in the application, and they require investigation and resolution to ensure the stability and reliability of the Java application.

55. What is Java Garbage Collection and how does it work?

Java Garbage Collection (GC) is the process by which Java programs perform automatic memory management. The Java runtime environment deletes objects when they are no longer in use to free up memory. This is done automatically by the garbage collector, which is part of the Java Virtual Machine (JVM).

GC is essential in long-running server applications to prevent memory leaks and ensure efficient memory usage.

56. What are the different garbage collection algorithms in Java?

Java provides several garbage collection algorithms:

- Serial GC: Uses a single thread to perform all garbage collection work, suitable for single-threaded applications.
- Parallel GC: Uses multiple threads for garbage collection, suitable for multi-threaded applications.
- CMS (Concurrent Mark-Sweep) GC: Performs most of the garbage collection concurrently with the application threads to minimize pause times.
- G1 (Garbage-First) GC: Divides the heap into regions and prioritizes reclaiming regions with the most garbage.

G1 GC is often used in large applications where predictability and low pause times are critical.

57. What is a memory leak in Java, and how can you detect it?

Memory leak in Java occurs when objects are no longer needed but are still referenced, preventing the garbage collector from reclaiming their memory. Tools like VisualVM, JProfiler, or YourKit can be used to detect memory leaks by monitoring heap usage and identifying objects that are not being garbage collected.

It is essential to identify and address memory leaks in high-availability systems to prevent them from gradually consuming memory resources. This proactive approach ensures that these systems do not experience memory exhaustion over time, which could potentially lead to performance degradation or system failures.

By leveraging tools like VisualVM, JProfiler, or YourKit, developers can effectively monitor heap usage and pinpoint any lingering objects that are impeding proper garbage collection. Taking prompt action to fix memory leaks is paramount for maintaining the stability and efficiency of such systems in the long run.

58. What are Soft, Weak, and Phantom references in Java?

SoftReference: Objects stored in SoftReference instances are cleared at the discretion of the garbage collector in response to memory demand. This means that when the system is running low on memory, objects held only by SoftReferences may be reclaimed to free up space for other data.

WeakReference: Objects stored in WeakReference instances are cleared as soon as they are no longer strongly reachable. This happens when there are no more strong references pointing to the object, making it eligible for garbage collection. WeakReferences are commonly used for implementing memory-sensitive caches where entries can be easily removed to prevent excessive memory consumption.

PhantomReference: Objects stored in PhantomReference instances are cleared after the garbage collector determines that they are no longer reachable. However, unlike WeakReferences, objects held by PhantomReferences are not immediately reclaimed when they become unreachable. Instead, they are enqueued for cleanup actions, allowing for additional processing before their final removal.

WeakReferences are particularly useful in scenarios such as caching mechanisms where objects can be discarded if memory resources are needed elsewhere.

This flexibility in managing memory usage efficiently makes WeakReferences a valuable tool in optimizing memory utilization in Java applications.

```
        // Strong Reference (Default)
        // Not collected
        Object strongRef = new Object();

        // Weak Reference
        //collected by the garbage collector if there are
        // no strong references to it. (Caching)
        WeakReference<Object> weakRef = new WeakReference<>(new Object());

        // Soft Reference (used in image caching libraries )
        //collected by the garbage collector if memory pressure increases.
        SoftReference<Object> softRef = new SoftReference<>(new Object());

        // Phantom Reference (Memory Management Frameworks)
        // weakest type of reference and
        // are mainly used to track when an object is being
        // finalized by the garbage collector.
        ReferenceQueue<Object> phantomQueue = new ReferenceQueue<>();
        PhantomReference<Object> phantomRef = new PhantomReference<>(new Object(), phantomQueue);

        // Let's simulate some memory pressure
        System.gc();

        // Print status of references
        System.out.println("Strong reference: " + strongRef); //java.lang.Object@5acf9800
        System.out.println("Weak reference: " + weakRef.get()); //null
        System.out.println("Soft reference: " + softRef.get()); //java.lang.Object@4617c264
        System.out.println("Phantom reference: " + phantomRef.get()); //null
```

59. How can you force garbage collection in Java?

You can suggest the JVM to perform garbage collection using System.gc() or Runtime.getRuntime().gc(). However, it's important to note that there's no guarantee that the garbage collector will run immediately upon invocation. Forcing GC may be utilized in testing scenarios to observe and analyze the garbage collection behavior in specific situations.

```
public static void main(String[] args) {
       Object obj = new Object();
       obj = null;
       System.gc();
   }
```

60. What are the common JVM options for tuning garbage collection?

JVM options for tuning GC include:

- -Xms and -Xmx for setting initial and maximum heap size.
- -XX:+UseG1GC to enable G1 Garbage Collector.
- -XX:MaxGCPauseMillis to set the maximum pause time goal for G1 GC.
- -XX:+PrintGCDetails to print detailed GC logs.

Tuning GC is essential for optimizing application performance, especially in large-scale applications.

61. Explain the internal Working of Java's G1 Garbage Collector and When to Use G1 GC?

Below are the key Concepts for G1 Garbage Collector:

- Regions: G1 divides the heap into numerous fixed-size regions (usually 1MB to 32MB). These regions are not limited to a specific generation (young or old).
- Remembered Sets (RSets): Each region has an associated RSet that tracks references from other regions pointing into it. This eliminates the need for full heap scans during garbage collection.
- Collection Set (CSet): The set of regions selected for garbage collection in a given cycle. G1 aims to collect regions with the most garbage first.
- Pause Time Goal: G1 GC prioritizes meeting user-defined pause time goals, making it suitable for applications with low latency requirements.

Collection Phases

1. Young Collection:

- Marking: G1 marks live objects starting from the roots (GC roots) using concurrent marking.
- Evacuation: Live objects in young regions are copied (evacuated) to survivor or old regions. This phase is done in parallel using multiple threads.
- Reclamation: Empty young regions are immediately made available for allocation.

2. Mixed Collection (Concurrent and Parallel Phases):

- Initial Marking (STW): Briefly pauses the application to mark the roots.
- Concurrent Marking: G1 concurrently traces live objects.
- Remark (STW): Finalizes the marking process by revisiting any objects that were modified during the

concurrent phase.

• Cleanup (Mostly STW): Identifies completely empty regions (candidates for immediate reclamation) and performs accounting related to liveness information.

• Copying/Evacuation (Parallel): Selects a set of old regions (based on the pause time goal) and copies live objects to new regions, freeing up the old regions.

Other Features

• Humongous Objects: Large objects (larger than half a region) are handled specially in "humongous regions."

• String Deduplication: G1 can optionally eliminate duplicate strings in the heap to save space.

• Adaptive Behavior: G1 GC continuously adapts its behavior based on application characteristics and pause time goals.

How G1 Achieves Its Goals

• Predictable Pauses: By collecting regions in smaller increments and focusing on the most garbage-filled regions, G1 GC aims to meet pause time goals.

• High Throughput: The parallel and concurrent nature of many phases allows for efficient garbage collection.

• Good Heap Utilization: The region-based approach helps with fragmentation issues.

When to Use G1 GC

• Large Heaps: G1 GC is designed for multi-gigabyte heaps.

• Low Latency Requirements: If your application needs to meet strict pause time goals.

• Mixed Workloads: G1 GC handles both young and old generation objects well.

62. How does the garbage collector know which objects are reachable?

The concept of reachability analysis in garbage collection is a fundamental technique used to manage memory in programming languages. When the garbage collector executes reachability analysis, it starts from certain designated root objects, such as local variables or static variables.

These root objects are essentially the starting points from which the garbage collector explores and traverses through the entire object graph to identify all reachable objects. As the garbage collector follows references from the root objects to other objects in memory, it marks each visited object as reachable.

Objects that are not encountered during this traversal process are considered unreachable or detached from the root objects. These unreachable objects are deemed unnecessary and can be safely removed from memory to free up resources. By conducting reachability analysis, the garbage collector can effectively distinguish between active objects that are still in use within the program and obsolete objects that can be safely discarded.

This proactive memory management strategy helps prevent memory leaks and ensures efficient utilization of memory resources in software applications.

63. What are some of the Useful Annotations used in Java?

Annotations in Java are like metadata tags that provide additional information about your code to the compiler or runtime environment. They're not part of the program's logic but offer instructions or hints.

Built-in Annotations

- @Override: Indicates that a method in a subclass is intended to override a method in its superclass. Helps prevent errors by ensuring the signature matches.
- @Deprecated: Marks a class, method, or field as deprecated, meaning it's discouraged from being used (often because it's obsolete or has a better alternative).
- @SuppressWarnings: Instructs the compiler to suppress specific warnings for the annotated element.
- @FunctionalInterface: Indicates that an interface is intended to be a functional interface (having a single abstract method) to be used with lambda expressions.
- @SafeVarargs: Used on a method or constructor that takes a variable number of arguments (varargs) to suppress warnings related to heap pollution (when a varargs parameter's array might contain objects of an incorrect type).

```
@Override
public String toString() { // Overrides the toString method
    return "This is my custom object";
}

@Deprecated
public void oldMethod() {  // Discourages use of this method
    // ...
}
```

64. What Are Meta-Annotations?

Think of meta-annotations as annotations that apply to other annotations. They provide information about how other annotations should behave or where they can be used. Here's a breakdown of the key meta-annotations:

@Retention

- Specifies how long an annotation should be retained in the code.
- RetentionPolicy.SOURCE: Discarded by the compiler.
- RetentionPolicy.CLASS: Retained in the compiled class files but not available at runtime.
- RetentionPolicy.RUNTIME: Retained at runtime and accessible through reflection.

@Target

- Specifies the types of elements (e.g., classes, methods, fields, parameters) that an annotation can be applied to.
- ElementType.TYPE: Classes, interfaces, enums.

- ElementType.METHOD: Methods.
- ElementType.FIELD: Fields (class variables).
- ElementType.PARAMETER: Method parameters.
- ElementType.CONSTRUCTOR: Constructors.

@Documented

Indicates that an annotation should be included in the Javadoc documentation for the annotated element.

@Inherited

Allows an annotation on a class to be inherited by its subclasses.

```
@Retention(RetentionPolicy.RUNTIME) // Retained at runtime
@Target(ElementType.METHOD)         // Can be applied to methods
public @interface MyAnnotation {
    // ... annotation elements
}
```

65. What Are Repeating Annotations?

Before Java 8, you could apply an annotation only once to an element. Repeating annotations (introduced in Java 8) allow you to apply the same annotation multiple times to the same element. Here's how:

Create a Container Annotation

- This annotation acts as a container to hold multiple instances of the repeating annotation.
- Annotate it with @Repeatable, specifying the repeating annotation's class.

Create the Repeating Annotation

This is the annotation you want to apply multiple times.

```
@Repeatable(Authors.class)
public @interface Author {
    String name();
}

public @interface Authors {
    Author[] value();
}


//Now you can use @Author multiple times
@Author(name = "John Doe")
@Author(name = "Jane Smith")
public class MyClass {
    // ...
}
```

Key Points

- Meta-annotations control the behavior and usage of other annotations.
- Repeating annotations allow for applying the same annotation multiple times, increasing flexibility.
- Both features enhance the expressiveness and capabilities of Java annotations.

66. Is It Possible to Extend Annotations and what is Indirect Extension?

You cannot directly extend an annotation in Java using the extends keyword, as you would with classes or interfaces. Annotations are a distinct type of metadata, not classes.

Indirect Extension

- You can achieve a kind of "inheritance" effect with annotations by combining them.
- Meta-annotations like @Inherited: If an annotation is marked with @Inherited, and you apply it to a class, then subclasses of that class will automatically inherit the annotation.
- Custom composed annotations: You can create a new annotation that combines multiple existing annotations. This way, applying your custom annotation is equivalent to applying all the combined annotations.

In below example, MyCustomAnnotation indirectly "extends" the Author annotation, as applying MyCustomAnnotation also applies Author.

So you can create custom annotations that combine multiple existing annotations, effectively achieving an "extension" effect.

```
@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.TYPE)
public @interface MyCustomAnnotation {
    Author author(); // Uses the previously defined @Author annotation
    int version() default 1;
}
```

```
static String readFirstLineFromFile(String path) throws IOException {
    try (BufferedReader br = new BufferedReader(new FileReader(path))) {
        return br.readLine();
    }
}
```

67. How to Limit the Elements in Which an Annotation Can Be Applied?

You can control the elements where an annotation can be applied using the @Target meta-annotation. Here's how it works:

The @Target Meta-Annotation

- @Target accepts an array of ElementType values, which specify the types of elements the annotation can

be used on.

Common ElementType values include:

- ElementType.TYPE: Classes, interfaces, enums.
- ElementType.METHOD: Methods.
- ElementType.FIELD: Fields (class variables).
- ElementType.PARAMETER: Method parameters.
- ElementType.CONSTRUCTOR: Constructors.
- ElementType.LOCAL_VARIABLE: Local variables.
- ElementType.ANNOTATION_TYPE: Annotations (for creating meta-annotations).
- ElementType.PACKAGE: Package declarations.
- ElementType.TYPE_PARAMETER: Type parameters (generics).
- ElementType.TYPE_USE: Type uses (introduced in Java 8).

```
import java.lang.annotation.*;

@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.METHOD) // This annotation can only be applied to methods
public @interface LogExecutionTime {
    // ... annotation elements (if any)
}
```

68. What is the Difference Between Primitive and Reference Types?

In programming, particularly in languages like Java, C#, and JavaScript, data types are broadly categorized into primitive types and reference types. Understanding the differences between these two is crucial for effective coding and memory management.

Primitive Types

Primitive types are the most basic data types provided by a programming language. They are:

- Simple and Immutable: They hold simple values and are immutable, meaning once a primitive value is created, it cannot be altered.
- Stored Directly: The actual value is stored directly in memory locations assigned to the variable.
- Fixed Size: They typically have a fixed size which can be directly represented in the memory. This size can vary between programming languages.

eg: int, char, boolean, byte, short, long, float, double

Reference Types

Reference types are more complex. They include structures that reference or point to memory locations where the actual data is stored.

- Contain References: Instead of holding the actual value, these types hold a reference (or address) to the memory location where the data is stored.
- Mutable: Generally, the objects these references point to can be modified.
- Dynamic Size: The size can be dynamic and not necessarily fixed since it can point to complex data structures like objects and arrays.

eg: All objects (instances of classes), arrays

Key Differences

Memory Allocation:

- Primitive Types: Stored directly in the stack. The memory allocation is straightforward and typically faster.
- Reference Types: Stored in the heap. The variable holds a reference (pointer) to the memory location. Accessing these can be slower due to the additional level of indirection.

Value vs. Reference Semantics:

- Primitive Types: When assigned to another variable, a copy of the value is made. Changes to one variable do not affect the other.
- Reference Types: When assigned to another variable, both variables point to the same memory location. Changes made through one variable reflect in the other.

Mutability:

- Primitive Types: Immutable in most languages (though some languages may offer ways to "simulate" mutability).
- Reference Types: Mutable, allowing modifications to the data they point to without changing the reference.

Default Values:

- Primitive Types: Usually have default values (e.g., 0 for integers, false for booleans).
- Reference Types: Typically default to null or undefined if not explicitly initialized.

```
// Primitive example
public class PrimitiveExample {
    public static void main(String[] args) {
        int a = 5;
        int b = a; // b is now a copy of a
        b = 10; // changing b does not affect a

        System.out.println("Value of a: " + a); // Output: Value of a: 5
        System.out.println("Value of b: " + b); // Output: Value of b: 10
    }
}
```

```
// Reference example
public class ReferenceExample {
    public static void main(String[] args) {
        // Creating an object of MyClass
        MyClass obj1 = new MyClass();
        obj1.name = "Alice";

        // obj2 is a reference to the same object as obj1
        MyClass obj2 = obj1;
        obj2.name = "Bob"; // modifying obj2 affects obj1

        System.out.println("Value of obj1.name: " + obj1.name); // Output: Value of obj1.name: Bob
        System.out.println("Value of obj2.name: " + obj2.name); // Output: Value of obj2.name: Bob
    }
}

class MyClass {
    String name;
}
```

69. What are the Different Primitive Types and the memory they occupy?

In Java, primitive types are the most basic data types provided by the language. Each primitive type has a fixed size and a default value. Here is a list of Java's primitive types along with their sizes and default values:

1. byte

- Size: 8 bits (1 byte)
- Range: -128 to 127
- Default Value: 0

2. short

- Size: 16 bits (2 bytes)
- Range: -32,768 to 32,767
- Default Value: 0

3. int

- Size: 32 bits (4 bytes)
- Range: -2^31 to 2^31 - 1 (-2,147,483,648 to 2,147,483,647)
- Default Value: 0

4. long

- Size: 64 bits (8 bytes)
- Range: -2^63 to 2^63 - 1 (-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807)
- Default Value: 0L

5. float

- Size: 32 bits (4 bytes)
- Range: Approximately ±3.40282347E+38F (6-7 significant decimal digits)
- Default Value: 0.0f

6. double

- Size: 64 bits (8 bytes)
- Range: Approximately ±1.79769313486231570E+308 (15 significant decimal digits)
- Default Value: 0.0d

7. char

- Size: 16 bits (2 bytes)
- Range: 0 to 65,535 (unsigned)
- Default Value: '\u0000' (null character)

8. boolean

- Size: The size is not precisely defined; it depends on the JVM. However, it typically takes up 1 bit of information, though it might be represented by 1 byte for simplicity in many implementations.
- Range: true or false
- Default Value: false

70. What is concept of Reflection in Java?

Reflection is a powerful feature in Java that allows you to inspect and manipulate the structure and behavior of classes, interfaces, fields, and methods at runtime. It's like a flashlight that lets you shine a light on the internal workings of your code, even if you don't know the names of those internal components beforehand.

How Reflection Works

• Introspection: Examining the structure of classes and objects. You can get information like class names, method signatures, field types, annotations, and more.

• Manipulation: Modifying the behavior of classes and objects at runtime. You can create new instances, invoke methods, access and modify fields, and even change access modifiers (within security constraints).

Common Use Cases of Reflection

1. Frameworks and Libraries:
   • Dependency Injection: Frameworks like Spring use reflection to instantiate and wire together components at runtime.
   • Testing Frameworks: JUnit and other testing tools use reflection to discover and run test methods.
   • ORM (Object-Relational Mapping): Libraries like Hibernate use reflection to map Java objects to database tables.
2. Tools and IDEs:
   • Debuggers: Use reflection to inspect objects and variables while your program is running.
   • IDEs: Use reflection to provide features like code completion, refactoring, and automatic generation of code.
3. Dynamic Loading and Configuration:
   • Plugins: You can load and configure plugins or modules at runtime without hardcoding their details.
   • Custom Annotations: Reflection helps process custom annotations to trigger specific behaviors.

```
import java.lang.reflect.Field;
import java.lang.reflect.Method;

class Company {
    private String name;

    public Company(String name) {
        this.name = name;
    }

    public String getName() {
        return name;
    }

    private void displayName() {
        System.out.println
                ("Private Method Invoked to display Company Name: " + name);
    }

}

public class ReflectionExample {

    // Using reflection to set a private field and invoke a private method
    public static void main(String[] args) throws Exception{
```

```
        // Create an instance of Company
        Company company = new Company("Youtube");
        System.out.println("Company Name: " + company.getName());

        // Change the name from Youtube to Ytube
        Field nameField = Company.class.getDeclaredField("name");
        nameField.setAccessible(true);
        nameField.set(company,"Ytube");
        System.out.println(company.getName());

        // Invoke the private method 'displayName'
        Method objMethod = Company.class.getDeclaredMethod("displayName");
        objMethod.setAccessible(true);
        objMethod.invoke(company);



    }
}
```

71. What Is the Difference Between an Inner Class and a Static Nested Class?

In Java, nested classes are classes defined within another class. There are two main types of nested classes: inner classes and static nested classes. Here's a detailed explanation of the differences between the two:

Inner Classes

Inner classes are non-static nested classes. They have the following characteristics:

- Associated with an Instance: Inner classes are tied to an instance of the outer class. This means they can access the outer class's instance variables and methods directly.

```
class OuterClass {
    class InnerClass {
        // Inner class code
    }
}
```

- Access to Outer Class Members: Inner classes have direct access to the outer class's instance members, including private ones.
- Instantiation: An inner class instance is created using an instance of the outer class.

```
OuterClass outer = new OuterClass();
OuterClass.InnerClass inner = outer.new InnerClass();
```

• Reference to Outer Class: An inner class has an implicit reference to its enclosing instance of the outer class.

Static Nested Classes

Static nested classes are static members of the outer class. They have the following characteristics:

• Not Associated with an Instance: Static nested classes do not have an implicit reference to an instance of the outer class. They can be instantiated without an instance of the outer class.

```
class OuterClass {
    static class StaticNestedClass {
        // Static nested class code
    }
}
```

• Access to Outer Class Members: Static nested classes can only access the static members of the outer class directly. To access instance members, they need an instance of the outer class.

• Instantiation: A static nested class instance is created without needing an instance of the outer class.

```
OuterClass.StaticNestedClass nested = new OuterClass.StaticNestedClass();
```

• No Reference to Outer Class: Static nested classes do not have an implicit reference to an outer class instance.

```
// Inner Class Example
public class OuterClass {
    private int outerField = 10;

    class InnerClass {
        void display() {
            System.out.println("Outer field: " + outerField);
        }
    }

    public static void main(String[] args) {
        OuterClass outer = new OuterClass();
        OuterClass.InnerClass inner = outer.new InnerClass();
        inner.display(); // Output: Outer field: 10
    }
}
```

```
// Static Nested Class Example:
public class OuterClass {
    private static int staticOuterField = 20;

    static class StaticNestedClass {
        void display() {
```

```
            System.out.println("Static outer field: " + staticOuterField);
        }
    }

    public static void main(String[] args) {
        OuterClass.StaticNestedClass nested = new OuterClass.StaticNestedClass();
        nested.display(); // Output: Static outer field: 20
    }
}
```

72. What is difference between equals() and ==?

In Java, equals() and == are used for comparison, but they serve different purposes and behave differently. Here's a detailed explanation of the differences between equals() and ==:

== Operator

Primitive Types: When used with primitive types, == compares the actual values.

```
int a = 5;
int b = 5;
System.out.println(a == b); // Output: true
```

Reference Types: When used with reference types (objects), == compares the references (memory addresses) to check if they point to the same object.

```
String s1 = new String("hello");
String s2 = new String("hello");
System.out.println(s1 == s2); // Output: false (different objects in memory)
```

equals() Method

1. Object Class: The equals() method is defined in the Object class. By default, it behaves like ==, comparing object references.

```
class MyClass {
    int value;
}

MyClass obj1 = new MyClass();
MyClass obj2 = new MyClass();
System.out.println(obj1.equals(obj2)); // Output: false (default behavior)
```

2. Overridden Method: Many classes, like String, Integer, and custom classes, override equals() to compare the actual content (state) of the objects.

```
String s1 = new String("hello");
String s2 = new String("hello");
System.out.println(s1.equals(s2)); // Output: true (content is the same)
```

3. Custom Classes: When you create your own classes, you can override equals() to define what it means for two instances of your class to be "equal".

```
class MyClass {
    int value;

    MyClass(int value) {
        this.value = value;
    }

    @Override
    public boolean equals(Object obj) {
        if (this == obj) return true;
        if (obj == null || getClass() != obj.getClass()) return false;
        MyClass myClass = (MyClass) obj;
        return value == myClass.value;
    }
}

MyClass obj1 = new MyClass(10);
MyClass obj2 = new MyClass(10);
System.out.println(obj1.equals(obj2)); // Output: true (values are the same)
```

Summary of Differences

1. Comparison Level:
   - ==: Compares references (memory addresses) for objects; compares actual values for primitives.
   - equals(): Compares the content/state of objects (if overridden); compares references if not overridden.
2. Default Behavior:
   - ==: Always checks for reference equality for objects.
   - equals(): By default (in Object class), checks for reference equality, but can be overridden for content equality.
3. Usage:
   - Use == for: Comparing primitive values. Checking if two references point to the same object.
   - Use equals() for: Comparing the content/state of two objects. Implementing logical equality in custom classes.

73. What Is an Anonymous Class and what is it's Use Case?

An anonymous class in Java is a type of inner class without a name. It is used to instantiate a class defined in

the method body, usually for implementing an interface or extending a class with minor modifications. Anonymous classes are often used for short-lived implementations that are not reused elsewhere.

Characteristics of Anonymous Classes

- No Name: As the name suggests, anonymous classes do not have a name. They are defined and instantiated in a single expression.
- Syntax: Typically created using new followed by the interface or class they extend or implement.

```
new InterfaceOrClass() {}
```

- Instantiation: They are usually instantiated where they are defined, often in the context of a method call or a variable initialization.
- Single Use: Anonymous classes are intended for use in a single context and are not reusable.

Use Cases for Anonymous Classes

1. Event Handling: Commonly used in GUI applications for handling events (e.g., button clicks).

```
import javax.swing.JButton;
import javax.swing.JFrame;

public class AnonymousClassExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Anonymous Class Example");
        JButton button = new JButton("Click Me");

        button.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                System.out.println("Button was clicked!");
            }
        });

        frame.add(button);
        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

2. Thread Creation: Simplifies thread creation without defining a separate class.

```
public class AnonymousClassExample {
    public static void main(String[] args) {
        new Thread(new Runnable() {
            @Override
            public void run() {
                System.out.println("Thread is running");
```

```
            }
        }).start();
    }
}
```

3. Customizing Behavior: Allows for quick customization of existing classes or interfaces.

```
import java.util.ArrayList;
import java.util.List;

public class AnonymousClassExample {
    public static void main(String[] args) {
        List<String> list = new ArrayList<String>() {
            @Override
            public boolean add(String s) {
                System.out.println("Adding element: " + s);
                return super.add(s);
            }
        };

        list.add("Hello");
        list.add("World");
    }
}
```

74. What Is a Marker Interface and examples of Marker Interfaces in Java?

Marker Interface in Java

A marker interface in Java is an interface with no methods or constants. It serves as a means to signal to the JVM or other code that a class implementing this interface has a particular property or should be treated in a special way. Marker interfaces are used to provide metadata to objects.

Purpose of Marker Interfaces

- Type Identification: Identifying the type of an object at runtime.
- Special Behavior: Indicating that a class should be processed in a special manner by certain Java APIs or libraries.
- Security and Serialization: Controlling the serialization mechanism or ensuring certain security measures.

Examples of Marker Interfaces in Java

1. Serializable: This marker interface is used to indicate that a class can be serialized, i.e., its state can be converted into a byte stream and subsequently restored.

```
import java.io.Serializable;
```

```
public class MyClass implements Serializable {
    private static final long serialVersionUID = 1L;
    private int id;
    private String name;

    // Constructor, getters, and setters
}
```

2. Cloneable: This marker interface is used to indicate that a class allows a clone of its objects to be created. Implementing this interface signals that the Object.clone() method can be used on instances of the class.

```
public class MyClass implements Cloneable {
    private int id;
    private String name;

    // Constructor, getters, and setters

    @Override
    protected Object clone() throws CloneNotSupportedException {
        return super.clone();
    }
}
```

3. Remote: This marker interface is used to indicate that an interface represents a remote object that can be invoked from a non-local virtual machine. It is a part of Java RMI (Remote Method Invocation).

```
import java.rmi.Remote;
import java.rmi.RemoteException;

public interface MyRemoteInterface extends Remote {
    void myRemoteMethod() throws RemoteException;
}
```

Implementation and Use Case

```
// Serializable Example:
import java.io.*;

public class SerializableExample {
    public static void main(String[] args) {
        MyClass obj = new MyClass(1, "John");

        // Serialize the object
        try (ObjectOutputStream out = new ObjectOutputStream(new FileOutputStream("object.ser"))) {
            out.writeObject(obj);
        } catch (IOException e) {
            e.printStackTrace();
```

```
        }

        // Deserialize the object
        try (ObjectInputStream in = new ObjectInputStream(new FileInputStream("object.ser"))) {
            MyClass deserializedObj = (MyClass) in.readObject();
            System.out.println("Deserialized object: " + deserializedObj.getName()); // Deserialized ob
        } catch (IOException | ClassNotFoundException e) {
            e.printStackTrace();
        }
    }
}

class MyClass implements Serializable {
    private static final long serialVersionUID = 1L;
    private int id;
    private String name;

    public MyClass(int id, String name) {
        this.id = id;
        this.name = name;
    }

    public int getId() {
        return id;
    }
public String getName() {
        return name;
    }
}
```

```
// Cloneable Example:
public class CloneableExample {
    public static void main(String[] args) {
        MyClass obj1 = new MyClass(1, "John");
        try {
            MyClass obj2 = (MyClass) obj1.clone();
            System.out.println("Cloned object: " + obj2.getName()); // Cloned object: John
        } catch (CloneNotSupportedException e) {
            e.printStackTrace();
        }
    }
}

class MyClass implements Cloneable {
    private int id;
    private String name;

    public MyClass(int id, String name) {
        this.id = id;
```

```
        this.name = name;
    }

    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }

    @Override
    protected Object clone() throws CloneNotSupportedException {
        return super.clone();
    }
}
```

75. What Is an Immutable Class and how to create it?

An immutable class in Java is a class whose instances cannot be modified after they are created. Once an object is created, its state (fields) cannot be changed. Immutable objects are inherently thread-safe and can be shared freely among multiple threads without synchronization concerns.

Characteristics of Immutable Classes

- Final Class: The class should be declared as final so that it cannot be subclassed.
- Private Final Fields: All fields should be declared as private and final to ensure they are not modified after initialization.
- No Setter Methods: The class should not provide any methods that modify fields or objects referred to by fields.

Example of an Immutable Class

```
import java.util.Date;

public final class ImmutablePerson {
    private final String name;
    private final int age;
    private final Date birthDate; // mutable object

    public ImmutablePerson(String name, int age, Date birthDate) {
        this.name = name;
        this.age = age;
        // Creating a defensive copy of the mutable Date object
        this.birthDate = new Date(birthDate.getTime());
    }
```

```
    public String getName() {
        return name;
    }

    public int getAge() {
        return age;
    }

    public Date getBirthDate() {
        // Returning a defensive copy of the mutable Date object
        return new Date(birthDate.getTime());
    }
}
```

Explanation of the Example

- Class Declaration: public final class ImmutablePerson: The class is declared as final.
- Private Final Fields: private final String name; private final int age; private final Date birthDate;
- Constructor: The constructor initializes all fields. For the mutable Date object, a defensive copy is made using new Date(birthDate.getTime()).
- Getter Methods: getName(), getAge(), and getBirthDate() methods provide access to the fields. For the Date field, getBirthDate() returns a new Date object to prevent modification of the internal birthDate field.

76. Is it possible to Override a Static Method?

In Java, it is not possible to override a static method. Static methods belong to the class rather than any instance of the class. Overriding is a concept related to instance methods, where a subclass provides a specific implementation for a method that is already defined in its superclass.

Static Method Behavior

- Class-Level Methods: Static methods are class-level methods, meaning they are associated with the class itself rather than any particular object instance.
- Hiding, Not Overriding: When a static method is defined in a superclass and a subclass defines a static method with the same name and parameter list, it is known as method hiding, not method overriding. The static method in the subclass hides the static method in the superclass.
- Compile-Time Binding: Static methods are bound at compile-time, whereas instance methods are bound at runtime. This is why you can't override static methods, but you can hide them.

Example of Static Method Hiding

```
class Parent {
    public static void staticMethod() {
        System.out.println("Static method in Parent class");
    }
```

```
    public void instanceMethod() {
        System.out.println("Instance method in Parent class");
    }
}

class Child extends Parent {
    // This hides the static method in Parent, it does not override it
    public static void staticMethod() {
        System.out.println("Static method in Child class");
    }

    // This overrides the instance method in Parent
    @Override
    public void instanceMethod() {
        System.out.println("Instance method in Child class");
    }
}

public class Test {
    public static void main(String[] args) {
        Parent parent = new Parent();
        Parent childAsParent = new Child();
        Child child = new Child();

        parent.staticMethod(); // Output: Static method in Parent class
        childAsParent.staticMethod(); // Output: Static method in Parent class
        child.staticMethod(); // Output: Static method in Child class

        parent.instanceMethod(); // Output: Instance method in Parent class
        childAsParent.instanceMethod(); // Output: Instance method in Child class
        child.instanceMethod(); // Output: Instance method in Child class
    }
}
```

Explanation

1. Static Method Calls:
   - Parent.staticMethod() prints "Static method in Parent class".
   - childAsParent.staticMethod() (where childAsParent is of type Parent) also prints "Static method in Parent class" because static methods are resolved at compile time based on the reference type, not the actual object.
   - child.staticMethod() (where child is of type Child) prints "Static method in Child class".
2. Instance Method Calls:
   - parent.instanceMethod() prints "Instance method in Parent class".
   - childAsParent.instanceMethod() prints "Instance method in Child class" because instance methods are resolved at runtime based on the actual object.

• child.instanceMethod() prints "Instance method in Child class".

77. How to Compare Two Enum Values?

In Java, enums are a special type of class used to define collections of constants. Comparing two enum values is straightforward and can be done using either the == operator or the equals() method. Both approaches are valid, but there are subtle differences in their usage and implications.

Comparing Enum Values Using ==

The == operator is the recommended way to compare enum values. It checks for reference equality, which is appropriate for enums because each enum constant is a singleton (there is only one instance of each enum constant).

```
public enum Color {
    RED, GREEN, BLUE;
}

public class EnumComparison {
    public static void main(String[] args) {
        Color color1 = Color.RED;
        Color color2 = Color.RED;
        Color color3 = Color.GREEN;

        if (color1 == color2) {
            System.out.println("color1 and color2 are the same");
        } else {
            System.out.println("color1 and color2 are different");
        }

        if (color1 == color3) {
            System.out.println("color1 and color3 are the same");
        } else {
            System.out.println("color1 and color3 are different");
        }
    }
}
// Output
// color1 and color2 are the same
// color1 and color3 are different
```

Comparing Enum Values Using equals()

The equals() method can also be used to compare enum values. It checks for logical equality, which in the case of enums is the same as reference equality. However, using equals() is less common because == is more concise and clearer for enums.

```
public class EnumComparison {
    public static void main(String[] args) {
```

```
        Color color1 = Color.RED;
        Color color2 = Color.RED;
        Color color3 = Color.GREEN;

        if (color1.equals(color2)) {
            System.out.println("color1 and color2 are the same");
        } else {
            System.out.println("color1 and color2 are different");
        }

        if (color1.equals(color3)) {
            System.out.println("color1 and color3 are the same");
        } else {
            System.out.println("color1 and color3 are different");
        }
    }
}

// Output
// color1 and color2 are the same
// color1 and color3 are different
```

Summary

- Using ==: This is the preferred way to compare enum values because it checks for reference equality and is more concise.
- Using equals(): This also works because it checks for logical equality, which is the same as reference equality for enums. However, it is less commonly used.

78. What is the purpose of the this keyword?

In Java, the this keyword is a reference to the current object within an instance method or constructor. It has several purposes, all related to distinguishing between instance variables and parameters or to explicitly referring to the current object.

Purpose of the this Keyword

1. To Refer to Instance Variables: When a local variable or parameter has the same name as an instance variable, this can be used to refer to the instance variable.

```
public class MyClass {
    private int value;

    public void setValue(int value) {
        this.value = value; // "this.value" refers to the instance variable
    }
}
```

2. To Call Another Constructor: Within a constructor, this() can be used to call another constructor in the same class.

```
public class MyClass {
    private int value;

    public MyClass() {
        this(0); // Calls the constructor with an int parameter
    }

    public MyClass(int value) {
        this.value = value;
    }
}
```

3. To Pass the Current Object as a Parameter: It can be used to pass the current object as a parameter to another method or constructor.

```
public class MyClass {
    public void doSomething() {
        AnotherClass.process(this); // Passes the current object to the process method
    }
}
```

4. To Return the Current Object: It can be used to return the current object from a method, enabling method chaining.

```
public class MyClass {
    public MyClass increment() {
        this.value++;
        return this; // Returns the current object
    }
}
```

Summary

The this keyword in Java is a reference to the current object and is primarily used to disambiguate between instance variables and parameters, or to explicitly refer to the current object. It's a fundamental tool for managing object-oriented programming concepts such as encapsulation, constructor chaining, and method chaining.

79. How does the final keyword affect class inheritance and method overriding?

In Java, the final keyword has different effects depending on whether it is applied to a class, method, or variable.

Final Class

- When a class is declared as final, it cannot be subclassed.
- Any attempt to extend a final class will result in a compilation error.

```
final class FinalClass {
    // Class implementation
}

// Error: Cannot inherit from final FinalClass
class Subclass extends FinalClass {
    // Subclass implementation
}
```

Final Method

- When a method is declared as final in a superclass, it cannot be overridden in subclasses.
- Subclasses are not allowed to provide a different implementation of a final method.

```
class Parent {
    public final void finalMethod() {
        // Method implementation
    }
}

class Child extends Parent {
    // Error: Cannot override final method finalMethod() in Parent
    public void finalMethod() {
        // Overridden method implementation
    }
}
```

Effects on Inheritance and Method Overriding

- Prevents Inheritance: Declaring a class as final prevents other classes from inheriting from it, ensuring that the behavior of the class cannot be modified or extended.
- Prevents Method Overriding: Declaring a method as final prevents subclasses from overriding it. This is useful when a method in a superclass is considered complete and should not be modified in subclasses.

When to Use final

- Use final for classes or methods when you want to prevent any modifications or extensions to them.
- It's often used for utility classes, immutable classes, or critical methods that should not be overridden.

80. What Are the Wrapper Classes and what is autoboxing and unboxing in Java?

In Java, the primitive data types (e.g., int, double, boolean) are not objects, which means they cannot be used in scenarios where objects are required, such as collections like ArrayList or methods that expect objects. Wrapper classes are a set of classes provided by Java to represent each of the primitive data types as objects. These classes provide utility methods and enable primitive types to be used in contexts where objects are required.

Wrapper Classes in Java

- Integer: Represents the int primitive type.
- Double: Represents the double primitive type.
- Boolean: Represents the boolean primitive type.
- Character: Represents the char primitive type.
- Byte: Represents the byte primitive type.
- Short: Represents the short primitive type.
- Long: Represents the long primitive type.
- Float: Represents the float primitive type.

Autoboxing and Unboxing

1. Autoboxing: Autoboxing is the automatic conversion of primitive data types into their corresponding wrapper class objects. It happens when the primitive data types are used in contexts where objects are required, such as adding them to collections or passing them to methods that expect objects.

```
int primitiveInt = 10;
Integer wrapperInt = primitiveInt; // Autoboxing
```

2. Unboxing: Unboxing is the automatic conversion of wrapper class objects back to their corresponding primitive data types. It happens when wrapper class objects are used in contexts where primitive data types are expected, such as performing arithmetic operations.

```
Integer wrapperInt = 20;
int primitiveInt = wrapperInt; // Unboxing
```

```
import java.util.ArrayList;

public class AutoboxingUnboxingExample {
    public static void main(String[] args) {
        // Autoboxing: converting int to Integer
        int primitiveInt = 42;
        Integer wrapperInt = primitiveInt;

        // Unboxing: converting Integer to int
        Integer wrapperInt2 = 50;
        int primitiveInt2 = wrapperInt2;

        // Autoboxing in collections
```

```
        ArrayList<Integer> list = new ArrayList<>();
        list.add(1); // Autoboxing
        list.add(2); // Autoboxing

        // Unboxing from collections
        int first = list.get(0); // Unboxing
        int second = list.get(1); // Unboxing

        System.out.println("Autoboxing: " + wrapperInt); // Autoboxing: 42
        System.out.println("Unboxing: " + primitiveInt2); // Unboxing: 50
        System.out.println("Autoboxing in collection: " + list); // Autoboxing in collection: [1, 2]
        System.out.println("Unboxing from collection: " + first + ", " + second); // Unboxing from coll
    }
}
```

In this example, autoboxing occurs when assigning the primitive int to the Integer object and when adding integers to the ArrayList. Unboxing occurs when assigning the Integer object to a primitive int and when retrieving integers from the ArrayList. Autoboxing and unboxing simplify the coding process by allowing primitive data types and their corresponding wrapper classes to be used interchangeably.

81. What Is a Var-Arg and how to use it? What are benefits and limitations?

A varargs (variable-length arguments) parameter allows a method to accept zero or more arguments of a specified type. It's declared by specifying three dots (...) after the parameter type in the method signature. Varargs provide flexibility in the number of arguments that can be passed to a method.

```
public void methodName(Type... parameterName) {
    // Method implementation
}
```

Usage: You can pass any number of arguments of the specified type (or none at all). Inside the method, the varargs parameter behaves like an array of the specified type.

```
public class VarargsExample {
    public static void main(String[] args) {
        printNumbers(1, 2, 3, 4, 5);
        printNumbers(); // Calling with no arguments
    }

    public static void printNumbers(int... numbers) {
        System.out.println("Printing numbers:");
        for (int num : numbers) {
            System.out.println(num);
        }
    }

  // Output
```

```
// Printing numbers:
// 1
// 2
// 3
// 4
// 5
// Printing numbers:
}
```

Benefits of Varargs

• Flexibility: Varargs allow methods to accept a variable number of arguments, providing flexibility in method calls.

• Simplified Syntax: Varargs simplify the syntax for methods that accept multiple arguments, avoiding the need for arrays or overloaded methods.

• Reduced Boilerplate Code: Varargs eliminate the need to create and manipulate arrays manually when passing multiple arguments.

Limitations of Varargs

• Positional Arguments Only: Varargs must be the last parameter in a method signature, and you cannot have more than one varargs parameter in a method.

• Performance Overhead: Varargs are implemented using arrays internally, which may result in a slight performance overhead compared to fixed-size parameter lists.

• Type Safety: Varargs can accept arguments of any length, potentially leading to type-related errors if not used carefully.

Core Java Interview Questions - Multithreading & Concurrency

1. What Is the Difference Between a Process and a Thread?

In Java programming, a process and a thread are both units of execution, but they operate at different levels of abstraction and have distinct characteristics.

A process can be thought of as an independent entity that runs in its own memory space and has its own resources, including memory, file handles, and system resources. Processes are heavyweight entities, and each process operates independently of other processes. Examples of processes include running multiple instances of a web browser or a text editor on your computer. Each instance operates as a separate process, with its own memory space and resources.

On the other hand, a thread is a lightweight unit of execution that exists within a process. Threads share the same memory space and resources as the process to which they belong. Multiple threads can exist within a single process and share resources such as memory and file handles. Threads are used to achieve concurrency within a program, allowing multiple tasks to be performed simultaneously. For example, in a web server application, multiple threads can handle incoming client requests concurrently, improving the server's responsiveness and efficiency.

Real-World Use Case Examples:

- Web Browser: When you open multiple tabs in a web browser, each tab typically runs as a separate process. This isolation ensures that if one tab crashes or experiences issues, it does not affect the other tabs or the browser itself. Within each tab, multiple threads handle tasks such as rendering web pages, processing user input, and downloading resources concurrently.
- Word Processing Application: In a word processing application like Microsoft Word, the application itself runs as a process. Within the application, various threads handle tasks such as user interface interactions, spell-checking, auto-saving, and printing. These threads operate concurrently, providing a smooth and responsive user experience.

In summary, while processes and threads both represent units of execution in Java, they differ in their level of isolation, resource allocation, and concurrency.

2. How can we create a Thread in Java?

You can create a thread instance by either extending the Thread class or implementing the Runnable interface. Once you have created a thread instance, you can start it by calling the start() method.

```
// Define a class that extends Thread
class MyThread extends Thread {
    public void run() {
        // Code to be executed by the thread
        System.out.println("Thread running: " + Thread.currentThread().getName());
    }
}

public class Main {
    public static void main(String[] args) {
        // Create an instance of MyThread
        MyThread myThread = new MyThread();
        // Start the thread
        myThread.start();
    }
}
```

```
// Define a class that implements the Runnable interface
class MyRunnable implements Runnable {
    public void run() {
        // Code to be executed by the thread
        System.out.println("Thread running: " + Thread.currentThread().getName());
    }
}

public class Main {
    public static void main(String[] args) {
        // Create an instance of MyRunnable
        MyRunnable myRunnable = new MyRunnable();
        // Create a Thread instance and pass MyRunnable as a parameter
        Thread thread = new Thread(myRunnable);
        // Start the thread
        thread.start();
    }
}
```

3. Describe the Different States of a Thread?

During its lifecycle, a thread in Java can exist in several states, each representing a different stage of its execution:

- New - When a thread is created but not yet started, it is in the "New" state. The start() method has not been called.
- Runnable - After calling start(), the thread moves to the "Runnable" state. It is eligible to run, but the scheduler has not yet selected it.
- Running - When the thread scheduler selects a thread for execution, it enters the "Running" state. The thread's code is actively being executed by the CPU.
- Blocked (or Waiting) - A thread enters this state when it is temporarily inactive, typically waiting for a resource or condition, such as I/O completion or acquiring a lock.
- Timed Waiting - Threads pause execution for a specified period, e.g., via sleep() or join() with a timeout.
- Terminated - The final state, indicating the thread has completed execution, either successfully or by being explicitly stopped.

Real-World Use Case:

Imagine you're building a chat application. Each user's interaction with the chat interface is handled by a separate thread. Here's how the different thread states might come into play:

- New: When a new user opens the chat application, a thread is created to handle their interactions.
- Runnable: Once the user logs in, their thread transitions to the "Runnable" state, awaiting execution by the thread scheduler.
- Running: When the scheduler selects the user's thread for execution, it enters the "Running" state, actively processing messages and user inputs.
- Blocked: If the user sends a message and it needs to be sent over the network, the thread might enter the "Blocked" state while waiting for the network I/O operation to complete.
- Terminated: Once the user logs out or closes the application, their thread enters the "Terminated" state, indicating the end of its lifecycle.

4. What Is a Daemon Thread and how to create a Daemon Thread? What are the real use cases of Daemon thread?

A daemon thread in Java is a special type of thread that runs in the background, providing services to other threads or performing tasks that are not critical to the application's lifecycle. Unlike user threads, daemon threads do not prevent the JVM from exiting when all user threads have finished executing. They are automatically terminated when all non-daemon threads have exited or when the JVM shuts down.

Real-World Use Cases:

- Garbage Collection: Daemon threads are commonly used by the JVM for tasks such as garbage collection, reclaiming unused memory while the application continues to execute.
- Monitoring Services: Daemon threads can monitor system resources or perform periodic maintenance tasks in the background, such as disk usage, network traffic, or performance metrics.
- Background Tasks: In server-side applications, daemon threads can handle background tasks such as logging, sending periodic heartbeats to a monitoring system, or cleaning up temporary files.

By using daemon threads, you can offload non-critical tasks to background threads, improving the overall responsiveness and efficiency of your Java applications.

To create a daemon thread in Java, you simply need to set the daemon flag to true before starting the thread.

```
public class DaemonThreadExample {
    public static void main(String[] args) {
        Thread daemonThread = new Thread(() -> {
            while (true) {
                System.out.println("Daemon thread is running...");
                try {
                    Thread.sleep(1000); // Simulate some task
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
            }
        });
        daemonThread.setDaemon(true); // Set the thread as daemon
        daemonThread.start(); // Start the daemon thread
    }
}
```

5. What Is the Difference Between the Runnable and Callable Interfaces?

The Runnable and Callable interfaces in Java both represent tasks that can be executed by threads, but they

differ in several key aspects:

Return Value

- Runnable does not return a result when its task completes. Its run() method has a void return type.
- Callable can return a result when its task completes, via its call() method.

Exception Handling

- In Runnable, any checked exceptions thrown by run() must be caught and handled within the method itself.
- Callable allows call() to throw checked exceptions, which can be caught and handled by the calling code.

Usage with Executors

- Runnable is primarily used with the Executor framework for executing tasks asynchronously in a thread pool.
- Callable is used with ExecutorService, which provides additional features such as submitting tasks that return values and handling exceptions.

```
package java.util.concurrent;

/**
 * A task that returns a result and may throw an exception.
 * Implementors define a single method with no arguments called
 * {@code call}.
 *
 * <p>The {@code Callable} interface is similar to {@link
 * java.lang.Runnable}, in that both are designed for classes whose
 * instances are potentially executed by another thread.  A
 * {@code Runnable}, however, does not return a result and cannot
 * throw a checked exception.
 *
 * <p>The {@link Executors} class contains utility methods to
 * convert from other common forms to {@code Callable} classes.
 *
 * @see Executor
 * @since 1.5
 * @author Doug Lea
 * @param <V> the result type of method {@code call}
 */
@FunctionalInterface
public interface Callable<V> {
    /**
     * Computes a result, or throws an exception if unable to do so.
     *
     * @return computed result
     * @throws Exception if unable to compute a result
     */
    V call() throws Exception;
}
```

6. What Is Java Memory Model (JMM) and its purpose?

The Java Memory Model (JMM) defines how Java programs interact with memory, including how threads interact through memory when executing concurrently. Its purpose is to ensure that concurrent Java programs behave predictably and consistently across different platforms and JVM implementations.

The Java memory model specifies how the Java virtual machine works with the computer's memory (RAM).

Each Java thread operates with its own thread stack, which maintains the call stack - a record of invoked

methods leading to the current execution point. Local variables for each method reside in the thread stack and are exclusive to that thread.

In contrast, the heap stores all objects created within the Java application, irrespective of the creating thread. This includes object versions of primitive types. Whether an object is assigned to a local or a member variable, it's stored in the heap.

7. What is Data Race Condition and Visibility problem?

Data Race Condition:

A data race condition occurs when two or more threads access shared data concurrently, and at least one of the threads modifies the data. This can lead to unpredictable behavior and incorrect results due to the non-deterministic interleaving of thread operations. Data races can occur when threads access shared variables without proper synchronization, leading to inconsistent or corrupted data.

Visibility Problem:

The visibility problem refers to the issue where changes made by one thread to shared variables may not be immediately visible to other threads. In Java, each thread has its own cache of variables, and changes made by one thread may not be immediately propagated to other threads' caches. This can result in one thread not seeing the most recent value of a shared variable, leading to incorrect behavior.

Real-World Use Case:

Consider a scenario where multiple threads are updating a shared counter variable. Without proper synchronization, such as using locks or volatile variables, data races can occur. For instance, one thread may read the counter's value while another thread is in the process of updating it, leading to inconsistent or incorrect results. Additionally, if one thread updates the counter, other threads may not immediately see the updated value due to visibility issues.

```
public class Counter {
    private int count;

    public void increment() {
        count++; // Not thread-safe
    }

    public int getCount() {
        return count; // Not thread-safe
    }
}
```

8. What is difference between Volatile, Atomic and Synchronized?

- Volatile: Indicates that a variable's value may be modified by multiple threads, and changes should be immediately visible to other threads. It does not provide atomicity or mutual exclusion, but ensures reads/writes are not reordered by the compiler or processor, addressing the visibility problem.
- Atomic: The java.util.concurrent.atomic package provides classes like AtomicInteger, AtomicLong, and AtomicReference, offering atomic operations without explicit synchronization, using compare-and-set (CAS) to ensure updates are performed atomically and without data races.
- Synchronized: Creates a mutually exclusive block of code that can only be executed by one thread at a time. It provides both atomicity and mutual exclusion, ensuring thread safety by preventing data races and ensuring consistent access to shared resources.

Real-World Use Cases:

- Volatile: Use when you have a variable shared among multiple threads and want changes to be immediately visible without the need for locking, e.g., a flag indicating whether a thread should continue running.
- Atomic: Use for atomic operations like incrementing, updating, or comparing variables in a thread-safe manner, e.g., a global counter or thread-safe caching.
- Synchronized: Use when only one thread should access a block of code or modify shared resources at a time, e.g., updating shared data structures or critical operations requiring exclusive access.

```
// Volatile
public class VolatileExample {
    private volatile boolean flag;

    public void setFlag(boolean value) {
        flag = value;
    }

    public boolean getFlag() {
        return flag;
    }
}

// Atomic
import java.util.concurrent.atomic.AtomicInteger;

public class AtomicExample {
    private AtomicInteger counter = new AtomicInteger(0);

    public void increment() {
        counter.incrementAndGet();
    }

    public int getCount() {
        return counter.get();
    }
}

// Synchronized
public class SynchronizedExample {
    private int count;

    public synchronized void increment() {
        count++;
    }

    public synchronized int getCount() {
        return count;
    }
}
```

9. Explain about Deadlock, Livelock, and Starvation conditions in Java Multithreading?

Deadlock

Deadlock occurs when two or more threads are permanently blocked, waiting for each other to release resources, leading to a circular dependency.

Example: Thread A locks Resource 1 and waits for Resource 2. Thread B locks Resource 2 and waits for Resource 1. Both threads wait indefinitely -> Deadlock occurs.

Prevention Strategies:

- Consistent Lock Ordering: Always acquire locks in a fixed order.
- Timeouts: Use tryLock() with a timeout to avoid indefinite waiting.
- Deadlock Detection: Implement a resource hierarchy or use a tool like ThreadMXBean to detect deadlocks.

Livelock

Livelock occurs when threads keep changing states in response to each other without making any actual progress.

Example: Two people trying to pass through a narrow hallway. Both step aside at the same time, then move back, repeating indefinitely. The system is not blocked but also not progressing.

Prevention Strategies:

- Random Delays: Introduce a random wait time before retrying.
- Priority Inversion Handling: Ensure a fallback mechanism to break the cycle.
- Use Message Passing: Use a queue-based or event-driven approach instead of retries.

Starvation

Starvation happens when a thread is perpetually denied access to resources because other high-priority threads keep executing.

Example: A low-priority thread is never scheduled because higher-priority threads keep running. This happens in priority-based scheduling or biased locking scenarios.

Prevention Strategies:

- Fair Scheduling Policies: Use ReentrantLock(true) for fair access to locks.
- Priority Adjustments: Dynamically adjust thread priorities if a thread is waiting too long.
- Avoid Starvation-Prone Algorithms: Implement fair resource allocation mechanisms like FIFO-based locks.

Real-World Use Cases

- Deadlock: In database transactions, two queries trying to lock rows in different order can cause deadlocks.
- Livelock: In distributed systems, multiple nodes retrying conflict resolution can lead to infinite state changes without progress.
- Starvation: In web servers, high-priority API requests may dominate CPU cycles, starving low-priority background tasks.

10. What is Java Executor Service and type of threadpools which can be created with Java Executor service?

The Java Executor Service is a higher-level concurrency utility that provides a simplified interface for managing threads and executing tasks asynchronously. It abstracts away the complexity of thread management.

Types of Thread Pools:

- FixedThreadPool: Maintains a fixed number of threads. If a task is submitted when all threads are busy, it is queued until a thread becomes available. Suitable for a predictable number of tasks with a limit on concurrent threads.

- CachedThreadPool: Dynamically adjusts the number of threads based on workload. Threads are created as needed and reused if available; unused threads are terminated after a specified idle timeout. Ideal for a large number of short-lived tasks or variable workloads.
- SingleThreadPool: Maintains a single thread, ensuring tasks execute sequentially in submission order. If the thread terminates due to an exception, a new one replaces it. Useful when tasks must execute sequentially.
- ScheduledThreadPool: Used for executing tasks at a specific time or with a fixed delay between executions, via schedule(), scheduleAtFixedRate(), and scheduleWithFixedDelay(). Ideal for periodic tasks like background jobs.

Real-World Use Case Example:

Suppose you're developing a web server application where multiple client requests need to be processed concurrently. You can use a FixedThreadPool to limit the maximum number of concurrent connections and ensure efficient resource utilization. Additionally, you can use a ScheduledThreadPool to schedule periodic tasks like log rotation or cache cleanup.

```
// Fixed Thread Pool
ExecutorService fixedThreadPool = Executors.newFixedThreadPool(3);

System.out.println("\nFixed Thread Pool:");
for (int i = 1; i <= 5; i++) {
    final int taskId = i;
    fixedThreadPool.execute(() -> {
        System.out.println("Fixed Thread Task " + taskId +
                " executed by Thread: " +
                Thread.currentThread().getName());
    });
}

// Cached Thread Pool
ExecutorService cachedThreadPool = Executors.newCachedThreadPool();

System.out.println("\nCached Thread Pool:");
for (int i = 1; i <= 5; i++) {
    final int taskId = i;
    cachedThreadPool.execute(() -> {
        System.out.println("Cached Thread Task " + taskId +
                " executed by Thread: " +
                Thread.currentThread().getName());
    });
}

// Single Thread Executor
ExecutorService singleThreadExecutor = Executors.newSingleThreadExecutor();

System.out.println("\nSingle Thread Executor:");
for (int i = 1; i <= 5; i++) {
    final int taskId = i;
    singleThreadExecutor.execute(() -> {
        System.out.println("Single Thread Task " + taskId +
                " executed by Thread: " +
                Thread.currentThread().getName());
    });
}

// Scheduled Thread Pool
ScheduledExecutorService scheduledThreadPool = Executors.newScheduledThreadPool(2);

scheduledThreadPool.schedule(() ->
        System.out.println("Scheduled Task"),
        5, TimeUnit.SECONDS);

scheduledThreadPool.scheduleAtFixedRate(() ->
        System.out.println("Scheduled Task"),
        0, 1, TimeUnit.SECONDS);
```

11. What Are Executor and ExecutorService and Differences Between them?

- Executor: A simple interface that decouples task submission from task execution. It defines a single method, execute(Runnable command), which executes the given command in a thread pool.
- ExecutorService: A subinterface of Executor that represents a more complete version, providing methods to manage the lifecycle of the thread pool, submit tasks for execution, and obtain Futures representing task results.

Differences:

- Lifecycle Management: Executor provides no lifecycle management. ExecutorService adds shutdown(), shutdownNow(), and awaitTermination().
- Task Submission and Execution: Both provide ways to submit tasks (execute(Runnable) in Executor; submit(Runnable) and submit(Callable<T>) in ExecutorService).
- Task Result Handling: Executor has no way to obtain a task's result directly. ExecutorService supports submit(Callable<T>), returning a Future for result retrieval.
- Termination: Executor provides no termination methods. ExecutorService provides shutdown(), shutdownNow(), and awaitTermination().

Use Case Scenarios:

- Use Executor when you only need to execute tasks asynchronously without managing the thread pool's lifecycle.
- Use ExecutorService when you need more control, such as managing lifecycle, submitting tasks with results, and awaiting termination.

12. What is Producer-Consumer problem and how to resolve it?

The Producer-Consumer problem is a classic synchronization problem in concurrent programming, where there are two types of threads: producers and consumers. Producers generate data or items, and consumers consume or process these items.

The challenge is to ensure that producers do not produce items when the buffer is full, and consumers do not consume items when the buffer is empty. It requires efficient coordination and synchronization between producers and consumers to avoid race conditions and ensure proper resource management.

One effective solution is using the BlockingQueue interface in Java, which provides a thread-safe queue implementation with blocking operations.

Real-World Use Case Example:

Consider a scenario where a web server application receives requests from clients and processes them concurrently. Producers enqueue incoming requests into a BlockingQueue, and multiple worker threads (consumers) dequeue requests from the queue and process them. This ensures efficient request handling without the risk of data corruption or race conditions.

```
import java.util.concurrent.BlockingQueue;
import java.util.concurrent.ArrayBlockingQueue;

public class ProducerConsumerExample {
    private static final int BUFFER_SIZE = 10;
    private static BlockingQueue<Integer> buffer = new ArrayBlockingQueue<>(BUFFER_SIZE);

    public static void main(String[] args) {
        Thread producerThread = new Thread(new Producer());
        Thread consumerThread = new Thread(new Consumer());

        producerThread.start();
        consumerThread.start();
    }

    static class Producer implements Runnable {
        public void run() {
            try {
                int item = 1;
                while (true) {
                    buffer.put(item++);
                    System.out.println("Produced item: " + (item - 1));
                    Thread.sleep(1000); // Simulate production time
                }
            } catch (InterruptedException e) {
                Thread.currentThread().interrupt();
            }
        }
    }

    static class Consumer implements Runnable {
        public void run() {
            try {
                while (true) {
                    int item = buffer.take();
                    System.out.println("Consumed item: " + item);
                    Thread.sleep(2000); // Simulate processing time
                }
            } catch (InterruptedException e) {
                Thread.currentThread().interrupt();
            }
        }
    }
}
```

13. What is ThreadLocal and what are the advantages of using ThreadLocal?

ThreadLocal is a class in Java that provides thread-local variables. Each thread accessing a ThreadLocal variable has its own, independently initialized copy of the variable. ThreadLocal variables are typically used to store data that is specific to a particular thread and should not be shared among multiple threads.

Advantages of Using ThreadLocal:

- Thread Isolation: Maintains separate copies of variables for each thread, preventing concurrent threads from accessing or modifying each other's data, enhancing thread safety.
- Thread Confinement: Supports confining data to a specific thread's execution context, encapsulating thread-specific data without needing synchronization or locks.

- Reduced Synchronization Overhead: Since each thread has its own copy, there is no need for explicit synchronization when accessing these variables, improving performance and scalability.
- Avoids Memory Leaks: ThreadLocal variables are garbage collected along with the thread that owns them, preventing memory leaks from shared resources across threads.
- Contextual Data Storage: Often used to store contextual data related to the current thread's execution context, such as user sessions, transaction contexts, or request-specific data in web applications.

Real-World Use Case Example:

In a web application framework, ThreadLocal variables are commonly used to store user authentication information or request context data. For example, in a servlet-based web application, a ThreadLocal variable can be used to store the current HttpServletRequest object, allowing components downstream to access request-specific information without passing it explicitly.

By leveraging ThreadLocal, developers can ensure that each thread has access to its own instance of the data, eliminating the need for complex parameter passing or synchronization mechanisms. This improves code readability, simplifies concurrency management, and enhances overall application performance.

```
public class ThreadLocalExample {
    private static ThreadLocal<Integer> threadLocal = new ThreadLocal<>();

    public static void main(String[] args) {
        // Set thread-local value for main thread
        threadLocal.set(100);

        // Create and start a new thread
        Thread thread = new Thread(() -> {
            // Access thread-local value in the new thread
            System.out.println("ThreadLocal value in new thread: " + threadLocal.get());
        });
        thread.start();
    }
}
```

14. What are Virtual Threads in Java 21?

Virtual Threads are lightweight, user-mode threads managed by the Java Virtual Machine (JVM). Unlike traditional OS threads (kernel threads), virtual threads are managed entirely by the JVM and can be created and scheduled more efficiently.

Virtual threads are lightweight compared to traditional OS threads, requiring fewer system resources for creation and management. This makes them suitable for applications with a large number of concurrent tasks.

Use virtual threads in high-throughput concurrent applications, especially those that consist of a great number of concurrent tasks that spend much of their time waiting. Server applications are examples of high-throughput applications because they typically handle many client requests that perform blocking I/O operations such as fetching resources.

Virtual threads are not faster threads; they do not run code any faster than platform threads. They exist to provide scale (higher throughput), not speed (lower latency).

```
public static void main(String[] args) {
    Thread virtualThread = Thread.startVirtualThread(() -> {
        System.out.println("Running task with virtual thread: "
                + Thread.currentThread().getName());
    });

    // Waiting for virtual threads to complete
    try {
        virtualThread.join();
    } catch (InterruptedException e) {
        e.printStackTrace();
    }
}
```

15. What is the Fork/Join Framework in Java?

The Fork/Join Framework is designed to efficiently parallelize divide-and-conquer algorithms, where a problem is recursively split into smaller independent subproblems, solved in parallel, and then combined to form the final result.

Key Components of the Fork/Join Framework

1. ForkJoinPool - A specialized implementation of ExecutorService optimized for recursive parallelism. It manages a pool of worker threads and efficiently distributes tasks using work stealing. Unlike fixed-thread pools, it dynamically adjusts workload distribution for optimal CPU utilization.

2. ForkJoinTask - A lightweight task that can be forked (split) into subtasks and joined (merged) once completed. ForkJoinTask is an abstract class, and its two main implementations are:
- RecursiveTask<T> -> Returns a result.
- RecursiveAction -> Performs an operation but does not return a result.

3. Work Stealing Algorithm - Instead of waiting for tasks, idle worker threads dynamically steal tasks from the tail of another thread's deque (double-ended queue). This improves load balancing and ensures efficient parallel execution.

How It Works (Workflow)

1. A main task is submitted to ForkJoinPool.
2. The task divides itself into smaller subtasks (forking).
3. Each subtask is executed independently in parallel.
4. Once all subtasks finish, their results are merged (joining) to get the final output.
5. If a worker thread runs out of tasks, it steals work from other busy threads.

Advantages of Fork/Join Framework

- Parallelism - Automates parallel execution of recursive tasks, reducing manual thread management.
- Load Balancing - Uses work-stealing to evenly distribute tasks across worker threads.
- Scalability - Leverages multiple processor cores for efficient computation of large datasets.

Real-World Use Cases

- Recursive Algorithms - Parallelizing Merge Sort, Quick Sort, Matrix Multiplication, Tree Traversal.
- Data Processing - Image processing, Matrix operations, Parallel reductions.
- Concurrent Computations - Tasks that can be broken into independent parallel sub-tasks.

The code below demonstrates how to leverage the Fork/Join Framework to parallelize the computation of the sum of array elements, improving performance by utilizing multiple threads efficiently.

```
import java.util.concurrent.RecursiveTask;
import java.util.concurrent.ForkJoinPool;

public class ForkJoinSumExample {
    // Define a task to compute the sum of elements in a given range of an array
    static class SumTask extends RecursiveTask<Integer> {
        private static final int THRESHOLD = 10; // Threshold for splitting tasks
        private int[] array;
        private int start, end;

        public SumTask(int[] array, int start, int end) {
            this.array = array;
            this.start = start;
            this.end = end;
        }

        protected Integer compute() {
            if (end - start <= THRESHOLD) {
                int sum = 0;
                for (int i = start; i < end; i++)
                    sum += array[i];
                return sum;
            } else {
                int mid = (start + end) / 2;
                SumTask leftTask = new SumTask(array, start, mid);
                SumTask rightTask = new SumTask(array, mid, end);

                leftTask.fork();
                rightTask.fork();

                return leftTask.join() + rightTask.join();
            }
        }
    }

    public static void main(String[] args) {
        int[] array = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};

        ForkJoinPool pool = ForkJoinPool.commonPool();

        int result = pool.invoke(new SumTask(array, 0, array.length));

        System.out.println("Sum: " + result);
    }
}
```

16. What Is the Purpose of the Wait, Notify and NotifyAll Methods?

The wait(), notify(), and notifyAll() methods in Java are used for inter-thread communication. They allow threads to wait for a certain condition to be met and then be notified when it is.

- wait(): Makes a thread wait until another thread notifies it. Typically called inside a synchronized block or method. Threads call wait() when they need to wait for a condition to be satisfied before proceeding with their execution - for example, a consumer thread may wait for a producer thread to produce data.
- notify(): Wakes up a single waiting thread that is waiting on the same object's monitor. Threads call notify() to signal other threads that a particular condition has been met or a resource is available for consumption - for example, a producer thread may notify a consumer thread when new data is available.
- notifyAll(): Wakes up all waiting threads that are waiting on the same object's monitor. Used when multiple threads are waiting for the same condition or resource, ensuring all waiting threads can compete

for the shared resource or condition; generally used to prevent potential deadlock situations.

Important Points:

- wait(), notify(), and notifyAll() must be called from within a synchronized block or method to ensure proper thread coordination and avoid race conditions.
- These methods are used to implement the classic producer-consumer pattern, where producers produce data and notify consumers when data is available for consumption.
- It's important to use these methods carefully to prevent potential issues such as deadlock or livelock.

```
public class WaitNotifyExample {
    private static final Object lock = new Object(); // Object used as a monitor for synchroni…
    private static boolean condition = false; // Shared condition variable

    public static void main(String[] args) {
        // Consumer thread
        Thread consumerThread = new Thread(() -> {
            synchronized (lock) {
                while (!condition) { // Wait until condition is true
                    try {
                        lock.wait(); // Wait for notification
                    } catch (InterruptedException e) {
                        e.printStackTrace();
                    }
                }
                // Condition is true, consume the resource
                System.out.println("Consumer: Resource consumed");
            }
        });

        // Producer thread
        Thread producerThread = new Thread(() -> {
            // Produce the resource
            System.out.println("Producer: Resource produced");
            synchronized (lock) {
                condition = true; // Set condition to true
                lock.notify(); // Notify waiting threads
            }
        });

        // Start consumer and producer threads
        consumerThread.start();
        producerThread.start();
    }
}
```

17. What Is the Purpose of Thread.yield()?

The purpose of the Thread.yield() method in Java is to hint to the thread scheduler that the current thread is willing to relinquish its current use of the CPU. When a thread calls yield(), it suggests that the scheduler should move it from the running state back to the runnable state, allowing other threads of the same or higher priority to execute.

However, the exact behavior of Thread.yield() is platform-dependent and not guaranteed. It's up to the thread scheduler to decide whether to act on this hint or not, so it may not have any effect at all. Generally, Thread.yield() is used in scenarios where a thread needs to wait for some condition but doesn't want to block completely (like with sleep() or wait()), allowing other threads a chance to run.

The output of the code below can be different every time you run it because the behavior of yield() is

non-deterministic and platform dependent as well.

```
Runnable r = () -> {
    int counter = 0;
    while (counter < 2) {
        System.out.println(Thread.currentThread()
                .getName());
        counter++;
        Thread.yield();
    }
};
new Thread(r).start();
new Thread(r).start();
```

18. What is the purpose of Semaphore?

The purpose of Semaphore is to control access to a shared resource or a pool of resources by multiple threads concurrently. It provides a way to limit the number of threads that can access the resource simultaneously, thus preventing resource contention and managing concurrent access in a controlled manner.

It is a synchronization construct that controls access to a shared resource by using a counter. If the counter is greater than zero, access is allowed. If it is zero, access is denied. The counter keeps track of permits that allow access to the shared resource. To access the resource, a thread must acquire a permit from the semaphore. When the thread no longer needs access to the shared resource, it releases the permit, allowing another thread to acquire it.

In Java, the Semaphore class in the java.util.concurrent package implements this mechanism, so you don't have to implement your own semaphores.

In the example below, each thread acquires a permit using semaphore.acquire() before accessing the shared resource and releases the permit using semaphore.release() after completing its work.

As a result, only 2 threads are allowed to access the resource concurrently, while other threads wait for a permit to become available.

```
import java.util.concurrent.Semaphore;

public class SemaphoreExample {
    private static final int NUM_THREADS = 5;
    private static final Semaphore semaphore = new Semaphore(2); // Allow only 2 threads to ac…

    public static void main(String[] args) {
        // Create and start multiple threads
        for (int i = 0; i < NUM_THREADS; i++) {
            Thread thread = new Thread(() -> {
                try {
                    semaphore.acquire(); // Acquire permit
                    System.out.println(Thread.currentThread().getName() + " is accessing the r…
                    Thread.sleep(1000); // Simulate resource usage
                    System.out.println(Thread.currentThread().getName() + " released the resou…
                    semaphore.release(); // Release permit
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
            });
            thread.start();
        }
    }
}
```

19. Approaches to write efficient and scalable code in Java concurrency?

Here are some key principles for writing efficient concurrent code:

Use High-Level Concurrency APIs

Example: Consider a web server handling incoming HTTP requests concurrently. Instead of managing threads manually, you can use Java's Executors to create a thread pool, where each thread processes a request independently.
Explanation: Executors abstract thread creation and management, making it easier to handle concurrent tasks without worrying about low-level details.

Choose the Right Concurrency Mechanisms

Example: In a financial trading system, multiple threads need to process trades concurrently. Here, you might use a ConcurrentHashMap to store and access trade data concurrently, ensuring thread safety.
Explanation: ConcurrentHashMap provides efficient concurrent access to shared data, preventing data corruption and ensuring consistency.

Avoid Unnecessary Synchronization

Example: In a multi-threaded logging system, instead of synchronizing access to the log file using locks, you can use a thread-safe logger implementation provided by a logging framework like Log4j.
Explanation: By avoiding unnecessary synchronization, you reduce contention and improve the system's scalability and performance.

Fine-Grained Locking

Example: In a banking application, multiple threads may concurrently access different accounts for transactions. Instead of locking the entire account database, you can use fine-grained locking to lock only the specific accounts being accessed.
Explanation: Fine-grained locking minimizes contention and allows multiple threads to access different parts of the system concurrently.

Reduce Lock Duration

Example: In a multi-threaded game, instead of holding a lock while rendering a frame, you can acquire the lock just before updating shared game state and release it immediately afterward.
Explanation: By reducing lock duration, you minimize the time threads spend waiting for locks, improving concurrency and responsiveness.

Concurrency Design Patterns

Example: In a messaging application, you can use the Producer-Consumer pattern to decouple message producers from consumers, allowing multiple threads to send and receive messages concurrently.
Explanation: Concurrency design patterns provide reusable solutions to common concurrency problems, promoting code maintainability and scalability.

Asynchronous and Non-Blocking IO

Example: In a web server, you can use non-blocking IO to handle multiple client connections concurrently without blocking threads, improving server scalability.
Explanation: Asynchronous and non-blocking IO enables efficient resource utilization and responsiveness in IO-bound applications.

Optimize Data Access

Example: In a high-frequency trading system, you can optimize access to order book data using a lock-free data structure like a ConcurrentSkipListMap, allowing multiple threads to access the order book concurrently with minimal contention.
Explanation: Optimizing data access patterns enhances concurrency and scalability in multi-threaded applications.

Race Conditions & Synchronization

Module 03 - Multithreading

Understanding data corruption in concurrent access and how to prevent it.

What is a Race Condition?

A race condition occurs when multiple threads access shared data simultaneously, and the outcome depends on the unpredictable timing of thread execution. This leads to data corruption and bugs that are hard to reproduce.

The Problem

Consider two threads incrementing a counter. Each increment seems atomic but is actually three CPU operations:

1. READ: Load current value from memory into CPU register
2. MODIFY: Add 1 to the register value
3. WRITE: Store the new value back to memory

If threads interleave these operations, updates get lost. For example, both threads READ counter as 0, each computes register = 0 + 1 = 1, and both WRITE 1 back - so counter ends up 1 instead of the expected 2.

```
public class Counter {
    private int count = 0;
    public void increment() {
        count++; // NOT ATOMIC!
    }
}
// Two threads call increment()
```

The synchronized Keyword

Java's synchronized keyword provides mutual exclusion - only one thread can execute a synchronized block at a time. This prevents race conditions by ensuring atomic access to shared data.

Synchronized Method

```
public class Counter {
    private int count = 0;

    // Only one thread can execute this at a time
    public synchronized void increment() {
        count++;  // Now atomic!
    }

    public synchronized int getCount() {
        return count;
    }
}
```

Synchronized Block

```
public class Counter {
    private int count = 0;
    private final Object lock = new Object();

    public void increment() {
        synchronized (lock) {  // Acquire lock
            count++;
        }  // Release lock
    }
}
```

How It Works

- Every object in Java has an intrinsic lock (monitor).
- synchronized(obj) acquires obj's lock before executing.
- Only one thread can hold a lock at a time.
- Other threads trying to acquire the same lock will BLOCK.
- The lock is automatically released when the block exits (even on exception).

What to Synchronize On

Bad: Wrong Lock Object

```
// WRONG: Different lock objects!
public void increment() {
    synchronized (new Object()) {
        count++;
    }
}
// Each call creates a new lock
```

Good: Same Lock Object

```
private final Object lock = new Object();

public void increment() {
    synchronized (lock) {
        count++;
    }
}
```

Lock Object Options

- this - The instance itself (used by synchronized methods).
- ClassName.class - For static synchronized methods.
- Private final object - Best practice for finer control.

Never Synchronize On

- String literals (interned and shared)
- Boxed primitives like Integer (cached instances)
- Objects that might be reassigned
- null (throws NullPointerException)

Atomic Variables

For simple operations like increment/decrement, java.util.concurrent.atomic classes provide lock-free thread safety using CPU-level atomic instructions (CAS - Compare And Swap).

```
import java.util.concurrent.atomic.AtomicInteger;

public class Counter {
    private AtomicInteger count = new AtomicInteger(0);

    public void increment() {
        count.incrementAndGet();  // Atomic, no lock needed
    }

    public int getCount() {
        return count.get();
    }
}
```

Common Atomic Classes

- AtomicInteger - Thread-safe int operations
- AtomicLong - Thread-safe long operations
- AtomicBoolean - Thread-safe boolean operations
- AtomicReference\<T\> - Thread-safe object reference

When to Use Atomic vs Synchronized

- Use Atomic when: Single variable, simple operations (increment, compare-and-set).
- Use Synchronized when: Multiple variables, compound operations, complex logic.

The volatile Keyword

volatile ensures visibility of changes across threads but does NOT provide atomicity.

```
public class Flag {
    private volatile boolean stopped = false;

    public void stop() {
        stopped = true;  // Visible to all threads immediately
    }

    public void run() {
        while (!stopped) {  // Reads fresh value each time
            doWork();
        }
    }
}
```

volatile Does NOT Fix This

```
private volatile int count = 0;

public void increment() {
    count++;  // STILL NOT THREAD-SAFE!
}
// count++ is still READ-MODIFY-WRITE
```

When to Use volatile

- Simple flags (boolean start/stop signals)
- Variables written by one thread, read by many
- Reference to immutable objects

Common Interview Questions

Q: What is a race condition?

A race condition occurs when the correctness of a program depends on the relative timing of multiple threads. It happens when threads access shared mutable state without proper synchronization.

Q: Difference between synchronized and volatile?

- synchronized: Provides atomicity AND visibility. Only one thread executes at a time.
- volatile: Provides visibility only. Changes are immediately visible, but compound operations are still non-atomic.

Q: Is i++ thread-safe?

No. i++ is a compound operation (read-increment-write). Even on volatile variables, it is not atomic. Use AtomicInteger.incrementAndGet() or a synchronized block.

Q: What happens if two threads synchronize on different objects?

No mutual exclusion occurs. Both threads can execute simultaneously because they hold different locks. Always synchronize on the SAME object to protect shared data.

Key Takeaways

- Race conditions occur when threads access shared data without synchronization.
- Even simple operations like i++ are not atomic (read-modify-write).
- synchronized provides mutual exclusion - only one thread at a time.
- Always synchronize on the SAME lock object to protect shared data.
- AtomicInteger and friends provide lock-free thread safety for simple operations.
- volatile provides visibility but NOT atomicity.
- Keep synchronized blocks as small as possible for performance.

Deadlock Detection & Prevention

Module 03 - Multithreading

Understanding why threads get stuck waiting forever and how to prevent it.

What is a Deadlock?

A deadlock is a situation where two or more threads are blocked forever, each waiting for a resource held by the other. The threads are stuck in a circular wait, and the program hangs indefinitely.

Classic Deadlock Scenario

- Thread 1 holds Lock A and waits for Lock B.
- Thread 2 holds Lock B and waits for Lock A.
- Neither can proceed - they will wait forever!

Four Conditions for Deadlock

All four of these conditions must be present simultaneously for a deadlock to occur. Breaking any one of them prevents deadlock.

1. Mutual Exclusion - Resources cannot be shared; only one thread can hold a lock at a time.
2. Hold and Wait - Thread holds one resource while waiting for another.
3. No Preemption - Resources cannot be forcibly taken from a thread.
4. Circular Wait - Chain of threads, each waiting for a resource held by the next thread in the chain.

Prevention Strategy

The most practical approach is to break the Circular Wait condition by always acquiring locks in a consistent, global order.

```
// Thread 1 runs method1()
synchronized (lockA) {
    // Thread 1 holds A
    synchronized (lockB) { // Waits for B
        doWork();
    }
}

// Thread 2 runs method2()
synchronized (lockB) {
    // Thread 2 holds B
    synchronized (lockA) { // Waits for A
        doWork();
    }
}
```

Deadlock Code Example

The Problem: Inconsistent Lock Order

```
public class DeadlockExample {
    private final Object lockA = new Object();
    private final Object lockB = new Object();

    public void method1() {
        synchronized (lockA) {           // Thread 1: Acquires A first
            System.out.println("Holding A, waiting for B");
            synchronized (lockB) {       // Thread 1: Waits for B
                doWork();
            }
        }
    }

    public void method2() {
        synchronized (lockB) {           // Thread 2: Acquires B first
            System.out.println("Holding B, waiting for A");
            synchronized (lockA) {       // Thread 2: Waits for A
                doWork();                // DEADLOCK if both run!
            }
        }
    }
}
```

The Solution: Consistent Lock Order

```
public class NoDeadlock {
    private final Object lockA = new Object();
    private final Object lockB = new Object();

    public void method1() {
        synchronized (lockA) {           // Always acquire A first
            synchronized (lockB) {
                doWork();
            }
        }
    }

    public void method2() {
        synchronized (lockA) {           // Also A first - same order!
            synchronized (lockB) {
                doWork();
            }
        }
    }
}
```

Key Insight

When both threads acquire locks in the same order (A then B), circular wait is impossible. Thread 2 cannot hold B while Thread 1 holds A, because Thread 2 must first acquire A.

Prevention Strategies

1. Lock Ordering

Define a global order for all locks and always acquire them in that order.

```
// Use object hash codes or IDs to determine order
public void transferMoney(Account from, Account to, int amount) {
    Account first = from.getId() < to.getId() ? from : to;
    Account second = from.getId() < to.getId() ? to : from;

    synchronized (first) {
        synchronized (second) {
            from.withdraw(amount);
            to.deposit(amount);
        }
    }
}
```

2. Lock Timeout with tryLock()

Use ReentrantLock.tryLock() with a timeout to avoid waiting forever.

```
import java.util.concurrent.locks.ReentrantLock;
import java.util.concurrent.TimeUnit;

public class TimeoutExample {
    private final ReentrantLock lockA = new ReentrantLock();
    private final ReentrantLock lockB = new ReentrantLock();

    public boolean doWork() {
        try {
            if (lockA.tryLock(1, TimeUnit.SECONDS)) {
                try {
                    if (lockB.tryLock(1, TimeUnit.SECONDS)) {
                        try {
                            // Do work with both locks
                            return true;
                        } finally {
                            lockB.unlock();
                        }
                    }
                } finally {
                    lockA.unlock();
                }
            }
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }
        return false;  // Could not acquire both locks
    }
}
```

3. Avoid Nested Locks

Restructure code to avoid holding multiple locks simultaneously.

Nested Locks (Risk):

```
synchronized (lockA) {
    synchronized (lockB) {
        doWork();
    }
}
```

Sequential Locks (Safer):

```
Data snapshot;
synchronized (lockA) {
    snapshot = copyData();
}
synchronized (lockB) {
    processData(snapshot);
}
```

Detecting Deadlocks

Using jstack (Command Line)

```
# Find the Java process ID
jps

# Get thread dump with deadlock detection
jstack <pid>

# Output shows:
# Found one Java-level deadlock:
# =============================
# "Thread-1":
#   waiting to lock monitor 0x00007f8b4c0062b8
#   which is held by "Thread-0"
```

Programmatic Detection

```
import java.lang.management.*;

public class DeadlockDetector {
    public static void detectDeadlock() {
        ThreadMXBean bean = ManagementFactory.getThreadMXBean();
        long[] deadlockedThreads = bean.findDeadlockedThreads();

        if (deadlockedThreads != null) {
            System.out.println("Deadlock detected!");
            for (long id : deadlockedThreads) {
                ThreadInfo info = bean.getThreadInfo(id);
                System.out.println(info.getThreadName() +
                    " blocked on " + info.getLockName());
            }
        }
    }
}
```

Common Interview Questions

Q: What is a deadlock?

A deadlock is a situation where two or more threads are blocked forever, each waiting for a resource held by the other. The threads form a circular dependency and cannot make progress.

Q: What are the four conditions for deadlock?

1. Mutual Exclusion: Resources cannot be shared.
2. Hold and Wait: Thread holds resources while waiting for others.
3. No Preemption: Resources cannot be forcibly taken.
4. Circular Wait: Circular chain of threads waiting for each other.

Q: How do you prevent deadlocks?

- Lock Ordering: Always acquire locks in the same global order.
- Lock Timeout: Use tryLock() with timeout.
- Avoid Nested Locks: Minimize holding multiple locks.

Key Takeaways

- Deadlock occurs when threads form a circular wait for locks.
- Four conditions must all be present for deadlock.

- Breaking any one condition prevents deadlock.
- Lock ordering is the most practical prevention strategy.
- tryLock() with timeout provides a fallback mechanism.
- jstack and ThreadMXBean can detect deadlocks.

Java Locks: Complete Guide

Core Java - Concurrency

From synchronized to StampedLock - understanding every lock type in Java 21+.

Lock Types Overview

Java provides multiple locking mechanisms, each designed for specific use cases. Choosing the right lock can make the difference between a system that scales and one that bottlenecks.

```
Lock Type   |   Package   |   Reentrancy   |   Fairness   |   Best For
`synchronized`   |   Built-in   |   Yes   |   No   |   Simple mutual exclusion
`ReentrantLock`   |   `java.util.concurrent`   |   Yes   |   Optional   |   Advanced
locking, tryLock, interruptible
`ReentrantReadWriteLock`   |   `java.util.concurrent`   |   Yes   |   Optional   |
Read-heavy workloads
`StampedLock`   |   `java.util.concurrent`   |   No   |   No   |   Maximum read
throughput
`Semaphore`   |   `java.util.concurrent`   |   N/A   |   Optional   |   Resource
pooling, rate limiting
```

Java 21+ Virtual Threads Note

With virtual threads, synchronized blocks can pin the carrier thread. For virtual thread compatibility:

- Prefer ReentrantLock over synchronized for long-held locks.
- Keep synchronized blocks extremely short.
- Avoid synchronized around blocking I/O.

1. synchronized (Intrinsic Lock)

The synchronized keyword uses the object's intrinsic monitor lock. It's simple but has limitations.

```
// Method-level synchronization
public synchronized void increment() {
    count++;
}

// Block-level synchronization (preferred)
public void increment() {
    synchronized (this) {
        count++;
    }
}

// Synchronizing on a dedicated lock object (best)
private final Object lock = new Object();
public void increment() {
    synchronized (lock) {
        count++;
    }
}
```

Advantages

- Simple syntax
- Automatic release on exit
- Reentrant by default
- Works with wait/notify

Limitations

- No tryLock() capability
- No fairness option
- Cannot interrupt a waiting thread
- Pins virtual threads (Java 21+)

Virtual Threads Warning (Java 21+)

When a virtual thread holds a synchronized lock and blocks, it pins the carrier thread. This defeats the scalability benefit of virtual threads. Use ReentrantLock instead for long-held locks.

2. ReentrantLock

ReentrantLock provides the same mutual exclusion as synchronized but with additional features like tryLock, timed waiting, and fairness.

```
private final ReentrantLock lock = new ReentrantLock();

public void doWork() {
    lock.lock();  // Blocks until acquired
    try {
        // Critical section
    } finally {
        lock.unlock();  // MUST unlock in finally!
    }
}

// Non-blocking attempt
public boolean tryDoWork() {
    if (lock.tryLock()) {
        try {
            // Critical section
            return true;
        } finally {
            lock.unlock();
        }
    }
    return false;  // Lock not available
}

// Timed attempt
public boolean timedDoWork() throws InterruptedException {
    if (lock.tryLock(1, TimeUnit.SECONDS)) {
        try {
            // Got lock within 1 second
        } finally {
            lock.unlock();
        }
    }
}
```

Fair vs Non-Fair

```
// Non-fair (default) - better throughput
ReentrantLock lock = new ReentrantLock();

// Fair - FIFO ordering, prevents starvation
ReentrantLock fairLock = new ReentrantLock(true);
```

When to Use ReentrantLock

- Need tryLock() for non-blocking attempts
- Need interruptible lock acquisition
- Need fairness guarantees
- Using with virtual threads (Java 21+)
- Need multiple Condition objects

3. ReentrantReadWriteLock

Optimizes for read-heavy workloads. Multiple readers can hold the lock simultaneously, but writers get exclusive access.

```
private final ReentrantReadWriteLock rwLock = new ReentrantReadWriteLock();
private final Lock readLock = rwLock.readLock();
private final Lock writeLock = rwLock.writeLock();

private Map<String, String> cache = new HashMap<>();

public String get(String key) {
    readLock.lock();
    try {
        return cache.get(key);  // Multiple readers OK
    } finally {
        readLock.unlock();
    }
}

public void put(String key, String value) {
    writeLock.lock();
    try {
        cache.put(key, value);  // Exclusive access
    } finally {
        writeLock.unlock();
    }
}
```

Lock Compatibility

```
Current Holders   |   Read Request   |   Write Request
None   |   Granted   |   Granted
Readers (1+)   |   Granted   |   Blocked (waits for all readers)
Writer (1)   |   Blocked   |   Blocked
```

Use Case: Configuration Cache

Perfect for caches where reads vastly outnumber writes. 100 readers can access simultaneously, but configuration updates get exclusive access.

4. StampedLock (Java 8+)

The highest-performance lock for read-heavy workloads. Supports optimistic reads that don't block writers.

```
private final StampedLock sl = new StampedLock();
private double x, y;

// Optimistic read - doesn't acquire lock!
public double distanceFromOrigin() {
    long stamp = sl.tryOptimisticRead();  // Non-blocking
    double currentX = x, currentY = y;

    if (!sl.validate(stamp)) {  // Check if write occurred
        stamp = sl.readLock();  // Fallback to read lock
        try {
            currentX = x;
            currentY = y;
        } finally {
            sl.unlockRead(stamp);
        }
    }
    return Math.sqrt(currentX * currentX + currentY * currentY);
}

// Write lock - exclusive
public void move(double deltaX, double deltaY) {
    long stamp = sl.writeLock();
    try {
        x += deltaX;
        y += deltaY;
    } finally {
        sl.unlockWrite(stamp);
    }
}
```

StampedLock Caveats

- Not reentrant - will deadlock if the same thread tries to acquire twice.
- Stamps must match - unlock with the same stamp returned by lock.
- More complex - optimistic reads need the validation pattern.
- Best for short read operations with rare writes.

Lock Mode Conversion

```
// Upgrade: optimistic -> read -> write
long stamp = sl.tryOptimisticRead();
// ... read data ...
if (!sl.validate(stamp)) {
    stamp = sl.readLock();  // Upgrade to read
}
// If need to write:
long writeStamp = sl.tryConvertToWriteLock(stamp);
if (writeStamp == 0L) {
    sl.unlockRead(stamp);
    writeStamp = sl.writeLock();  // Acquire write lock
}
// ... write data ...
sl.unlockWrite(writeStamp);
```

5. Semaphore

Controls access to a resource with a limited number of permits. Unlike locks, permits are not tied to threads - any thread can release.

```
// Connection pool with max 10 connections
private final Semaphore semaphore = new Semaphore(10);
private final Queue<Connection> pool = new ConcurrentLinkedQueue<>();

public Connection getConnection() throws InterruptedException {
    semaphore.acquire();  // Blocks if no permits
    return pool.poll();
}

public void releaseConnection(Connection conn) {
    pool.offer(conn);
    semaphore.release();  // Return permit
}

// Rate limiting example
private final Semaphore rateLimiter = new Semaphore(100);  // 100 concurrent requests

public Response handleRequest(Request req) {
    if (rateLimiter.tryAcquire()) {
        try {
            return processRequest(req);
        } finally {
            rateLimiter.release();
        }
    }
    return Response.tooManyRequests();
}
```

Common Use Cases

- Connection Pooling: Limit concurrent database connections.
- Rate Limiting: Control request throughput.
- Resource Bounding: Limit memory usage by capping concurrent operations.
- Producer-Consumer: Coordinate between producers and consumers.

Which Lock Should You Use?

- synchronized - Simple mutual exclusion with automatic release. Use when: simple cases, short critical sections, not using virtual threads.
- ReentrantLock - Advanced locking with tryLock and fairness. Use when: need tryLock, interruptibility, fairness, or virtual thread compatibility.
- ReadWriteLock - Shared reads, exclusive writes. Use when: read-heavy workloads (90%+ reads), cache implementations.
- StampedLock - Optimistic reads for maximum throughput. Use when: extreme read performance needed, short operations, no reentrancy needed.
- Semaphore - Permit-based resource control. Use when: limiting concurrent access count, connection pools, rate limiting.

Decision Flowchart

- Need to limit concurrent access count? -> Semaphore
- Read-heavy with rare writes? -> ReadWriteLock or StampedLock
- Need tryLock or fairness? -> ReentrantLock
- Using virtual threads with blocking I/O? -> ReentrantLock
- Simple case, short lock hold? -> synchronized

Key Takeaways

- synchronized is simple but limited - no tryLock, no fairness, pins virtual threads.
- ReentrantLock offers tryLock, interruptibility, and fairness options.
- ReadWriteLock allows concurrent reads, exclusive writes - ideal for caches.
- StampedLock provides optimistic reads for maximum throughput but is not reentrant.
- Semaphore controls permit count - perfect for pools and rate limiting.
- Always unlock in a finally block with explicit locks.
- For Java 21+ virtual threads, prefer ReentrantLock over synchronized.

Virtual Threads

1. The Problem Statement

Traditional Java applications use platform threads (OS threads) for concurrency. Each platform thread consumes ~1MB of stack memory and is managed by the operating system. When your application handles thousands of concurrent requests - each waiting on database calls, HTTP responses, or file I/O - those threads sit idle, blocked, doing nothing. You quickly hit limits: memory exhaustion, context-switching overhead, and poor throughput. This is the "thread-per-request" bottleneck that has plagued Java servers for decades.

2. What are Virtual Threads

Virtual Threads (introduced in Java 21 via Project Loom) are lightweight threads managed by the JVM, not the OS. They use only ~1KB of memory and can number in the millions. When a virtual thread encounters a blocking operation (like a database call), it yields instead of blocking - the underlying carrier thread immediately picks up another virtual thread. This means your application can handle massive concurrency without the resource overhead of traditional threads.

3. Platform Threads vs Virtual Threads

Platform Threads:

- Memory: ~1MB per thread
- Limited to thousands (constrained by OS)
- High blocking cost - thread sits completely idle during I/O
- Managed by the Operating System
- Best suited for CPU-intensive tasks

Virtual Threads:

- Memory: ~1KB per thread (1000x lighter)
- Can scale to millions (managed by JVM)
- Near-zero blocking cost - yields to carrier thread during I/O
- Managed by the JVM runtime
- Best suited for I/O-bound operations

4. When to Use Virtual Threads

- High-concurrency I/O operations (HTTP clients, database queries)
- REST API servers handling thousands of simultaneous requests
- Microservices with frequent network calls
- Applications where threads spend most time waiting (not computing)
- Replacing reactive/async code with simpler synchronous style

When NOT to use:

CPU-bound tasks (parallel computation, data processing) - platform threads are still better here since virtual threads don't add parallelism, only concurrency.

5. Key Differences

```
Aspect   |   Platform   |   Virtual
Memory   |   ~1MB   |   ~1KB
Max Count   |   Thousands   |   Millions
Managed   |   OS   |   JVM
Blocking   |   High Cost   |   Near Zero
Best For   |   CPU Tasks   |   I/O Tasks
```

Collections Under the Hood

Java Collections

Visualize how ArrayList, LinkedList, HashMap, and TreeMap work internally.

Why Understanding Internals Matters

Knowing how collections work internally helps you make better decisions about which collection to use. In interviews, you'll often be asked about time complexity, resizing behavior, and collision handling.

What You'll Learn

- ArrayList: Dynamic array resizing and element shifting
- LinkedList: Node linking and pointer manipulation
- HashMap: Hashing, bucket arrays, and collision resolution
- TreeMap: Red-Black tree structure and balancing

ArrayList: Dynamic Array

ArrayList uses a resizable array internally. When you add elements beyond capacity, it creates a new array with 1.5x capacity and copies all elements.

Java 8+ Implementation Detail

ArrayList uses lazy initialization. The internal array starts as an empty shared instance and grows to capacity 10 on the first add() call.

Key Operations

```
Operation   |   Time   |   Why?
`get(index)`   |   O(1)   |   Direct array access: `elementData[index]`
`add(element)`   |   O(1)*   |   Append to end. *Amortized - resize is O(n) but rare
`add(index, element)`   |   O(n)   |   Must shift all elements after index to the right
`remove(index)`   |   O(n)   |   Must shift all elements after index to the left
`contains(element)`   |   O(n)   |   Linear search through array
```

Resizing Formula

```
int newCapacity = oldCapacity + (oldCapacity >> 1);  // 1.5x growth
```

- Default initial capacity: 10 (on first add)
- Growth: 10 -> 15 -> 22 -> 33 -> 49 -> ...
- Resize copies all elements via Arrays.copyOf()

```
// Actual ArrayList source (simplified)
public class ArrayList<E> {
    transient Object[] elementData;  // The backing array
    private int size;                // Number of elements

    private static final int DEFAULT_CAPACITY = 10;

    public boolean add(E e) {
        ensureCapacityInternal(size + 1);  // Grow if needed
        elementData[size++] = e;           // O(1)
        return true;
    }

    public void add(int index, E element) {
        rangeCheckForAdd(index);
        ensureCapacityInternal(size + 1);
        System.arraycopy(elementData, index,    // Shift right - O(n)
                         elementData, index + 1,
                         size - index);
        elementData[index] = element;
        size++;
    }
}
```

LinkedList: Doubly Linked Nodes

LinkedList uses doubly-linked nodes. Each node holds data plus pointers to previous and next nodes. This makes insertion/deletion at ends O(1), but random access O(n).

LinkedList Implements Deque

Since Java 1.6, LinkedList implements Deque, making it suitable for queue (FIFO) or stack (LIFO) with O(1) operations at both ends.

Node Structure

```
private static class Node<E> {
    E item;           // The data
    Node<E> next;     // Reference to next node
    Node<E> prev;     // Reference to previous node
}

transient Node<E> first;  // Head of list
transient Node<E> last;   // Tail of list
transient int size;
```

Key Operations

```
Operation   |   Time   |   Why?
`addFirst()` / `addLast()`   |   O(1)   |   Just update first/last pointers
`removeFirst()` / `removeLast()`   |   O(1)   |   Just update first/last pointers
`get(index)`   |   O(n)   |   Must traverse from first or last
`add(index, element)`   |   O(n)   |   O(n) traverse + O(1) insert
```

When to Use LinkedList

- Frequent insertions/deletions at both ends (Queue/Deque)
- You have an Iterator and insert/remove during traversal
- You don't need random access by index

HashMap: Hash Table with Chaining

HashMap uses an array of buckets (table[]). Keys are hashed to determine bucket index. Collisions are handled by chaining (linked list, or tree for 8+ entries).

How put(key, value) Works

1. Hash: (h = key.hashCode()) ^ (h >>> 16) - spreads bits
2. Index: hash & (table.length - 1) - bucket position
3. Insert: Add to bucket (chain if collision)
4. Resize: If size > threshold, double table and rehash

```
// HashMap constants (Java 8+)
static final int DEFAULT_INITIAL_CAPACITY = 16;   // Must be power of 2
static final float DEFAULT_LOAD_FACTOR = 0.75f;
static final int TREEIFY_THRESHOLD = 8;           // List -> Tree
static final int UNTREEIFY_THRESHOLD = 6;         // Tree -> List
static final int MIN_TREEIFY_CAPACITY = 64;       // Min table size for tree

// Hash function - spreads bits to reduce collisions
static final int hash(Object key) {
    int h;
    return (key == null) ? 0 : (h = key.hashCode()) ^ (h >>> 16);
}

// Bucket index (fast modulo for power of 2)
int index = hash & (table.length - 1);
```

Key Operations

```
Operation   |   Average   |   Worst (pre-Java 8)   |   Worst (Java 8+)
`get(key)`   |   O(1)   |   O(n)   |   O(log n)
`put(key, value)`   |   O(1)   |   O(n)   |   O(log n)
`remove(key)`   |   O(1)   |   O(n)   |   O(log n)
```

Treeification (Java 8+)

When a bucket exceeds 8 entries AND table size >= 64:

- Linked list converts to Red-Black tree.
- Worst case improves from O(n) to O(log n).
- Converts back to list when entries drop to 6.

TreeMap: Red-Black Tree

TreeMap uses a Red-Black tree - a self-balancing binary search tree. Keys are always sorted, and all operations are guaranteed O(log n).

Red-Black Tree Properties

- Every node is RED or BLACK.
- Root is always BLACK.
- All leaves (null) are BLACK.
- RED nodes have only BLACK children.
- Every path from root to leaf has the same BLACK count.

These rules ensure height is at most 2 * log(n+1).

Key Operations

```
Operation   |   Time   |   Why?
`get(key)`   |   O(log n)   |   Binary search down tree
`put(key, value)`   |   O(log n)   |   Find position + rebalance
`firstKey()` / `lastKey()`   |   O(log n)   |   Traverse to leftmost/rightmost
`subMap(from, to)`   |   O(log n)   |   Returns view backed by tree
```

When to Use TreeMap

- Need keys in sorted order
- Need range queries: subMap(), headMap(), tailMap()
- Need navigation: floorKey(), ceilingKey(), higherKey()

Quick Comparison

```
Collection   |   Structure   |   Order   |   Null Keys   |   Best For
ArrayList   |   Dynamic Array   |   Insertion   |   Yes   |   Random access
LinkedList   |   Doubly Linked   |   Insertion   |   Yes   |   Queue/Deque
HashMap   |   Hash Table   |   None   |   Yes (1)   |   Fast lookup
TreeMap   |   Red-Black Tree   |   Sorted   |   No   |   Sorted map
```

Key Takeaways

- ArrayList: Best for random access. Avoid inserts in middle.
- LinkedList: Best for queue/stack. Avoid get(index).
- HashMap: Best for fast lookup. Override hashCode() properly.
- TreeMap: Best for sorted keys. Keys must be Comparable.

Choosing the Right Collection

Java Collections

The Collection Selection Problem

Java provides 15+ collection implementations. Choosing the wrong one can cause performance issues, bugs, or unnecessary complexity. This chapter helps you make the right choice quickly.

Key Questions to Ask

1. Do you need key-value pairs or just values?
2. Do you need elements in sorted order?
3. Are duplicates allowed?
4. Do you need thread safety?
5. What operations are most frequent? (add, get, search, iterate)

Time Complexity Comparison

```
Collection   |   Add   |   Remove   |   Get/Contains   |   Iterate
ArrayList   |   O(1)*   |   O(n)   |   O(1)/O(n)   |   O(n)
LinkedList   |   O(1)   |   O(1)**   |   O(n)   |   O(n)
HashSet   |   O(1)   |   O(1)   |   O(1)   |   O(n)
LinkedHashSet   |   O(1)   |   O(1)   |   O(1)   |   O(n)
TreeSet   |   O(log n)   |   O(log n)   |   O(log n)   |   O(n)
HashMap   |   O(1)   |   O(1)   |   O(1)   |   O(n)
LinkedHashMap   |   O(1)   |   O(1)   |   O(1)   |   O(n)
TreeMap   |   O(log n)   |   O(log n)   |   O(log n)   |   O(n)
ArrayDeque   |   O(1)   |   O(1)   |   O(n)   |   O(n)
PriorityQueue   |   O(log n)   |   O(log n)   |   O(n)   |   O(n)
ConcurrentHashMap   |   O(1)   |   O(1)   |   O(1)   |   O(n)
CopyOnWriteArrayList   |   O(n)   |   O(n)   |   O(1)   |   O(n)
```

* Amortized, O(n) when resizing. ** O(1) at ends, O(n) by index.

Quick Reference Cards

ArrayList - implements List
Default choice for lists. Fast random access, slow inserts in middle.
get(i) O(1) · add() O(1)* · contains() O(n)

LinkedList - implements List, Deque
Best for queue/deque operations. Avoid for random access.
addFirst() O(1) · addLast() O(1) · get(i) O(n)

HashSet - implements Set
Fast uniqueness check. No order, no duplicates.
add() O(1) · contains() O(1) · remove() O(1)

TreeSet - implements SortedSet, NavigableSet
Sorted unique elements. Range queries.
add() O(log n) · first() O(log n) · subSet() O(log n)

HashMap - implements Map
Default choice for key-value. Fastest lookup.
put() O(1) · get() O(1) · containsKey() O(1)

TreeMap - implements SortedMap, NavigableMap
Sorted keys. Range queries on keys.
put() O(log n) · firstKey() O(log n) · subMap() O(log n)

ArrayDeque - implements Deque
Best stack/queue. Faster than LinkedList.
push() O(1) · pop() O(1) · offer() O(1)

ConcurrentHashMap - implements ConcurrentMap
Thread-safe map. Better than synchronized HashMap.
putIfAbsent() O(1) · computeIfAbsent() O(1)

Decision Flowchart

Follow this flowchart to quickly narrow down your choice:

```
Need key-value pairs?
|-- YES -> Need sorted keys?
|          |-- YES -> TreeMap
|          `-- NO -> Need insertion order?
|                   |-- YES -> LinkedHashMap
|                   `-- NO -> Thread safe?
|                             |-- YES -> ConcurrentHashMap
|                             `-- NO -> HashMap (default)
|
`-- NO -> Allow duplicates?
          |-- YES -> Need FIFO/LIFO?
          |          |-- YES -> ArrayDeque (stack/queue)
          |          `-- NO -> Need random access?
          |                    |-- YES -> ArrayList (default)
          |                    `-- NO -> LinkedList
          |
          `-- NO (Set) -> Need sorted?
                          |-- YES -> TreeSet
                          `-- NO -> Need insertion order?
                                    |-- YES -> LinkedHashSet
                                    `-- NO -> HashSet (default)
```

Common Mistakes to Avoid

1. Using LinkedList for Random Access

linkedList.get(5000) is O(n) - must traverse 5000 nodes!

Fix: Use ArrayList for index-based access.

2. Using ArrayList as a Queue

arrayList.remove(0) shifts all elements - O(n)!

Fix: Use ArrayDeque for FIFO operations.

3. Using TreeMap When Order Doesn't Matter

TreeMap is O(log n) for everything. HashMap is O(1).

Fix: Use HashMap unless you need sorted keys.

4. Using Vector or Hashtable

These are legacy classes with unnecessary synchronization overhead.

Fix: Use ArrayList/HashMap or their concurrent alternatives.

5. Not Overriding hashCode() with equals()

HashSet/HashMap won't work correctly if equals() is overridden but hashCode() isn't.

Fix: Always override both together.

Key Takeaways

Default Choices

- List: ArrayList (unless you need queue operations)
- Set: HashSet (unless you need order/sorting)
- Map: HashMap (unless you need order/sorting)
- Queue/Stack: ArrayDeque (faster than LinkedList)
- Thread-safe Map: ConcurrentHashMap (not Hashtable)

When to Use Alternatives

- LinkedHashMap/Set: When insertion order matters (LRU cache)
- TreeMap/Set: When you need sorted order or range queries
- LinkedList: Only for frequent add/remove at both ends
- PriorityQueue: When you need elements in priority order
- CopyOnWriteArrayList: Read-heavy, write-rare concurrent scenarios

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Collections in Action: Real-World Patterns

Java Collections

7 essential patterns every Java developer should know, visualized step by step.

From Theory to Practice

Knowing how collections work internally is only half the battle. The real skill is recognizing which pattern to apply when you see a problem. This chapter covers 7 patterns that appear constantly in interviews and production code.

Patterns Covered

1. Frequency Counting - HashMap for counting occurrences
2. Deduplication - HashSet for removing duplicates
3. LRU Cache - LinkedHashMap with eviction policy
4. Top-K Elements - PriorityQueue (min-heap) for finding top items
5. Grouping - Map of Lists for categorizing data
6. Stack Problems - ArrayDeque for bracket matching and undo
7. Range Queries - TreeMap for sorted lookups and ranges

Pattern 1: Frequency Counting

Use a HashMap<K, Integer> to count how often each element appears. This is one of the most common

interview patterns.

```
// Classic frequency count
Map<String, Integer> freq = new HashMap<>();
for (String word : words) {
    freq.merge(word, 1, Integer::sum);  // Java 8+
}

// Alternative using getOrDefault
for (String word : words) {
    freq.put(word, freq.getOrDefault(word, 0) + 1);
}

// With Streams (Collectors.groupingBy + counting)
Map<String, Long> freq = words.stream()
    .collect(Collectors.groupingBy(w -> w, Collectors.counting()));
```

When to Use

- Word frequency / character frequency in strings
- Vote counting, histogram building
- Finding most/least common elements
- Anagram detection (compare character frequency maps)

Pattern 2: Deduplication

Use a HashSet to track seen elements and filter out duplicates in O(1) per check.

```
// Remove duplicates from a list
List<String> unique = new ArrayList<>(new LinkedHashSet<>(list));
// LinkedHashSet preserves insertion order

// Deduplication during stream processing
Set<String> seen = new HashSet<>();
for (Event event : eventStream) {
    if (seen.add(event.getId())) {  // returns false if already present
        process(event);             // only process first occurrence
    }
}

// With Streams
List<String> unique = list.stream().distinct().toList();
```

Important: hashCode() and equals()

For custom objects, HashSet.add() uses hashCode() to find the bucket and equals() to check for duplicates. If you override equals(), you must also override hashCode().

Pattern 3: LRU Cache

A Least Recently Used cache evicts the entry that hasn't been accessed for the longest time. LinkedHashMap with accessOrder=true does the heavy lifting.

```
public class LRUCache<K, V> extends LinkedHashMap<K, V> {
    private final int capacity;

    public LRUCache(int capacity) {
        super(capacity, 0.75f, true);  // accessOrder = true
        this.capacity = capacity;
    }

    @Override
    protected boolean removeEldestEntry(Map.Entry<K, V> eldest) {
        return size() > capacity;  // evict when over capacity
    }
}

// Usage
LRUCache<String, String> cache = new LRUCache<>(100);
cache.put("user:1", "Alice");
cache.get("user:1");  // moves to most-recently-used
```

How accessOrder Works

When accessOrder=true, every get() or put() moves that entry to the tail of the internal linked list. The head is always the least recently used entry. When removeEldestEntry() returns true, the head entry is evicted.

Pattern 4: Top-K Elements

Use a PriorityQueue (min-heap) of size K. For each element: if heap has room, add it. If the new element is larger than the min, remove min and add new.

```
// Find top K largest elements from a stream
public static List<Integer> topK(int[] nums, int k) {
    PriorityQueue<Integer> minHeap = new PriorityQueue<>();  // min at top

    for (int num : nums) {
        if (minHeap.size() < k) {
            minHeap.offer(num);
        } else if (num > minHeap.peek()) {
            minHeap.poll();   // remove smallest
            minHeap.offer(num); // add new larger element
        }
    }

    return new ArrayList<>(minHeap);  // contains top K elements
}

// Time: O(n log k) - much better than sorting O(n log n) for small k
// Space: O(k)
```

Why Min-Heap for Top-K?

A min-heap of size K keeps the K largest elements seen so far. The root (min) acts as a gatekeeper: anything smaller is rejected. This is O(n log k) vs. sorting which is O(n log n).

Pattern 5: Grouping

Use Map<K, List<V>> to group elements by a key. Java 8+ Collectors.groupingBy() makes this clean.

```
// Manual grouping
Map<String, List<Employee>> byDept = new HashMap<>();
for (Employee emp : employees) {
    byDept.computeIfAbsent(emp.getDept(), k -> new ArrayList<>())
          .add(emp);
}

// Stream grouping (same result, cleaner code)
Map<String, List<Employee>> byDept = employees.stream()
    .collect(Collectors.groupingBy(Employee::getDept));

// Grouping with downstream collector
Map<String, Long> countByDept = employees.stream()
    .collect(Collectors.groupingBy(Employee::getDept, Collectors.counting()));

// Partitioning (special case: boolean grouping)
Map<Boolean, List<Employee>> seniorVsJunior = employees.stream()
    .collect(Collectors.partitioningBy(e -> e.getYears() > 5));
```

computeIfAbsent() is Key

map.computeIfAbsent(key, k -> new ArrayList<>()) creates the list only if the key is absent, then returns it. This avoids the boilerplate null-check-then-create pattern.

Pattern 6: Stack Problems

Use ArrayDeque as a stack for bracket matching, expression evaluation, undo/redo, and depth-first traversal.

```
// Bracket matching
public static boolean isValid(String s) {
    Deque<Character> stack = new ArrayDeque<>();
    Map<Character, Character> pairs = Map.of(')', '(', ']', '[', '}', '{');

    for (char c : s.toCharArray()) {
        if ("([{".indexOf(c) >= 0) {
            stack.push(c);               // opening: push
        } else if (pairs.containsKey(c)) {
            if (stack.isEmpty() || stack.pop() != pairs.get(c)) {
                return false;            // no match
            }
        }
    }
    return stack.isEmpty();  // all brackets matched
}
```

ArrayDeque, not Stack

The java.util.Stack class is legacy - it extends Vector, adding unnecessary synchronization. Always use ArrayDeque for stack operations. It's faster and has a cleaner API.

Pattern 7: Range Queries

Use TreeMap when you need sorted keys and operations like "find all entries between A and B" or "find the closest key."

```
TreeMap<Integer, String> map = new TreeMap<>();

// Navigation methods
map.floorKey(50);      // largest key <= 50
map.ceilingKey(50);    // smallest key >= 50
map.lowerKey(50);      // largest key < 50 (strictly less)
map.higherKey(50);     // smallest key > 50 (strictly greater)

// Range queries
map.subMap(20, true, 60, true);  // all entries with key 20..60 inclusive
map.headMap(50, true);           // all entries with key <= 50
map.tailMap(50, true);           // all entries with key >= 50

// Practical: Find events in a time window
TreeMap<LocalDateTime, Event> events = new TreeMap<>();
var inRange = events.subMap(startTime, true, endTime, true);
```

Real-World TreeMap Use Cases

- Scheduling systems - find next available time slot
- Rate limiting - count requests in a sliding time window
- Financial systems - find closest price point
- Leaderboards - find rank range queries

Pattern Quick Reference

```
Pattern   |   Collection   |   Key Method   |   Time
Frequency Count   |   HashMap   |   `merge()` / `getOrDefault()`   |   O(n)
Deduplication   |   HashSet   |   `add()` returns boolean   |   O(n)
LRU Cache   |   LinkedHashMap   |   `removeEldestEntry()`   |   O(1) per op
Top-K   |   PriorityQueue   |   `offer()` / `poll()`   |   O(n log k)
Grouping   |   HashMap + List   |   `computeIfAbsent()`   |   O(n)
Stack Problems   |   ArrayDeque   |   `push()` / `pop()`   |   O(n)
Range Queries   |   TreeMap   |   `subMap()` / `floorKey()`   |   O(log n)
```

Key Takeaways

- Most interview collection problems boil down to one of these 7 patterns.
- merge(), computeIfAbsent(), and getOrDefault() eliminate boilerplate.
- PriorityQueue (min-heap) of size K is the standard Top-K approach.
- LinkedHashMap with accessOrder=true is the simplest LRU cache.
- Always use ArrayDeque over Stack for stack operations.
- TreeMap shines when you need sorted keys or range lookups.

Java Collections Interview Revision Guide

This comprehensive guide is designed to help you refresh and strengthen your understanding of Java Collections concepts in preparation for technical interviews.

In this chapter, you'll learn:

- Fail-Fast vs Fail-Safe Iterators
- ArrayList vs LinkedList

- HashMap vs Hashtable
- HashSet vs TreeSet
- How HashMap works internally
- HashMap Initial Capacity and Load Factor
- EnumSet and EnumMap
- Sorting using Comparable and Comparator
- Key Interfaces in Java Collections
- Collection vs Collections
- Custom Collections
- HashMap vs LinkedHashMap
- HashMap vs ConcurrentHashMap
- How TreeMap Works
- PriorityQueue vs Regular Queue
- Deep Copy vs Shallow Copy
- Handling Duplicates in Collections
- Synchronized Collections vs Concurrent Collections

What Is the Difference Between Fail-Safe and Fail-Fast Iterators?

Fail-Safe and Fail-Fast iterators are concepts used to describe the behavior of iterators in the face of concurrent modifications.

Fail-Fast Iterators:

- Definition: Fail-Fast systems abort operation as-fast-as-possible, exposing failures immediately and stopping the whole operation. They immediately throw a ConcurrentModificationException if they detect any structural modifications to the collection while iterating.
- Behavior: These iterators directly access the collection's data structure, making them highly sensitive to modifications.
- Example: Iterators of most Java Collections Framework classes such as ArrayList, HashSet, HashMap are Fail-Fast.

```
import java.util.ArrayList;
import java.util.Arrays;
import java.util.Iterator;
import java.util.List;

public class FailFastTest {
    public void failFast() {
        ArrayList<Integer> numbers = new ArrayList<>(Arrays.asList(1, 2, 3));
        Iterator<Integer> iterator = numbers.iterator();
        while (iterator.hasNext()) {
            Integer number = iterator.next();
            numbers.add(50); // This line will throw error
        }
    }
}
```

Fail-Safe Iterators:

- Definition: Fail-Safe iterators do not throw exceptions when the underlying collection is modified. Instead, they operate on a clone or a snapshot of the collection's data.
- Behavior: These iterators maintain a copy of the collection's state at the time the iterator was created.
- Example: Iterators of concurrent collections such as CopyOnWriteArrayList, ConcurrentHashMap in Java are Fail-Safe.

```
import java.util.Arrays;
import java.util.Iterator;
import java.util.concurrent.CopyOnWriteArrayList;

public class FailSafeTest {
    public void failSafe() {
        CopyOnWriteArrayList<String> list = new CopyOnWriteArrayList<>(Arrays.asList("a", "b",…
        Iterator<String> iterator = list.iterator();
        while (iterator.hasNext()) {
            if (iterator.next().equals("b")) {
                list.remove("b"); // This will not cause an exception
            }
        }
        System.out.println(list); // ["a", "c"]
    }
}
```

What is the Difference Between LinkedList and ArrayList?

LinkedList and ArrayList both are data structures in Java used to store data.

ArrayList:

- Underlying Data Structure: Uses a dynamic array to store elements. Provides fast random access to elements using an index.
- Time Complexity: Uses O(1) for get and set operations and appending element to last. But it takes O(n) to add or remove an element in between.
- Use Case: Best suited for applications where frequent access to elements using an index is required. Ideal when the size of the list is relatively stable, and insertion or deletion operations are infrequent.

LinkedList:

- Underlying Data Structure: Uses a doubly linked list to store elements. Does not support efficient random access. Elements are accessed sequentially.
- Time Complexity: Uses O(n) for get and set operations because it requires traversal from the beginning or end to the desired index. O(1) for adding or removing elements at the beginning or end, but O(n) for accessing elements at arbitrary positions.
- Use Case: Best suited for applications where frequent insertions and deletions are required, especially at the beginning or middle of the list.

```
import java.util.ArrayList;
import java.util.List;

public class ArrayListExample {
    public static void main(String[] args) {
        // Create an ArrayList
        List<String> arrayList = new ArrayList<>();

        // Add elements to the ArrayList
        arrayList.add("Apple");
        arrayList.add("Banana");
        arrayList.add("Cherry");

        // Access elements by index
        System.out.println("Element at index 1: " + arrayList.get(1)); // Element at index 1: …

        // Iterate through the ArrayList
        System.out.println("ArrayList elements:");
        for (String fruit : arrayList) {
            System.out.println(fruit);
        }
        // Output:
        // ArrayList elements:
        // Apple
        // Banana
        // Cherry

        // Remove an element
        arrayList.remove(1); // Removes "Banana"
        System.out.println("After removal:");
        for (String fruit : arrayList) {
            System.out.println(fruit);
        }
        // After removal:
        // Apple
        // Cherry
    }
}
```

```
import java.util.LinkedList;

public class LinkedListExample {
    public static void main(String[] args) {
        // Create a LinkedList
        LinkedList<String> linkedList = new LinkedList<>();

        // Add elements to the LinkedList
        linkedList.add("Apple");
        linkedList.add("Banana");
        linkedList.add("Cherry");

        // Access elements by index (not efficient)
        System.out.println("Element at index 1: " + linkedList.get(1)); // Banana

        // Iterate through the LinkedList
        System.out.println("LinkedList elements:");
        for (String fruit : linkedList) {
            System.out.println(fruit);
        }
        // LinkedList elements:
        // Apple
        // Banana
        // Cherry

        // Remove an element
        linkedList.remove(1); // Removes "Banana"
        System.out.println("After removal:");
        for (String fruit : linkedList) {
            System.out.println(fruit);
        }
        // After removal:
        // Apple
        // Cherry

        // Add element at the beginning and the end
        linkedList.addFirst("Mango");
        linkedList.addLast("Orange");
        System.out.println("After adding at the beginning and end:");
        for (String fruit : linkedList) {
            System.out.println(fruit);
        }
        // After adding at the beginning and end:
        // Mango
        // Apple
        // Cherry
        // Orange
    }
}
```

What is the Difference Between HashMap and HashTable?

1. Thread Safety

- HashMap: Not thread-safe. Concurrent modification can lead to inconsistent state.
- Hashtable: Thread-safe. All methods are synchronized.

2. Synchronization

- HashMap: No internal synchronization. Needs external synchronization for thread safety.

- Hashtable: Methods are internally synchronized.

3. Performance

- HashMap: Faster because it does not have synchronization overhead.
- Hashtable: Slower due to synchronization overhead.

4. Null Keys and Values

- HashMap: Allows one null key and multiple null values.
- Hashtable: Does not allow null keys or null values.

5. Iterator Type

- HashMap: Fail-fast iterator, which throws ConcurrentModificationException if the map is modified during iteration.
- Hashtable: Fail-safe enumeration, which does not throw ConcurrentModificationException.

6. Use Case

- HashMap: Suitable for non-threaded applications where thread safety is not a concern.
- Hashtable: Suitable for threaded applications where thread safety is required.

7. Legacy

- HashMap: Considered modern and more versatile.
- Hashtable: Considered legacy; use ConcurrentHashMap for thread-safe operations.

```
// HashMap allows one null key and any number of null values.
Map<String, Integer> hashMap = new HashMap<>();
hashMap.put(null, null);

// Hashtable does not allow null keys or values
Map<String, Integer> hashtable = new Hashtable<>();
hashtable.put(null, null); // exception
```

What Is the Difference Between HashSet and TreeSet?

HashSet and TreeSet are both implementations of the Set interface in Java, but they have different underlying mechanisms, performance characteristics, and usage scenarios.

HashSet:

- Underlying Data Structure: Uses a hash table to store elements. Relies on hashing for its operations, meaning it uses the hashCode method to determine the storage location of elements.
- Time Complexity: O(1) average time complexity for add, remove, and contains operations.
- Ordering: Does not maintain any order of elements. The order of elements when iterating over the set is unpredictable and can change over time.
- Use Case: Suitable when you need fast lookups and do not care about the order of elements.

TreeSet:

- Underlying Data Structure: Uses a Red-Black tree (a self-balancing binary search tree) to store elements.
- Time Complexity: O(log n) for add, remove, and contains operations.
- Ordering: Maintains elements in a sorted order. The elements are sorted according to their natural ordering (using Comparable) or by a specified Comparator.
- Use Case: Suitable when you need to maintain a sorted order of elements.

```
// HashSet Example
import java.util.HashSet;
import java.util.Set;

public class HashSetExample {
    public static void main(String[] args) {
        Set<String> hashSet = new HashSet<>();

        // Add elements to the HashSet
        hashSet.add("Apple");
        hashSet.add("Banana");
        hashSet.add("Cherry");

        // Check if an element exists
        System.out.println("Contains 'Apple': " + hashSet.contains("Apple")); // true

        // Remove an element
        hashSet.remove("Banana");

        // Iterate over the HashSet
        System.out.println("HashSet elements:");
        for (String fruit : hashSet) {
            System.out.println(fruit);
        }
        // Output:
        // HashSet elements:
        // Apple
        // Cherry
    }
}
```

```
// TreeSet example
import java.util.Set;
import java.util.TreeSet;

public class TreeSetExample {
    public static void main(String[] args) {
        Set<String> treeSet = new TreeSet<>();

        // Add elements to the TreeSet
        treeSet.add("Apple");
        treeSet.add("Banana");
        treeSet.add("Cherry");

        // Check if an element exists
        System.out.println("Contains 'Apple': " + treeSet.contains("Apple")); // true

        // Remove an element
        treeSet.remove("Banana");

        // Iterate over the TreeSet
        System.out.println("TreeSet elements:");
        for (String fruit : treeSet) {
            System.out.println(fruit);
        }
        // Output:
        // TreeSet elements:
        // Apple
        // Cherry
    }
}
```

How Is HashMap Implemented in Java (explain in detail)?

A HashMap is a data structure that allows you to store and retrieve key-value pairs efficiently. Think of it like a digital dictionary where you can look up values (like definitions) using unique keys (like words).

The HashMap class implements a hash map data structure using a resizable array of power-of-two size. When adding an element, the key's hashCode is calculated and its lower bits determine the array index (bucket). Accessing elements by array index is very fast (O(1)).

However, hash collisions occur when different keys produce the same index. In Java's HashMap, each bucket points to a red-black tree (a linked list before Java 8) of colliding elements. The HashMap traverses this tree to insert or find key-value pairs, replacing existing pairs with matching keys.

Retrieving an element involves calculating the key's hashCode, finding the bucket, and traversing the tree to match the key. HashMap operations generally have O(1) complexity but can degrade to O(log n) with many collisions. Using a good hash function helps maintain performance.

When the internal array fills up, the HashMap doubles its size, triggering a costly rehashing process. Thus, it's advisable to estimate the initial size to avoid frequent resizing.

Here is an implementation of HashMap in Java:

```
// Entry.java
class Entry<K, V> {
    K key;
    V value;
    Entry<K, V> next;

    public Entry(K key, V value) {
        this.key = key;
        this.value = value;
        this.next = null;
    }
}

// MyHashMap.java
public class MyHashMap<K, V> {
    private Entry<K, V>[] buckets;
    private static final int INITIAL_CAPACITY = 16;
    private int size = 0;

    public MyHashMap() {
        this.buckets = new Entry[INITIAL_CAPACITY];
    }

    // Hash function to calculate the index for a given key
    private int getBucketIndex(K key) {
        return Math.abs(key.hashCode()) % buckets.length;
    }

    // Add or update a key-value pair
    public void put(K key, V value) {
        int bucketIndex = getBucketIndex(key);
        Entry<K, V> newEntry = new Entry<>(key, value);

        // If bucket is empty, add the new entry
        if (buckets[bucketIndex] == null) {
            buckets[bucketIndex] = newEntry;
            size++;
        } else {
            // Traverse the bucket and update the value if key is found, otherwise append the …
            Entry<K, V> current = buckets[bucketIndex];
            Entry<K, V> previous = null;
            while (current != null) {
                if (current.key.equals(key)) {
                    current.value = value; // Update value
                    return;
                }
                previous = current;
                current = current.next;
            }
            previous.next = newEntry; // Append the new entry at the end
            size++;
        }
    }

    // Retrieve a value by key
    public V get(K key) {
        int bucketIndex = getBucketIndex(key);
        Entry<K, V> current = buckets[bucketIndex];

        // Traverse the bucket to find the key
        while (current != null) {
            if (current.key.equals(key)) {
                return current.value;
            }
            current = current.next;
        }
        return null; // Key not found
    }
```

What Is the Use of the Initial Capacity and Load Factor Parameters of a HashMap and Their Default Values?

Initial Capacity:

- Purpose: Sets the initial size of the internal array that stores the key-value pairs.
- Default Value: 16.
- Effect: If the specified initial capacity is not a power of two, it is adjusted to the next power of two. For example, an initial capacity of 10 will be adjusted to 16.
- Usage: Helps to minimize the number of resizings if the number of elements to be stored is known in advance, improving performance.

Load Factor:

- Purpose: Determines when to increase the size of the internal array to maintain performance.
- Default Value: 0.75.
- Effect: When the number of elements exceeds the product of the load factor and the current capacity, the array size is doubled, and the elements are rehashed.
- Usage: Balances time complexity of operations and space usage. A lower load factor reduces the likelihood of collisions but increases memory usage, while a higher load factor saves memory but may increase collisions and degrade performance.

Choosing the right initialCapacity can be challenging. Guava's Maps.newHashMapWithExpectedSize() and Sets.newHashSetWithExpectedSize() methods help create maps and sets sized to minimize resizing.

Initial Capacity: Determines the starting size of the internal array. Load Factor: Controls the threshold for resizing the array. Default Settings: 16 for initial capacity and 0.75 for load factor, providing a good balance of performance and memory usage for general use cases.

```
// Creating a HashMap with initial capacity of 32 and load factor of 0.5
HashMap<Integer, String> map = new HashMap<>(32, 0.5f);
```

Explain EnumSet and EnumMap

EnumSet and EnumMap are specialized data structures in Java that are specifically designed to work with enumerations (enums). Both provide efficient and type-safe ways to handle collections and mappings involving enums.

EnumSet:

- Efficiency: EnumSet is implemented internally as a bit vector, which makes it very fast and efficient in terms of both time and space compared to other Set implementations like HashSet or TreeSet.
- Type-Safety: It can only contain elements of the specified enum type, ensuring type safety.
- Iteration Order: Elements are maintained in their natural order (the order in which the enum constants are declared).
- Null Handling: EnumSet does not allow null elements.

```
import java.util.EnumSet;

enum Day {
    MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY, SUNDAY
}

public class EnumSetExample {
    public static void main(String[] args) {
        EnumSet<Day> weekend = EnumSet.of(Day.SATURDAY, Day.SUNDAY);
        System.out.println("Weekend days: " + weekend); // [SATURDAY, SUNDAY]

        EnumSet<Day> workdays = EnumSet.range(Day.MONDAY, Day.FRIDAY);
        System.out.println("Workdays: " + workdays); // [MONDAY, TUESDAY, WEDNESDAY, THURSDAY,…

        EnumSet<Day> allDays = EnumSet.allOf(Day.class);
        System.out.println("All days: " + allDays); // [MONDAY, TUESDAY, WEDNESDAY, THURSDAY, …
    }
}
```

EnumMap:

- Efficiency: EnumMap is implemented using arrays, which makes it very fast and memory-efficient compared to other Map implementations like HashMap or TreeMap.
- Type-Safety: It ensures type safety by requiring the keys to be of the specified enum type.
- Iteration Order: Entries are maintained in the natural order of their keys (the order in which the enum constants are declared).
- Null Handling: EnumMap does not allow null keys, but it allows null values.

```
import java.util.EnumMap;

enum Day {
    MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY, SUNDAY
}

public class EnumMapExample {
    public static void main(String[] args) {
        EnumMap<Day, String> activityMap = new EnumMap<>(Day.class);
        activityMap.put(Day.MONDAY, "Gym");
        activityMap.put(Day.TUESDAY, "Swimming");
        activityMap.put(Day.WEDNESDAY, "Running");

        for (Day day : Day.values()) {
            System.out.println(day + ": " + activityMap.getOrDefault(day, "No activity"));
        }
        // MONDAY: Gym
        // TUESDAY: Swimming
        // WEDNESDAY: Running
        // THURSDAY: No activity
        // FRIDAY: No activity
        // SATURDAY: No activity
        // SUNDAY: No activity
    }
}
```

How to Use Comparable and Comparator Interfaces to Sort Collections?

Sorting collections in Java boils down to two heavyweight contenders: Comparable and Comparator. Both

serve the same end goal but take completely different approaches.

Comparable Interface: The "Natural Order" Champion

Comparable is all about defining the natural ordering of your objects. Think of it as teaching your class how to compare itself with other instances of the same type.

Key Characteristics:

- Single sorting logic: One way to sort, baked right into the class.
- Method signature: int compareTo(T other).
- Return values: Negative integer -> current object is "less than" the parameter. Zero -> objects are equal. Positive integer -> current object is "greater than" the parameter.

When to Use Comparable:

- When there's an obvious, universally accepted way to sort your objects.
- You want sorting to be the class's default behavior.
- Examples: sorting students by ID, products by price, dates chronologically.

```
import java.util.*;

class Student implements Comparable<Student> {
    private String name;
    private int grade;
    private int studentId;

    public Student(String name, int grade, int studentId) {
        this.name = name;
        this.grade = grade;
        this.studentId = studentId;
    }

    // Getters
    public String getName() { return name; }
    public int getGrade() { return grade; }
    public int getStudentId() { return studentId; }

    @Override
    public int compareTo(Student other) {
        // Natural order: sort by student ID (ascending)
        return Integer.compare(this.studentId, other.studentId);
    }

    @Override
    public String toString() {
        return String.format("%s (ID: %d, Grade: %d)", name, studentId, grade);
    }
}

public class ComparableExample {
    public static void main(String[] args) {
        List<Student> students = Arrays.asList(
            new Student("Alice", 85, 103),
            new Student("Bob", 90, 101),
            new Student("Charlie", 80, 102)
        );

        Collections.sort(students); // Uses natural order (by student ID)

        students.forEach(System.out::println);
        // Output:
        // Bob (ID: 101, Grade: 90)
        // Charlie (ID: 102, Grade: 80)
        // Alice (ID: 103, Grade: 85)
    }
}
```

Comparator Interface: The "Custom Logic" Powerhouse

Comparator is your go-to when you need multiple ways to sort the same object or when you can't modify the original class. It's like having a bunch of different sorting algorithms in your toolkit.

Key Characteristics:

- Multiple sorting strategies: Define as many as you need.
- Method signature: int compare(T o1, T o2).
- External to the class: Keep your sorting logic separate.
- Flexibility: Can be implemented as separate classes, anonymous classes, or lambda expressions.

When to Use Comparator:

- Multiple sorting criteria for the same class.
- Can't modify the original class (third-party libraries).
- Want to keep sorting logic separate from business logic.
- Need different sorting behavior in different contexts.

```
import java.util.*;

class Student {
    private String name;
    private int grade;
    private int studentId;

    public Student(String name, int grade, int studentId) {
        this.name = name;
        this.grade = grade;
        this.studentId = studentId;
    }

    // Getters
    public String getName() { return name; }
    public int getGrade() { return grade; }
    public int getStudentId() { return studentId; }

    @Override
    public String toString() {
        return String.format("%s (ID: %d, Grade: %d)", name, studentId, grade);
    }
}

// Approach 1: Separate Comparator class
class NameComparator implements Comparator<Student> {
    @Override
    public int compare(Student s1, Student s2) {
        return s1.getName().compareTo(s2.getName());
    }
}

class GradeComparator implements Comparator<Student> {
    @Override
    public int compare(Student s1, Student s2) {
        return Integer.compare(s2.getGrade(), s1.getGrade()); // Descending order
    }
}

public class ComparatorExample {
    public static void main(String[] args) {
        List<Student> students = Arrays.asList(
            new Student("Alice", 85, 103),
            new Student("Bob", 90, 101),
            new Student("Charlie", 80, 102),
            new Student("Diana", 95, 104)
        );

        System.out.println("=== Original List ===");
        students.forEach(System.out::println);

        // Approach 1: Using separate Comparator class
        System.out.println("\n=== Sorted by Name (Ascending) ===");
        Collections.sort(students, new NameComparator());
        students.forEach(System.out::println);

        // Approach 2: Lambda expression (most common in modern Java)
        System.out.println("\n=== Sorted by Grade (Descending) ===");
        students.sort((s1, s2) -> Integer.compare(s2.getGrade(), s1.getGrade()));
        students.forEach(System.out::println);

        // Approach 3: Method reference with Comparator utility methods
        System.out.println("\n=== Sorted by Student ID (Ascending) ===");
        students.sort(Comparator.comparing(Student::getStudentId));
        students.forEach(System.out::println);

        // Approach 4: Chaining comparators for complex sorting
```

```
// Modern Java Comparator Utilities (Java 8+)

// Basic comparisons
Comparator.comparing(Student::getName)                    // Sort by name
Comparator.comparing(Student::getGrade).reversed()        // Sort by grade (descending)

// Chaining comparators
Comparator.comparing(Student::getGrade).reversed()
          .thenComparing(Student::getName)                // Grade desc, then name asc

// Null-safe comparisons
Comparator.comparing(Student::getName,
          Comparator.nullsFirst(String::compareTo))       // Handle null names

// Natural order
Comparator.naturalOrder()                                 // For Comparable objects
Comparator.reverseOrder()                                 // Reverse natural order
```

Performance Notes:

- Comparable: Slightly faster since it's built into the object.
- Comparator: Minimal overhead, especially with method references.
- Lambda expressions: JVM optimizes these heavily, so performance is excellent.
- Anonymous classes: Slight memory overhead due to class file generation.

What Are the Main Interfaces in Java Collections?

Java Collections Framework provides several interfaces that define various types of collections:

Collection Interface:

- The root interface of the collection hierarchy. It defines the basic operations for all types of collections, such as adding, removing, and querying elements.
- Key methods: add(), remove(), size(), iterator(), clear(), isEmpty(), contains().

List Interface:

- An ordered collection (also known as a sequence). Lists can contain duplicate elements and allow positional access and insertion of elements.
- Extends Collection.
- Key methods: get(int index), set(int index, E element), add(int index, E element), remove(int index), indexOf(Object o), subList(int fromIndex, int toIndex).

Set Interface:

- A collection that cannot contain duplicate elements. It models the mathematical set abstraction.
- Extends Collection.
- Key methods: Same as Collection, but with the constraint that no duplicates are allowed.

SortedSet Interface:

- A Set that maintains its elements in ascending order. Implements the NavigableSet interface.
- Extends Set.
- Key methods: first(), last(), headSet(E toElement), tailSet(E fromElement), subSet(E fromElement, E toElement).

NavigableSet Interface:

- A SortedSet with navigation methods reporting closest matches for given search targets.
- Extends SortedSet.
- Key methods: lower(E e), floor(E e), ceiling(E e), higher(E e), pollFirst(), pollLast().

What Are the Differences Between Collection and Collections in Java?

In Java, Collection and Collections are often confused due to their similar names, but they serve different purposes and have distinct functionalities:

Collection:

- Type: Interface
- Package: java.util
- Purpose: Collection is the root interface in the Java Collections Framework. It represents a group of objects, known as elements. Some of its subinterfaces include List, Set, and Queue.
- Usage: Collection defines the basic operations that can be performed on a collection of elements, such as adding, removing, and querying elements.
- Key Methods: add(E e), remove(Object o), size(), iterator(), isEmpty(), clear(), contains(Object o).

```
Collection<String> collection = new ArrayList<>();
collection.add("Element1");
collection.add("Element2");
```

Collections:

- Type: Utility Class
- Package: java.util
- Purpose: Collections is a utility class that consists of static methods that operate on or return collections. It provides methods for manipulating and creating collections.
- Usage: Collections provides static methods for various collection operations, such as sorting, searching, reversing, and synchronizing collections. It also provides methods to create empty or singleton collections.
- Key Methods: sort(), reverse(), shuffle(), unmodifiableCollection(), synchronizedCollection(), binarySearch().

```
List<String> list = new ArrayList<>();
list.add("Element3");
list.add("Element1");
list.add("Element2");

Collections.sort(list);
System.out.println(list); // Output: [Element1, Element2, Element3]
```

How Do You Implement a Custom Collection in Java?

Implementing a custom collection in Java involves defining a new class that encapsulates the data structure you want to create and implementing relevant interfaces from the Java Collections Framework (JCF). Here's a step-by-step guide on how to create a simple custom collection, such as a custom list, by implementing the List interface.

1. Define the Class and Implement the Interface:

```
import java.util.*;

public class CustomList<E> implements List<E> {
    private Object[] elements;
    private int size = 0;
    private static final int DEFAULT_CAPACITY = 10;

    public CustomList() {
        elements = new Object[DEFAULT_CAPACITY];
    }
}
```

2. Implement Required Methods:

The List interface contains many methods that need to be implemented. Here, we'll implement the core methods: size(), add(), get(), and remove().

```
@Override
public int size() {
    return size;
}

@Override
public boolean isEmpty() {
    return size == 0;
}

@Override
public boolean add(E e) {
    ensureCapacity();
    elements[size++] = e;
    return true;
}

@Override
public E get(int index) {
    checkIndex(index);
    return (E) elements[index];
}

@Override
public E remove(int index) {
    checkIndex(index);
    E oldValue = (E) elements[index];

    int numMoved = size - index - 1;
    if (numMoved > 0) {
        System.arraycopy(elements, index + 1, elements, index, numMoved);
    }
    elements[--size] = null; // Clear to let GC do its work
    return oldValue;
}

private void ensureCapacity() {
    if (size == elements.length) {
        int newCapacity = elements.length * 2;
        elements = Arrays.copyOf(elements, newCapacity);
    }
}

private void checkIndex(int index) {
    if (index < 0 || index >= size) {
        throw new IndexOutOfBoundsException("Index: " + index + ", Size: " + size);
    }
}
```

3. Implement Remaining Methods:

The List interface has many other methods. Implementing them all can be tedious, but it ensures your collection behaves correctly as a list. Here are some more essential methods you might want to implement:

```
@Override
public void add(int index, E element) {
    checkIndexForAdd(index);
    ensureCapacity();

    System.arraycopy(elements, index, elements, index + 1, size - index);
    elements[index] = element;
    size++;
}

@Override
public boolean contains(Object o) {
    return indexOf(o) >= 0;
}

@Override
public int indexOf(Object o) {
    if (o == null) {
        for (int i = 0; i < size; i++) {
            if (elements[i] == null) {
                return i;
            }
        }
    } else {
        for (int i = 0; i < size; i++) {
            if (o.equals(elements[i])) {
                return i;
            }
        }
    }
    return -1;
}

@Override
public Object[] toArray() {
    return Arrays.copyOf(elements, size);
}

@Override
public <T> T[] toArray(T[] a) {
    if (a.length < size) {
        return (T[]) Arrays.copyOf(elements, size, a.getClass());
    }
    System.arraycopy(elements, 0, a, 0, size);
    if (a.length > size) {
        a[size] = null;
    }
    return a;
}
```

What is the Difference Between HashMap and LinkedHashMap?

HashMap:

- Order of Entries: HashMap does not maintain any order of its entries. The order of elements can change over time, especially when rehashing occurs after resizing the map.
- Performance: HashMap generally provides constant-time performance for the basic operations (get and put), assuming the hash function disperses the elements properly among the buckets.
- Internal Structure: HashMap uses a hash table to store the map entries. Each entry is stored in a bucket

which is determined by the hash code of the key.
- Use Cases: Suitable for general-purpose use where the order of the entries is not important.

```
import java.util.HashMap;
import java.util.Map;

public class HashMapExample {
    public static void main(String[] args) {
        Map<String, Integer> map = new HashMap<>();
        map.put("one", 1);
        map.put("two", 2);
        map.put("three", 3);

        System.out.println(map); // Output can be in any order
    }
}
```

LinkedHashMap:

- Order of Entries: LinkedHashMap maintains a doubly-linked list running through all of its entries. This allows it to maintain the order in which keys were inserted (insertion-order).
- Performance: LinkedHashMap has a slightly lower performance compared to HashMap due to the overhead of maintaining the linked list. However, it still provides constant-time performance for basic operations, assuming the hash function disperses the elements properly.
- Internal Structure: Like HashMap, LinkedHashMap uses a hash table to store the map entries. Additionally, each entry maintains pointers to the next and previous entries in the doubly-linked list.
- Use Cases: Suitable when you need to maintain the insertion order of the elements.

```
import java.util.LinkedHashMap;
import java.util.Map;

public class LinkedHashMapExample {
    public static void main(String[] args) {
        // Creating a LinkedHashMap
        Map<String, Integer> linkedMap = new LinkedHashMap<>();

        // Adding key-value pairs
        linkedMap.put("one", 1);
        linkedMap.put("two", 2);
        linkedMap.put("three", 3);
        linkedMap.put("four", 4);

        // Printing the map
        System.out.println(linkedMap); // {one=1, two=2, three=3, four=4}

        // Accessing an entry to check if the order is maintained
        System.out.println("Value for key 'two': " + linkedMap.get("two"));

        // Iterating through the LinkedHashMap
        System.out.println("Iterating over LinkedHashMap:");
        for (Map.Entry<String, Integer> entry : linkedMap.entrySet()) {
            System.out.println(entry.getKey() + " -> " + entry.getValue());
        }

        // Using LinkedHashMap with access-order enabled (for LRU caching)
        Map<String, Integer> accessOrderMap = new LinkedHashMap<>(16, 0.75f, true);
        accessOrderMap.put("one", 1);
        accessOrderMap.put("two", 2);
        accessOrderMap.put("three", 3);
        accessOrderMap.put("four", 4);

        // Accessing some elements
        accessOrderMap.get("one");
        accessOrderMap.get("three");

        // Printing after access to show LRU order (least recently used goes to end)
        System.out.println("LinkedHashMap with Access Order: " + accessOrderMap);
        // Output: {two=2, four=4, one=1, three=3}
    }
}
```

How Does ConcurrentHashMap Work and How Is It Different From HashMap?

ConcurrentHashMap and HashMap are both implementations of the Map interface in Java, but they are designed for different use cases, especially when it comes to handling concurrent access.

ConcurrentHashMap:

- Concurrency Level: ConcurrentHashMap divides the map into segments or buckets. Each segment is a smaller, independently synchronized map, allowing multiple threads to access different segments concurrently without contention.
- Locking Mechanism: Instead of locking the entire map, ConcurrentHashMap locks only the specific segment that a key belongs to. This fine-grained locking mechanism significantly improves performance in multi-threaded environments.
- Performance: ConcurrentHashMap provides better performance in a multi-threaded environment

compared to HashMap when synchronized externally, due to its advanced locking mechanism.

```
import java.util.concurrent.ConcurrentHashMap;
import java.util.Map;

public class ConcurrentHashMapExample {
    public static void main(String[] args) {
        Map<String, Integer> map = new ConcurrentHashMap<>();
        map.put("one", 1);
        map.put("two", 2);
        map.put("three", 3);

        map.forEach((key, value) -> System.out.println(key + ": " + value));

        // Concurrent modification example
        map.computeIfAbsent("four", k -> 4);
        System.out.println("four: " + map.get("four"));
        // Output:
        // one: 1
        // two: 2
        // three: 3
        // four: 4
    }
}
```

HashMap:

- Simple Structure: HashMap uses an array of buckets. Each bucket is a linked list (or a tree, in case of many hash collisions), where each entry contains a key-value pair.
- No Internal Synchronization: HashMap does not handle synchronization internally. When accessed by multiple threads concurrently, external synchronization is needed to avoid data corruption or inconsistent results.
- Performance: In a single-threaded environment, HashMap is faster than ConcurrentHashMap due to the lack of synchronization overhead.
- Iteration: Iterators on a HashMap are fail-fast, meaning that if the map is modified after the iterator is created, a ConcurrentModificationException is thrown.

```
import java.util.HashMap;
import java.util.Map;

public class HashMapExample {
    public static void main(String[] args) {
        Map<String, Integer> map = new HashMap<>();
        map.put("one", 1);
        map.put("two", 2);
        map.put("three", 3);

        map.forEach((key, value) -> System.out.println(key + ": " + value));
        // Output:
        // one: 1
        // two: 2
        // three: 3
    }
}
```

Explain the Internal Structure and Working of a TreeMap

Internal Structure:

1. Red-Black Tree:

- Internally, TreeMap is implemented using a Red-Black tree, which is a type of self-balancing binary search tree.
- Red-Black tree properties ensure that the tree remains approximately balanced, providing O(log n) time complexity for basic operations like get, put, remove, etc.

2. Nodes:

- Each node in the Red-Black tree contains a key-value pair.
- Every node also contains references to its parent, left child, and right child nodes.
- Additionally, each node has a color attribute, which can either be red or black.

Working of TreeMap:

1. Insertion (put method):

- When a new key-value pair is added, the put method traverses the tree to find the correct location for the new node, maintaining the sorted order of keys.
- After inserting the new node, the tree may need to be rebalanced to maintain the Red-Black tree properties. This involves recoloring and possibly rotating nodes.

2. Deletion (remove method):

- When a key is removed, the remove method locates the node and removes it from the tree.
- After deletion, the tree is rebalanced to ensure it remains a valid Red-Black tree. This may also involve recoloring and rotating nodes.

3. Searching (get method):

- The get method searches for a specific key by traversing the tree from the root, moving left or right based on the comparison of the key with the current node's key.
- The search operation has a time complexity of O(log n) due to the balanced nature of the Red-Black tree.

4. Iteration:

- TreeMap supports in-order traversal, which means iterating over the entries will return them in the sorted order of keys.
- The iterators are fail-fast, meaning they throw ConcurrentModificationException if the map is modified after the iterator is created.

How Does a PriorityQueue Work and How Is It Different From a Regular Queue?

PriorityQueue uses a binary heap. Elements are ordered based on their priority (natural order or a provided comparator). In a min-heap, the smallest element is always at the head.

Differences from a Regular Queue

Order of Elements:

- PriorityQueue: Elements are ordered by priority.
- Regular Queue (e.g., LinkedList): Elements follow FIFO order.

Head Element:

- PriorityQueue: Head is the element with the highest priority.

- Regular Queue: Head is the oldest element (first inserted).

Use Cases:

- PriorityQueue: Suitable for scheduling tasks, simulations, or any scenario where order is determined by priority.
- Regular Queue: Suitable for simple task queues, buffers, or any scenario requiring FIFO order.

```
import java.util.PriorityQueue;
import java.util.LinkedList;
import java.util.Queue;

PriorityQueue<Integer> pq = new PriorityQueue<>();
pq.offer(3);
pq.offer(1);
pq.offer(2);

System.out.println(pq.poll()); // Outputs: 1 (smallest element)

Queue<Integer> queue = new LinkedList<>();
queue.offer(3);
queue.offer(1);
queue.offer(2);

System.out.println(queue.poll()); // Outputs: 3 (first inserted element)
```

What Is the Difference Between Deep Copy and Shallow Copy in Collections?

Shallow Copy:

A shallow copy of an object is a new object which is a copy of the original object, but the internal objects referenced by the original object are not copied. Instead, the references to the original objects are copied. This means that both the original object and its shallow copy share references to the same objects.

Characteristics of Shallow Copy:

1. Copies Object Structure Only:

- The outer object is copied, but the inner objects (objects referred by the fields) are not copied.

2. Shared References:

- Any changes made to the mutable objects within the copied object will reflect in the original object, and vice versa.

```
package com.example.hungrycoder;

class ABC {
    // instance variable of the class ABC
    int x = 30;
}

public class ShallowCopyTest {
    // main method
    public static void main(String argvs[]) {
        // creating an object of the class ABC
        ABC obj1 = new ABC();

        // it will copy the reference, not value
        ABC obj2 = obj1;

        // updating the value to 6
        // using the reference variable obj2
        obj2.x = 6;

        // printing the value of x using reference variable obj1
        System.out.println("The value of x is: " + obj1.x); // The value of x is: 6
    }
}
```

Deep Copy:

A deep copy of an object is a new object which is a copy of the original object along with all the objects referenced by the original object, recursively. This means that the deep copy has its own copies of the objects, and changes to the inner objects in the copied structure do not affect the original object and vice versa.

Characteristics of Deep Copy:

1. Copies Entire Structure:

- Both the outer object and all objects it references (directly or indirectly) are recursively copied.

2. Independent Copies:

- Changes to the copied objects do not affect the original objects, and vice versa.

```
class ABC {
    // instance variable of the class ABC
    int x = 30;
}

public class DeepCopyExample {
    // main method
    public static void main(String argvs[]) {
        // creating an object of the class ABC
        ABC obj1 = new ABC();

        // it will copy the reference, not value
        ABC obj2 = new ABC();

        // updating the value to 6
        // using the reference variable obj2
        obj2.x = 6;

        // printing the value of x using reference variable obj1
        System.out.println("The value of x is: " + obj1.x); // The value of x is: 30
    }
}
```

How Do You Handle Duplicates in Collections Such as List, Set, and Map?

Handling duplicates in collections in Java varies depending on the type of collection you are using (List, Set, or Map).

List

A List in Java allows duplicate elements. However, if you need to handle or remove duplicates, you can do so manually using various approaches.

Removing Duplicates from a List

1. Using a Set to Filter Duplicates:

```
import java.util.ArrayList;
import java.util.LinkedHashSet;
import java.util.List;
import java.util.Set;

public class RemoveDuplicatesFromList {
    public static void main(String[] args) {
        List<String> listWithDuplicates = new ArrayList<>();
        listWithDuplicates.add("apple");
        listWithDuplicates.add("banana");
        listWithDuplicates.add("apple");
        listWithDuplicates.add("orange");

        // Using LinkedHashSet to maintain insertion order
        Set<String> set = new LinkedHashSet<>(listWithDuplicates);
        List<String> listWithoutDuplicates = new ArrayList<>(set);

        System.out.println("List without duplicates: " + listWithoutDuplicates);
        // List without duplicates: [apple, banana, orange]
    }
}
// This approach maintains the insertion order by using a LinkedHashSet.
```

2. Using Streams (Java 8+):

```
import java.util.ArrayList;
import java.util.List;
import java.util.stream.Collectors;

public class RemoveDuplicatesWithStreams {
    public static void main(String[] args) {
        List<String> listWithDuplicates = new ArrayList<>();
        listWithDuplicates.add("apple");
        listWithDuplicates.add("banana");
        listWithDuplicates.add("apple");
        listWithDuplicates.add("orange");

        List<String> listWithoutDuplicates = listWithDuplicates.stream()
                .distinct()
                .collect(Collectors.toList());

        System.out.println("List without duplicates: " + listWithoutDuplicates);
        // List without duplicates: [apple, banana, orange]
    }
}
```

Set

A Set in Java inherently does not allow duplicate elements. If you add a duplicate element to a Set, it will not be added. Different implementations of Set (e.g., HashSet, TreeSet, LinkedHashSet) have different behaviors regarding order and performance.

```
import java.util.HashSet;
import java.util.Set;

public class SetExample {
    public static void main(String[] args) {
        Set<String> set = new HashSet<>();
        set.add("apple");
        set.add("banana");
        set.add("apple"); // Duplicate element

        System.out.println("Set: " + set); // Set: [banana, apple]
    }
}
```

Map

A Map in Java does not allow duplicate keys, but it allows duplicate values. If you insert an entry with a key that already exists, the old value is replaced with the new value.

```
import java.util.ArrayList;
import java.util.HashMap;
import java.util.List;
import java.util.Map;

public class MapWithListValues {
    public static void main(String[] args) {
        Map<String, List<String>> map = new HashMap<>();

        addValueToMap(map, "fruit", "apple");
        addValueToMap(map, "fruit", "banana");
        addValueToMap(map, "fruit", "apple"); // Duplicate value for the same key

        System.out.println("Map: " + map); // Map: {fruit=[apple, banana, apple]}
    }

    public static void addValueToMap(Map<String, List<String>> map, String key, String value) {
        map.computeIfAbsent(key, k -> new ArrayList<>()).add(value);
    }
}
```

What Is the Difference Between Synchronized Collections and Concurrent Collections in Java?

In Java, both synchronized collections and concurrent collections are used to handle multithreaded access to data structures.

Synchronized Collections

Synchronized collections are part of the legacy collections framework and are designed to provide basic thread safety by synchronizing the entire collection. This means that all methods that access or modify the collection are synchronized, which can lead to contention and reduced performance under high concurrency.

Key Points:

1. Implementation: Synchronized collections are usually created using the Collections.synchronized* methods.

2. Synchronization Mechanism: Methods are synchronized using intrinsic locks (the synchronized keyword).

This ensures that only one thread can access the method at a time.

3. Performance: Synchronizing on the entire collection can cause significant performance bottlenecks, especially in highly concurrent environments.

4. Iteration: Iterators must be used within a synchronized block to avoid ConcurrentModificationException.

5. Use Cases: Suitable for low to moderate concurrency scenarios where thread safety is needed, but high performance is not critical.

```
import java.util.Collections;
import java.util.List;
import java.util.ArrayList;

public class SynchronizedCollectionsExample {
    public static void main(String[] args) {
        List<String> synchronizedList = Collections.synchronizedList(new ArrayList<>());
        synchronizedList.add("one");
        synchronizedList.add("two");

        synchronized (synchronizedList) {
            for (String item : synchronizedList) {
                System.out.println(item);
            }
        }
    }
    // Output:
    // one
    // two
}
```

Concurrent Collections

Concurrent collections are part of the java.util.concurrent package and are designed for high concurrency. They use advanced concurrency control mechanisms to allow multiple threads to access and modify the collection concurrently without the need for explicit synchronization on the part of the user.

Key Points:

1. Implementation: Some common concurrent collections include: ConcurrentHashMap, ConcurrentLinkedQueue, CopyOnWriteArrayList, ConcurrentSkipListMap, ConcurrentSkipListSet.

2. Synchronization Mechanism: These collections use sophisticated algorithms and non-blocking techniques to achieve thread safety. For example, ConcurrentHashMap uses a segmented locking technique, and CopyOnWriteArrayList creates a new copy of the underlying array on each write operation.

3. Performance: Designed for high concurrency and scalability. They minimize lock contention and are more efficient under heavy multi-threaded access.

4. Iteration: Iterators are weakly consistent, meaning they can tolerate concurrent modifications without throwing ConcurrentModificationException and may reflect some, all, or none of the changes made to the collection after the iterator was created.

5. Use Cases: Suitable for high concurrency scenarios where thread safety and performance are critical.

```
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.ConcurrentMap;

public class ConcurrentCollectionsExample {
    public static void main(String[] args) {
        ConcurrentMap<String, String> concurrentMap = new ConcurrentHashMap<>();
        concurrentMap.put("one", "1");
        concurrentMap.put("two", "2");

        for (String key : concurrentMap.keySet()) {
            System.out.println(key + ": " + concurrentMap.get(key));
        }
    }
    // Output:
    // one: 1
    // two: 2
}
```

SQL Query Execution Flow

SQL Fundamentals

Understanding how databases process your queries from parsing to results.

The Big Picture

When you write a SQL query and hit execute, the database engine performs a series of sophisticated operations to return your results. Understanding this process is essential for writing efficient queries, debugging performance issues, and acing technical interviews.

Why This Matters in Interviews

- "Explain the order of SQL query execution"
- "How does the query optimizer work?"
- "What is the difference between a query plan and execution plan?"
- "How would you analyze and optimize a slow query?"

The Five Phases of Query Execution

1. Parser - Validates syntax and creates AST
2. Analyzer - Validates schema and resolves references
3. Optimizer - Generates and selects best plan
4. Executor - Runs the plan and retrieves data
5. Formatter - Formats and returns results

Phase 1: Query Parsing

The parser is the first component that processes your SQL query. It converts the text string into a structured format that the database can work with.

What Happens During Parsing

1. Lexical Analysis: Breaks SQL into tokens (keywords, identifiers, operators, literals)
2. Syntax Validation: Checks that tokens follow SQL grammar rules
3. AST Generation: Creates an Abstract Syntax Tree representing the query structure

```
SELECT name FROM users WHERE id = 1

Lexer -> Tokens: SELECT, name, FROM, users, WHERE, id, =, 1
Parser -> Abstract Syntax Tree (AST)
```

Parser Catches These Errors

- Syntax errors: SELEC * FROM users (misspelled keyword)
- Missing clauses: SELECT FROM users (no columns specified)
- Unbalanced quotes: WHERE name = 'John
- Invalid operators: WHERE age === 25

Note: The parser does NOT check if tables or columns exist - that's the analyzer's job.

Phase 2: Semantic Analysis

The semantic analyzer (also called the binder) validates that your query makes logical sense by checking it against the database schema.

Analyzer Responsibilities

- Schema Validation: Verifies tables and columns exist
- Type Checking: Ensures data types are compatible
- Reference Resolution: Resolves aliases and qualified names
- Permission Checking: Validates user has access to referenced objects

```
-- This query will fail at semantic analysis, not parsing
SELECT employee_name    -- Column doesn't exist (should be 'name')
FROM employees
WHERE salary = 'high'  -- Type mismatch: salary is numeric

-- Error: Unknown column 'employee_name' in field list
```

Query Rewriting Happens Here

The analyzer may transform your query into an equivalent but more efficient form:

- View expansion: Replaces view references with underlying query
- Subquery flattening: Converts some subqueries to joins
- Predicate pushdown: Moves WHERE conditions closer to data source

Phase 3: Query Optimization

The query optimizer is the "brain" of the database engine. It generates multiple execution strategies and selects the one with the lowest estimated cost.

How the Optimizer Works

```
Parsed Query
  -> Generate Candidate Plans
       Plan 1: Index Scan
       Plan 2: Full Table Scan
       Plan 3: Hash Join
  -> Cost Estimation
  -> Select Lowest Cost
  -> Chosen Execution Plan
```

Cost Estimation Factors

```
Factor   |   What It Measures   |   Impact
I/O Cost   |   Disk reads and writes required   |   Usually the dominant factor
CPU Cost   |   Processing time for operations   |   Important for complex calculations
Cardinality   |   Estimated number of rows   |   Affects join strategy selection
Selectivity   |   Percentage of rows matching condition   |   Determines index
usefulness
```

Common Optimizer Decisions

- Index Selection - Which index to use, or whether to do a full table scan
- Join Strategy - Nested loop, hash join, or merge join
- Join Order - Which table to process first in multi-table joins

```
-- View the execution plan in PostgreSQL
EXPLAIN ANALYZE
SELECT * FROM orders o
JOIN customers c ON o.customer_id = c.id
WHERE c.country = 'USA';

-- Output shows:
-- Hash Join (cost=245.00..1890.00 rows=5000)
--   -> Seq Scan on orders (cost=0.00..1520.00 rows=50000)
--   -> Hash (cost=180.00..180.00 rows=2000)
--        -> Index Scan on customers (cost=0.00..180.00 rows=2000)
```

Why Statistics Matter

The optimizer relies heavily on table statistics to make good decisions. Stale statistics can lead to poor plans. Always run ANALYZE (PostgreSQL) or UPDATE STATISTICS (SQL Server) after major data changes.

Phase 4: Execution Engine

The execution engine takes the optimized plan and actually retrieves the data. It coordinates with the storage engine and manages resources like memory and locks.

Execution Process

1. Plan Initialization: Allocates resources, opens cursors
2. Data Retrieval: Fetches data from storage (disk or buffer cache)
3. Operation Execution: Applies filters, joins, aggregations
4. Result Buffering: Collects results for return to client

Key Execution Components

- Buffer Pool - In-memory cache for frequently accessed data pages. Reduces disk I/O dramatically.
- Lock Manager - Handles concurrency control to ensure transaction isolation.
- Transaction Log - Records all changes for durability and recovery purposes.

Iterator Model (Volcano Model)

Most databases use the iterator model where each operator in the plan implements three methods:

- Open(): Initialize the operator
- Next(): Return the next row (called repeatedly)
- Close(): Clean up resources

Data flows from bottom operators (scans) up through the tree to the top (final result).

SQL Logical Execution Order

SQL queries are NOT executed in the order you write them. Understanding the logical execution order helps you write correct and efficient queries.

1. FROM / JOIN
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT
6. DISTINCT
7. ORDER BY
8. LIMIT / OFFSET

```
Order   |   Clause   |   What Happens
1   |   FROM   |   Tables are identified and joined (Cartesian product created)
2   |   WHERE   |   Rows are filtered based on conditions
3   |   GROUP BY   |   Remaining rows are grouped
4   |   HAVING   |   Groups are filtered
5   |   SELECT   |   Columns are selected/expressions evaluated
6   |   DISTINCT   |   Duplicate rows removed
7   |   ORDER BY   |   Results are sorted
8   |   LIMIT   |   Result set is truncated
```

Common Gotcha: Column Aliases

Because SELECT is processed after WHERE and GROUP BY, you can't use column aliases in those clauses:

```
-- This FAILS
SELECT salary * 12 AS annual_salary
FROM employees
WHERE annual_salary > 100000;  -- Error: unknown column

-- This WORKS (ORDER BY is processed after SELECT)
SELECT salary * 12 AS annual_salary
FROM employees
ORDER BY annual_salary;  -- OK!
```

Top 10 Interview Questions

Master these frequently asked questions about SQL query execution:

1. What is the order of SQL query execution?

Logically: FROM/JOIN -> WHERE -> GROUP BY -> HAVING -> SELECT -> DISTINCT -> ORDER BY -> LIMIT/OFFSET. This is why you can't reference a SELECT alias in WHERE (it doesn't exist yet) but you can

in ORDER BY (it runs after SELECT). Note this is the *logical* order the engine reasons about - the optimizer may physically execute operations in a different sequence as long as the result is equivalent.

2. What is the difference between a query plan and an execution plan?

A query plan (or "plan") is the strategy the optimizer *chooses* before running the query - a tree of operations (scans, joins, sorts) with estimated costs, produced by tools like EXPLAIN. An execution plan is what you get from EXPLAIN ANALYZE - the plan annotated with *actual* runtime numbers (real rows returned, real time taken, real I/O) after the query has actually run. In short: estimated vs. actual.

3. How does the query optimizer choose between execution plans?

It enumerates candidate plans (different index choices, join orders, join algorithms), estimates the cost of each using table statistics (row counts, data distribution, selectivity), and picks the plan with the lowest estimated cost - usually dominated by I/O cost, with CPU cost as a secondary factor. This is a cost-based optimizer (CBO); it depends heavily on statistics being up to date.

4. What is a full table scan and when is it actually good?

A full table scan reads every row in a table sequentially, ignoring indexes. It's actually the *better* choice when: the table is small, the query needs a large percentage of the rows (low selectivity - e.g., >15-20%), or there's no useful index. Sequential I/O is often faster than the random I/O of jumping between an index and the table (index lookups), so for low-selectivity queries a scan can beat an index seek.

5. What is the difference between index seek and index scan?

An index seek uses the index's B-tree structure to jump directly to the matching rows (like a binary search) - fast, used for highly selective queries (e.g., WHERE id = 5). An index scan reads the entire index sequentially, in order, checking each entry - used when a large portion of rows match, or when the index is scanned for ordering rather than filtering. A seek is O(log n) to find the start point; a scan is proportional to the number of entries scanned.

6. Explain the three join algorithms databases use.

- Nested Loop Join: For each row in the outer table, scan the inner table (or probe its index) for matches. Good for small tables or when one side has a very selective index; O(n × m) worst case without an index.
- Hash Join: Build an in-memory hash table from the smaller (build) input keyed on the join column, then probe it with rows from the larger (probe) input. Efficient for large, unsorted datasets with equality joins; roughly O(n + m).
- Merge Join (Sort-Merge): Both inputs are sorted (or already sorted via an index) on the join key, then merged like the merge step of merge-sort. Efficient when both inputs are already sorted; O(n + m) given sorted inputs, O(n log n + m log m) if sorting is required first.

7. What does the parser check vs what does the analyzer check?

The parser checks pure syntax - is the SQL grammatically valid? (missing keywords, unbalanced quotes, bad operators) - producing an Abstract Syntax Tree without knowing anything about the actual schema. The analyzer/binder checks semantics against the schema - do the referenced tables and columns actually exist, are the data types compatible, does the user have permission, and it resolves aliases/qualified names. Parsing happens first; semantic errors (like a typo'd column name) only surface at the analysis phase.

8. What is the buffer pool and why is it critical for performance?

The buffer pool (or buffer cache) is an in-memory area that caches frequently/recently accessed data pages from disk. Since disk I/O is orders of magnitude slower than memory access, serving reads and writes from the buffer pool instead of hitting disk every time is the single biggest performance lever a database has. A "cold" buffer pool (after a restart) means early queries are slow until relevant pages are cached ("warmed up").

9. How do you analyze and fix a slow query?

1. Run EXPLAIN ANALYZE (or the engine's equivalent) to get the actual execution plan with real row counts and timings.
2. Look for mismatches between estimated and actual row counts (stale statistics), unexpected full table scans, or expensive operations (nested loops over large tables, sorts, hash joins spilling to disk).
3. Check whether appropriate indexes exist on filter/join columns, and whether the optimizer is actually using them.
4. Update statistics (ANALYZE / UPDATE STATISTICS) if they're stale.
5. Consider rewriting the query (avoiding functions on indexed columns, reducing subqueries, pushing filters earlier) or adding/adjusting indexes.
6. Re-run and compare the new plan and timing.

10. What is query rewriting and give examples?

Query rewriting is when the analyzer/optimizer transforms your query into a logically equivalent but more efficient form before execution. Examples:

- View expansion: A query against a view is rewritten to reference the view's underlying base-table query directly.
- Subquery flattening/unnesting: A correlated subquery is converted into an equivalent JOIN, which is often cheaper to execute.
- Predicate pushdown: A WHERE condition on a joined or aggregated result is pushed down closer to the base table scan, so filtering happens before more expensive operations like joins or grouping, reducing the number of rows processed downstream.

Database Indexing Visualized

SQL Fundamentals

Understanding how indexes work and when to use them for optimal query performance.

The Big Picture

Imagine you have a 1000-page book and need to find every mention of "database". Without an index, you'd have to read every single page. With an index at the back, you simply look up "database" and get the exact page numbers. Database indexes work the same way - they're the single most important tool for query optimization.

Why This Matters in Interviews

- "Explain how database indexes work"
- "What is the difference between clustered and non-clustered index?"
- "When should you NOT create an index?"
- "Does column order matter in a composite index?"

The Book Index Analogy

- Without Index (Full Table Scan): Reading every page of a 1000-page book to find mentions of "database". Slow but guaranteed to find everything.
- With Index (Index Seek): Looking up "database" in the back index, getting page numbers 42, 156, 891. Jump directly to those pages.

Index Scan vs Full Table Scan

Compare performance: Finding employees in the Engineering department.

```
Index Scan   |   Full Table Scan
Method   |   Using `idx_department` index   |   No index available
Rows Scanned   |   3   |   10,000
Execution Time   |   0.2ms   |   450ms
Page Reads   |   12   |   2,500
Complexity   |   O(log n)   |   O(n)
```

How B-Tree Index Works

Most database indexes use a B-Tree (Balanced Tree) data structure. It keeps data sorted and allows searches, insertions, and deletions in logarithmic time O(log n).

B-Tree Properties

- Self-Balancing: All leaf nodes are at the same depth.
- Sorted Order: Keys within each node are sorted.
- Multiple Keys per Node: Each node can hold multiple keys (reduces tree height).
- Leaf Nodes Linked: Leaves are linked for efficient range scans.

```
Root Node:      [30, 60]
               /    |    \
Branch:   [10, 20] [40, 50] [70, 80]
          /    \    /    \    /    \
Leaf:   5,8  15,18 35,38 55,58 65,68 75,85  (row pointers)
```

Search Algorithm

1. Start at Root: Compare search key with root node keys.
2. Navigate Down: Follow the appropriate child pointer based on comparison.
3. Repeat: Continue until reaching a leaf node.
4. Scan Leaf: Find the key in the leaf node or determine it doesn't exist.

Why B-Tree is Efficient

- Height is Low: A B-Tree with 1 million keys typically has height 3-4.
- Each Level = 1 Disk Read: Only 3-4 disk reads to find any record.
- Range Queries: Linked leaves allow efficient range scans.

Types of Database Indexes

Different index types serve different purposes. Understanding when to use each type is crucial for optimal performance.

Clustered Index

Determines physical order of data in table. Only one per table. Usually on primary key.

```
-- Usually created automatically on PK
CREATE TABLE users (
    id INT PRIMARY KEY -- Clustered
);
```

Non-Clustered Index

Separate structure pointing to data rows. Multiple allowed per table. Like a book's back index.

```
CREATE INDEX idx_email ON users(email);
```

Composite Index

Index on multiple columns. Column order matters! Follows leftmost prefix rule.

```
CREATE INDEX idx_name_dept ON employees(last_name, department);
```

Covering Index

Contains all columns needed by query. Avoids table lookup entirely (index-only scan).

```
CREATE INDEX idx_covering ON orders(customer_id) INCLUDE (order_date, total);
```

Unique Index

Enforces uniqueness constraint. Automatically created for PRIMARY KEY and UNIQUE constraints.

```
CREATE UNIQUE INDEX idx_email ON users(email);
```

Partial Index

Indexes only rows matching a condition. Smaller and faster for specific queries.

```
CREATE INDEX idx_active ON orders(created_at) WHERE status = 'active';
```

Clustered vs Non-Clustered: Key Differences

```
Aspect   |   Clustered Index   |   Non-Clustered Index
Data Storage   |   Data rows stored in index order   |   Separate structure with
pointers to data
Per Table   |   Only ONE allowed   |   Multiple allowed (typically up to 999)
Leaf Nodes   |   Contain actual data rows   |   Contain pointers to data rows
Speed   |   Fastest for range queries   |   Requires extra lookup (unless covering)
Size   |   Doesn't add storage (IS the table)   |   Additional storage required
```

Composite Index: Column Order Matters!

One of the most misunderstood aspects of indexing. The order of columns in a composite index determines which queries can use it efficiently.

The Leftmost Prefix Rule

An index on (A, B, C) can efficiently support queries on:

- A - Yes
- A, B - Yes
- A, B, C - Yes
- B - No (doesn't use index efficiently)
- B, C - No
- C - No

```
-- Index on (last_name, first_name, department)
CREATE INDEX idx_employee_name
ON employees(last_name, first_name, department);

-- Uses index efficiently
SELECT * FROM employees WHERE last_name = 'Smith';
SELECT * FROM employees WHERE last_name = 'Smith' AND first_name = 'John';

-- Cannot use index efficiently (skips last_name)
SELECT * FROM employees WHERE first_name = 'John';
SELECT * FROM employees WHERE department = 'Engineering';
```

Column Order Strategy

- Equality columns first: Columns used with = should come before range columns.
- High selectivity first: Columns that filter out more rows should come first.
- Consider query patterns: Order based on your most common queries.

When to Use Indexes (and When NOT To)

Indexes aren't free - they consume storage and slow down writes. Understanding when to use them is crucial.

DO Create Indexes When

- Column is frequently used in WHERE clauses.
- Column is used in JOIN conditions.
- Column is used in ORDER BY / GROUP BY.
- Column has high selectivity (many unique values).
- Table is large and queries return small result sets.
- Read operations far outnumber writes.

DON'T Create Indexes When

- Table is very small (full scan may be faster).
- Column has low selectivity (few unique values).
- Table has heavy INSERT/UPDATE/DELETE load.
- Queries return a large percentage of table rows.
- Column is rarely used in queries.
- Column values change frequently.

Index Selectivity

Selectivity = Number of Distinct Values / Total Number of Rows

```
Column   |   Distinct Values   |   Total Rows   |   Selectivity   |   Index Useful?
email   |   100,000   |   100,000   |   1.0 (100%)   |   Excellent
last_name   |   5,000   |   100,000   |   0.05 (5%)   |   Good
country   |   50   |   100,000   |   0.0005   |   Maybe
gender   |   2   |   100,000   |   0.00002   |   Poor
is_active (boolean)   |   2   |   100,000   |   0.00002   |   Poor (use partial index)
```

The Write Penalty

Every index must be updated when data changes. More indexes = slower writes.

```
INSERT Statement
  -> Write to Table
  -> Update Index 1
  -> Update Index 2
  -> Update Index 3
  -> Update Index N...
  -> Transaction Complete
```

Index Maintenance Operations

- INSERT: New entry added to every index on the table.
- UPDATE: If indexed column changes, old entry removed, new entry added.
- DELETE: Entry removed from every index.
- Page Splits: When a B-Tree node fills up, it splits into two (expensive).

Finding the Right Balance

Rule of thumb for OLTP systems: 5-10 indexes per table maximum. For OLAP/analytics: more indexes may be acceptable since writes are batched.

Analyzing Index Usage

Learn to identify missing indexes and unused indexes using database tools.

Check If Index Is Being Used

```
-- PostgreSQL: Check index usage stats
SELECT
    indexrelname AS index_name,
    idx_scan AS times_used,
    idx_tup_read AS tuples_read,
    idx_tup_fetch AS tuples_fetched
FROM pg_stat_user_indexes
WHERE schemaname = 'public'
ORDER BY idx_scan DESC;

-- Find unused indexes (idx_scan = 0)
SELECT indexrelname, idx_scan
FROM pg_stat_user_indexes
WHERE idx_scan = 0;
```

Identify Missing Indexes

```
-- SQL Server: Missing index recommendations
SELECT
    d.statement AS table_name,
    d.equality_columns,
    d.inequality_columns,
    d.included_columns,
    s.user_seeks,
    s.avg_user_impact
FROM sys.dm_db_missing_index_details d
JOIN sys.dm_db_missing_index_groups g ON d.index_handle = g.index_handle
JOIN sys.dm_db_missing_index_group_stats s ON g.index_group_handle = s.group_handle
ORDER BY s.avg_user_impact DESC;
```

Use EXPLAIN to Verify Index Usage

```
EXPLAIN ANALYZE
SELECT * FROM employees
WHERE department = 'Engineering';

-- Look for: "Index Scan" or "Index Only Scan" = Good
-- Watch out for: "Seq Scan" on large tables = May need index
```

Top 10 Interview Questions

Master these frequently asked questions about database indexing:

1. What is a database index and how does it work?

An index is a separate data structure (typically a B-Tree) that stores a sorted copy of one or more columns' values along with pointers back to the full row. It lets the database find matching rows via a fast O(log n) tree search instead of scanning every row (O(n)) - the same principle as a book's back index letting you jump to page numbers instead of reading cover to cover.

2. Explain the difference between clustered and non-clustered index.

A clustered index determines the physical storage order of the table's rows - the table data itself IS the index's leaf level, so there can only be one per table (usually on the primary key). A non-clustered index is a separate structure whose leaf nodes hold pointers (row locators) back to the actual data rows, which live elsewhere in their own order; a table can have many non-clustered indexes. Looking up via a non-clustered index that isn't "covering" requires an extra lookup step back to the table.

3. What is a composite index and does column order matter?

A composite (multi-column) index is built on two or more columns together. Column order matters enormously because of the leftmost prefix rule: an index on (A, B, C) can be used efficiently for lookups on A, (A, B), or (A, B, C), but not for a lookup on B or C alone, since the tree is sorted primarily by A first. Put equality-filtered, high-selectivity columns first.

4. What is a covering index?

A covering index includes every column a query needs - either as part of the key or as included/non-key columns - so the database can satisfy the entire query from the index alone without a separate lookup into the base table (an "index-only scan"). This avoids the extra I/O of jumping from index to table.

5. When should you NOT create an index?

Skip indexing when: the table is very small (a full scan is cheap and may beat the overhead of using an index); the column has low selectivity (few distinct values, like a boolean or a 2-3 value status column); the table sees heavy write traffic relative to reads (every index slows down INSERT/UPDATE/DELETE); the column is rarely queried; or queries typically return a large fraction of the table's rows anyway (a scan beats many random index lookups).

6. How does index selectivity affect performance?

Selectivity = distinct values / total rows. High selectivity (close to 1.0, e.g. an email column) means an index lookup narrows results to very few rows - ideal for indexing. Low selectivity (e.g. a boolean flag) means the index barely narrows anything down, so the optimizer may reasonably choose a full table scan instead, and the index adds write overhead for little read benefit (a partial index on the minority value can still help in that case).

7. What is index fragmentation and how do you fix it?

Fragmentation happens as INSERTs, UPDATEs, DELETEs, and especially B-Tree page splits leave index

pages partially empty or out of physical/logical order over time, which increases the number of page reads needed for the same logical search and hurts range-scan performance. It's fixed by rebuilding the index (recreating it from scratch, fully compacted) or reorganizing it (a lighter-weight defragmentation of existing pages), typically run as periodic maintenance.

8. Explain the difference between unique and non-unique indexes.

A unique index enforces that no two rows can have the same value in the indexed column(s) (nulls are often treated as distinct, depending on the engine) - it's both a performance structure and a data-integrity constraint, and is created automatically for PRIMARY KEY and UNIQUE column constraints. A non-unique (regular) index allows duplicate values and exists purely to speed up lookups, with no constraint behavior.

9. What happens to indexes during INSERT/UPDATE/DELETE?

INSERT adds a new entry to every index defined on the table, not just the primary structure. UPDATE removes and re-inserts the entry in any index whose indexed column(s) changed value (if unrelated columns change, non-covering indexes on those columns are untouched). DELETE removes the corresponding entry from every index. This is the "write penalty" - more indexes mean more work per write, and B-Tree node page splits during inserts are a particularly expensive case.

10. How do you identify missing or unused indexes?

For unused indexes, query the engine's usage statistics view (e.g. PostgreSQL's pg_stat_user_indexes, filtering for idx_scan = 0) to find indexes that are never used by the query planner - candidates for removal since they only add write overhead. For missing indexes, use the engine's recommendation tooling (e.g. SQL Server's sys.dm_db_missing_index_* views) or manually inspect EXPLAIN ANALYZE output on slow queries, looking for full/sequential scans on large tables where a WHERE, JOIN, or ORDER BY column has no supporting index.

SQL JOINs Explained

SQL Fundamentals

Master all JOIN types with visual explanations and practical examples.

The Big Picture

JOINs are the foundation of relational databases - they allow you to combine data from multiple tables based on related columns. Understanding JOINs is essential for writing efficient queries and is one of the most frequently tested topics in SQL interviews.

Why This Matters in Interviews

- "What is the difference between INNER JOIN and LEFT JOIN?"
- "When would you use a FULL OUTER JOIN?"
- "Explain SELF JOIN with an example"
- "How do you optimize queries with multiple JOINs?"

The Six Types of JOINs

- INNER JOIN - Returns only matching rows from both tables
- LEFT JOIN - All rows from left table, matching from right
- RIGHT JOIN - All rows from right table, matching from left

- FULL OUTER JOIN - All rows from both tables, NULL where no match
- CROSS JOIN - Cartesian product - every row combined
- SELF JOIN - Table joined to itself

Example Dataset

Employees (Left Table)

```
id   |   name   |   dept_id
1   |   Alice   |   1
2   |   Bob   |   2
3   |   Charlie   |   1
4   |   Diana   |   NULL
```

Departments (Right Table)

```
id   |   dept_name
1   |   Engineering
2   |   Marketing
3   |   Finance
```

```
SELECT e.name, d.dept_name FROM employees e INNER JOIN departments d ON e.dept_id = d.id;
```

```
name   |   dept_name
Alice   |   Engineering
Bob   |   Marketing
Charlie   |   Engineering
```

Result: 3 rows.

INNER JOIN

INNER JOIN returns only the rows where there is a match in both tables. If a row in the left table has no matching row in the right table (or vice versa), it is excluded from the result.

```
-- INNER JOIN: Only matching rows
SELECT e.name, e.salary, d.dept_name
FROM employees e
INNER JOIN departments d ON e.dept_id = d.id;

-- Equivalent using WHERE (older syntax)
SELECT e.name, e.salary, d.dept_name
FROM employees e, departments d
WHERE e.dept_id = d.id;
```

When to Use INNER JOIN

- You only want rows that have related data in both tables.
- Foreign key relationships are mandatory (NOT NULL).
- Reporting where incomplete data should be excluded.

LEFT JOIN (LEFT OUTER JOIN)

LEFT JOIN returns all rows from the left table, and the matching rows from the right table. If there is no match, NULL values are returned for the right table columns.

```
-- LEFT JOIN: All from left, matching from right
SELECT e.name, e.salary, d.dept_name
FROM employees e
LEFT JOIN departments d ON e.dept_id = d.id;

-- Find employees without a department
SELECT e.name
FROM employees e
LEFT JOIN departments d ON e.dept_id = d.id
WHERE d.id IS NULL;
```

Common Use Case: Finding Missing Relationships

LEFT JOIN with WHERE right_table.column IS NULL is a common pattern to find rows in the left table that have no corresponding rows in the right table:

- Customers who have never placed an order
- Products that have never been sold
- Employees not assigned to any project

RIGHT JOIN (RIGHT OUTER JOIN)

RIGHT JOIN is the opposite of LEFT JOIN. It returns all rows from the right table, and the matching rows from the left table. If there is no match, NULL values are returned for the left table columns.

```
-- RIGHT JOIN: All from right, matching from left
SELECT e.name, d.dept_name
FROM employees e
RIGHT JOIN departments d ON e.dept_id = d.id;

-- Find departments with no employees
SELECT d.dept_name
FROM employees e
RIGHT JOIN departments d ON e.dept_id = d.id
WHERE e.id IS NULL;
```

RIGHT JOIN vs LEFT JOIN

In practice, RIGHT JOIN is rarely used because you can always rewrite it as a LEFT JOIN by swapping the table order. Most developers prefer LEFT JOIN for consistency and readability.

```
-- These two queries are equivalent:
SELECT * FROM A RIGHT JOIN B ON A.id = B.a_id;
SELECT * FROM B LEFT JOIN A ON A.id = B.a_id;
```

FULL OUTER JOIN

FULL OUTER JOIN returns all rows from both tables. Where there is a match, the row contains data from both tables. Where there is no match, the row contains NULL for the missing side.

```
-- FULL OUTER JOIN: All rows from both tables
SELECT e.name, d.dept_name
FROM employees e
FULL OUTER JOIN departments d ON e.dept_id = d.id;

-- MySQL doesn't support FULL OUTER JOIN directly
-- Use UNION of LEFT and RIGHT JOIN instead:
SELECT e.name, d.dept_name
FROM employees e
LEFT JOIN departments d ON e.dept_id = d.id
UNION
SELECT e.name, d.dept_name
FROM employees e
RIGHT JOIN departments d ON e.dept_id = d.id;
```

When to Use FULL OUTER JOIN

- Data reconciliation between two systems.
- Finding all mismatches in both directions.
- Audit reports showing complete picture.
- Comparing two datasets for differences.

CROSS JOIN (Cartesian Product)

CROSS JOIN returns the Cartesian product of both tables - every row from the first table is combined with every row from the second table. If Table A has 10 rows and Table B has 5 rows, the result has 50 rows.

```
-- CROSS JOIN: Every combination
SELECT colors.name, sizes.name
FROM colors
CROSS JOIN sizes;

-- Equivalent syntax (implicit cross join)
SELECT colors.name, sizes.name
FROM colors, sizes;

-- Practical example: Generate all date-product combinations
SELECT d.date, p.product_name,
       COALESCE(s.quantity, 0) AS quantity
FROM dates d
CROSS JOIN products p
LEFT JOIN sales s ON d.date = s.date AND p.id = s.product_id;
```

Caution: CROSS JOIN Can Be Dangerous

CROSS JOIN can produce massive result sets. A cross join between two tables with 1000 rows each produces 1,000,000 rows. Use with caution and only when you specifically need all combinations.

- Always verify table sizes before using CROSS JOIN.
- Consider adding a WHERE clause to limit results.
- Common legitimate uses: generating test data, creating schedules, matrix reports.

SELF JOIN

A SELF JOIN is when a table is joined to itself. This is useful for hierarchical data (like employee-manager relationships) or comparing rows within the same table.

```
-- SELF JOIN: Find employees and their managers
SELECT
    e.name AS employee_name,
    m.name AS manager_name
FROM employees e
LEFT JOIN employees m ON e.manager_id = m.id;

-- Find employees who earn more than their manager
SELECT e.name, e.salary, m.name AS manager, m.salary AS manager_salary
FROM employees e
JOIN employees m ON e.manager_id = m.id
WHERE e.salary > m.salary;

-- Find pairs of employees in the same department
SELECT e1.name, e2.name, e1.dept_id
FROM employees e1
JOIN employees e2 ON e1.dept_id = e2.dept_id
WHERE e1.id < e2.id;  -- Avoid duplicates and self-pairs
```

Common SELF JOIN Use Cases

- Organizational hierarchies (employee-manager)
- Finding duplicates or similar records
- Comparing sequential records (current vs previous)
- Graph traversal (friends of friends)
- Bill of materials (parts containing other parts)

Joining Multiple Tables

Real-world queries often require joining three or more tables. The key is to chain JOINs logically, connecting each table through their relationships.

```
-- Join 3 tables: orders, customers, and products
SELECT
    c.name AS customer,
    o.order_date,
    p.product_name,
    oi.quantity,
    oi.price
FROM orders o
JOIN customers c ON o.customer_id = c.id
JOIN order_items oi ON o.id = oi.order_id
JOIN products p ON oi.product_id = p.id
WHERE o.order_date >= '2024-01-01'
ORDER BY o.order_date DESC;
```

JOIN Order and Performance

The order in which you write JOINs can affect readability, but most modern query optimizers will reorder JOINs for optimal performance. However, understanding join order helps with:

- Readability: Start from the main entity and join related tables.
- Debugging: Build queries incrementally, adding one JOIN at a time.
- Hints: Some databases allow forcing join order with hints.

Best Practice: Qualify All Column Names

When joining multiple tables, always use table aliases and qualify column names to avoid ambiguity and improve readability:

```
-- Bad: Ambiguous column names
SELECT name, date, amount FROM orders JOIN customers...

-- Good: Qualified column names with aliases
SELECT c.name, o.order_date, o.amount
FROM orders o
JOIN customers c...
```

JOIN vs Subquery

Many queries can be written using either JOINs or subqueries. Understanding when to use each approach is important for writing efficient and readable SQL.

Using JOIN - Generally preferred when you need columns from multiple tables in the result.

```
SELECT e.name, d.dept_name
FROM employees e
JOIN departments d
  ON e.dept_id = d.id
WHERE d.location = 'NYC';
```

Using Subquery - Useful for existence checks or when you only need data from one table.

```
SELECT e.name
FROM employees e
WHERE e.dept_id IN (
  SELECT id
  FROM departments
  WHERE location = 'NYC'
);
```

```
Aspect   |   JOIN   |   Subquery
Performance   |   Usually optimized well by database   |   Correlated subqueries can be
slow
Readability   |   Better for combining multiple tables   |   Better for simple existence
checks
Result Columns   |   Can select from all joined tables   |   Limited to outer query
tables
Duplicates   |   May produce duplicates (use DISTINCT)   |   IN/EXISTS naturally handles
duplicates
```

Top 10 Interview Questions

Master these frequently asked questions about SQL JOINs:

1. What is the difference between INNER JOIN and LEFT JOIN?

INNER JOIN returns only rows where a match exists in both tables - unmatched rows from either side are excluded entirely. LEFT JOIN returns every row from the left table regardless of whether a match exists in the right table; when there's no match, the right table's columns come back as NULL. So LEFT JOIN is a superset: it includes everything INNER JOIN would return, plus the left table's unmatched rows.

2. When would you use a FULL OUTER JOIN?

When you need every row from both tables regardless of whether a match exists on either side - useful for data reconciliation between two systems, finding mismatches in both directions, audit reports needing the

complete picture, or comparing two datasets to see what's in A but not B, in B but not A, and what matches. Note MySQL has no native FULL OUTER JOIN; it's emulated with UNION of a LEFT JOIN and a RIGHT JOIN.

3. Explain SELF JOIN with a practical example.

A SELF JOIN joins a table to itself, typically using two aliases to treat it as if it were two tables. The classic example is an employee-manager hierarchy stored in one employees table with a manager_id column referencing employees.id: SELECT e.name, m.name AS manager FROM employees e LEFT JOIN employees m ON e.manager_id = m.id;. It's also used to find duplicate/similar records or compare each row against other rows in the same table (e.g., employees earning more than their manager).

4. How do you find rows in one table that have no match in another?

Use a LEFT JOIN from the table you want "all rows" from, then filter with WHERE <right_table>.<column> IS NULL - this isolates the rows where the LEFT JOIN couldn't find a match and had to fill in NULLs. Example: SELECT e.name FROM employees e LEFT JOIN departments d ON e.dept_id = d.id WHERE d.id IS NULL; finds employees with no department. This same pattern finds customers with no orders, or products never sold.

5. What is a CROSS JOIN and when is it useful?

A CROSS JOIN produces the full Cartesian product of two tables - every row of the first paired with every row of the second, with no join condition. It's useful for intentionally generating all combinations: test data, scheduling/calendar grids, matrix reports, or pairing a date range with a product list to ensure every date-product combination appears (even with zero sales) before left-joining in actual data. It's dangerous on large tables since row counts multiply (1000 x 1000 = 1,000,000 rows).

6. How does JOIN order affect query performance?

In most modern databases, the query optimizer is free to reorder joins internally based on cost estimates (table sizes, available indexes, selectivity) regardless of the order you wrote them in - so the SQL-level order mainly affects readability, not execution order. That said, join order can matter for: debugging (build incrementally, one JOIN at a time), using optimizer hints to force a specific order when you know better than the optimizer, and in some engines/configurations where the optimizer's search space is limited and it takes the written order more literally.

7. What happens when you JOIN on columns with NULL values?

NULL never equals NULL in standard SQL comparison semantics, so a row with a NULL in the join column will never match another row via a standard ON a.col = b.col condition - even if both sides are NULL. This means rows with NULL foreign keys are naturally excluded from INNER JOIN matches and show up as the "no match" side in a LEFT/RIGHT JOIN. If you need NULL-aware matching, you'd need something like a.col IS NOT DISTINCT FROM b.col (PostgreSQL) or explicit IS NULL handling.

8. What is the difference between JOIN and subquery? When to use which?

A JOIN combines columns from multiple tables directly in the result set and is generally well-optimized by the query planner; it's the right choice when you need actual columns from more than one table in your output. A subquery (especially with IN/EXISTS) is often clearer and sometimes faster for pure existence/filtering checks where you don't need columns from the other table - and it naturally avoids the duplicate-row problem a JOIN can introduce on one-to-many relationships, since IN/EXISTS just test membership rather than multiplying rows.

9. How do you join three or more tables?

Chain JOIN clauses sequentially, connecting each new table through a shared key with the tables already joined: FROM orders o JOIN customers c ON o.customer_id = c.id JOIN order_items oi ON o.id = oi.order_id JOIN products p ON oi.product_id = p.id. Always alias every table and qualify column names (o.order_date,

not just order_date) to avoid ambiguity, since multiple joined tables commonly share column names like id or name.

10. Why is RIGHT JOIN rarely used in practice?

Because any RIGHT JOIN can be rewritten as an equivalent LEFT JOIN simply by swapping the order of the two tables (A RIGHT JOIN B ? B LEFT JOIN A), and most teams standardize on LEFT JOIN for consistency and readability - it keeps the "which side has all rows" convention always on the left, so you don't have to mentally flip direction depending on which keyword was used.

Key Takeaways

- INNER JOIN returns only matching rows from both tables.
- LEFT JOIN returns all rows from left table, with NULLs for non-matches.
- RIGHT JOIN is the opposite of LEFT JOIN (rarely used in practice).
- FULL OUTER JOIN returns all rows from both tables.
- CROSS JOIN produces Cartesian product - use with caution.
- SELF JOIN connects a table to itself for hierarchical data.
- LEFT JOIN with WHERE ... IS NULL finds rows without matches.
- Always use table aliases when joining multiple tables.
- JOINs are generally preferred over subqueries for combining data.
- The optimizer usually determines the best join order automatically.

Transactions and ACID Properties

SQL Fundamentals

Understanding how databases ensure data integrity and consistency.

The Big Picture

A transaction is a sequence of database operations that are treated as a single unit of work. Either all operations succeed (commit), or none of them take effect (rollback). This is crucial for maintaining data integrity, especially in concurrent environments where multiple users access the database simultaneously.

Why This Matters in Interviews

- "What are ACID properties? Explain each one."
- "What happens if a transaction fails midway?"
- "Explain the difference between COMMIT and ROLLBACK"
- "How does the database ensure durability?"

The Classic Example: Bank Transfer

Consider transferring $100 from Alice's account to Bob's account. This requires two operations: debit Alice and credit Bob. What happens if the system crashes after debiting Alice but before crediting Bob? Without transactions, $100 would simply disappear. Transactions ensure this never happens.

ACID Properties

ACID is an acronym describing the four key properties that guarantee reliable transaction processing:

Atomicity - All or Nothing

A transaction is an indivisible unit. Either all operations complete successfully, or none of them take effect. There is no partial completion.

Example: Transfer $100: Debit Alice AND Credit Bob must both succeed, or neither happens.

Consistency - Valid State to Valid State

A transaction brings the database from one valid state to another. All constraints, rules, and triggers must be satisfied after the transaction.

Example: Total money before = Total money after. Account balances cannot go negative if constrained.

Isolation - Concurrent Independence

Concurrent transactions execute as if they were running sequentially. One transaction cannot see the intermediate state of another.

Example: Two simultaneous transfers don't interfere with each other's balance calculations.

Durability - Permanent Once Committed

Once a transaction is committed, its changes are permanent and survive system failures, crashes, or power outages.

Example: After COMMIT, even if server crashes immediately, the transfer is preserved.

Transaction Lifecycle

A transaction goes through several states from start to finish:

```
BEGIN -> Active
Active -> (last operation) -> Partially Committed
Active -> (error/abort) -> Failed
Partially Committed -> (COMMIT success) -> Committed
Partially Committed -> (COMMIT failure) -> Failed
Failed -> (ROLLBACK) -> Aborted
```

```
State   |   Description   |   Can Transition To
Active   |   Transaction is executing operations   |   Partially Committed, Failed
Partially Committed   |   All operations completed, waiting for commit   |   Committed,
Failed
Committed   |   Changes are permanently saved   |   End (terminal state)
Failed   |   Error occurred, cannot proceed   |   Aborted
Aborted   |   Transaction rolled back, changes undone   |   End (terminal state)
```

Transaction Syntax

Here's how to work with transactions in SQL:

```
-- Basic transaction structure
BEGIN TRANSACTION;  -- or just BEGIN, or START TRANSACTION

-- Your SQL operations here
UPDATE accounts SET balance = balance - 100 WHERE id = 1;
UPDATE accounts SET balance = balance + 100 WHERE id = 2;

COMMIT;  -- Make changes permanent
-- or
ROLLBACK;  -- Undo all changes
```

Transaction with Error Handling

```
-- PostgreSQL example with error handling
BEGIN;

UPDATE accounts SET balance = balance - 100 WHERE id = 1;

-- Check if debit was successful
IF (SELECT balance FROM accounts WHERE id = 1) < 0 THEN
    ROLLBACK;
    RAISE EXCEPTION 'Insufficient funds';
END IF;

UPDATE accounts SET balance = balance + 100 WHERE id = 2;

COMMIT;
```

Savepoints

Savepoints allow partial rollbacks within a transaction:

```
BEGIN;

INSERT INTO orders (customer_id, total) VALUES (1, 500);
SAVEPOINT order_created;

INSERT INTO order_items (order_id, product_id) VALUES (1, 101);
INSERT INTO order_items (order_id, product_id) VALUES (1, 999);  -- This fails!

-- Roll back to savepoint (keeps order, removes items)
ROLLBACK TO SAVEPOINT order_created;

-- Try again with valid data
INSERT INTO order_items (order_id, product_id) VALUES (1, 102);

COMMIT;
```

How Databases Ensure Durability

Durability guarantees that committed transactions survive system failures. Databases achieve this through several mechanisms:

Write-Ahead Logging (WAL)

```
Transaction -> Write to WAL Log -> Acknowledge Commit -> Write to Data Files -> Checkpoint
Crash -> Recovery: Replay WAL
```

- WAL (Write-Ahead Log): Before any data modification, the change is first written to a log file.
- Sequential Writes: Log writes are sequential (fast), data file writes can be random (slower).
- Checkpoints: Periodically, all dirty pages are flushed to disk and a checkpoint record is written.
- Crash Recovery: After a crash, the database replays the WAL from the last checkpoint.

Why WAL Works

The key insight is that as long as the log is durable, data can always be reconstructed. Writing to the log is fast (sequential), and the actual data files can be updated lazily in the background.

Common Transaction Scenarios

Successful Transfer

Alice sends $100 to Bob. Both UPDATE statements succeed. COMMIT is issued. Changes are permanent.

Result: Alice has $900, Bob has $600.

Insufficient Funds

Alice (balance $50) tries to send $100. Constraint violation detected. ROLLBACK is issued.

Result: Both accounts unchanged.

System Crash Mid-Transaction

Alice debited, but system crashes before crediting Bob. On restart, uncommitted transaction is rolled back.

Result: Both accounts unchanged.

System Crash After Commit

Transfer completes, COMMIT written to WAL, then crash. On restart, committed changes are recovered.

Result: Transfer preserved (Durability).

Auto-Commit Mode

By default, many databases run in auto-commit mode where each statement is automatically wrapped in its own transaction:

```
-- With auto-commit ON (default in most databases)
UPDATE accounts SET balance = balance - 100 WHERE id = 1;  -- Immediately committed
UPDATE accounts SET balance = balance + 100 WHERE id = 2;  -- Immediately committed
-- DANGER: If second statement fails, first is already committed!

-- With explicit transaction (auto-commit effectively OFF)
BEGIN;
UPDATE accounts SET balance = balance - 100 WHERE id = 1;
UPDATE accounts SET balance = balance + 100 WHERE id = 2;
COMMIT;  -- Both committed together, or neither
```

Best Practice

For operations that must succeed or fail together, always use explicit transactions. Never rely on auto-commit for multi-statement operations that should be atomic.

Top 10 Interview Questions

Master these frequently asked questions about transactions:

1. What are ACID properties? Explain each one.

- Atomicity: A transaction is indivisible - either all its operations complete, or none do. No partial completion is ever left in place.
- Consistency: A transaction moves the database from one valid state to another, never violating constraints, rules, or triggers (e.g., total money in the system before and after a transfer must match).
- Isolation: Concurrent transactions behave as if executed sequentially - one transaction never observes another's intermediate, uncommitted state.
- Durability: Once committed, a transaction's changes are permanent and survive crashes or power loss, typically guaranteed via write-ahead logging.

2. What happens if a transaction fails midway through execution?

The transaction enters a "Failed" state and the database rolls it back (either automatically or via an explicit ROLLBACK), undoing every operation performed since BEGIN so the database returns to the state it was in before the transaction started. No partial changes are left behind - this is Atomicity in action. If a savepoint was set, the rollback can instead target just that savepoint, undoing only the operations after it while preserving earlier work in the same transaction.

3. Explain the difference between COMMIT and ROLLBACK.

COMMIT makes all changes made during the transaction permanent and visible to other transactions - once committed, Durability guarantees the changes survive even a crash. ROLLBACK discards all changes made since the transaction began (or since a named savepoint), returning the data to its prior state as if the transaction never happened. Every transaction ends in exactly one of these two outcomes.

4. What is a savepoint and when would you use it?

A savepoint is a named marker set within a transaction that you can later roll back to without aborting the entire transaction. It's useful when a transaction has multiple logical steps and you want the ability to undo just a later step while keeping earlier work - e.g., creating an order succeeds and is kept via SAVEPOINT order_created, but if inserting a specific order item fails, you ROLLBACK TO SAVEPOINT order_created to discard just the failed items and retry, instead of discarding the whole order.

5. How does the database ensure durability?

Primarily through Write-Ahead Logging (WAL): before any data file is modified, the change is first written to a sequential log file, and only after that log write succeeds is the transaction acknowledged as committed. Because sequential log writes are fast and durable, the database can commit quickly while updating the actual (random-access) data files lazily in the background. If a crash happens, recovery replays the WAL from the last checkpoint to reconstruct any committed changes that hadn't yet been flushed to the data files.

6. What is the difference between optimistic and pessimistic locking?

Pessimistic locking acquires a lock on data as soon as it's read, blocking other transactions from modifying (or sometimes even reading) it until the lock is released - it assumes conflicts are likely, trading concurrency for safety. Optimistic locking assumes conflicts are rare: transactions read data without locking it, and at commit time the database checks (often via a version number or timestamp column) whether the data changed since it was read; if it did, the transaction is rejected/retried, otherwise it commits. Optimistic locking scales better under low contention; pessimistic locking is safer under high contention.

7. What is a distributed transaction and what is 2PC?

A distributed transaction spans multiple separate databases or services (e.g., updating data in two different

database servers) that must all commit or all roll back together, even though no single database can coordinate across the boundary on its own. Two-Phase Commit (2PC) is the classic protocol for this: in the prepare phase, a coordinator asks every participant to get ready to commit and lock in that they *can* commit; only if all participants vote yes does the coordinator send the commit phase instruction to all of them (if any votes no, it tells all to abort instead). This guarantees atomicity across systems, at the cost of blocking and coordinator-failure risk.

8. What is auto-commit mode and why should you be careful with it?

Auto-commit is the default mode in most databases where every individual SQL statement is implicitly wrapped in and immediately committed as its own transaction. The danger is for multi-statement operations that need to be atomic together (like debit-then-credit in a transfer): if the first statement auto-commits successfully and the second one then fails, you're left with a corrupted, half-done state - the debit happened but the credit never did. The fix is to wrap related statements in an explicit BEGIN ... COMMIT/ROLLBACK block so they succeed or fail as one unit.

9. How do long-running transactions affect the database?

They hold locks (or prevent old row versions from being cleaned up, in MVCC systems) for extended periods, which increases contention and can block other transactions waiting on the same rows/tables, hurting overall throughput. They also keep the transaction log / undo space from being reclaimed (since the data might still be rolled back), which can bloat storage and, in some systems, degrade performance for all other queries (e.g., PostgreSQL vacuum being held up by a long-running transaction). Best practice is to keep transactions short and focused, doing any slow, non-transactional work (like external API calls) outside the transaction boundary.

10. What is a deadlock and how do databases handle it?

A deadlock in the database sense is the same core problem as in multithreading: Transaction A holds a lock on row/table X and waits for a lock on Y, while Transaction B holds a lock on Y and waits for X - neither can proceed. Databases handle this with automatic deadlock detection (periodically checking the lock wait-for graph for cycles): when a cycle is found, the database picks one transaction as the "victim," aborts it with an error, and rolls it back, releasing its locks so the other transaction(s) can proceed. Applications are expected to catch that error and retry the aborted transaction.

Key Takeaways

- A transaction is a unit of work that is atomic - all or nothing.
- ACID: Atomicity, Consistency, Isolation, Durability.
- Atomicity ensures partial failures don't corrupt data.
- Consistency maintains database constraints and rules.
- Isolation prevents concurrent transactions from interfering.
- Durability guarantees committed data survives crashes.
- Use BEGIN/COMMIT/ROLLBACK to control transactions explicitly.
- Savepoints allow partial rollbacks within a transaction.
- Write-Ahead Logging (WAL) is how databases ensure durability.
- Never rely on auto-commit for multi-statement atomic operations.

Transaction Isolation Levels

SQL Fundamentals

Understanding concurrency control and the trade-offs between consistency and performance.

The Big Picture

When multiple transactions run concurrently, they can interfere with each other in unexpected ways. Isolation levels define how much one transaction is isolated from the effects of other concurrent transactions. Higher isolation provides more consistency but reduces concurrency and performance.

Why This Matters in Interviews

- "What are the four transaction isolation levels?"
- "Explain dirty read, non-repeatable read, and phantom read"
- "Which isolation level prevents phantom reads?"
- "What is the default isolation level in PostgreSQL/MySQL?"

Read Phenomena (Concurrency Problems)

There are three main problems that can occur when transactions run concurrently:

Dirty Read

Transaction reads data written by another transaction that has not yet committed. If that transaction rolls back, you have read data that never officially existed.

Non-Repeatable Read

Transaction reads the same row twice and gets different values because another transaction modified and committed the row between the two reads.

Phantom Read

Transaction executes the same query twice and gets different sets of rows because another transaction inserted or deleted rows that match the query condition.

Visual Timeline Examples

Dirty Read Example

```
-- Transaction 1                    -- Transaction 2
BEGIN;
UPDATE accounts
  SET balance = 900
  WHERE id = 1;
                                    BEGIN;
                                    SELECT balance FROM accounts
                                      WHERE id = 1;
                                    -- Returns 900 (uncommitted!)
ROLLBACK;
                                    -- T2 has read data that never existed!
```

Non-Repeatable Read Example

```
-- Transaction 1                    -- Transaction 2
BEGIN;
SELECT balance FROM accounts
  WHERE id = 1;
-- Returns 1000
                                    BEGIN;
                                    UPDATE accounts
                                      SET balance = 500
                                      WHERE id = 1;
                                    COMMIT;
SELECT balance FROM accounts
  WHERE id = 1;
-- Returns 500 (different!)
```

Phantom Read Example

```
-- Transaction 1                    -- Transaction 2
BEGIN;
SELECT COUNT(*) FROM employees
  WHERE dept = 'Sales';
-- Returns 5
                                    BEGIN;
                                    INSERT INTO employees
                                      (name, dept)
                                      VALUES ('New', 'Sales');
                                    COMMIT;
SELECT COUNT(*) FROM employees
  WHERE dept = 'Sales';
-- Returns 6 (phantom row appeared!)
```

The Four Isolation Levels

SQL standard defines four isolation levels, from lowest to highest isolation:

READ UNCOMMITTED (Lowest)

Transactions can see uncommitted changes from other transactions. Maximum concurrency but minimum consistency. Rarely used in practice. Allows Dirty Read, Non-Repeatable Read, and Phantom Read.

READ COMMITTED (Default for many DBs)

Transactions only see committed changes. Each query sees a fresh snapshot. Good balance of consistency and performance. Prevents Dirty Read; allows Non-Repeatable Read and Phantom Read.

REPEATABLE READ (Default for MySQL)

Transaction sees a consistent snapshot from start. Same query returns same rows. MySQL InnoDB also prevents phantoms at this level. Prevents Dirty Read and Non-Repeatable Read; allows Phantom Read (except in MySQL InnoDB).

SERIALIZABLE (Highest)

Transactions execute as if they were running one at a time. Complete isolation but significantly reduced concurrency. Prevents Dirty Read, Non-Repeatable Read, and Phantom Read.

Summary Table

```
Isolation Level   |   Dirty Read   |   Non-Repeatable Read   |   Phantom Read
READ UNCOMMITTED   |   Possible   |   Possible   |   Possible
```

```
READ COMMITTED   |   Prevented   |   Possible   |   Possible
REPEATABLE READ   |   Prevented   |   Prevented   |   Possible*
SERIALIZABLE   |   Prevented   |   Prevented   |   Prevented
```

* MySQL InnoDB prevents phantom reads at REPEATABLE READ using next-key locking.

Setting Isolation Level

You can set the isolation level for a session or a specific transaction:

```
-- Set for session (all subsequent transactions)
SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;

-- Set for next transaction only
SET TRANSACTION ISOLATION LEVEL REPEATABLE READ;

-- PostgreSQL: Set within transaction
BEGIN;
SET TRANSACTION ISOLATION LEVEL SERIALIZABLE;
-- ... your queries ...
COMMIT;

-- MySQL: Check current isolation level
SELECT @@transaction_isolation;

-- PostgreSQL: Check current isolation level
SHOW transaction_isolation;
```

Database Default Isolation Levels

```
Database   |   Default Isolation Level
PostgreSQL   |   READ COMMITTED
MySQL (InnoDB)   |   REPEATABLE READ
SQL Server   |   READ COMMITTED
Oracle   |   READ COMMITTED
SQLite   |   SERIALIZABLE
H2 Database   |   READ COMMITTED
```

How Isolation Is Implemented: MVCC

Most modern databases use Multi-Version Concurrency Control (MVCC) to implement isolation without heavy locking:

How MVCC Works

- Multiple Versions: Each row can have multiple versions, each tagged with transaction IDs.
- Snapshot: When a transaction starts, it gets a snapshot of which transactions are committed.
- Visibility Rules: A row version is visible if it was created by a committed transaction before the snapshot.
- No Read Locks: Readers don't block writers, writers don't block readers.

MVCC Benefits

- Readers never block writers.
- Writers never block readers.
- High concurrency for read-heavy workloads.
- Consistent snapshots without locking.

MVCC Trade-offs

- Old versions accumulate (requires VACUUM in PostgreSQL).
- More storage space needed for multiple versions.
- Write-write conflicts still require locking or conflict detection.

Choosing the Right Isolation Level

```
Use Case   |   Recommended Level   |   Reason
Simple reads, high throughput   |   READ COMMITTED   |   Good balance of consistency and
performance
Reports that need consistent data   |   REPEATABLE READ   |   Entire report sees same
snapshot
Financial transactions   |   SERIALIZABLE   |   Maximum consistency, prevent all
anomalies
Analytics on stale data OK   |   READ UNCOMMITTED   |   Maximum throughput, approximate
results
Inventory/booking systems   |   SERIALIZABLE   |   Prevent double-booking via phantoms
```

Rule of Thumb

Start with the database default (usually READ COMMITTED). Only increase isolation level when you have a specific consistency requirement that demands it, as higher isolation reduces concurrency and can cause more conflicts/retries.

Top 10 Interview Questions

Master these frequently asked questions about isolation levels:

1. What are the four transaction isolation levels?

From lowest to highest isolation: READ UNCOMMITTED, READ COMMITTED, REPEATABLE READ, and SERIALIZABLE. Each level progressively prevents more of the three read phenomena (dirty read, non-repeatable read, phantom read) at the cost of reduced concurrency.

2. Explain dirty read, non-repeatable read, and phantom read.

A dirty read happens when a transaction reads another transaction's uncommitted changes, which may vanish if that transaction rolls back. A non-repeatable read happens when a transaction reads the same row twice within itself and gets different values because another transaction updated and committed that row in between. A phantom read happens when a transaction re-runs the same query and gets a different set of rows because another transaction inserted or deleted rows matching the query's condition in between.

3. Which isolation level prevents phantom reads?

SERIALIZABLE is the only standard level that fully prevents phantom reads. REPEATABLE READ prevents them in some implementations (notably MySQL InnoDB, via next-key locking) but the SQL standard only guarantees phantom-read prevention at SERIALIZABLE.

4. What is the default isolation level in PostgreSQL and MySQL?

PostgreSQL defaults to READ COMMITTED. MySQL (InnoDB) defaults to REPEATABLE READ, and uniquely among mainstream databases also prevents phantom reads at that level via next-key locking.

5. What is MVCC and how does it relate to isolation levels?

MVCC (Multi-Version Concurrency Control) is the mechanism most modern databases use to implement isolation without heavy locking: instead of blocking readers behind writers, each row can have multiple

versions tagged by the transaction that created them, and each transaction sees a consistent snapshot based on which versions were committed before it started. This lets readers avoid blocking writers and vice versa, while still delivering the consistency guarantees that each isolation level promises.

6. When would you use READ UNCOMMITTED isolation level?

Rarely in practice - only when you need maximum throughput and can tolerate seeing uncommitted, potentially-rolled-back data, such as rough real-time dashboards/analytics or monitoring queries where approximate numbers are acceptable and you don't want your query to wait on or be blocked by other transactions' locks.

7. How does SERIALIZABLE isolation level affect performance?

It significantly reduces concurrency because it must guarantee transactions behave as if executed strictly one at a time - implementations achieve this either through heavy locking (blocking concurrent access to the same data) or, more commonly in MVCC databases, through conflict detection that aborts one of the conflicting transactions at commit time. This means SERIALIZABLE transactions are more likely to be blocked or forced to retry due to serialization failures under contention, trading throughput for correctness.

8. What is the difference between locking and MVCC for isolation?

Lock-based isolation makes readers and writers block each other directly - a reader may need a shared lock that blocks a writer's exclusive lock, and vice versa - which guarantees consistency but limits concurrency. MVCC avoids this by keeping multiple versions of each row: readers are given a consistent snapshot of already-committed data and never need to wait on writers, and writers create new versions without blocking readers, only needing locks against other concurrent writers to the same row.

9. How do you choose the right isolation level for an application?

Start from the database's default (usually READ COMMITTED) and only raise it when you have a specific, identified consistency requirement - e.g. REPEATABLE READ for a report that must see one consistent snapshot throughout, or SERIALIZABLE for financial transactions or booking/inventory systems where double-spending or double-booking via phantom reads must be impossible. Weigh this against the performance and retry-rate cost: higher isolation means more blocking or more aborted/retried transactions under contention.

10. What happens when two transactions conflict under SERIALIZABLE?

The database detects that allowing both transactions to commit would produce a result inconsistent with some serial (one-at-a-time) execution order, and it aborts one of them with a serialization failure error rather than letting the conflicting commit through. The application is expected to catch this error and retry the aborted transaction from the beginning.

Key Takeaways

- Three read phenomena: dirty read, non-repeatable read, phantom read.
- Four isolation levels: READ UNCOMMITTED, READ COMMITTED, REPEATABLE READ, SERIALIZABLE.
- Higher isolation = more consistency but less concurrency.
- READ COMMITTED is the default for most databases (PostgreSQL, Oracle, SQL Server).
- MySQL InnoDB defaults to REPEATABLE READ.
- MVCC enables high concurrency by maintaining multiple row versions.
- SERIALIZABLE prevents all anomalies but may cause more conflicts.
- Choose isolation level based on specific consistency requirements.
- Higher isolation levels may require retry logic for conflicts.
- Know your database's default and adjust only when necessary.

Subqueries and CTEs

SQL Fundamentals

Master nested queries, common table expressions, and recursive patterns.

The Big Picture

Subqueries and CTEs (Common Table Expressions) allow you to break complex queries into manageable pieces. A subquery is a query nested inside another query, while a CTE is a named temporary result set that you can reference within a statement. Both are essential for writing readable, maintainable SQL.

Why This Matters in Interviews

- "What is the difference between a subquery and a CTE?"
- "When would you use a correlated subquery?"
- "Explain how recursive CTEs work"
- "Rewrite this subquery as a JOIN"

Types of Subqueries

Scalar Subquery - Returns a single value (one row, one column). Can be used anywhere a single value is expected.

```
SELECT name, (SELECT MAX(salary) FROM employees) AS max_sal FROM employees;
```

Row Subquery - Returns a single row with multiple columns. Used with row constructors for comparisons.

```
WHERE (dept_id, salary) = (SELECT dept_id, MAX(salary) FROM ...)
```

Table Subquery - Returns multiple rows and columns. Used in FROM clause (derived table) or with IN/EXISTS.

```
SELECT * FROM (SELECT ... ) AS derived_table WHERE ...
```

Correlated Subquery - References columns from outer query. Executes once per row of outer query - can be slow!

```
WHERE salary > (SELECT AVG(salary) FROM employees e2 WHERE e2.dept = e1.dept)
```

Subquery Examples

Subquery in WHERE Clause

```
-- Find employees who earn more than average
SELECT name, salary
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);

-- Find employees in departments located in NYC
SELECT name, dept_id
FROM employees
WHERE dept_id IN (
    SELECT id FROM departments WHERE location = 'NYC'
);
```

Subquery in FROM Clause (Derived Table)

```
-- Get department statistics and filter
SELECT dept_name, avg_salary
FROM (
    SELECT
        d.name AS dept_name,
        AVG(e.salary) AS avg_salary
    FROM departments d
    JOIN employees e ON d.id = e.dept_id
    GROUP BY d.name
) AS dept_stats
WHERE avg_salary > 75000;
```

Subquery in SELECT Clause (Scalar)

```
-- Show each employee with their department's average
SELECT
    name,
    salary,
    (SELECT AVG(salary)
     FROM employees e2
     WHERE e2.dept_id = e1.dept_id) AS dept_avg
FROM employees e1;
```

EXISTS vs IN

```
-- Using IN - good for small result sets
SELECT name FROM employees
WHERE dept_id IN (SELECT id FROM departments WHERE budget > 1000000);

-- Using EXISTS - often better for large tables
SELECT name FROM employees e
WHERE EXISTS (
    SELECT 1 FROM departments d
    WHERE d.id = e.dept_id AND d.budget > 1000000
);

-- NOT EXISTS - find employees without orders
SELECT name FROM customers c
WHERE NOT EXISTS (
    SELECT 1 FROM orders o WHERE o.customer_id = c.id
);
```

IN vs EXISTS Performance

- IN: Subquery executes once, result cached. Better when subquery result is small.
- EXISTS: Stops at first match (short-circuit). Better when outer table is small or subquery result is large.
- IN with NULLs: Be careful - NULL IN (...) returns NULL, not FALSE!

Common Table Expressions (CTEs)

A CTE is a named temporary result set defined using the WITH clause. It exists only for the duration of the query and makes complex queries more readable.

```
-- Basic CTE syntax
WITH cte_name AS (
    SELECT ... FROM ...
)
SELECT * FROM cte_name;

-- Multiple CTEs
WITH
    high_earners AS (
        SELECT * FROM employees WHERE salary > 100000
    ),
    dept_counts AS (
        SELECT dept_id, COUNT(*) AS cnt
        FROM high_earners
        GROUP BY dept_id
    )
SELECT d.name, dc.cnt
FROM departments d
JOIN dept_counts dc ON d.id = dc.dept_id;
```

CTE vs Subquery

Subquery (Derived Table) - Defined inline where used. Can be harder to read when nested.

```
SELECT *
FROM (
    SELECT dept_id,
           AVG(salary) AS avg_sal
    FROM employees
    GROUP BY dept_id
) AS dept_avg
WHERE avg_sal > 50000;
```

CTE - Defined at the top, named, can be referenced multiple times.

```
WITH dept_avg AS (
    SELECT dept_id,
           AVG(salary) AS avg_sal
    FROM employees
    GROUP BY dept_id
)
SELECT * FROM dept_avg
WHERE avg_sal > 50000;
```

When to Use CTEs

- Breaking complex queries into logical steps.
- When you need to reference the same subquery multiple times.
- Recursive queries (hierarchies, graphs).

- Improving readability of deeply nested queries.

Recursive CTEs

Recursive CTEs can reference themselves, making them perfect for hierarchical or graph data like org charts, bill of materials, or folder structures.

```
-- Recursive CTE structure
WITH RECURSIVE cte_name AS (
    -- Base case (anchor member)
    SELECT ... FROM table WHERE condition

    UNION ALL

    -- Recursive case (recursive member)
    SELECT ... FROM table
    JOIN cte_name ON ...
)
SELECT * FROM cte_name;
```

Org Chart Example

```
-- Find all employees under a manager (org hierarchy)
WITH RECURSIVE org_tree AS (
    -- Base case: start with the CEO (no manager)
    SELECT
        id,
        name,
        manager_id,
        0 AS level
    FROM employees
    WHERE manager_id IS NULL

    UNION ALL

    -- Recursive case: find employees of current level
    SELECT
        e.id,
        e.name,
        e.manager_id,
        ot.level + 1
    FROM employees e
    JOIN org_tree ot ON e.manager_id = ot.id
)
SELECT * FROM org_tree ORDER BY level, name;
```

The recursion proceeds level by level: iteration 0 finds the root (WHERE manager_id IS NULL), iteration 1 finds the root's direct reports, iteration 2 finds their direct reports, and so on until an iteration produces no new rows, at which point the recursion stops.

Subquery vs JOIN

Many queries can be written using either subqueries or JOINs. Understanding when to use each is important:

Using Subquery:

```
SELECT name
FROM employees
WHERE dept_id IN (
    SELECT id
    FROM departments
    WHERE location = 'NYC'
);
```

Using JOIN:

```
SELECT e.name
FROM employees e
JOIN departments d
  ON e.dept_id = d.id
WHERE d.location = 'NYC';
```

```
Aspect   |   Subquery   |   JOIN
Readability   |   Can be clearer for existence checks   |   Clearer when combining data from multiple tables
Performance   |   Modern optimizers often convert to JOIN   |   Generally efficient, especially with indexes
Duplicates   |   IN/EXISTS naturally handles duplicates   |   May produce duplicates (need DISTINCT)
Columns   |   Cannot easily select from subquery table   |   Can select from all joined tables
```

Rule of Thumb

- Use JOIN when you need columns from multiple tables.
- Use IN/EXISTS subquery for existence checks.
- Use CTE to improve readability of complex queries.
- Modern query optimizers often produce the same execution plan either way.

Common Patterns

Find Records with No Match

```
-- Customers with no orders (3 equivalent approaches)

-- 1. LEFT JOIN + IS NULL
SELECT c.* FROM customers c
LEFT JOIN orders o ON c.id = o.customer_id
WHERE o.id IS NULL;

-- 2. NOT EXISTS
SELECT * FROM customers c
WHERE NOT EXISTS (
    SELECT 1 FROM orders o WHERE o.customer_id = c.id
);

-- 3. NOT IN (careful with NULLs!)
SELECT * FROM customers
WHERE id NOT IN (
    SELECT customer_id FROM orders WHERE customer_id IS NOT NULL
);
```

Find Maximum Per Group

```
-- Highest paid employee per department

-- Using correlated subquery
SELECT * FROM employees e1
WHERE salary = (
    SELECT MAX(salary) FROM employees e2
    WHERE e2.dept_id = e1.dept_id
);

-- Using CTE with window function (preferred)
WITH ranked AS (
    SELECT *,
           RANK() OVER (PARTITION BY dept_id ORDER BY salary DESC) AS rn
    FROM employees
)
SELECT * FROM ranked WHERE rn = 1;
```

Running Difference

```
-- Calculate change from previous row using CTE
WITH with_prev AS (
    SELECT
        date,
        value,
        LAG(value) OVER (ORDER BY date) AS prev_value
    FROM metrics
)
SELECT
    date,
    value,
    value - prev_value AS change
FROM with_prev;
```

Top 10 Interview Questions

Master these frequently asked questions about subqueries and CTEs:

1. What is the difference between a subquery and a CTE?

A subquery is a query nested inline inside another query (in WHERE, SELECT, or FROM), defined at the point of use and not named. A CTE (WITH name AS (...)) is defined up front, given a name, and can be referenced one or more times later in the same statement, which makes complex queries more readable and avoids repeating the same subquery logic. Functionally, a non-recursive CTE is often treated by the optimizer much like a subquery/derived table - the main practical difference is readability and reusability, plus CTEs support recursion, which subqueries don't.

2. What is a correlated subquery and why can it be slow?

A correlated subquery references a column from the outer query (e.g. WHERE e2.dept_id = e1.dept_id), which means, conceptually, it must be re-evaluated once for every row processed by the outer query rather than being computed a single time. For large outer tables this can turn into effectively an O(n × m) operation, which is why they're often rewritten as JOINs or window functions for better performance, though modern optimizers can sometimes rewrite/decorrelate simple cases automatically.

3. What is the difference between IN and EXISTS?

IN evaluates the subquery once, materializes its full result set, and checks whether the outer value appears in it - efficient when the subquery's result set is small. EXISTS is a boolean check that stops as soon as it

finds one matching row (short-circuits), making it efficient when the outer table is small or when the subquery itself would produce a huge result set. A key gotcha: IN behaves unexpectedly with NULLs (x IN (1, NULL) can yield NULL instead of a clean true/false), whereas EXISTS/NOT EXISTS doesn't have this problem.

4. How do recursive CTEs work?

A recursive CTE has two parts combined with UNION ALL: an anchor/base case (a plain query that produces the starting rows, e.g. the root of a hierarchy with no parent) and a recursive case that joins the CTE back to itself to find the "next level" based on the previous iteration's rows. The database repeatedly executes the recursive case, feeding each iteration's output back in as input to the next, until an iteration produces zero new rows, at which point the recursion terminates and all accumulated rows are returned.

5. When would you use a subquery vs a JOIN?

Use a JOIN when you need actual columns from more than one table in your result set - it's the natural tool for combining data. Use a subquery (particularly with IN/EXISTS/NOT EXISTS) when you only need to filter the outer table based on a condition in another table and don't need to display any of that other table's columns - it avoids the risk of duplicate rows that a JOIN can introduce on a one-to-many relationship.

6. Why should you be careful with NOT IN and NULL values?

If the subquery feeding a NOT IN can return even a single NULL value, the entire NOT IN expression evaluates to NULL (not true or false) for every comparison, because SQL's NULL comparison semantics propagate - this silently causes the query to return zero rows instead of the expected result. The safe fix is to either filter out NULLs explicitly in the subquery (WHERE column IS NOT NULL) or use NOT EXISTS, which doesn't have this problem since it's a row-existence check rather than a value-membership check.

7. Can a CTE improve query performance?

Generally, a plain (non-recursive, non-materialized) CTE does not inherently improve performance on its own - most optimizers treat it like an inlined subquery/derived table and may even re-evaluate it multiple times if referenced more than once, unless the engine materializes it. Its main benefit is readability and maintainability, letting you name and reuse a complex intermediate result. Some databases (e.g. PostgreSQL with MATERIALIZED, or when a CTE is referenced multiple times) can use CTEs to avoid recomputing the same subquery, which can help performance in those specific cases - but it's not a performance feature by default.

8. How do you find the top N records per group using a subquery?

The classic correlated-subquery approach for top-1 per group compares each row's value to the max within its group: SELECT * FROM employees e1 WHERE salary = (SELECT MAX(salary) FROM employees e2 WHERE e2.dept_id = e1.dept_id). For top-N (N > 1) or ties handled cleanly, the preferred modern approach uses a window function inside a CTE: rank rows with ROW_NUMBER()/RANK() OVER (PARTITION BY dept_id ORDER BY salary DESC), then filter the CTE's result with WHERE rn <= N.

9. What is a derived table?

A derived table is a subquery used in the FROM clause and given an alias, effectively creating an unnamed, temporary, in-memory "table" that the rest of the query can select from and filter further - e.g. SELECT * FROM (SELECT dept_id, AVG(salary) AS avg_sal FROM employees GROUP BY dept_id) AS dept_avg WHERE avg_sal > 50000. It's functionally very similar to a CTE, but defined inline at the point of use rather than named up front.

10. How do you prevent infinite loops in recursive CTEs?

Most databases impose a default recursion depth limit (e.g. PostgreSQL defaults don't hard-limit by default but SQL Server defaults to 100 and allows OPTION (MAXRECURSION n)) that will error out or stop a runaway recursive CTE caused by cyclic data (e.g. a corrupted hierarchy where an employee is indirectly

their own manager). Beyond relying on that safety net, well-designed recursive CTEs should include an explicit termination condition - such as tracking visited IDs in an array/path column and adding a WHERE NOT (id = ANY(path)) guard in the recursive member - to detect and stop cycles in the actual data rather than depending solely on an engine-imposed depth cap.

Key Takeaways

- Subqueries can return scalar values, single rows, or tables.
- Correlated subqueries reference outer query - execute once per row.
- CTEs (WITH clause) improve readability and allow reuse.
- Recursive CTEs have base case + recursive case for hierarchies.
- EXISTS stops at first match (efficient); IN loads full result.
- NOT IN with NULLs can give unexpected results - use NOT EXISTS.
- Many subqueries can be rewritten as JOINs and vice versa.
- Use JOIN when you need columns from multiple tables.
- Use CTE + window functions for "top N per group" problems.
- Modern optimizers often produce same plan for equivalent queries.

Window Functions Deep Dive

SQL Fundamentals

Master powerful analytics with OVER, PARTITION BY, and frame specifications.

The Big Picture

Window functions perform calculations across a set of rows that are related to the current row. Unlike GROUP BY which collapses rows, window functions keep all rows while adding computed values. They are essential for analytics, rankings, running totals, and comparing values across rows.

Why This Matters in Interviews

- "What is a window function and how is it different from GROUP BY?"
- "Explain ROW_NUMBER vs RANK vs DENSE_RANK"
- "How do you get the top N records per group?"
- "What is the difference between ROWS and RANGE?"

Window Function Syntax

```
function_name(arguments) OVER (
    [PARTITION BY partition_expression]
    [ORDER BY sort_expression]
    [frame_clause]
)
```

- OVER: Required keyword that defines this as a window function.
- PARTITION BY: Divides rows into groups (like GROUP BY but keeps rows).
- ORDER BY: Defines the order within each partition.
- Frame Clause: Defines which rows to include in the calculation.

Types of Window Functions

- Ranking Functions - ROW_NUMBER, RANK, DENSE_RANK, NTILE: assign positions to rows based on order.
- Value Functions - LEAD, LAG, FIRST_VALUE, LAST_VALUE, NTH_VALUE: access values from other rows.
- Aggregate Functions - SUM, AVG, COUNT, MIN, MAX: used with OVER for running totals and moving averages.

```
SELECT name, department, salary,
       ROW_NUMBER() OVER (
           PARTITION BY department
           ORDER BY salary DESC
       ) AS row_num
FROM employees;
```

ROW_NUMBER vs RANK vs DENSE_RANK

These three ranking functions handle ties differently. Understanding the difference is crucial:

```
Employee   |   Salary   |   ROW_NUMBER   |   RANK   |   DENSE_RANK
Alice   |   $90,000   |   1   |   1   |   1
Bob   |   $80,000   |   2   |   2   |   2
Charlie   |   $80,000   |   3   |   2   |   2
Diana   |   $70,000   |   4   |   4   |   3
```

Key Differences

- ROW_NUMBER: Always unique, arbitrary for ties (1, 2, 3, 4).
- RANK: Same rank for ties, skips numbers (1, 2, 2, 4).
- DENSE_RANK: Same rank for ties, no gaps (1, 2, 2, 3).

```
-- All three ranking functions in one query
SELECT
    name,
    salary,
    ROW_NUMBER() OVER (ORDER BY salary DESC) AS row_num,
    RANK() OVER (ORDER BY salary DESC) AS rank,
    DENSE_RANK() OVER (ORDER BY salary DESC) AS dense_rank
FROM employees;
```

Top N Per Group Pattern

One of the most common interview questions: "Get the top 3 highest paid employees per department."

```
-- Top 3 employees per department by salary
WITH ranked AS (
    SELECT
        name,
        department,
        salary,
        ROW_NUMBER() OVER (
            PARTITION BY department
            ORDER BY salary DESC
        ) AS rn
    FROM employees
)
SELECT name, department, salary
FROM ranked
WHERE rn <= 3;
```

ROW_NUMBER vs RANK for Top N

Use ROW_NUMBER when you want exactly N rows per group, even with ties. Use RANK or DENSE_RANK when you want to include all tied values (may return more than N rows).

LAG and LEAD Functions

Access data from previous (LAG) or subsequent (LEAD) rows without self-joins:

```
-- Compare each sale to previous sale
SELECT
    sale_date,
    amount,
    LAG(amount) OVER (ORDER BY sale_date) AS prev_amount,
    amount - LAG(amount) OVER (ORDER BY sale_date) AS change
FROM sales;

-- With default value for first row
SELECT
    sale_date,
    amount,
    LAG(amount, 1, 0) OVER (ORDER BY sale_date) AS prev_amount
FROM sales;

-- Look ahead with LEAD
SELECT
    employee_name,
    hire_date,
    LEAD(employee_name) OVER (ORDER BY hire_date) AS next_hire
FROM employees;
```

LAG/LEAD Syntax

```
LAG(column, offset, default) OVER (ORDER BY ...)
LEAD(column, offset, default) OVER (ORDER BY ...)
```

- offset: Number of rows back (LAG) or forward (LEAD). Default is 1.
- default: Value to return when there is no row. Default is NULL.

Running Totals and Moving Averages

Aggregate functions with OVER can calculate cumulative and moving computations:

```
-- Running total of sales
SELECT
    sale_date,
    amount,
    SUM(amount) OVER (
        ORDER BY sale_date
        ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
    ) AS running_total
FROM sales;

-- 3-day moving average
SELECT
    sale_date,
    amount,
    AVG(amount) OVER (
        ORDER BY sale_date
        ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
    ) AS moving_avg_3day
FROM sales;

-- Running total per department
SELECT
    department,
    sale_date,
    amount,
    SUM(amount) OVER (
        PARTITION BY department
        ORDER BY sale_date
    ) AS dept_running_total
FROM sales;
```

Frame Specification: ROWS vs RANGE

The frame clause defines which rows are included in the window calculation:

```
ROWS | RANGE BETWEEN start AND end

Where start/end can be:
  UNBOUNDED PRECEDING  -- From the first row of partition
  n PRECEDING          -- n rows before current
  CURRENT ROW          -- Current row
  n FOLLOWING          -- n rows after current
  UNBOUNDED FOLLOWING  -- To the last row of partition
```

```
Aspect   |   ROWS   |   RANGE
Definition   |   Physical row positions   |   Logical value ranges
With Ties   |   Processes rows individually   |   Includes all rows with same ORDER BY
value
Performance   |   Generally faster   |   May be slower
Use Case   |   "Last 3 rows"   |   "All rows with same date"
```

```
-- ROWS: Exactly 3 physical rows
SUM(amount) OVER (
    ORDER BY sale_date
    ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
)

-- RANGE: All rows with same date value as current
SUM(amount) OVER (
    ORDER BY sale_date
    RANGE BETWEEN CURRENT ROW AND CURRENT ROW
)
```

Default Frame

When you specify ORDER BY without a frame clause, the default is:

```
RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
```

This is why SUM with just ORDER BY gives a running total by default.

Window Function vs GROUP BY

GROUP BY (Aggregation) - Collapses rows into groups. You lose individual row details.

```
SELECT
    department,
    AVG(salary) AS avg_salary
FROM employees
GROUP BY department;

-- Result: One row per department
-- Engineering | 85000
-- Sales       | 75000
```

Window Function - Keeps all rows while adding computed columns.

```
SELECT
    name,
    department,
    salary,
    AVG(salary) OVER (
        PARTITION BY department
    ) AS dept_avg
FROM employees;

-- Result: All rows preserved
-- Alice | Engineering | 90000 | 85000
-- Bob   | Engineering | 80000 | 85000
```

When to Use Each

- GROUP BY: When you want summary statistics only (report totals, averages per group).
- Window Function: When you need individual rows WITH aggregate context (employee salary vs department average).

Practical Examples

Calculate Percentage of Total

```
SELECT
    product_name,
    sales,
    ROUND(
        100.0 * sales / SUM(sales) OVER (),
        2
    ) AS pct_of_total
FROM products;
```

Find Gaps in Sequences

```
SELECT
    id,
    LEAD(id) OVER (ORDER BY id) AS next_id,
    LEAD(id) OVER (ORDER BY id) - id AS gap
FROM orders
WHERE LEAD(id) OVER (ORDER BY id) - id > 1;
```

Year-over-Year Comparison

```
SELECT
    year,
    revenue,
    LAG(revenue) OVER (ORDER BY year) AS prev_year,
    ROUND(
        100.0 * (revenue - LAG(revenue) OVER (ORDER BY year))
        / LAG(revenue) OVER (ORDER BY year),
        2
    ) AS yoy_growth_pct
FROM annual_revenue;
```

Assign Row to Buckets (Quartiles)

```
SELECT
    name,
    salary,
    NTILE(4) OVER (ORDER BY salary) AS salary_quartile
FROM employees;
```

Top 10 Interview Questions

Master these frequently asked questions about window functions:

1. What is a window function and how is it different from GROUP BY?

A window function performs a calculation across a set of related rows (the "window," defined by OVER, optionally with PARTITION BY/ORDER BY) but returns a value for every individual row rather than collapsing rows into groups. GROUP BY reduces N rows into one row per group with aggregated values, losing per-row detail; a window function keeps all N rows and simply adds a computed column (e.g. each employee's row alongside their department's average salary).

2. Explain the difference between ROW_NUMBER, RANK, and DENSE_RANK.

All three assign a position to each row based on an ORDER BY, but they differ on ties. ROW_NUMBER always gives a unique, sequential number even for tied values (1, 2, 3, 4 - arbitrary which tied row gets which number). RANK gives tied rows the same rank but then skips subsequent numbers accordingly (1, 2, 2, 4).

DENSE_RANK gives tied rows the same rank but doesn't skip numbers afterward (1, 2, 2, 3).

3. How do you get the top N records per group?

Wrap the table in a CTE that computes ROW_NUMBER() OVER (PARTITION BY group_column ORDER BY sort_column DESC), then filter the CTE's result with WHERE rn <= N in the outer query. Use ROW_NUMBER for exactly N rows per group even with ties, or RANK/DENSE_RANK if you want all tied rows included (which may return more than N rows for a group).

4. What are LEAD and LAG functions used for?

They let you access a value from another row in the result set - relative to the current row, based on the ORDER BY - without writing a self-join. LAG looks backward (the previous row's value, useful for period-over-period comparisons like "change from yesterday"), and LEAD looks forward (the next row's value, useful for finding gaps or "what happens next"). Both accept an optional offset (how many rows back/forward, default 1) and a default value to use when there's no such row (e.g. the very first row has no LAG).

5. How do you calculate a running total?

Use an aggregate function like SUM with OVER (ORDER BY sort_column) - when ORDER BY is present without an explicit frame, most databases default the frame to RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW, which naturally produces a running/cumulative total up through the current row. You can make this explicit and ROWS-based with ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW, and combine with PARTITION BY to get a running total that resets per group (e.g. per department).

6. What is the difference between ROWS and RANGE frame specifications?

ROWS defines the frame by physical row positions - e.g. "the 2 rows immediately before this one, plus this one" - regardless of whether their ORDER BY values are equal. RANGE defines the frame by logical value ranges relative to the current row's ORDER BY value - e.g. "all rows with the same date as the current row" get included together, even if that's more than a fixed number of physical rows. ROWS is generally faster and more predictable; RANGE is useful when ties on the ORDER BY column should be treated as a single logical group.

7. Can you use window functions in a WHERE clause?

No - window functions are evaluated after the WHERE clause (and after GROUP BY/HAVING) in SQL's logical processing order, so their results aren't yet available when WHERE is applied. To filter on a window function's result, you must compute it in a subquery or CTE first, then filter in an outer query's WHERE clause against that computed column (this is exactly the pattern used for "top N per group").

8. What is NTILE and when would you use it?

NTILE(n) divides the ordered rows in a partition into n roughly equal-sized buckets and returns each row's bucket number (1 through n). It's commonly used for percentile/quantile analysis - e.g. NTILE(4) to split employees into salary quartiles, or NTILE(100) for percentile ranking - letting you bucket continuous data into discrete groups for reporting or further analysis.

9. How do window functions affect query performance?

Window functions typically require the database to sort (and/or partition) the data according to the PARTITION BY/ORDER BY clause, which has a cost (often O(n log n) for the sort) - and using multiple different PARTITION BY/ORDER BY combinations in one query may force multiple separate sort passes. That said, they're usually far more efficient than the equivalent hand-written correlated subqueries or self-joins they replace, since the database can compute the window calculation in a single pass per sort rather than re-scanning the table once per outer row.

10. How do you calculate percentage of total using window functions?

Divide each row's value by the sum of that value across the whole result set (or partition), computed with an empty or partitioned OVER: 100.0 * sales / SUM(sales) OVER () for percentage of grand total, or 100.0 * sales / SUM(sales) OVER (PARTITION BY category) for percentage within each category. The window function recomputes the denominator (the total) for every row without collapsing the result, so each row shows both its own value and its share of the relevant total side by side.

Key Takeaways

- Window functions calculate across rows without collapsing them (unlike GROUP BY).
- OVER clause is required - defines the window for calculation.
- PARTITION BY divides data into groups (like GROUP BY but keeps rows).
- ROW_NUMBER: unique sequential; RANK: gaps for ties; DENSE_RANK: no gaps.
- LAG looks back, LEAD looks forward in the ordered result.
- Running totals use SUM with ORDER BY and appropriate frame.
- ROWS counts physical rows, RANGE uses logical value ranges.
- Default frame with ORDER BY is RANGE UNBOUNDED PRECEDING to CURRENT ROW.
- Top N per group pattern: ROW_NUMBER + PARTITION BY in a CTE, then filter.
- Window functions cannot be used in WHERE clause - use CTE or subquery.

Query Optimization Techniques

SQL Fundamentals

Learn to identify and fix performance bottlenecks in SQL queries.

The Big Picture

Query optimization is the art of making database queries run faster while returning the same results. A poorly written query can take seconds or minutes; an optimized version might take milliseconds. Understanding optimization is crucial for building scalable applications and is heavily tested in interviews.

Why This Matters in Interviews

- "How do you identify slow queries?"
- "What is the N+1 problem and how do you solve it?"
- "How do you read an EXPLAIN plan?"
- "What makes a query slow and how would you optimize it?"

Key Optimization Areas

- Indexing - Create indexes on columns used in WHERE, JOIN, and ORDER BY clauses.
- Query Rewriting - Restructure queries to use indexes effectively and reduce work.
- Schema Design - Normalize or denormalize based on read/write patterns.
- Batching - Combine multiple queries into fewer round trips.
- Caching - Cache query results to avoid repeated database hits.
- EXPLAIN Analysis - Use execution plans to understand query behavior.

The N+1 Query Problem

The N+1 problem is one of the most common performance issues. It occurs when code executes 1 query to fetch N records, then N additional queries to fetch related data for each record. This is extremely common with ORMs like Hibernate, ActiveRecord, and Entity Framework.

N+1 Problem (Slow) - 1 query for orders + 100 queries for customers = 101 queries

```
-- Query 1: Get all orders
SELECT * FROM orders;

-- Then for EACH order (N times):
SELECT * FROM customers
WHERE id = 1; -- order 1
SELECT * FROM customers
WHERE id = 2; -- order 2
-- ... repeated 100 times!
```

JOIN Solution (Fast) - 1 single query fetches everything

```
-- Single query with JOIN
SELECT
    o.*,
    c.name AS customer_name,
    c.email AS customer_email
FROM orders o
JOIN customers c
    ON o.customer_id = c.id;

-- All data in ONE round trip!
```

Alternative Solutions

```
-- Solution 2: Batch with IN clause (when JOIN not ideal)
SELECT * FROM orders;
-- Collect all customer_ids: [1, 2, 3, 4, 5...]
SELECT * FROM customers
WHERE id IN (1, 2, 3, 4, 5, ...);
-- Only 2 queries total!

-- ORM Solutions:
-- JPA/Hibernate: @BatchSize, JOIN FETCH, EntityGraph
-- ActiveRecord: includes(:customer)
-- Entity Framework: Include(o => o.Customer)
```

Reading EXPLAIN Plans

EXPLAIN shows how the database will execute a query. Understanding it is essential for optimization. Key things to look for: table scans vs index usage, join methods, and estimated costs.

```
-- Basic EXPLAIN
EXPLAIN SELECT * FROM employees WHERE dept_id = 5;

-- EXPLAIN with more details (PostgreSQL)
EXPLAIN ANALYZE SELECT * FROM employees WHERE dept_id = 5;

-- MySQL format
EXPLAIN FORMAT=JSON SELECT * FROM employees WHERE dept_id = 5;
```

What to Look For

```
Indicator   |   Good   |   Bad
Access Type   |   const, eq_ref, ref, range   |   ALL (full table scan)
Key   |   Index name shown   |   NULL (no index used)
Rows   |   Small number   |   Large number (scanning many rows)
Extra   |   Using index   |   Using filesort, Using temporary
```

Example EXPLAIN Output

Optimized query: Index Scan using idx_employees_dept on employees - cost 0.15..8.20, filter dept_id = 5, rows: 25.

Unoptimized query: Seq Scan on employees (full table scan) - cost 0.00..1520.00, filter dept_id = 5, rows: 50000 scanned, 25 returned.

Common Query Anti-patterns

1. SELECT * (Selecting All Columns) - Fetches unnecessary data, prevents covering index usage, increases network traffic.
2. Functions on Indexed Columns - WHERE YEAR(created_at) = 2024 prevents index usage. Use range instead.
3. Implicit Type Conversion - WHERE varchar_col = 123 forces conversion on every row, preventing index use.
4. Leading Wildcard in LIKE - WHERE name LIKE '%smith' cannot use index. Only trailing wildcard can.
5. OR Conditions on Different Columns - WHERE a = 1 OR b = 2 often prevents index usage. Consider UNION instead.

Anti-pattern Fixes

```
-- BAD: Function on indexed column
SELECT * FROM orders
WHERE YEAR(created_at) = 2024;

-- GOOD: Use range comparison
SELECT * FROM orders
WHERE created_at >= '2024-01-01'
  AND created_at < '2025-01-01';

-- BAD: Leading wildcard
SELECT * FROM users WHERE email LIKE '%@gmail.com';

-- GOOD: Consider full-text search or reversed index
-- Or store email_domain separately

-- BAD: OR on different columns
SELECT * FROM products
WHERE category_id = 5 OR brand_id = 10;

-- GOOD: Use UNION
SELECT * FROM products WHERE category_id = 5
UNION
SELECT * FROM products WHERE brand_id = 10;
```

Index Optimization Tips

When to Create Indexes

- Columns frequently used in WHERE clauses.
- Columns used in JOIN conditions.
- Columns used in ORDER BY (to avoid sorting).
- Columns with high selectivity (many unique values).

When NOT to Create Indexes

- Small tables (full scan is fast enough).
- Columns with low selectivity (e.g., boolean, status with few values).
- Heavily updated tables (indexes slow down writes).
- Columns rarely used in queries.

Composite Index Column Order

```
-- Index on (a, b, c) can be used for:
WHERE a = 1                      -- Yes (leftmost)
WHERE a = 1 AND b = 2            -- Yes
WHERE a = 1 AND b = 2 AND c = 3  -- Yes (all columns)
WHERE b = 2                      -- No (skipped a)
WHERE a = 1 AND c = 3            -- Partial (only a used)
WHERE c = 3                      -- No

-- Rule: Index is used left-to-right until a gap
```

Covering Index

A covering index includes all columns needed by a query, so the database can answer entirely from the index without reading the table. This is the fastest possible access.

```
-- Create covering index
CREATE INDEX idx_covering
ON orders (customer_id, status)
INCLUDE (total, created_at);

-- This query uses only the index
SELECT status, total, created_at
FROM orders
WHERE customer_id = 123;
```

Efficient Pagination

OFFSET-based pagination becomes slow for large offsets because the database must scan and discard rows:

OFFSET Pagination (Slow)

```
-- Page 1000 (slow!)
SELECT * FROM products
ORDER BY id
LIMIT 20 OFFSET 19980;
-- Must scan 20,000 rows!
```

Keyset Pagination (Fast)

```
-- Use last seen ID
SELECT * FROM products
WHERE id > 19980  -- last_id
ORDER BY id
LIMIT 20;
-- Index seek, very fast!
```

Keyset Pagination Limitations

- Cannot jump to arbitrary page (no "go to page 500").
- Requires stable, unique sort key.
- More complex with multi-column sorts.
- Best for infinite scroll or "load more" patterns.

Top 10 Interview Questions

Master these frequently asked questions about query optimization:

1. What is the N+1 query problem and how do you solve it?

It's when code runs 1 query to fetch a list of N records, then issues N additional queries to fetch related data for each record individually (common with ORM lazy loading), resulting in N+1 total round trips instead of 1 or 2. It's solved by fetching the related data in bulk: either a single JOIN that pulls everything at once, or a batched WHERE id IN (...) query collecting all the related IDs first - ORMs expose this via eager-loading features (JOIN FETCH in JPA/Hibernate, includes in ActiveRecord, Include() in Entity Framework).

2. How do you identify slow queries?

Enable and review the database's slow query log (e.g. MySQL's slow query log, PostgreSQL's log_min_duration_statement), which records queries exceeding a configured duration threshold. Beyond logs, use EXPLAIN ANALYZE on suspect queries to see actual execution time and row counts, check database-level monitoring/APM tooling for query latency percentiles, and look at usage statistics views (e.g.

pg_stat_statements in PostgreSQL) that rank queries by total or average execution time.

3. How do you read an EXPLAIN plan?

Read it looking for: the access method per table (an index seek/scan is good; a full "Seq Scan"/"ALL" on a large table is a red flag), whether a key/index is actually being used, the estimated vs. actual row counts (EXPLAIN ANALYZE shows both - a big mismatch suggests stale statistics), the join algorithm chosen, and warning signs in the "Extra" column like "Using filesort" or "Using temporary" (MySQL) that indicate expensive sort/temp-table operations. Start from the most expensive/outermost operation and work inward to find the bottleneck.

4. Why does putting a function on an indexed column prevent index usage?

An index stores the raw column values in sorted order; wrapping the column in a function (WHERE YEAR(created_at) = 2024) means the database would need to apply that function to every stored value before it could compare - which defeats the purpose of a sorted structure the optimizer could otherwise binary-search. Since the index wasn't built on the function's output, the optimizer typically falls back to a full table scan. The fix is to rewrite the condition as a plain range comparison on the raw column (created_at >= '2024-01-01' AND created_at < '2025-01-01'), or build a functional/expression index on that exact function if the database supports it.

5. What is a covering index?

An index that includes every column a particular query needs - either as indexed key columns or as additional included columns - so the query can be fully satisfied by reading the index alone, without a separate lookup back into the base table ("index-only scan"). This avoids the extra I/O cost of jumping from the index to the table to fetch other columns, making it the fastest possible way to answer a specific, known query pattern.

6. Why is OFFSET pagination slow for large offsets?

OFFSET n tells the database to still compute and then discard the first n matching rows before returning the next page - so page 1000 with OFFSET 19980 requires scanning and throwing away nearly 20,000 rows every single time, even though you only want 20 of them. The cost grows linearly with the offset. Keyset pagination avoids this by filtering directly on the last-seen sort key (WHERE id > last_id ORDER BY id LIMIT 20), letting an index seek jump straight to the right starting point regardless of how deep into the result set you are.

7. How does composite index column order affect query performance?

Per the leftmost prefix rule, a composite index on (a, b, c) can be used efficiently for a query filtering on a, or a and b, or all three - but not for a query that filters on b or c alone, since the index is physically sorted by a first. A query like WHERE a = 1 AND c = 3 can only use the index to narrow down by a (then must scan/filter the rest for c), getting partial benefit. Choosing the right order (equality columns before range columns, higher-selectivity columns first) determines which of your actual query patterns the index can actually accelerate.

8. When should you NOT create an index?

Skip indexing when the table is small enough that a full scan is already fast, the column has low selectivity (few distinct values, like a boolean), the table has heavy write (INSERT/UPDATE/DELETE) traffic relative to reads (since every index adds write overhead and must be kept in sync), or the column is rarely referenced in actual queries - in all these cases the index's maintenance cost outweighs any read benefit.

9. What is the difference between EXISTS and IN for performance?

IN evaluates its subquery fully and caches the complete result set before checking membership, which works well when that result set is small. EXISTS is a short-circuiting existence check - it stops scanning as soon as it finds one matching row - which tends to be more efficient when the outer table driving the check is

relatively small or when the subquery's result set would otherwise be very large. In practice, most modern optimizers rewrite one into the other (semi-join) when it's advantageous, so the difference matters most on older/simpler optimizers or unusual data distributions.

10. How do you optimize a query that uses OR conditions?

An OR across two different, separately-indexed columns (WHERE category_id = 5 OR brand_id = 10) often prevents the optimizer from efficiently using either index, since it would need to combine two index scans and de-duplicate - many optimizers instead fall back to a full scan. Rewriting it as a UNION of two separate single-condition queries (each of which can use its own index cleanly) is often faster, with UNION deduplicating any rows matching both conditions, or UNION ALL if duplicates are acceptable/impossible and you want to skip the dedup cost.

Key Takeaways

- N+1 problem: 1 + N queries instead of 1 JOIN - extremely common with ORMs.
- Use EXPLAIN to understand query execution plans.
- Look for full table scans (type: ALL) and missing indexes.
- Avoid functions on indexed columns in WHERE clause.
- Composite index column order matters - leftmost prefix rule.
- SELECT only needed columns, not SELECT *.
- Leading wildcards (LIKE '%text') prevent index usage.
- Use keyset pagination instead of OFFSET for large datasets.
- Covering indexes are fastest - include all needed columns.
- Profile before optimizing - measure, don't guess.

Aggregations and GROUP BY

SQL Fundamentals

Master data summarization with aggregate functions, grouping, and filtering.

The Big Picture

Aggregate functions compute a single result from a set of input values. Combined with GROUP BY, they allow you to summarize data by categories. Understanding aggregation is essential for reporting, analytics, and answering business questions with SQL.

Why This Matters in Interviews

- "What is the difference between WHERE and HAVING?"
- "How does NULL affect aggregate functions?"
- "Why can you not use non-aggregated columns without GROUP BY?"
- "What is the execution order of SELECT, WHERE, GROUP BY, HAVING?"

Aggregate Functions

Aggregate functions operate on a set of rows and return a single value: COUNT, SUM, AVG, MIN, MAX, and others like STRING_AGG, ARRAY_AGG.

```
-- Basic aggregate functions
SELECT
    COUNT(*) AS total_rows,
    COUNT(email) AS has_email,
    COUNT(DISTINCT department) AS unique_depts,
    SUM(salary) AS total_payroll,
    AVG(salary) AS avg_salary,
    MIN(hire_date) AS earliest_hire,
    MAX(salary) AS highest_salary
FROM employees;
```

COUNT(*) vs COUNT(column)

- COUNT(*) counts all rows, including those with NULL values.
- COUNT(column) counts only rows where that column is NOT NULL.
- COUNT(DISTINCT column) counts unique non-NULL values.

GROUP BY Clause

GROUP BY divides rows into groups based on column values. Aggregate functions then operate on each group separately:

```
-- Sales by department
SELECT
    department,
    COUNT(*) AS employee_count,
    SUM(salary) AS dept_payroll,
    AVG(salary) AS avg_salary
FROM employees
GROUP BY department;

-- Multiple grouping columns
SELECT
    department,
    job_title,
    COUNT(*) AS count,
    AVG(salary) AS avg_salary
FROM employees
GROUP BY department, job_title
ORDER BY department, avg_salary DESC;
```

Common Error: Non-Aggregated Columns

Every column in SELECT must either be in GROUP BY or wrapped in an aggregate function:

```
-- ERROR: name is not aggregated or grouped
SELECT department, name, AVG(salary)
FROM employees
GROUP BY department;

-- CORRECT: Include name in GROUP BY
SELECT department, name, AVG(salary)
FROM employees
GROUP BY department, name;
```

WHERE vs HAVING

Both filter rows, but they operate at different stages:

WHERE - Filters individual rows BEFORE grouping. Cannot use aggregate functions.

```
-- Filter rows before grouping
SELECT department, AVG(salary)
FROM employees
WHERE status = 'active'
GROUP BY department;
```

HAVING - Filters groups AFTER aggregation. Can use aggregate functions.

```
-- Filter groups after aggregation
SELECT department, AVG(salary)
FROM employees
GROUP BY department
HAVING AVG(salary) > 50000;
```

Using Both Together

```
-- Find departments with average salary > 60K among active employees
SELECT
    department,
    COUNT(*) AS active_count,
    AVG(salary) AS avg_salary
FROM employees
WHERE status = 'active'        -- Filter rows first
GROUP BY department
HAVING AVG(salary) > 60000    -- Then filter groups
   AND COUNT(*) >= 5           -- Multiple HAVING conditions
ORDER BY avg_salary DESC;
```

```
Aspect   |   WHERE   |   HAVING
When Applied   |   Before GROUP BY   |   After GROUP BY
Operates On   |   Individual rows   |   Grouped rows
Aggregate Functions   |   Cannot use   |   Can use
Performance   |   Filters early (faster)   |   Filters late (after aggregation)
```

Query Execution Order

Understanding execution order is crucial for writing correct queries:

FROM -> WHERE -> GROUP BY -> HAVING -> SELECT -> ORDER BY -> LIMIT

Key Implications

- WHERE cannot reference SELECT aliases (WHERE runs before SELECT).
- HAVING can use aggregates (GROUP BY creates them first).
- ORDER BY can reference SELECT aliases (ORDER BY runs after SELECT).
- Column aliases from SELECT work in ORDER BY but not WHERE or HAVING.

NULL Handling in Aggregates

Understanding how NULL values affect aggregates is critical:

```
-- Sample data: [100, 200, NULL, 300]

COUNT(*)        -- 4 (counts all rows)
COUNT(value)    -- 3 (ignores NULL)
SUM(value)      -- 600 (ignores NULL)
AVG(value)      -- 200 (600/3, ignores NULL)
MIN(value)      -- 100 (ignores NULL)
MAX(value)      -- 300 (ignores NULL)
```

NULL Gotchas

- All aggregates except COUNT(*) ignore NULL values.
- AVG divides by count of non-NULL values, not total rows.
- If all values are NULL, most aggregates return NULL (not 0).
- GROUP BY treats all NULLs as equal (groups them together).

Handling NULLs Explicitly

```
-- Replace NULL with 0 for calculation
SELECT AVG(COALESCE(bonus, 0)) AS avg_bonus
FROM employees;

-- Return 0 instead of NULL when no rows
SELECT COALESCE(SUM(amount), 0) AS total
FROM orders
WHERE customer_id = 999;  -- no matching rows

-- Count NULLs specifically
SELECT
    COUNT(*) - COUNT(email) AS null_email_count
FROM customers;
```

Advanced Grouping

ROLLUP - Hierarchical Subtotals

```
-- Subtotals by region, then grand total
SELECT
    region,
    department,
    SUM(sales) AS total_sales
FROM sales_data
GROUP BY ROLLUP(region, department);

-- Result includes:
-- East    | Sales      | 10000  (detail)
-- East    | Marketing  | 8000   (detail)
-- East    | NULL       | 18000  (region subtotal)
-- West    | Sales      | 12000  (detail)
-- West    | NULL       | 12000  (region subtotal)
-- NULL    | NULL       | 30000  (grand total)
```

CUBE - All Combinations

```
-- All possible grouping combinations
SELECT
    region,
    department,
    SUM(sales) AS total_sales
FROM sales_data
GROUP BY CUBE(region, department);

-- Includes: by region+dept, by region, by dept, grand total
```

GROUPING SETS - Custom Combinations

```
-- Specific grouping combinations only
SELECT
    region,
    department,
    SUM(sales) AS total_sales
FROM sales_data
GROUP BY GROUPING SETS(
    (region, department),  -- detail level
    (region),              -- region subtotals
    ()                     -- grand total
);
```

Common Patterns

Top N Per Group

```
-- Top 3 products by sales in each category
WITH ranked AS (
    SELECT
        category,
        product_name,
        SUM(sales) AS total_sales,
        ROW_NUMBER() OVER (
            PARTITION BY category
            ORDER BY SUM(sales) DESC
        ) AS rn
    FROM sales
    GROUP BY category, product_name
)
SELECT * FROM ranked WHERE rn <= 3;
```

Conditional Aggregation

```
-- Count by status in single query (pivot-style)
SELECT
    department,
    COUNT(*) AS total,
    COUNT(CASE WHEN status = 'active' THEN 1 END) AS active,
    COUNT(CASE WHEN status = 'inactive' THEN 1 END) AS inactive,
    SUM(CASE WHEN status = 'active' THEN salary ELSE 0 END) AS active_payroll
FROM employees
GROUP BY department;

-- PostgreSQL FILTER syntax (cleaner)
SELECT
    department,
    COUNT(*) AS total,
    COUNT(*) FILTER (WHERE status = 'active') AS active
FROM employees
GROUP BY department;
```

Running Totals with GROUP BY

```
-- Daily sales with running total
SELECT
    sale_date,
    SUM(amount) AS daily_total,
    SUM(SUM(amount)) OVER (ORDER BY sale_date) AS running_total
FROM sales
GROUP BY sale_date
ORDER BY sale_date;
```

Top 10 Interview Questions

Master these frequently asked questions about aggregations:

1. What is the difference between WHERE and HAVING?

WHERE filters individual rows before any grouping happens, and cannot reference aggregate functions since they haven't been computed yet. HAVING filters groups after GROUP BY has aggregated the rows, and is specifically designed to use aggregate functions (e.g. HAVING AVG(salary) > 50000) to decide which groups survive. You can use both together: WHERE to cheaply exclude rows up front, HAVING to filter on the aggregated result.

2. How does NULL affect aggregate functions?

Every aggregate function except COUNT(*) silently ignores NULL values in its input - SUM, AVG, MIN, MAX, and COUNT(column) all skip NULLs. This means AVG divides by the count of non-NULL values, not the total row count, and if every value in the group is NULL, most aggregates return NULL rather than 0. GROUP BY treats all NULLs as a single equal group rather than excluding them.

3. Why can you not use non-aggregated columns in SELECT without GROUP BY?

Because once you aggregate, each output row represents a whole group of input rows collapsed into one - any column not wrapped in an aggregate function and not part of the GROUP BY list would have multiple different values within that group, and the database has no defined rule for which one to show. SQL requires every selected column to be either part of the grouping key (one value per group, unambiguous) or explicitly aggregated (reduced to one value per group) to guarantee a deterministic result.

4. What is the execution order of a SQL query with GROUP BY?

FROM (identify/join tables) -> WHERE (filter rows) -> GROUP BY (form groups) -> HAVING (filter groups) ->

SELECT (compute output columns/aggregates) -> ORDER BY (sort) -> LIMIT (truncate). This is why WHERE can't use aggregates or SELECT aliases (they don't exist yet) while HAVING can use aggregates and ORDER BY can use SELECT aliases.

5. What is the difference between COUNT(*), COUNT(column), and COUNT(DISTINCT column)?

COUNT(*) counts every row in the group regardless of NULLs. COUNT(column) counts only the rows where that specific column is NOT NULL. COUNT(DISTINCT column) counts the number of unique non-NULL values in that column, collapsing duplicates.

6. How do you calculate percentage of total using GROUP BY?

Compute each group's aggregate normally, then divide by the grand total - the grand total can come from a subquery, a CTE, or (more elegantly) a window function with an empty OVER () that computes the overall sum alongside the grouped rows: SELECT category, SUM(sales), 100.0 * SUM(sales) / SUM(SUM(sales)) OVER () AS pct FROM sales GROUP BY category; - the outer SUM(...) OVER () sums the already-grouped per-category totals into one grand total available on every output row.

7. What is conditional aggregation?

It's using a CASE WHEN expression (or the cleaner PostgreSQL FILTER (WHERE ...) syntax) inside an aggregate function to count or sum only rows matching a specific condition, letting you compute multiple conditional metrics in a single pass/single query instead of running separate queries per condition - e.g. COUNT(CASE WHEN status = 'active' THEN 1 END) AS active_count alongside COUNT(CASE WHEN status = 'inactive' THEN 1 END) AS inactive_count in the same SELECT, producing a pivot-style summary row per group.

8. What are ROLLUP, CUBE, and GROUPING SETS?

All three extend GROUP BY to produce multiple levels of subtotals in a single query instead of running several separate aggregate queries and UNION-ing them. ROLLUP(a, b) produces a hierarchy of subtotals: detail rows by (a, b), subtotals by a alone, and a grand total. CUBE(a, b) produces every possible combination of groupings (by a,b, by a, by b, and overall). GROUPING SETS lets you specify exactly which grouping combinations you want, when you don't need the full hierarchy or full cross-product.

9. How do you find duplicates using GROUP BY?

Group by the column(s) that should be unique, then filter with HAVING COUNT(*) > 1: SELECT email, COUNT(*) FROM users GROUP BY email HAVING COUNT(*) > 1; - this returns every value that appears more than once, along with how many times, which is the standard pattern for duplicate detection.

10. Can you use aggregate functions without GROUP BY?

Yes - without a GROUP BY, the entire result set is treated as a single implicit group, so an aggregate function like SELECT COUNT(*), AVG(salary) FROM employees; returns one row summarizing the whole table. The restriction on mixing non-aggregated columns only kicks in if you try to select a plain column alongside the aggregate in that case (e.g. SELECT name, COUNT(*) FROM employees; would error, since name has no single value across all rows).

Key Takeaways

- Aggregate functions (COUNT, SUM, AVG, MIN, MAX) summarize data.
- COUNT(*) counts all rows; COUNT(column) ignores NULLs.
- GROUP BY creates groups; aggregates compute per group.
- Every SELECT column must be grouped or aggregated.
- WHERE filters rows before grouping; HAVING filters groups after.
- HAVING can use aggregates; WHERE cannot.

- Execution order: FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY.
- All aggregates except COUNT(*) ignore NULL values.
- ROLLUP, CUBE, GROUPING SETS for advanced subtotals.
- Conditional aggregation with CASE WHEN or FILTER.

Database Normalization

SQL Fundamentals

Organize data to eliminate redundancy and ensure integrity.

The Big Picture

Normalization is the process of organizing database tables to minimize redundancy and dependency. It involves dividing large tables into smaller, related tables and defining relationships between them. The goal is to isolate data so that additions, deletions, and modifications can be made in one table and propagated through the rest via relationships.

Why This Matters in Interviews

- "What are the normal forms and what problem does each solve?"
- "Explain the difference between 2NF and 3NF"
- "When would you denormalize a database?"
- "What are update, insert, and delete anomalies?"

Why Normalize?

Unnormalized data leads to three types of anomalies:

Update Anomaly - Same data exists in multiple rows. Updating one occurrence but not others leads to inconsistency.

Insert Anomaly - Cannot insert data without having other unrelated data. Example: cannot add department without an employee.

Delete Anomaly - Deleting data unintentionally removes other important data. Example: deleting last employee deletes department info.

Example: Unnormalized Data

```
order_id   |   customer_name   |   customer_email   |   product   |   price
1001   |   John Smith   |   john@email.com   |   Laptop   |   999
1002   |   John Smith   |   john@email.com   |   Mouse   |   29
1003   |   Jane Doe   |   jane@email.com   |   Keyboard   |   79
```

John's info is stored redundantly across multiple rows.

The Normal Forms

1NF - First Normal Form (Atomic Values)

Must have atomic (indivisible) values in each column, no repeating groups or arrays.

Rule: Eliminate repeating groups; each cell has a single value.

2NF - Second Normal Form (Full Dependency)

Must be in 1NF. All non-key columns depend on the entire primary key, not just part of it.

Rule: Remove partial dependencies on composite keys.

3NF - Third Normal Form (No Transitive Dependency)

Must be in 2NF. Non-key columns depend only on the primary key, not on other non-key columns.

Rule: Remove transitive dependencies (A -> B -> C).

BCNF - Boyce-Codd Normal Form (Strict 3NF)

Must be in 3NF. Every determinant must be a candidate key. Handles edge cases 3NF misses.

Rule: For X -> Y, X must be a superkey.

First Normal Form (1NF)

A table is in 1NF if:

- Each column contains atomic (indivisible) values.
- Each column contains values of the same type.
- Each row is unique (has a primary key).
- No repeating groups or arrays.

Violation Example

```
-- NOT in 1NF: phone_numbers contains multiple values
| student_id | name  | phone_numbers           |
|------------|-------|-------------------------|
| 1          | Alice | 555-1234, 555-5678      |
| 2          | Bob   | 555-9999                |

-- IN 1NF: Split into separate rows or table
| student_id | name  | phone_number |
|------------|-------|--------------|
| 1          | Alice | 555-1234     |
| 1          | Alice | 555-5678     |
| 2          | Bob   | 555-9999     |
```

```
-- OR better: Separate phone_numbers table
CREATE TABLE students (student_id INT PRIMARY KEY, name VARCHAR);
CREATE TABLE phone_numbers (
    student_id INT REFERENCES students,
    phone_number VARCHAR
);
```

Second Normal Form (2NF)

A table is in 2NF if:

- It is in 1NF.
- All non-key columns are fully dependent on the entire primary key.
- No partial dependencies (relevant only for composite primary keys).

Key Insight

2NF only matters when you have a composite primary key. If your primary key is a single column, and you are in 1NF, you are automatically in 2NF.

Violation Example

```
-- NOT in 2NF: course_name depends only on course_id, not student_id
-- Primary Key: (student_id, course_id)
| student_id | course_id | course_name | grade |
|------------|-----------|-------------|-------|
| 1          | CS101     | Intro CS    | A     |
| 1          | CS102     | Data Struct | B     |
| 2          | CS101     | Intro CS    | B     |  <- course_name repeated
```

```
-- IN 2NF: Split into two tables
CREATE TABLE enrollments (
    student_id INT,
    course_id VARCHAR,
    grade CHAR(1),
    PRIMARY KEY (student_id, course_id)
);

CREATE TABLE courses (
    course_id VARCHAR PRIMARY KEY,
    course_name VARCHAR
);
```

Third Normal Form (3NF)

A table is in 3NF if:

- It is in 2NF.
- No transitive dependencies - non-key columns depend only on the primary key.
- No non-key column depends on another non-key column.

Violation Example

```
-- NOT in 3NF: dept_name depends on dept_id, not employee_id
-- employee_id -> dept_id -> dept_name (transitive!)
| employee_id | name  | dept_id | dept_name   |
|-------------|-------|---------|-------------|
| 1           | Alice | D1      | Engineering |
| 2           | Bob   | D1      | Engineering |  <- dept_name repeated
| 3           | Carol | D2      | Sales       |
```

```
-- IN 3NF: Split into two tables
CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    name VARCHAR,
    dept_id VARCHAR REFERENCES departments
);

CREATE TABLE departments (
    dept_id VARCHAR PRIMARY KEY,
    dept_name VARCHAR
);
```

3NF Summary (Bill Kent)

"Every non-key attribute must provide a fact about the key, the whole key, and nothing but the key."

- The key: 1NF - attributes depend on a key.
- The whole key: 2NF - no partial dependencies.
- Nothing but the key: 3NF - no transitive dependencies.

Denormalization

Sometimes we intentionally violate normal forms for performance. This is called denormalization:

```
Aspect   |   Normalized   |   Denormalized
Data Redundancy   |   Minimal   |   Increased
Write Performance   |   Faster (less data)   |   Slower (update multiple places)
Read Performance   |   Slower (many JOINs)   |   Faster (fewer JOINs)
Data Integrity   |   Easier to maintain   |   Risk of inconsistency
Storage   |   Less space   |   More space
```

When to Denormalize

- Read-heavy workloads: When queries vastly outnumber writes.
- Reporting/Analytics: Pre-computed aggregates, materialized views.
- Caching: Store computed values to avoid repeated calculations.
- Performance critical paths: When JOINs are too slow.

```
-- Denormalized: Store customer_name in orders for faster reads
CREATE TABLE orders (
    order_id INT PRIMARY KEY,
    customer_id INT REFERENCES customers,
    customer_name VARCHAR,  -- Denormalized!
    total DECIMAL
);

-- Now this query doesn't need a JOIN:
SELECT order_id, customer_name, total
FROM orders
WHERE order_id = 1001;
```

Denormalization Trade-offs

When you denormalize, you take on the responsibility of keeping redundant data in sync. Use database triggers, application logic, or eventual consistency patterns to maintain data integrity.

Top 10 Interview Questions

Master these frequently asked questions about normalization:

1. What is database normalization and why is it important?

Normalization is the process of structuring a database's tables and relationships to minimize data redundancy and avoid the update/insert/delete anomalies that redundancy causes. It's important because it keeps data consistent (a fact stored once can only be true in one place), reduces storage waste, and makes the database easier to maintain correctly as it grows - at the cost of needing more JOINs to reassemble related data at query time.

2. Explain the first three normal forms.

1NF requires atomic values - no repeating groups or multi-valued columns, each cell holds one value. 2NF requires 1NF plus no partial dependencies - every non-key column must depend on the *entire* primary key,

not just part of a composite key. 3NF requires 2NF plus no transitive dependencies - a non-key column can't depend on another non-key column; it must depend directly and only on the primary key.

3. What is a partial dependency?

A partial dependency occurs when a non-key column depends on only part of a composite primary key rather than the whole key - e.g. in a table keyed on (student_id, course_id), if course_name depends only on course_id (and not on student_id at all), that's a partial dependency, and it violates 2NF. It's resolved by splitting the table so the partially-dependent column moves to a table keyed on just the column it actually depends on.

4. What is a transitive dependency?

A transitive dependency is when a non-key column depends on another non-key column, which in turn depends on the primary key - forming a chain like employee_id -> dept_id -> dept_name, where dept_name depends on the key only indirectly, through dept_id. This violates 3NF and is fixed by moving dept_name into its own departments table keyed on dept_id.

5. What are update, insert, and delete anomalies?

An update anomaly happens when the same fact is duplicated across multiple rows and updating it in one place but not others leaves the data inconsistent. An insert anomaly happens when you can't add a piece of data without also having unrelated data available (e.g. can't record a new department until at least one employee is hired into it, if department data only lives inside the employee table). A delete anomaly happens when deleting one row unintentionally destroys other information that happened to be stored alongside it (e.g. deleting the last employee in a department wipes out the department's name/info too). Normalization eliminates all three by storing each fact in exactly one place.

6. What is the difference between 3NF and BCNF?

3NF allows an edge case: a non-key attribute can still be a "determinant" (something that functionally determines another attribute) as long as it isn't itself transitively dependent on the key in a prohibited way, if the table has overlapping candidate keys. BCNF is the stricter rule: for every functional dependency X -> Y in the table, X must be a candidate key (a superkey) - no exceptions. In other words, BCNF requires that every determinant be a key, catching some redundancy cases that 3NF's definition technically permits.

7. When would you denormalize a database?

When read performance matters more than write performance or storage efficiency and JOINs across normalized tables have become a measurable bottleneck - common in read-heavy reporting/analytics systems, materialized views/pre-computed aggregates, or caching frequently-joined data (like storing a customer's name directly on an order row to avoid a JOIN on every read). The trade-off is accepting responsibility for keeping the duplicated data in sync, typically via triggers, application logic, or an eventual-consistency process.

8. How do you identify if a table is not in 3NF?

Look for any non-key column whose value can be determined entirely by another non-key column rather than directly by the primary key - a telltale sign is a column's value repeating in lockstep with another non-key column across multiple rows (e.g. dept_name always matching whatever dept_id is, independent of which employee row you're looking at). If updating that one non-key column in one row would require also updating it in other rows to stay consistent, there's a transitive dependency and the table isn't in 3NF.

9. What is a candidate key?

A candidate key is any minimal set of one or more columns that can uniquely identify a row in a table - "minimal" meaning you can't drop any column from the set and still have uniqueness. A table can have multiple candidate keys (e.g. both an auto-increment id and a email column could each uniquely identify a user); one of them is chosen as the primary key, and the rest remain candidate keys (sometimes enforced

with UNIQUE constraints).

10. Is it always necessary to normalize to 3NF?

No - 3NF is a widely used practical balance point for most transactional (OLTP) systems, but it's not a universal requirement. Analytics/reporting (OLAP) systems, data warehouses, and read-heavy caching layers often intentionally use denormalized or star-schema designs for performance, accepting redundancy because reads vastly outnumber writes and JOIN cost matters more than storage or write-consistency overhead. The right level of normalization depends on the actual read/write pattern of the system.

Key Takeaways

- Normalization eliminates redundancy and prevents anomalies.
- Three anomalies: Update, Insert, Delete - caused by redundant data.
- 1NF: Atomic values, no repeating groups.
- 2NF: No partial dependencies (on part of composite key).
- 3NF: No transitive dependencies (non-key depending on non-key).
- BCNF: Every determinant must be a candidate key.
- "The key, the whole key, nothing but the key" summarizes 1NF-3NF.
- Most production databases aim for 3NF as a balance.
- Denormalization trades integrity for read performance.
- Denormalize only with clear performance requirements.

Constraints and Data Integrity

SQL Fundamentals

Enforce business rules and maintain data quality at the database level.

The Big Picture

Constraints are rules enforced by the database to maintain data integrity. They prevent invalid data from being inserted, updated, or deleted. By defining constraints at the database level, you ensure data quality regardless of which application or user accesses the database.

Why This Matters in Interviews

- "What are the different types of constraints in SQL?"
- "Explain the difference between PRIMARY KEY and UNIQUE"
- "What are foreign key actions (CASCADE, SET NULL, etc.)?"
- "When would you use a CHECK constraint vs application validation?"

Types of Constraints

PRIMARY KEY - Uniquely identifies each row. Combines NOT NULL and UNIQUE. Only one per table.

```
id INT PRIMARY KEY
```

FOREIGN KEY - References primary key in another table. Enforces referential integrity between tables.

```
FOREIGN KEY (dept_id) REFERENCES departments(id)
```

UNIQUE - Ensures all values in column are different. Allows one NULL (varies by database).

```
email VARCHAR(255) UNIQUE
```

NOT NULL - Prevents NULL values. Column must always have a value on insert/update.

```
name VARCHAR(100) NOT NULL
```

CHECK - Validates data against a boolean expression. Enforces domain rules.

```
CHECK (age >= 0 AND age <= 150)
```

DEFAULT - Provides automatic value when none specified. Often used with timestamps.

```
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
```

PRIMARY KEY

A primary key uniquely identifies each row in a table. It combines two constraints: NOT NULL and UNIQUE.

```
-- Single column primary key
CREATE TABLE users (
    id INT PRIMARY KEY,
    name VARCHAR(100)
);

-- Auto-increment primary key (common pattern)
CREATE TABLE users (
    id SERIAL PRIMARY KEY,  -- PostgreSQL
    name VARCHAR(100)
);

CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,  -- MySQL
    name VARCHAR(100)
);

-- Composite primary key
CREATE TABLE order_items (
    order_id INT,
    product_id INT,
    quantity INT,
    PRIMARY KEY (order_id, product_id)
);
```

PRIMARY KEY vs UNIQUE

- PRIMARY KEY: Only one per table, cannot be NULL, defines row identity.
- UNIQUE: Multiple per table, allows NULL (usually one), enforces uniqueness.

FOREIGN KEY

Foreign keys enforce referential integrity by ensuring values in one table correspond to values in another.

```
-- Basic foreign key
CREATE TABLE orders (
    id INT PRIMARY KEY,
    customer_id INT,
    total DECIMAL(10,2),
    FOREIGN KEY (customer_id) REFERENCES customers(id)
);

-- With referential actions
CREATE TABLE orders (
    id INT PRIMARY KEY,
    customer_id INT,
    FOREIGN KEY (customer_id) REFERENCES customers(id)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

Referential Actions

```
Action   |   ON DELETE   |   ON UPDATE
CASCADE   |   Delete child rows when parent deleted   |   Update child FKs when parent
PK changes
SET NULL   |   Set FK to NULL when parent deleted   |   Set FK to NULL when parent PK
changes
SET DEFAULT   |   Set FK to default value   |   Set FK to default value
RESTRICT   |   Prevent delete if children exist   |   Prevent update if children exist
NO ACTION   |   Same as RESTRICT (checked at end of statement)   |   Same as RESTRICT
```

Choosing Referential Actions

- CASCADE: When child data is meaningless without parent (order items when order deleted).
- SET NULL: When child can exist independently (employee's manager_id when manager leaves).
- RESTRICT: When deletion should be explicit (prevent accidental data loss).

CHECK Constraint

CHECK constraints validate data against a boolean expression before allowing insert or update:

```
-- Simple CHECK constraint
CREATE TABLE employees (
    id INT PRIMARY KEY,
    name VARCHAR(100),
    age INT CHECK (age >= 18 AND age <= 120),
    salary DECIMAL CHECK (salary > 0)
);

-- Named CHECK constraint
CREATE TABLE products (
    id INT PRIMARY KEY,
    price DECIMAL(10,2),
    discount DECIMAL(5,2),
    CONSTRAINT chk_price_positive CHECK (price > 0),
    CONSTRAINT chk_discount_range CHECK (discount >= 0 AND discount <= 100)
);

-- CHECK with multiple columns
CREATE TABLE events (
    id INT PRIMARY KEY,
    start_date DATE,
    end_date DATE,
    CHECK (end_date >= start_date)
);

-- CHECK with IN list
CREATE TABLE orders (
    id INT PRIMARY KEY,
    status VARCHAR(20) CHECK (status IN ('pending', 'shipped', 'delivered'))
);
```

CHECK vs Application Validation

- CHECK constraints: Enforced at database level, cannot be bypassed, simpler expressions.
- Application validation: More complex logic, better error messages, can be bypassed.
- Best practice: Use both! Database for critical rules, application for UX.

Managing Constraints

Adding Constraints to Existing Tables

```
-- Add PRIMARY KEY
ALTER TABLE users ADD PRIMARY KEY (id);

-- Add FOREIGN KEY
ALTER TABLE orders
ADD CONSTRAINT fk_customer
FOREIGN KEY (customer_id) REFERENCES customers(id);

-- Add UNIQUE constraint
ALTER TABLE users ADD CONSTRAINT uq_email UNIQUE (email);

-- Add CHECK constraint
ALTER TABLE products ADD CONSTRAINT chk_price CHECK (price > 0);

-- Add NOT NULL (varies by database)
ALTER TABLE users ALTER COLUMN name SET NOT NULL;  -- PostgreSQL
ALTER TABLE users MODIFY name VARCHAR(100) NOT NULL;  -- MySQL
```

Removing Constraints

```
-- Drop constraint by name
ALTER TABLE orders DROP CONSTRAINT fk_customer;
ALTER TABLE users DROP CONSTRAINT uq_email;

-- Drop PRIMARY KEY
ALTER TABLE users DROP PRIMARY KEY;  -- MySQL
ALTER TABLE users DROP CONSTRAINT users_pkey;  -- PostgreSQL

-- Temporarily disable constraints (for bulk loads)
ALTER TABLE orders DISABLE TRIGGER ALL;  -- PostgreSQL
SET FOREIGN_KEY_CHECKS = 0;  -- MySQL
```

Top 10 Interview Questions

Master these frequently asked questions about constraints:

1. What is the difference between PRIMARY KEY and UNIQUE?

A PRIMARY KEY uniquely identifies every row and implicitly combines NOT NULL + UNIQUE - a table can have only one. A UNIQUE constraint also enforces distinct values, but a table can have multiple UNIQUE constraints, and most databases allow a UNIQUE column to contain NULL (typically just one NULL, since NULL isn't considered equal to another NULL). In short: PRIMARY KEY defines the row's core identity; UNIQUE is for additional uniqueness rules (like email addresses) that aren't the table's primary identifier.

2. What are the different referential actions for foreign keys?

CASCADE propagates the delete/update to child rows automatically. SET NULL sets the child's foreign key to NULL when the parent is deleted/changed. SET DEFAULT sets it to a predefined default value. RESTRICT blocks the delete/update entirely if dependent child rows exist. NO ACTION behaves like RESTRICT in most databases, differing mainly in exactly when the check is enforced (end of statement vs immediately).

3. When would you use a CHECK constraint vs application validation?

Use a CHECK constraint for simple, critical data-integrity rules that must hold true no matter which application, script, or user touches the data directly (e.g. price > 0), since it can't be bypassed even by direct database access. Use application-level validation for more complex business logic, cross-system checks, and to give users friendlier, more specific error messages - application code can be bypassed by direct DB access, so critical invariants belong in both places, with the database as the non-negotiable safety net.

4. Can a foreign key reference a non-primary key column?

Yes - a foreign key must reference a column (or column set) that has a UNIQUE constraint (or is the PRIMARY KEY) in the parent table, since the referenced value must be guaranteed unique for the relationship to be well-defined, but it doesn't strictly have to be that table's primary key as long as uniqueness is enforced on it.

5. What happens if you try to insert NULL into a PRIMARY KEY column?

It fails with a constraint violation error - PRIMARY KEY implicitly includes a NOT NULL constraint, so NULL is never a valid value for any column that's part of a primary key, whether it's a single-column or composite key.

6. How do you add a NOT NULL constraint to an existing column with data?

First ensure every existing row already has a non-NULL value in that column (update any NULLs to a valid

default first), then run the database-specific ALTER TABLE syntax - e.g. ALTER TABLE users ALTER COLUMN name SET NOT NULL; in PostgreSQL, or ALTER TABLE users MODIFY name VARCHAR(100) NOT NULL; in MySQL. If any row still has NULL in that column, the ALTER TABLE will fail.

7. What is a composite primary key and when would you use one?

A composite primary key is a primary key made up of two or more columns together, where the combination (not any single column alone) uniquely identifies a row. It's commonly used for junction/association tables representing a many-to-many relationship, like order_items keyed on (order_id, product_id), where neither column alone is unique but the pair always is.

8. What is the difference between RESTRICT and NO ACTION for foreign keys?

They're functionally very similar - both prevent a parent delete/update if dependent child rows exist - but differ in timing semantics per the SQL standard: RESTRICT checks and blocks immediately, while NO ACTION defers the check until the end of the statement (which matters if other triggers or cascades within the same statement might resolve the conflict before the check happens). In many databases (e.g. MySQL), the two are treated as effectively identical in practice.

9. Can you have a foreign key that references the same table?

Yes - this is a self-referencing foreign key, commonly used for hierarchical data like an employee-manager relationship: manager_id INT REFERENCES employees(id) within the employees table itself, where each row's manager_id points to another row's id in that same table.

10. How do you temporarily disable constraints for bulk data loading?

Database-specific commands can suspend constraint/trigger checking during a bulk load for performance, then re-enable and optionally re-validate afterward - e.g. SET FOREIGN_KEY_CHECKS = 0; in MySQL, or ALTER TABLE orders DISABLE TRIGGER ALL; in PostgreSQL (which disables the triggers enforcing some constraint checks). This should be done carefully and constraints re-enabled (with data re-validated) immediately after the load, since it temporarily removes the database's data-integrity safety net.

Key Takeaways

- Constraints enforce data integrity at the database level.
- PRIMARY KEY = UNIQUE + NOT NULL, only one per table.
- FOREIGN KEY enforces referential integrity between tables.
- Referential actions: CASCADE, SET NULL, RESTRICT, NO ACTION.
- UNIQUE allows NULL (usually one), can have multiple per table.
- CHECK validates data against boolean expressions.
- DEFAULT provides automatic values when none specified.
- Named constraints are easier to manage and debug.
- Use both database and application validation.
- Constraints can be added/removed with ALTER TABLE.

Stored Procedures and Functions

SQL Fundamentals

Encapsulate business logic in reusable database programs.

The Big Picture

Stored procedures and functions are precompiled SQL programs stored in the database. They encapsulate business logic, reduce network traffic, improve security, and provide code reuse. Understanding when to use procedures versus functions is a common interview topic.

Why This Matters in Interviews

- "What is the difference between a stored procedure and a function?"
- "When would you use a stored procedure vs inline SQL?"
- "What are the advantages and disadvantages of stored procedures?"
- "Can you call a stored procedure from a SELECT statement?"

Procedures vs Functions

Stored Procedure

- Action-oriented.
- May or may not return a value.
- Can have IN, OUT, INOUT parameters.
- Can modify database state (INSERT, UPDATE, DELETE).
- Called with CALL or EXECUTE.
- Cannot be used in SELECT statements.
- Can call other procedures and functions.

Function

- Value-oriented.
- Must return a value.
- Only IN parameters (typically).
- Should not modify database state.
- Called within SQL expressions.
- Can be used in SELECT, WHERE, etc.
- Can call other functions (not procedures).

```
Aspect   |   Stored Procedure   |   Function
Return Value   |   Optional (via OUT params or result sets)   |   Required (single value
or table)
Use in SELECT   |   No   |   Yes
Transaction Control   |   Can use COMMIT/ROLLBACK   |   Cannot (usually)
Side Effects   |   Allowed (modify data)   |   Should be avoided (pure)
Invocation   |   `CALL procedure_name()`   |   `SELECT function_name()`
```

Creating Stored Procedures

```
-- PostgreSQL / Standard SQL
CREATE OR REPLACE PROCEDURE transfer_funds(
    IN from_account INT,
    IN to_account INT,
    IN amount DECIMAL(10,2)
)
LANGUAGE plpgsql
AS $$
BEGIN
    -- Deduct from source account
    UPDATE accounts
    SET balance = balance - amount
    WHERE id = from_account;

    -- Add to destination account
    UPDATE accounts
    SET balance = balance + amount
    WHERE id = to_account;

    COMMIT;
END;
$$;

-- Call the procedure
CALL transfer_funds(1, 2, 500.00);
```

```
-- MySQL syntax
DELIMITER //
CREATE PROCEDURE get_employee_by_dept(
    IN dept_id INT,
    OUT emp_count INT
)
BEGIN
    SELECT COUNT(*) INTO emp_count
    FROM employees
    WHERE department_id = dept_id;

    SELECT * FROM employees
    WHERE department_id = dept_id;
END //
DELIMITER ;

-- Call with OUT parameter
CALL get_employee_by_dept(10, @count);
SELECT @count;
```

Parameter Modes

- IN: Input only (default). Value passed into procedure, changes not visible to caller.
- OUT: Output only. Procedure sets value, returned to caller.
- INOUT: Both input and output. Value passed in, can be modified and returned.

Creating Functions

```
-- PostgreSQL scalar function
CREATE OR REPLACE FUNCTION calculate_tax(
    price DECIMAL(10,2),
    tax_rate DECIMAL(4,2) DEFAULT 0.08
)
RETURNS DECIMAL(10,2)
LANGUAGE plpgsql
AS $$
BEGIN
    RETURN price * tax_rate;
END;
$$;

-- Use in SELECT
SELECT
    product_name,
    price,
    calculate_tax(price) AS tax,
    price + calculate_tax(price) AS total
FROM products;
```

```
-- Table-returning function (PostgreSQL)
CREATE OR REPLACE FUNCTION get_high_value_orders(
    min_amount DECIMAL(10,2)
)
RETURNS TABLE (
    order_id INT,
    customer_name VARCHAR,
    total DECIMAL(10,2)
)
LANGUAGE plpgsql
AS $$
BEGIN
    RETURN QUERY
    SELECT o.id, c.name, o.total
    FROM orders o
    JOIN customers c ON o.customer_id = c.id
    WHERE o.total >= min_amount;
END;
$$;

-- Use as a table source
SELECT * FROM get_high_value_orders(1000.00);
```

```
-- MySQL function
DELIMITER //
CREATE FUNCTION get_full_name(
    first_name VARCHAR(50),
    last_name VARCHAR(50)
)
RETURNS VARCHAR(101)
DETERMINISTIC
BEGIN
    RETURN CONCAT(first_name, ' ', last_name);
END //
DELIMITER ;

-- Use in query
SELECT get_full_name(first_name, last_name) AS name
FROM employees;
```

Control Flow Statements

```
-- IF-THEN-ELSE
IF quantity > 100 THEN
    discount := 0.20;
ELSIF quantity > 50 THEN
    discount := 0.10;
ELSE
    discount := 0;
END IF;

-- CASE statement
CASE status
    WHEN 'pending' THEN
        process_pending();
    WHEN 'approved' THEN
        process_approved();
    ELSE
        log_unknown_status();
END CASE;

-- LOOP with EXIT
LOOP
    FETCH cur INTO row_data;
    EXIT WHEN NOT FOUND;
    -- Process row
END LOOP;

-- WHILE loop
WHILE counter < 10 LOOP
    counter := counter + 1;
END LOOP;

-- FOR loop
FOR i IN 1..10 LOOP
    total := total + i;
END LOOP;
```

Error Handling

```
-- PostgreSQL exception handling
CREATE OR REPLACE PROCEDURE safe_transfer(
    from_acc INT, to_acc INT, amount DECIMAL
)
LANGUAGE plpgsql
AS $$
DECLARE
    from_balance DECIMAL;
BEGIN
    -- Check balance
    SELECT balance INTO from_balance
    FROM accounts WHERE id = from_acc;

    IF from_balance < amount THEN
        RAISE EXCEPTION 'Insufficient funds: % available', from_balance;
    END IF;

    -- Perform transfer
    UPDATE accounts SET balance = balance - amount WHERE id = from_acc;
    UPDATE accounts SET balance = balance + amount WHERE id = to_acc;

EXCEPTION
    WHEN others THEN
        RAISE NOTICE 'Transfer failed: %', SQLERRM;
        ROLLBACK;
END;
$$;
```

```
-- MySQL error handling
DELIMITER //
CREATE PROCEDURE safe_insert(
    IN p_email VARCHAR(255)
)
BEGIN
    DECLARE EXIT HANDLER FOR 1062
    BEGIN
        SELECT 'Duplicate email address' AS error_message;
    END;

    INSERT INTO users (email) VALUES (p_email);
    SELECT 'User created successfully' AS message;
END //
DELIMITER ;
```

When to Use Stored Procedures

- Complex Business Logic - Multi-step operations that should be atomic and consistent across applications.
- Security - Grant execute permission without exposing table access. Prevent SQL injection.
- Performance - Reduce network round trips. Execution plan cached and reused.
- Data Validation - Enforce rules beyond what constraints can express.
- Batch Operations - Process large amounts of data efficiently server-side.
- Audit Logging - Automatically log changes with triggers calling procedures.

When NOT to Use Stored Procedures

- Simple CRUD: ORM or parameterized queries are often simpler.
- Portability needed: Syntax varies significantly between databases.

- Version control: Harder to track changes vs application code.
- Testing: More difficult to unit test than application code.
- Debugging: Limited debugging tools compared to application IDEs.

Top 10 Interview Questions

Master these frequently asked questions about stored procedures and functions:

1. What is the difference between a stored procedure and a function?

A function must return a value and is designed to be used inline within SQL expressions (SELECT, WHERE, etc.) - it should be side-effect-free and typically can't commit/rollback transactions. A stored procedure is action-oriented: it may or may not return a value (often via OUT parameters or result sets), can freely modify data and manage transactions, but cannot be called from within a SELECT - it's invoked explicitly with CALL/EXECUTE.

2. What are the advantages of stored procedures?

Reduced network round trips (multiple statements execute server-side as one call), improved security (grant execute permission on a procedure without exposing direct table access, and parameterization reduces SQL injection risk), reusability and consistency (centralized business logic shared across every application/client that calls it), and performance (the execution plan is precompiled/cached and reused across calls).

3. What are the disadvantages of stored procedures?

Poor portability (syntax differs significantly across PostgreSQL, MySQL, SQL Server, Oracle), harder version control and code review compared to application code in a normal source repo, more difficult to unit test in isolation, and generally weaker debugging tooling compared to full application-language IDEs/debuggers. Logic split between application code and database procedures can also become harder to reason about as a whole system.

4. Can you call a stored procedure from a SELECT statement?

No - stored procedures are invoked with CALL (or EXECUTE, depending on the database) as a standalone statement, not embedded inside a SELECT. Only functions, which are required to return a value and designed for use in expressions, can be called from within a SELECT, WHERE, or other SQL expression context.

5. What are IN, OUT, and INOUT parameters?

IN parameters pass a value into the procedure; the procedure can use it but any changes to the parameter inside don't propagate back to the caller (this is the default and the only mode functions typically support). OUT parameters are output-only: the caller passes a placeholder variable, and the procedure sets its value, which the caller can read after the call. INOUT parameters do both - the caller's initial value is passed in, the procedure can read and modify it, and the final value is returned to the caller's variable.

6. How do you handle errors in stored procedures?

Most procedural SQL dialects provide exception/handler blocks: PostgreSQL's PL/pgSQL uses EXCEPTION WHEN ... THEN ... blocks within the procedure body (e.g. WHEN others THEN RAISE NOTICE ...; ROLLBACK;), while MySQL uses DECLARE ... HANDLER FOR <condition> to define what happens when a specific error code (like 1062 for duplicate key) occurs. You can also proactively raise custom errors with RAISE EXCEPTION (PostgreSQL) to enforce business-rule violations before they'd otherwise cause a less meaningful database error.

7. What is the difference between DETERMINISTIC and non-deterministic functions?

A DETERMINISTIC function always returns the same output given the same input arguments, with no

dependency on changing external state (e.g. a function that just concatenates two strings). A non-deterministic function can return different results on different calls with identical arguments (e.g. one that reads the current timestamp or a random value, or queries continuously-changing table data). Marking a function DETERMINISTIC when it genuinely is lets the optimizer cache/reuse results and is required by some databases (like MySQL) for certain contexts (e.g. functions used in generated/indexed columns).

8. How do stored procedures help with security?

They let you grant users or application roles permission to EXECUTE a specific procedure without granting them direct SELECT/INSERT/UPDATE access to the underlying tables - the procedure acts as a controlled gateway that only performs the specific, vetted operations it was written to do. Because procedures use parameterized inputs rather than string-concatenated SQL, they also naturally reduce SQL injection risk compared to hand-built dynamic queries in application code.

9. What is a cursor and when would you use one?

A cursor is a database object that lets you iterate through a result set row by row within procedural code (e.g. inside a stored procedure), rather than operating on the whole set at once as normal SQL does. You'd use one when you need row-by-row procedural logic that can't be expressed as a single set-based SQL statement - e.g. complex conditional processing per row, or calling other procedures per row - though in general, set-based SQL operations should be preferred over cursors whenever possible since cursors are typically much slower.

10. Should business logic be in stored procedures or application code?

It depends on the trade-offs: stored procedures favor performance (fewer round trips), security (controlled access), and logic that must be enforced consistently regardless of which client touches the database - but at the cost of portability, testability, and version-control friendliness compared to application code. Many teams favor keeping business logic in application code (easier to test, review, deploy, and keep portable across databases) and reserve stored procedures for performance-critical batch operations, security-sensitive gateways, or logic that genuinely must live at the database layer (like triggers for audit logging).

Key Takeaways

- Procedures perform actions; functions return values.
- Functions can be used in SELECT; procedures cannot.
- Procedures can have IN, OUT, INOUT parameters.
- Functions should be pure (no side effects).
- Use procedures for complex multi-step operations.
- Error handling with EXCEPTION/HANDLER blocks.
- Control flow: IF, CASE, LOOP, WHILE, FOR.
- Benefits: Security, performance, consistency.
- Drawbacks: Portability, testing, debugging.
- Consider trade-offs vs ORM/application code.

JDBC (Java Database Connectivity) Interview Revision Guide

This comprehensive guide is designed to help you refresh and strengthen your understanding of JDBC concepts in preparation for technical interviews.

In this chapter, you'll learn:

- What is JDBC and Why It's Used in Java Applications
- JDBC Architecture and Core Components
- Steps to Establish a Database Connection
- Types of JDBC Drivers: Type 1, 2, 3, and 4
- Statement vs PreparedStatement: Key Differences
- Preventing SQL Injection with PreparedStatement
- Performing Batch Updates for Better Performance
- ResultSet Interface: Common Methods and Usage
- Transaction Management: Commit, Rollback, and Auto-Commit
- ACID Properties in JDBC Transactions
- DataSource and Connection Pooling
- Handling SQL Exceptions Effectively
- Best Practices for Resource Management (try-with-resources)

What Is JDBC, and Why Is It Used in Java Applications?

JDBC (Java Database Connectivity) is an API (Application Programming Interface) in Java that enables Java applications to interact with databases. It provides a standard method for connecting to a database, sending SQL queries, and processing the results.

What Is JDBC?

1. Java API: JDBC is a part of the Java Standard Edition platform, consisting of a set of interfaces and classes that allow Java applications to interact with relational databases.
2. Standardization: It provides a standard API for database-agnostic interaction, meaning the same code can be used to connect to different types of databases (like MySQL, PostgreSQL, Oracle, etc.) with minimal changes.
3. Components: JDBC includes four main components:
- JDBC Drivers: Vendor-specific implementations that enable Java applications to communicate with a specific database.
- Connection: Represents a connection to a specific database.
- Statement: Used for executing a static SQL statement and returning the results it produces.
- ResultSet: Represents the result set of a query.

Why Is JDBC Used in Java Applications?

1. Database Interaction: JDBC provides a way for Java applications to perform database operations like querying, updating, and managing data.
2. Platform Independence: Java applications can connect to any database that has a JDBC driver, making the applications database-independent and highly portable.
3. Standardized API: It abstracts the complexities of database-specific communication, providing a uniform interface for interacting with different databases.
4. Performance: JDBC allows efficient processing of SQL statements, batch updates, and retrieval of large datasets.

How JDBC Works

```
// Establish a Connection
Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/mydatabase", "usern…

// Create a Statement: Create a statement to execute SQL queries.
Statement stmt = conn.createStatement();

// Execute Queries: Execute SQL queries and retrieve results.
ResultSet rs = stmt.executeQuery("SELECT * FROM mytable");
while (rs.next()) {
    System.out.println("Column Value: " + rs.getString("columnname"));
}

// Close Connections: Close the connection and other resources to free up database resources.
rs.close();
stmt.close();
conn.close();
```

Example Full Code:

Here is a complete example demonstrating how to connect to a MySQL database, execute a query, and process the results:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class JDBCExample {
    public static void main(String[] args) {
        String jdbcUrl = "jdbc:mysql://localhost:3306/mydatabase";
        String username = "root";
        String password = "password";

        Connection conn = null;
        Statement stmt = null;
        ResultSet rs = null;

        try {
            // Load the JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Establish the connection
            conn = DriverManager.getConnection(jdbcUrl, username, password);

            // Create a statement
            stmt = conn.createStatement();

            // Execute a query
            rs = stmt.executeQuery("SELECT * FROM mytable");

            // Process the result set
            while (rs.next()) {
                System.out.println("ID: " + rs.getInt("id"));
                System.out.println("Name: " + rs.getString("name"));
            }
        } catch (Exception e) {
            e.printStackTrace();
        } finally {
            // Close resources
            try {
                if (rs != null) rs.close();
                if (stmt != null) stmt.close();
                if (conn != null) conn.close();
            } catch (Exception e) {
                e.printStackTrace();
            }
        }
    }
}
```

Explain the Steps Involved in Establishing a Connection to a Database Using JDBC

Establishing a connection to a database using JDBC involves several key steps:

1. Load the JDBC Driver

The first step is to load the JDBC driver class. The driver class connects your application to the database. This is done using the Class.forName() method.

2. Establish a Connection

Next, you create a connection to the database using the DriverManager.getConnection() method. This method requires the database URL, username, and password.

3. Create a Statement

Once you have a connection, you can create a statement object. This object is used to execute SQL queries.

4. Execute a Query

Using the statement object, you can execute SQL queries. For example, to execute a SELECT query, you use the executeQuery method, which returns a ResultSet object containing the result of the query.

5. Process the ResultSet

If you executed a query that returns results (such as a SELECT query), you need to process the ResultSet to retrieve the data.

6. Close Resources

Finally, you should close all the resources to free up database connections and avoid potential memory leaks. This includes closing the ResultSet, Statement, and Connection.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class JDBCExample {
    public static void main(String[] args) {
        String jdbcUrl = "jdbc:mysql://localhost:3306/mydatabase";
        String username = "root";
        String password = "password";

        Connection conn = null;
        Statement stmt = null;
        ResultSet rs = null;

        try {
            // Step 1: Load the JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Step 2: Establish the connection
            conn = DriverManager.getConnection(jdbcUrl, username, password);

            // Step 3: Create a statement
            stmt = conn.createStatement();

            // Step 4: Execute a query
            String sql = "SELECT * FROM mytable";
            rs = stmt.executeQuery(sql);

            // Step 5: Process the ResultSet
            while (rs.next()) {
                int id = rs.getInt("id");
                String name = rs.getString("name");
                int age = rs.getInt("age");
                System.out.println("ID: " + id + ", Name: " + name + ", Age: " + age);
            }
        } catch (Exception e) {
            e.printStackTrace();
        } finally {
            // Step 6: Close resources
            try {
                if (rs != null) rs.close();
                if (stmt != null) stmt.close();
                if (conn != null) conn.close();
            } catch (Exception e) {
                e.printStackTrace();
            }
        }
    }
}
```

What Are the Different Types of JDBC Drivers, and How Do They Differ?

JDBC drivers facilitate communication between Java applications and databases. There are four main types:

1. Type 1: JDBC-ODBC Bridge Driver

- Description: Uses the JDBC-ODBC bridge to connect to databases that support ODBC (Open Database

Connectivity).
- Architecture: Translates JDBC calls into ODBC calls, which are then executed by the ODBC driver.
- Platform: Typically platform-dependent and requires native ODBC libraries.
- Advantages: Allows Java applications to connect to any database that has an ODBC driver; provides JDBC access to non-Java databases.
- Disadvantages: Requires an ODBC driver installed on the client machine; limited in performance and portability.
- Example: sun.jdbc.odbc.JdbcOdbcDriver (Deprecated in Java 8).

2. Type 2: Native-API Driver

- Description: Converts JDBC calls directly into native API calls of the database.
- Architecture: Written partly in Java and partly in the native language of the database.
- Platform: Platform-dependent, as it relies on native code libraries.
- Advantages: Offers better performance compared to the Type 1 driver; does not require ODBC, as it communicates directly with the database.
- Disadvantages: Requires different drivers for different platforms; less portable compared to Type 4 drivers.
- Example: Oracle OCI (Oracle Call Interface) driver.

3. Type 3: Network Protocol Driver (Middleware Driver)

- Description: Uses middleware to convert JDBC calls into a database-independent network protocol.
- Architecture: The middleware server translates JDBC calls into the database-specific protocol.
- Platform: Platform-independent on the client side; requires middleware server.
- Advantages: Provides database independence; allows clients to connect to multiple types of databases through a single driver.
- Disadvantages: Relies on middleware server, which may introduce additional latency; configuration and maintenance of middleware server required.
- Example: JDBC-Net pure Java driver.

4. Type 4: Thin Driver (Direct-to-Database Pure Java Driver)

- Description: Converts JDBC calls directly into the network protocol used by the database.
- Architecture: Pure Java implementation that communicates directly with the database server.
- Platform: Platform-independent; does not require any additional software installation.
- Advantages: Highest performance among JDBC drivers; platform-independent and easy to deploy.
- Disadvantages: May not support all database features; may require client-side installation of additional JAR files.
- Example: Oracle Thin driver, MySQL Connector/J.

What Is the Purpose of PreparedStatement in JDBC, and How Does It Differ from Statement?

The PreparedStatement interface in JDBC is used to execute parameterized SQL queries. It is a subinterface of the Statement interface and provides several advantages over it.

Purpose of PreparedStatement:

1. Parameterized Queries: PreparedStatement allows you to execute parameterized SQL queries, where placeholders (?) are used for dynamic values. This helps prevent SQL injection attacks and improves performance by reusing compiled queries.
2. Improved Performance: Prepared statements are precompiled by the database server, which can lead to improved performance when executing the same query multiple times with different parameters.
3. Readability and Maintainability: Using parameterized queries with PreparedStatement makes your code

more readable and maintainable, as it separates SQL logic from data.
4. Automatic Escaping: PreparedStatement automatically handles escaping of special characters in parameters, reducing the risk of SQL injection vulnerabilities.

Differences from Statement:

1. Precompilation: PreparedStatement is precompiled by the database server when it is created, whereas Statement is compiled each time it is executed. This can result in better performance for PreparedStatement when the same query is executed multiple times with different parameters.
2. Parameterized Queries: PreparedStatement supports parameterized queries, allowing you to pass dynamic values to SQL statements. Statement does not support parameterized queries directly, which makes it more susceptible to SQL injection attacks.
3. Execution Flexibility: Statement is suitable for executing static SQL queries without parameters, while PreparedStatement is more flexible and efficient for executing parameterized queries.

Example:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;

public class PreparedStatementExample {
    public static void main(String[] args) {
        String jdbcUrl = "jdbc:mysql://localhost:3306/mydatabase";
        String username = "username";
        String password = "password";
        String name = "John"; // Example parameter value

        try (Connection conn = DriverManager.getConnection(jdbcUrl, username, password)) {
            String sql = "SELECT * FROM mytable WHERE name = ?";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, name); // Set parameter value

            ResultSet rs = pstmt.executeQuery();

            while (rs.next()) {
                int id = rs.getInt("id");
                String resultName = rs.getString("name");
                int age = rs.getInt("age");
                System.out.println("ID: " + id + ", Name: " + resultName + ", Age: " + age);
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}
```

In this example, the PreparedStatement object is created with a parameterized query (SELECT * FROM mytable WHERE name = ?). The parameter value is set using setString() before executing the query. This approach ensures better security and performance compared to executing a plain Statement.

Describe How You Can Use JDBC to Perform Batch Updates

Using JDBC, you can perform batch updates to a database efficiently by executing multiple SQL statements in a single batch. This can significantly reduce the overhead of database communication and improve performance.

1. Establish a Connection

First, you need to establish a connection to your database using JDBC. You typically do this by loading the appropriate database driver and then creating a connection object using the DriverManager.getConnection() method.

```
Connection connection = DriverManager.getConnection(url, username, password);
```

2. Create a Statement or PreparedStatement Object

You can create either a Statement or a PreparedStatement object to execute your SQL statements. PreparedStatement is preferred when using batch updates because it allows you to parameterize your SQL queries, which can improve performance and security.

```
PreparedStatement preparedStatement = connection.prepareStatement("INSERT INTO table_name (col…
```

3. Add Batch Updates

You can add multiple SQL statements to the batch using the addBatch() method of the Statement or PreparedStatement object.

```
preparedStatement.setInt(1, value1);
preparedStatement.setString(2, value2);
preparedStatement.addBatch();
```

Repeat this step for each SQL statement you want to add to the batch.

4. Execute Batch Updates

Once you have added all the statements to the batch, you can execute them using the executeBatch() method. This will execute all the statements in the batch and return an array of integers representing the number of rows affected by each statement.

```
int[] updateCounts = preparedStatement.executeBatch();
```

5. Handle Results

After executing the batch, you can iterate over the array of update counts to handle the results of each SQL statement.

```
for (int updateCount : updateCounts) {
    // Handle update count
}
```

6. Close Resources

Finally, don't forget to close the PreparedStatement, Connection, and any other resources you used to interact with the database.

```
preparedStatement.close();
connection.close();
```

What Are the Common Methods Available in the ResultSet Interface, and How Are They Used?

The ResultSet interface in Java provides methods to retrieve and manipulate data from a database query

result set.

1. next() - Moves the cursor to the next row in the result set. It returns true if there is a next row, and false if there are no more rows.

```
while (resultSet.next()) {
    // Process current row
}
```

2. getString(int columnIndex) / getString(String columnLabel) - Retrieves the value of the specified column in the current row as a String.

```
String name = resultSet.getString("name");
```

3. getInt(int columnIndex) / getInt(String columnLabel) - Retrieves the value of the specified column in the current row as an int.

```
int age = resultSet.getInt("age");
```

4. getDouble(int columnIndex) / getDouble(String columnLabel) - Retrieves the value of the specified column in the current row as a double.

```
double salary = resultSet.getDouble("salary");
```

5. getDate(int columnIndex) / getDate(String columnLabel) - Retrieves the value of the specified column in the current row as a java.sql.Date object.

```
Date hireDate = resultSet.getDate("hire_date");
```

6. getObject(int columnIndex) / getObject(String columnLabel) - Retrieves the value of the specified column in the current row as a generic Object.

```
Object value = resultSet.getObject("column_name");
```

7. wasNull() - Indicates whether the last column read was NULL.

```
if (resultSet.wasNull()) {
    // Handle NULL value
}
```

8. close() - Closes the ResultSet object, releasing any resources it holds.

```
resultSet.close();
```

Explain How Transactions Are Managed in JDBC

In JDBC, transactions are managed using the Connection object. JDBC provides support for both manual and automatic transaction management.

1. Starting a Transaction

To start a transaction, you set the auto-commit mode of the Connection object to false. By default, auto-commit mode is enabled, meaning each SQL statement is treated as a separate transaction and is automatically committed immediately after it is executed.

```
connection.setAutoCommit(false);
```

2. Executing SQL Statements

Once the auto-commit mode is disabled, you can execute multiple SQL statements as part of a single transaction using the same Connection object.

```
Statement statement = connection.createStatement();
statement.executeUpdate("INSERT INTO table_name (column1, column2) VALUES (value1, value2)");
statement.executeUpdate("UPDATE table_name SET column1 = value WHERE condition");
// More SQL statements
```

3. Committing a Transaction

If all SQL statements within the transaction execute successfully and you want to make the changes permanent in the database, you commit the transaction using the commit() method.

```
connection.commit();
```

4. Rolling Back a Transaction

If an error occurs during the execution of the SQL statements or if you want to discard the changes made in the transaction, you can roll back the transaction using the rollback() method.

```
connection.rollback();
```

5. Re-enabling Auto-Commit

Once the transaction is completed (either committed or rolled back), you can re-enable the auto-commit mode by setting it back to true.

```
connection.setAutoCommit(true);
```

By managing transactions manually, you have more control over the atomicity, consistency, isolation, and durability (ACID properties) of your database operations. It allows you to group multiple SQL statements into a single logical unit of work and ensures that either all the statements within the transaction are executed successfully and committed, or none of them are. This helps maintain data integrity and consistency in your database.

What Is the Role of DataSource in JDBC, and How Does It Improve Database Connection Management?

The DataSource interface in JDBC plays a significant role in managing database connections efficiently. It provides a standard interface for obtaining a database connection and helps improve connection management in several ways:

1. Connection Pooling: One of the primary benefits of using a DataSource is connection pooling. Instead of creating a new database connection every time one is requested, a pool of pre-initialized connections is maintained by the DataSource. This minimizes the overhead of creating and destroying connections, as connections are reused from the pool.
2. Optimized Connection Creation: DataSource implementations typically initialize a set of database connections when they are created. These connections are ready for use and are kept alive within the connection pool. This eliminates the need to establish a new connection every time a database operation is performed, resulting in faster response times.
3. Connection Configuration: DataSource allows for configuring various properties related to database connections, such as maximum number of connections in the pool, timeout settings, and authentication credentials. These configurations can be adjusted based on application requirements and database

constraints.
4. Support for Connection Pooling Libraries: Many JDBC drivers come with built-in support for DataSource implementations that integrate with popular connection pooling libraries such as Apache Commons DBCP (Database Connection Pooling) and HikariCP. These libraries offer advanced connection pooling features like connection validation, idle connection eviction, and dynamic resizing of connection pools.
5. Abstraction of Connection Management: DataSource abstracts the details of database connection management from the application code. Applications obtain database connections through the DataSource interface without needing to know the specifics of how connections are managed internally. This simplifies application development and makes it easier to switch between different database implementations or connection pooling mechanisms without changing application code.

In summary, the DataSource interface in JDBC improves database connection management by providing a standardized way to obtain database connections, supporting connection pooling, optimizing connection creation, allowing for flexible configuration, and abstracting the complexities of connection management from application code. This leads to better performance, scalability, and reliability of database operations in Java applications.

What Is the Difference Between execute(), executeQuery(), and executeUpdate() Methods?

These three methods in JDBC are used to execute SQL statements, but they serve different purposes and return different types of results.

1. executeQuery()

- Returns a ResultSet containing the data retrieved by the query.
- Used exclusively for SELECT statements.
- Throws SQLException if used with non-SELECT statements.

```
Statement stmt = conn.createStatement();
ResultSet rs = stmt.executeQuery("SELECT id, name, salary FROM employees");

while (rs.next()) {
    System.out.println(rs.getInt("id") + " - " + rs.getString("name"));
}
```

2. executeUpdate()

- Returns an int representing the number of rows affected.
- Used for DML statements: INSERT, UPDATE, DELETE.
- Also used for DDL statements: CREATE, ALTER, DROP (returns 0).

```
Statement stmt = conn.createStatement();

// INSERT example
int rowsInserted = stmt.executeUpdate("INSERT INTO employees (name, salary) VALUES ('John', 50…
System.out.println("Rows inserted: " + rowsInserted); // Output: Rows inserted: 1

// UPDATE example
int rowsUpdated = stmt.executeUpdate("UPDATE employees SET salary = 55000 WHERE name = 'John'"…
System.out.println("Rows updated: " + rowsUpdated); // Output: Rows updated: 1

// DELETE example
int rowsDeleted = stmt.executeUpdate("DELETE FROM employees WHERE name = 'John'");
System.out.println("Rows deleted: " + rowsDeleted); // Output: Rows deleted: 1

// DDL example (returns 0)
int ddlResult = stmt.executeUpdate("CREATE TABLE temp_table (id INT)");
System.out.println("DDL result: " + ddlResult); // Output: DDL result: 0
```

3. execute()

- Returns boolean: true if result is a ResultSet, false if it's an update count.
- Used when SQL type is unknown at compile time or for stored procedures returning multiple results.
- More flexible but requires additional handling.

```
Statement stmt = conn.createStatement();
String sql = "SELECT * FROM employees"; // Could be dynamic

boolean hasResultSet = stmt.execute(sql);

if (hasResultSet) {
    // It's a SELECT query
    ResultSet rs = stmt.getResultSet();
    while (rs.next()) {
        System.out.println(rs.getString("name"));
    }
} else {
    // It's an UPDATE/INSERT/DELETE/DDL
    int updateCount = stmt.getUpdateCount();
    System.out.println("Rows affected: " + updateCount);
}
```

What Are ResultSet Types and Concurrency Modes in JDBC?

When creating a Statement or PreparedStatement, you can specify the type and concurrency mode of the ResultSet. These determine how you can navigate through the result set and whether you can update it.

ResultSet Types

a) TYPE_FORWARD_ONLY

- Cannot scroll (only forward).
- Not sensitive to DB changes.
- Default and most efficient.
- Use when you only need to read data once.

b) TYPE_SCROLL_INSENSITIVE

- Can scroll (forward, backward, random access).
- Does not reflect DB changes after ResultSet creation.

- Snapshot of data at query time.

c) TYPE_SCROLL_SENSITIVE

- Can scroll (forward, backward, random access).
- Reflects DB changes made by others.
- More overhead, depends on driver support.

ResultSet Concurrency Modes

a) CONCUR_READ_ONLY

- Cannot update data.
- Default mode.
- Best for simple read operations.

b) CONCUR_UPDATABLE

- Can update, insert, and delete rows via ResultSet.
- Requires proper SQL query and driver support.

```
// Default: Forward-only, Read-only (most efficient)
Statement stmt1 = conn.createStatement();

// Scrollable, Read-only
Statement stmt2 = conn.createStatement(
    ResultSet.TYPE_SCROLL_INSENSITIVE,
    ResultSet.CONCUR_READ_ONLY
);

// Scrollable, Updatable
Statement stmt3 = conn.createStatement(
    ResultSet.TYPE_SCROLL_SENSITIVE,
    ResultSet.CONCUR_UPDATABLE
);
```

Common Interview Question: Why is TYPE_FORWARD_ONLY the default?

It's the most memory-efficient option. The database driver doesn't need to cache all rows in memory, and it can stream results row by row. Scrollable ResultSets require caching all results, which consumes more memory and may not be supported efficiently by all database drivers.

How Can You Handle SQL Exceptions in JDBC Effectively?

Handling SQL exceptions effectively in JDBC is crucial for robust error handling and ensuring the stability of database operations.

1. Use try-with-resources

When working with JDBC resources such as Connection, Statement, and ResultSet, use try-with-resources to automatically close these resources after use. This ensures that resources are properly released even if an exception occurs.

```
try (Connection connection = DriverManager.getConnection(url, username, password);
     Statement statement = connection.createStatement()) {
    // Database operations
} catch (SQLException e) {
    // Exception handling
}
```

2. Catch Specific Exceptions

Catch specific SQLExceptions and handle them appropriately based on the nature of the exception. Common SQL exceptions include SQLException, SQLTimeoutException, SQLIntegrityConstraintViolationException, etc. Handling specific exceptions allows for more granular error handling.

```
try {
    // Database operations
} catch (SQLTimeoutException e) {
    // Handle timeout exception
} catch (SQLIntegrityConstraintViolationException e) {
    // Handle constraint violation exception
} catch (SQLException e) {
    // Handle other SQL exceptions
}
```

3. Log Exceptions

Log SQL exceptions along with relevant information such as error codes, error messages, and SQL statements. This helps in troubleshooting and debugging issues during development and in production environments.

```
catch (SQLException e) {
    logger.error("SQL Exception occurred: ", e);
}
```

4. Rollback Transactions

If an exception occurs during a transaction, roll back the transaction to ensure data consistency and integrity. This helps avoid partially completed transactions that could leave the database in an inconsistent state.

```
try {
    // Start transaction
    connection.setAutoCommit(false);
    // Database operations
    connection.commit();
} catch (SQLException e) {
    // Rollback transaction
    connection.rollback();
    // Exception handling
} finally {
    // Reset auto-commit mode
    connection.setAutoCommit(true);
}
```

5. Provide Meaningful Error Messages

When catching and handling SQL exceptions, provide meaningful error messages or notifications to users or system administrators. This improves the usability of the application and helps users understand the cause of the error.

```
catch (SQLException e) {
    // Display error message to user
    System.out.println("An error occurred while processing your request. Please try again late…
}
```

JPA (Java Persistence API) Interview Revision Guide

This comprehensive guide is designed to help you refresh and strengthen your understanding of JPA (Java Persistence API) concepts in preparation for technical interviews.

In this chapter, you'll learn:

Core JPA Concepts

- What is JPA and How It Simplifies Database Operations
- EntityManager vs EntityManagerFactory
- Persistence Context: Transaction-Scoped vs Extended
- Configuring JPA with persistence.xml

Entity Mapping

- Defining Entities with @Entity and @Table
- Key Annotations: @Id, @Column, @GeneratedValue, @Temporal, @Enumerated, @Lob, @Transient
- Entity Relationships: @OneToOne, @OneToMany, @ManyToOne, @ManyToMany
- Join Columns and Join Tables

Querying

- JPQL vs SQL: Key Differences
- Named Queries with @NamedQuery
- Pagination using setFirstResult() and setMaxResults()

Operations & Transactions

- CRUD Operations: persist(), find(), merge(), remove()
- Transaction Management: begin(), commit(), rollback()
- Entity Lifecycle Callbacks: @PrePersist, @PostPersist, @PreUpdate, @PostUpdate, @PreRemove, @PostRemove

Performance & Concurrency

- Lazy vs Eager Fetching (FetchType.LAZY vs FetchType.EAGER)
- Optimistic Locking with @Version

What Is JPA, and How Does It Simplify Database Operations in Java Applications?

Java Persistence API (JPA) is a specification for object-relational mapping (ORM) in Java. It provides a framework for managing relational data in Java applications by mapping Java objects to database tables, and vice versa. JPA simplifies database operations through several key features:

Key Features of JPA

1. Annotations and XML Configuration: JPA allows you to define the mapping between Java classes and database tables using annotations or XML configuration. This declarative approach makes it easier to manage and maintain the mapping information.
2. Entity Classes: In JPA, a simple Java class annotated with @Entity becomes an entity class that maps to a database table. Each instance of the class corresponds to a row in the table.
3. Entity Manager: The EntityManager interface provides the core functionality for performing CRUD (Create, Read, Update, Delete) operations on the database. It handles the lifecycle of entities and transactions.
4. JPQL (Java Persistence Query Language): JPQL is a query language similar to SQL but operates on

the entity objects rather than database tables. This allows you to write database queries using Java syntax and object-oriented principles.
5. Automatic Schema Generation: JPA can automatically generate the database schema based on the entity mappings. This reduces the need for manual SQL scripts to create and update tables.
6. Transaction Management: JPA supports both local and global transactions. It integrates seamlessly with Java Transaction API (JTA) to manage transactions across multiple resources.
7. Cascading and Fetching: JPA provides options for cascading operations (e.g., cascading persist, remove, etc.) and fetching strategies (e.g., eager or lazy loading) to optimize performance and manage relationships between entities.

Here is a simple example of a JPA entity class and basic CRUD operations:

```
import javax.persistence.*;

@Entity
public class User {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;
    private String email;

    // Getters and setters
}

public class UserService {
    @PersistenceContext
    private EntityManager entityManager;

    public void createUser(String name, String email) {
        User user = new User();
        user.setName(name);
        user.setEmail(email);
        entityManager.persist(user);
    }

    public User getUser(Long id) {
        return entityManager.find(User.class, id);
    }

    public void updateUser(Long id, String name, String email) {
        User user = entityManager.find(User.class, id);
        if (user != null) {
            user.setName(name);
            user.setEmail(email);
            entityManager.merge(user);
        }
    }

    public void deleteUser(Long id) {
        User user = entityManager.find(User.class, id);
        if (user != null) {
            entityManager.remove(user);
        }
    }
}
```

Explain the Difference Between EntityManager and

EntityManagerFactory in JPA

In JPA, EntityManager and EntityManagerFactory are core components used for managing the persistence of entities in a relational database. They serve different purposes and are used at different stages of the application lifecycle.

EntityManagerFactory

1. Definition: EntityManagerFactory is a factory class responsible for creating EntityManager instances.
2. Lifecycle: EntityManagerFactory is a heavyweight object designed to be created once and used throughout the application. It is typically created at application startup and closed when the application is shutting down.
3. Configuration: It is configured using persistence units defined in the persistence.xml file. This configuration includes database connection details, JPA provider properties, and class mappings.
4. Resource Management: Since it is an expensive object to create, EntityManagerFactory should be reused rather than recreated frequently. It handles resource-intensive tasks such as connection pooling and caching.
5. Thread Safety: EntityManagerFactory is thread-safe and can be shared across multiple threads without synchronization issues.

```
EntityManagerFactory emf = Persistence.createEntityManagerFactory("my-persistence-unit");
```

EntityManager

1. Definition: EntityManager is the primary interface used to interact with the persistence context. It provides methods for CRUD operations, querying, and transaction management.
2. Lifecycle: EntityManager is a lightweight, short-lived object. It is created by EntityManagerFactory and typically used within the scope of a single transaction or request.
3. Persistence Context: Each EntityManager instance is associated with a persistence context, which is a set of managed entity instances. The persistence context ensures that each entity is managed and synchronized with the database.
4. Transaction Management: EntityManager manages the lifecycle of entities within a transaction. It handles operations such as persisting, merging, removing, and querying entities.
5. Thread Safety: EntityManager is not thread-safe. Each thread or transaction should use its own EntityManager instance.

```
EntityManager em = emf.createEntityManager();
em.getTransaction().begin();
// perform operations
em.getTransaction().commit();
em.close();
```

How Do You Define an Entity in JPA, and What Are the Key Annotations Used?

Defining an entity in JPA involves creating a plain Java class and using specific annotations to map it to a database table.

Defining an Entity

1. Create a Java Class: The class should be a simple POJO (Plain Old Java Object) with properties, getters, and setters.
2. Annotate the Class with @Entity: This annotation specifies that the class is an entity and will be mapped to a database table.

3. Define the Primary Key with @Id: Every entity must have a primary key, which is annotated with @Id.
4. Specify the Table Name with @Table (Optional): If the table name is different from the class name, use the @Table annotation to specify the table name.
5. Map Fields to Columns with @Column (Optional): Use the @Column annotation to customize the mapping between class fields and table columns.
6. Generate Primary Key Values with @GeneratedValue (Optional): To automatically generate primary key values, use the @GeneratedValue annotation.

Key Annotations:

```
import javax.persistence.*;
import java.util.Date;

@Entity // Marks the class as an entity.
@Table(name = "users") // Specifies the table name
public class User {

    @Id // Specifies the primary key field.
    @GeneratedValue(strategy = GenerationType.IDENTITY) // Specifies the primary key generatio…
    private Long id;

    @Column(name = "user_name", nullable = false, length = 50)
    private String name;

    @Column(name = "email", nullable = false, unique = true) // Maps a field to a column in th…
    private String email;

    @Temporal(TemporalType.DATE) // Specifies the temporal type (DATE, TIME, or TIMESTAMP)
    private Date birthDate;

    @Enumerated(EnumType.STRING) // Specifies that a field is an enumeration type.
    private Status status;

    @Lob // Specifies that a field should be treated as a large object (BLOB or CLOB).
    private byte[] profilePicture;

    @Transient // Indicates that a field is not persistent and should not be mapped to a datab…
    private String temporaryData;

    // Getters and setters

    public Long getId() {
        return id;
    }

    public void setId(Long id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }

    public Date getBirthDate() {
        return birthDate;
    }

    public void setBirthDate(Date birthDate) {
        this.birthDate = birthDate;
    }

    public Status getStatus() {
        return status;
    }
```

Describe the Concept of Persistence Context in JPA and How Do You Configure a JPA Application Using persistence.xml?

Persistence Context in JPA

Concept:

- The persistence context in JPA is essentially a set of entity instances in which for any persistent entity identity there is a unique entity instance. The EntityManager maintains a context that manages the lifecycle and state of entities.
- It ensures that changes to entity instances are tracked and synchronized with the database.

Types:

1. Transaction-Scoped Persistence Context: Tied to a JTA transaction. The persistence context is created and associated with the current transaction and is closed when the transaction ends.
2. Extended Persistence Context: Used primarily in stateful session beans. The persistence context spans multiple transactions and remains open for the duration of the extended scope.

Benefits:

- Change Tracking: Automatically tracks changes made to entities and synchronizes them with the database upon transaction commit.
- Identity Management: Ensures that within a persistence context, there is only one instance of an entity for a specific database row.
- Lazy Loading: Supports lazy loading of relationships, optimizing performance by loading related entities on demand.

Configuring a JPA Application Using persistence.xml

The persistence.xml file is the configuration file for JPA applications. It defines the persistence units and their properties.

Structure of persistence.xml:

1. Basic Structure: The file is placed in the META-INF directory of the classpath. Contains one or more <persistence-unit> elements that define configurations for different persistence contexts.

2. Example Configuration:

```
<?xml version="1.0" encoding="UTF-8"?>
<persistence xmlns="http://xmlns.jcp.org/xml/ns/persistence"
             xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
             xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/persistence
                                 http://xmlns.jcp.org/xml/ns/persistence/persistence_2_1.xsd"
             version="2.1">
    <persistence-unit name="myPersistenceUnit" transaction-type="RESOURCE_LOCAL">
        <provider>org.hibernate.jpa.HibernatePersistenceProvider</provider>
        <class>com.example.User</class>
        <properties>
            <property name="javax.persistence.jdbc.url" value="jdbc:mysql://localhost:3306/myd…
            <property name="javax.persistence.jdbc.user" value="root"/>
            <property name="javax.persistence.jdbc.password" value="password"/>
            <property name="javax.persistence.jdbc.driver" value="com.mysql.cj.jdbc.Driver"/>

            <property name="hibernate.dialect" value="org.hibernate.dialect.MySQL5Dialect"/>
            <property name="hibernate.hbm2ddl.auto" value="update"/>
            <property name="hibernate.show_sql" value="true"/>
            <property name="hibernate.format_sql" value="true"/>
        </properties>
    </persistence-unit>
</persistence>
```

Elements and Attributes:

1. \<persistence\>: Root element.
2. \<persistence-unit\>: Represents a persistence unit with attributes such as name and transaction-type (either JTA or RESOURCE_LOCAL).
3. \<provider\>: Specifies the JPA provider (e.g., Hibernate).
4. \<class\>: Lists the entity classes managed by this persistence unit.
5. \<properties\>: Contains configuration properties:

- javax.persistence.jdbc.url: Database connection URL.
- javax.persistence.jdbc.user: Database username.
- javax.persistence.jdbc.password: Database password.
- javax.persistence.jdbc.driver: JDBC driver class.

Provider-specific properties, e.g., Hibernate:

- hibernate.dialect: Specifies the SQL dialect.
- hibernate.hbm2ddl.auto: Schema generation settings (e.g., update, create, create-drop, validate).
- hibernate.show_sql: Displays generated SQL statements.
- hibernate.format_sql: Formats SQL statements for readability.

What Are the Different Types of Relationships in JPA, and How Are They Mapped Using Annotations?

In JPA, relationships between entities can be mapped using annotations to define how entities interact with each other in terms of database relationships. The primary types of relationships are:

- One-to-One
- One-to-Many
- Many-to-One
- Many-to-Many

1. One-to-One Relationship

A one-to-one relationship occurs when one entity is associated with exactly one instance of another entity.

Annotations Used: @OneToOne, @JoinColumn

```
@Entity
public class Employee {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @OneToOne
    @JoinColumn(name = "parking_space_id")
    private ParkingSpace parkingSpace;

    // getters and setters
}

@Entity
public class ParkingSpace {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String location;

    @OneToOne(mappedBy = "parkingSpace")
    private Employee employee;

    // getters and setters
}
```

- @OneToOne: Defines a one-to-one relationship.
- @JoinColumn: Specifies the foreign key column.

2. One-to-Many Relationship

A one-to-many relationship occurs when one entity is associated with multiple instances of another entity.

Annotations Used: @OneToMany, @JoinColumn

```
@Entity
public class Department {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @OneToMany(mappedBy = "department")
    private List<Employee> employees;

    // getters and setters
}

@Entity
public class Employee {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @ManyToOne
    @JoinColumn(name = "department_id")
    private Department department;

    // getters and setters
}
```

- @OneToMany: Defines a one-to-many relationship.
- @ManyToOne: Defines the other side of the relationship.
- @JoinColumn: Specifies the foreign key column in the child entity.

3. Many-to-One Relationship

A many-to-one relationship occurs when multiple instances of one entity are associated with a single instance of another entity.

Annotations Used: @ManyToOne, @JoinColumn

```
@Entity
public class Employee {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @ManyToOne
    @JoinColumn(name = "department_id")
    private Department department;

    // getters and setters
}

@Entity
public class Department {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @OneToMany(mappedBy = "department")
    private List<Employee> employees;

    // getters and setters
}
```

- @ManyToOne: Defines a many-to-one relationship.
- @JoinColumn: Specifies the foreign key column.

4. Many-to-Many Relationship

A many-to-many relationship occurs when multiple instances of one entity are associated with multiple instances of another entity.

Annotations Used: @ManyToMany, @JoinTable, @JoinColumn

```
@Entity
public class Student {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @ManyToMany
    @JoinTable(
        name = "student_course",
        joinColumns = @JoinColumn(name = "student_id"),
        inverseJoinColumns = @JoinColumn(name = "course_id")
    )
    private List<Course> courses;

    // getters and setters
}

@Entity
public class Course {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String title;

    @ManyToMany(mappedBy = "courses")
    private List<Student> students;

    // getters and setters
}
```

- @ManyToMany: Defines a many-to-many relationship.
- @JoinTable: Specifies the join table that holds the foreign keys.
- @JoinColumn: Specifies the join columns in the join table.

Explain the Role of the @Entity and @Table Annotations in JPA

@Entity Annotation

Role: The @Entity annotation specifies that the class is an entity and is mapped to a database table. It marks the class as a JPA entity, meaning it will be managed by the JPA provider (e.g., Hibernate).

Usage:

- It is placed at the class level.
- Every class annotated with @Entity must have a primary key field, which is annotated with @Id.

```
import javax.persistence.Entity;
import javax.persistence.Id;

@Entity
public class User {
    @Id
    private Long id;
    private String name;
    private String email;

    // getters and setters
}
```

@Table Annotation

Role: The @Table annotation specifies the details of the table that the entity is mapped to. While @Entity indicates that a class is an entity, @Table allows you to customize the table mapping.

Usage:

- It is placed at the class level, typically in conjunction with the @Entity annotation.
- It is optional; if not used, the default table name is the unqualified name of the entity class.

Attributes:

- name: Specifies the name of the table. If not specified, the class name is used by default.
- catalog: Specifies the catalog of the table.
- schema: Specifies the schema of the table.
- uniqueConstraints: Specifies unique constraints that should be applied to the table.
- indexes: Specifies the indexes on the table.

```
import javax.persistence.Entity;
import javax.persistence.Id;
import javax.persistence.Table;

@Entity
@Table(name = "users", schema = "public")
public class User {
    @Id
    private Long id;
    private String name;
    private String email;

    // getters and setters
}
```

What Is JPQL, and How Does It Differ from SQL? Provide an Example Query

JPQL (Java Persistence Query Language) is a query language used to perform database queries on entities mapped to the database tables in a JPA application. It is similar to SQL but operates on entity objects rather than directly on database tables.

Differences Between JPQL and SQL:

1. Object-Oriented Queries: JPQL operates on entity objects and their relationships, allowing developers to write queries using entity names, attributes, and relationships directly. SQL operates on database tables and columns, requiring knowledge of the underlying database schema.

2. Platform Independence: JPQL queries are platform-independent, meaning they are written in terms of entity objects and are translated by the JPA provider (e.g., Hibernate, EclipseLink) into native SQL queries for the underlying database. SQL queries are specific to the database platform and may need to be rewritten when switching between different database vendors.
3. Type Safety: JPQL queries are type-safe, as they are validated against entity mappings at compile-time. SQL queries are strings passed directly to the database, potentially leading to runtime errors if there are syntax errors or mismatches between the query and the database schema.
4. Support for Inheritance: JPQL supports querying entities and their subclasses, including polymorphic queries. SQL does not directly support querying across inheritance hierarchies.

Example Query:

Suppose we have an entity class named Product mapped to a database table. We want to retrieve the names of all products with a price greater than 100.

```
// JPQL:
Query query = entityManager.createQuery(
    "SELECT p.name FROM Product p WHERE p.price > :price");
query.setParameter("price", 100);
List<String> productNames = query.getResultList();
```

```
-- Equivalent SQL Query
SELECT name FROM Product WHERE price > 100;
```

Describe How You Can Perform CRUD Operations Using JPA

CRUD operations (Create, Read, Update, Delete) can be performed in JPA using the EntityManager interface.

1. Create (Persist) Operation

To create a new entity and persist it to the database:

```
EntityManager entityManager = entityManagerFactory.createEntityManager();
entityManager.getTransaction().begin();

User user = new User();
user.setName("John Doe");
user.setEmail("john@example.com");

entityManager.persist(user);

entityManager.getTransaction().commit();
entityManager.close();
```

2. Read (Retrieve) Operation

To retrieve an entity from the database:

```
EntityManager entityManager = entityManagerFactory.createEntityManager();
User user = entityManager.find(User.class, userId);
entityManager.close();
```

3. Update (Merge) Operation

To update an existing entity:

```
EntityManager entityManager = entityManagerFactory.createEntityManager();
entityManager.getTransaction().begin();

User user = entityManager.find(User.class, userId);
if (user != null) {
    user.setName("Jane Doe");
    entityManager.merge(user);
}

entityManager.getTransaction().commit();
entityManager.close();
```

4. Delete (Remove) Operation

To delete an existing entity:

```
EntityManager entityManager = entityManagerFactory.createEntityManager();
entityManager.getTransaction().begin();

User user = entityManager.find(User.class, userId);
if (user != null) {
    entityManager.remove(user);
}

entityManager.getTransaction().commit();
entityManager.close();
```

What Is the Purpose of the @OneToMany and @ManyToOne Annotations, and How Are They Used?

@OneToMany Annotation

Purpose: The @OneToMany annotation is used to define a one-to-many relationship between two entities. It indicates that one instance of the source entity (parent) can be associated with multiple instances of the target entity (child).

Usage:

- It is placed on the field or getter method in the source entity class.
- It is typically used in conjunction with the mappedBy attribute to specify the corresponding many-to-one relationship in the target entity.

```
@Entity
public class Department {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @OneToMany(mappedBy = "department")
    private List<Employee> employees;

    // getters and setters
}
```

@ManyToOne Annotation

Purpose: The @ManyToOne annotation is used to define a many-to-one relationship between two entities. It indicates that multiple instances of the source entity (child) can be associated with one instance of the target entity (parent).

Usage:

- It is placed on the field or getter method in the child entity class.
- It is typically used in conjunction with the @JoinColumn annotation to specify the foreign key column in the child table.

```
@Entity
public class Employee {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @ManyToOne
    @JoinColumn(name = "department_id")
    private Department department;

    // getters and setters
}
```

Explain How You Can Handle Transactions in JPA

In JPA, transactions are used to group multiple database operations into an atomic unit of work, ensuring data consistency and integrity.

1. Obtain an EntityManager

```
EntityManager entityManager = entityManagerFactory.createEntityManager();
```

2. Begin a Transaction

Before performing any database operations, start a transaction using the begin() method of the EntityTransaction interface.

```
EntityTransaction transaction = entityManager.getTransaction();
transaction.begin();
```

3. Perform Database Operations

```
// Example: persisting a new entity
User user = new User();
user.setName("John Doe");
user.setEmail("john@example.com");

entityManager.persist(user);
```

4. Commit the Transaction

```
transaction.commit();
```

5. Rollback on Failure

```
transaction.rollback();
```

6. Close the EntityManager

```
entityManager.close();
```

Best Practices

1. Transaction Scope: Keep transactions as short as possible to minimize the time the database locks resources.
2. Exception Handling: Use try-catch-finally blocks to handle exceptions and ensure proper transaction management.
3. Rollback on Failure: Always rollback the transaction if an error occurs to maintain data consistency.
4. Closing EntityManager: Close the EntityManager after each transaction to release database connections.
5. Propagation: Understand transaction propagation rules if you have nested transactional methods.

What Are the Lifecycle Callbacks in JPA, and How Can You Use Them?

Entity lifecycle callbacks allow you to execute custom logic at various stages of an entity's lifecycle. These callbacks provide hooks to perform actions such as initialization, validation, auditing, or cleanup.

Standard Lifecycle Callbacks:

1. @PrePersist: Executed before an entity is inserted into the database. Typically used for initializing fields or performing validation checks.
2. @PostPersist: Executed after an entity is inserted into the database. Useful for triggering additional actions after persistence, such as sending notifications or updating related entities.
3. @PreUpdate: Executed before an entity is updated in the database. Often used for auditing purposes or to enforce business rules before updating.
4. @PostUpdate: Executed after an entity is updated in the database. Similar to @PostPersist, but specifically for update operations.
5. @PreRemove: Executed before an entity is removed from the database. Commonly used for cleanup tasks or for enforcing referential integrity constraints.
6. @PostRemove: Executed after an entity is removed from the database. Useful for cascading deletion actions or logging.

Usage:

To use lifecycle callbacks, you annotate the methods you want to be invoked with the appropriate lifecycle annotation. These methods should be defined in your entity class.

```
@Entity
public class User {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String username;
    private String email;

    @PrePersist
    public void prePersist() {
        // Custom logic before persisting the entity
        System.out.println("Preparing to persist user: " + username);
    }

    @PostPersist
    public void postPersist() {
        // Custom logic after persisting the entity
        System.out.println("User persisted: " + username);
    }

    // Other entity methods, getters, and setters
}
```

In this example, the prePersist() method is annotated with @PrePersist and is called before the entity is persisted, and postPersist() is annotated with @PostPersist and is called after the entity is persisted. You can define any custom logic within these methods.

Describe How You Can Implement Optimistic Locking in JPA

Optimistic locking in JPA is a concurrency control mechanism used to prevent concurrent updates to the same entity from causing data inconsistency. It works by allowing multiple transactions to read and potentially update the same entity concurrently, but it detects and prevents conflicts at the time of transaction commit.

Implementation of Optimistic Locking in JPA:

1. Version Attribute: To implement optimistic locking, add a version attribute to the entity class. This attribute is typically a numeric or timestamp field that is incremented or updated automatically by the persistence provider.
2. @Version Annotation: Annotate the version attribute with @Version to indicate that it serves as the version for optimistic locking.
3. Handling Concurrent Updates: When multiple transactions attempt to update the same entity concurrently, JPA checks the version attribute's value. If the version value has changed since the entity was loaded into memory, it indicates that another transaction has modified the entity concurrently. This situation is referred to as a version conflict.
4. Detecting Version Conflict: JPA detects the version conflict during transaction commit. If a conflict is detected, JPA throws an OptimisticLockException.

Example Implementation:

```
@Entity
public class Product {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;
    private double price;

    @Version
    private int version; // Version attribute for optimistic locking

    // Getters and setters
}
```

What Is the Purpose of the @NamedQuery Annotation, and How Is It Used?

The @NamedQuery annotation in JPA is used to define named queries at the entity level. Named queries are pre-defined JPQL queries that can be executed multiple times with different parameters. They are particularly useful for complex queries that are used frequently within an application.

Purpose:

1. Encapsulation of Query Logic: Named queries encapsulate complex JPQL queries within the entity class, promoting code organization and maintainability.
2. Improved Performance: Named queries are parsed and validated at application startup, reducing the overhead of query compilation and validation during runtime.
3. Type Safety: Named queries provide compile-time type checking of query strings, reducing the likelihood of syntax errors or query mismatches.

Usage:

To use @NamedQuery, you annotate a method in the entity class with @NamedQueries, which is then followed by one or more @NamedQuery annotations. Each @NamedQuery annotation defines a named query with a unique name and a JPQL query string.

Example:

Consider an entity class User with a named query to retrieve users by their email:

```
@Entity
@NamedQueries({
    @NamedQuery(name = "User.findByEmail", query = "SELECT u FROM User u WHERE u.email = :emai…
})
public class User {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;
    private String email;

    // Getters and setters
}
```

How Can You Achieve Pagination in JPA?

Pagination in JPA allows you to retrieve a subset of query results, typically to display data in chunks or pages in a user interface. You can achieve pagination using the setFirstResult() and setMaxResults() methods of the Query interface or CriteriaQuery.

Using JPQL:

1. Create a JPQL Query: Construct a JPQL query to retrieve the desired data.
2. Set Pagination Parameters: Use the setFirstResult() method to specify the starting index of the result set. Use the setMaxResults() method to specify the maximum number of results to return.
3. Execute the Query: Execute the query to retrieve the paginated data.

```
EntityManager entityManager = entityManagerFactory.createEntityManager();

String jpql = "SELECT u FROM User u ORDER BY u.id";
Query query = entityManager.createQuery(jpql);

int pageNumber = 1;
int pageSize = 10;

query.setFirstResult((pageNumber - 1) * pageSize);
query.setMaxResults(pageSize);

List<User> users = query.getResultList();
```

Using Criteria API:

1. Create a CriteriaQuery: Use the Criteria API to construct a CriteriaQuery object for the desired entity type.
2. Apply Pagination Parameters: Use the setFirstResult() method of the CriteriaQuery object to specify the starting index of the result set. Use the setMaxResults() method to specify the maximum number of results to return.
3. Execute the Query: Execute the query to retrieve the paginated data.

```
EntityManager entityManager = entityManagerFactory.createEntityManager();
CriteriaBuilder criteriaBuilder = entityManager.getCriteriaBuilder();
CriteriaQuery<User> criteriaQuery = criteriaBuilder.createQuery(User.class);

Root<User> root = criteriaQuery.from(User.class);
criteriaQuery.select(root);

int pageNumber = 1;
int pageSize = 10;

criteriaQuery.orderBy(criteriaBuilder.asc(root.get("id")));
TypedQuery<User> typedQuery = entityManager.createQuery(criteriaQuery);

typedQuery.setFirstResult((pageNumber - 1) * pageSize);
typedQuery.setMaxResults(pageSize);

List<User> users = typedQuery.getResultList();
```

Explain the Difference Between Lazy and Eager Fetching in JPA

Lazy and eager fetching are strategies used to define how related entities are loaded from the database when querying for an entity. These strategies determine whether associated entities are loaded immediately

along with the parent entity or deferred until explicitly accessed.

Lazy Fetching:

- Purpose: Lazy fetching loads associated entities only when they are explicitly accessed by the application code.
- Deferred Loading: Associated entities are not loaded when the parent entity is fetched from the database.
- Reduced Overhead: Lazy fetching reduces the amount of data retrieved from the database, improving performance and reducing memory consumption.
- Suitable for Large Associations: Lazy fetching is suitable for associations with many related entities or when the associated data is not always needed.
- FetchType.LAZY: In JPA, lazy fetching is typically configured using the FetchType.LAZY annotation attribute.

Eager Fetching:

- Purpose: Eager fetching loads associated entities immediately along with the parent entity during the initial fetch operation.
- Immediate Loading: Associated entities are loaded along with the parent entity, resulting in a single database query.
- Increased Overhead: Eager fetching may result in loading unnecessary data, especially for large associations, leading to performance overhead.
- Suitable for Small Associations: Eager fetching is suitable for associations with a small number of related entities or when the associated data is frequently accessed.
- FetchType.EAGER: In JPA, eager fetching is typically configured using the FetchType.EAGER annotation attribute.

Example:

Consider a Department entity with a one-to-many association to Employee entities:

```
@Entity
public class Department {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @OneToMany(mappedBy = "department", fetch = FetchType.LAZY)
    private List<Employee> employees;

    // Getters and setters
}
```

In this example, the employees association is configured with lazy fetching (FetchType.LAZY), meaning that when a Department entity is fetched, its associated Employee entities are not loaded immediately. If the application accesses the employees collection, the associated Employee entities will be loaded from the database at that time.

```
@Entity
public class Department {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @OneToMany(mappedBy = "department", fetch = FetchType.EAGER)
    private List<Employee> employees;

    // Getters and setters
}
```

In this example, the employees association is configured with eager fetching (FetchType.EAGER), meaning that when a Department entity is fetched, its associated Employee entities are loaded immediately along with the parent entity.

JDBC, JPA and Hibernate - Under the Hood

JDBC, JPA and Hibernate

Understanding the architecture from your code to the database.

The Big Picture

When you write userRepository.findById(1), what actually happens? Your code travels through multiple layers before reaching the database: Your Code -> Spring Data JPA -> JPA API -> Hibernate -> JDBC -> Database Driver -> Database. Understanding these layers helps you write better queries, debug performance issues, and make informed architectural decisions.

Why So Many Layers?

- JDBC - The foundation. Direct database access, but verbose and error-prone.
- JPA - A specification (interface). Defines how ORM should work, but has no implementation.
- Hibernate - The most popular JPA implementation. Does the actual work.
- Spring Data JPA - Reduces boilerplate further. Generates implementations from interface methods.

The Persistence Stack (layer by layer)

1. Your Application Code (userRepository.findById(1)) - Entry point.
2. Spring Data JPA - Generates query from method name.
3. JPA (Jakarta Persistence API) (EntityManager.find(User.class, 1)) - Specification.
4. Hibernate - Session, first-level cache, dirty checking - Implementation.
5. JDBC - Connection, PreparedStatement, ResultSet - Foundation.
6. Database Driver - MySQL Connector/J, PostgreSQL JDBC, etc. - Driver.
7. Database - MySQL, PostgreSQL, Oracle, etc. - Storage.

Each layer adds abstraction and functionality. Understanding what happens at each level helps you debug issues and optimize performance.

JDBC: The Foundation

JDBC (Java Database Connectivity) is the lowest-level API for database access in Java. Everything else is built on top of it. Even when you use JPA, Hibernate eventually calls JDBC to execute SQL.

Core JDBC Components

- DriverManager - Manages database drivers and creates connections.
- Connection - A session with the database. Expensive to create.
- Statement / PreparedStatement - Executes SQL. PreparedStatement prevents SQL injection.
- ResultSet - Holds query results. Cursor-based iteration.

```
// Raw JDBC - verbose but you see exactly what happens
String sql = "SELECT id, name, email FROM users WHERE id = ?";

try (Connection conn = dataSource.getConnection();
     PreparedStatement stmt = conn.prepareStatement(sql)) {

    stmt.setLong(1, userId);  // Set parameter (1-indexed)

    try (ResultSet rs = stmt.executeQuery()) {
        if (rs.next()) {
            User user = new User();
            user.setId(rs.getLong("id"));
            user.setName(rs.getString("name"));
            user.setEmail(rs.getString("email"));
            return user;
        }
    }
}
// Connection, Statement, ResultSet all auto-closed
```

Why We Moved Beyond JDBC

- Manual mapping of ResultSet to objects (boilerplate).
- No caching - every query hits the database.
- No automatic dirty checking - manual UPDATE statements.
- No relationship management - manual JOINs and foreign keys.
- SQL is database-specific (MySQL vs PostgreSQL differences).

JPA: The Specification

JPA (Jakarta Persistence API, formerly Java Persistence API) is a specification, not an implementation. It defines interfaces and annotations that ORM frameworks must implement.

Why a Specification Matters

Before JPA, you were locked into your ORM vendor. If you used Hibernate, your code was full of Hibernate-specific APIs. Switching to EclipseLink meant rewriting everything. JPA provides a standard API so you can switch implementations without changing your code.

Before JPA (Vendor Lock-in):

```
// Hibernate-specific code
Session session = sessionFactory.openSession();
User user = session.get(User.class, 1L);
session.close();

// TopLink-specific code (different API)
UnitOfWork uow = session.acquireUnitOfWork();
User user = (User) uow.readObject(User.class);
```

With JPA (Portable):

```
// Works with ANY JPA implementation
EntityManager em = emf.createEntityManager();
User user = em.find(User.class, 1L);
em.close();

// Switch from Hibernate to EclipseLink?
// Just change the dependency. Code stays the same.
```

Key JPA Concepts

- Entity - A Java class mapped to a database table (@Entity).
- EntityManager - The main API for CRUD operations.
- Persistence Context - A set of managed entities (first-level cache).
- JPQL - Java Persistence Query Language (object-oriented SQL).
- EntityManagerFactory - Creates EntityManagers (expensive, create once).

```
@Entity
@Table(name = "users")
public class User {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(nullable = false)
    private String name;

    @OneToMany(mappedBy = "user", fetch = FetchType.LAZY)
    private List<Order> orders;
}

// Using EntityManager
EntityManager em = emf.createEntityManager();
em.getTransaction().begin();

User user = em.find(User.class, 1L);  // SELECT from database
user.setName("New Name");             // No SQL yet - just marks dirty
// Hibernate tracks changes automatically

em.getTransaction().commit();  // Now: UPDATE users SET name = ? WHERE id = ?
em.close();
```

Hibernate: The Implementation

Hibernate is the most popular JPA implementation. It implements all JPA interfaces and adds powerful features on top.

What Hibernate Adds Beyond JPA

```
Feature   |   JPA Standard   |   Hibernate Extra
Caching   |   First-level cache only   |   Second-level cache (EHCache, Infinispan)
Batch Operations   |   Basic   |   `@BatchSize`, `StatelessSession`
Fetching   |   LAZY/EAGER   |   `@Fetch` strategies, entity graphs
Query Language   |   JPQL   |   HQL (superset), Criteria API extensions
ID Generation   |   Basic strategies   |   Custom generators, pooled optimizers
```

Hibernate Session vs JPA EntityManager

```
// JPA standard way
EntityManager em = emf.createEntityManager();
User user = em.find(User.class, 1L);

// Hibernate native way (same thing internally)
Session session = sessionFactory.openSession();
User user = session.get(User.class, 1L);

// Getting Hibernate Session from JPA EntityManager
Session session = em.unwrap(Session.class);
```

Which API Should You Use?

Use JPA APIs (EntityManager) for portability. Only use Hibernate-specific APIs (Session) when you need features not in JPA, like StatelessSession for bulk operations or specific caching configurations.

Connection Pooling Deep Dive

Creating a database connection is expensive: TCP handshake, authentication, session setup. This takes 50-200ms. Without pooling, a request taking 5ms to execute spends 95% of its time just getting a connection.

HikariCP: The Default Choice

Spring Boot uses HikariCP by default. It's the fastest connection pool available.

```
# application.properties - HikariCP configuration
spring.datasource.hikari.maximum-pool-size=10
spring.datasource.hikari.minimum-idle=5
spring.datasource.hikari.idle-timeout=300000
spring.datasource.hikari.connection-timeout=20000
spring.datasource.hikari.max-lifetime=1200000

# Pool sizing formula (from HikariCP wiki):
# connections = ((core_count * 2) + effective_spindle_count)
# For SSD with 4 cores: (4 * 2) + 1 = 9 connections
```

Connection Pool Sizing

More connections is NOT better. Too many connections cause:

- Database CPU contention (context switching).
- Memory pressure on database server.
- Lock contention and deadlocks.

Start with 10 connections and measure before increasing. Most apps need fewer than you think.

The N+1 Query Problem

The N+1 problem occurs when fetching a collection of entities, then accessing a lazy-loaded relationship on each one. You get 1 query for the parent + N queries for each child.

```
// Entity with lazy-loaded orders
@Entity
public class User {
    @Id private Long id;
    private String name;

    @OneToMany(mappedBy = "user", fetch = FetchType.LAZY)
    private List<Order> orders;  // LAZY by default for collections
}

// N+1 Problem - DON'T DO THIS
List<User> users = userRepository.findAll();  // 1 query: SELECT * FROM users
for (User user : users) {
    user.getOrders().size();  // N queries: SELECT * FROM orders WHERE user_id = ?
}
// Total: 1 + N queries (if 100 users = 101 queries!)
```

Solutions

1. JOIN FETCH (Best for known access patterns)

```
@Query("SELECT u FROM User u JOIN FETCH u.orders")
List<User> findAllWithOrders();
// 1 query: SELECT u.*, o.* FROM users u JOIN orders o ON u.id = o.user_id
```

2. @EntityGraph (Declarative)

```
@EntityGraph(attributePaths = {"orders"})
List<User> findAll();
```

3. @BatchSize (Hibernate-specific)

```
@OneToMany(mappedBy = "user")
@BatchSize(size = 25)  // Loads 25 users' orders in one query
private List<Order> orders;
// Instead of N queries, uses N/25 queries with IN clause
```

Detecting N+1 in Development

Enable SQL logging to see the problem:

- spring.jpa.show-sql=true (basic)
- logging.level.org.hibernate.SQL=DEBUG (SQL only)
- logging.level.org.hibernate.orm.jdbc.bind=TRACE (with parameters)

Key Takeaways

- JDBC is the foundation - all ORM frameworks use it underneath.
- JPA is a specification (interfaces), Hibernate is an implementation.
- Use JPA APIs for portability, Hibernate APIs only when needed.
- Connection pooling is essential - creating connections is expensive (50-200ms).
- HikariCP is the default and fastest pool, start with 10 connections.
- N+1 is the most common JPA performance bug - use JOIN FETCH or @BatchSize.
- Always enable SQL logging in development to catch N+1 early.

JPA and Hibernate in Action

JDBC, JPA and Hibernate

6 essential patterns visualized step by step.

Patterns Covered

- Entity Lifecycle - Transient, Managed, Detached, Removed states.
- Lazy vs Eager Loading - When SQL actually fires.
- Dirty Checking - How Hibernate detects changes automatically.
- Optimistic Locking - Handling concurrent updates with @Version.
- Batch Operations - Performance comparison single vs batch.
- Query Types - JPQL vs Criteria vs Native SQL.

Pattern 1: Entity Lifecycle

Every JPA entity exists in one of four states. Understanding these states tells you when changes sync to the database.

The state machine: Transient (new User()) -> Managed (em.persist() / em.find()) -> Detached (em.detach()) -> back to Managed via em.merge(), or Managed -> Removed (em.remove()).

```
State   |   In Context   |   Tracked   |   Synced to DB
Transient   |   No   |   No   |   No
Managed   |   Yes   |   Yes   |   On flush
Detached   |   No   |   No   |   No
Removed   |   Yes   |   Yes   |   DELETE on flush
```

Common Mistake

Modifying a detached entity and expecting it to save. You must call em.merge(entity) to re-attach it.

Pattern 2: Lazy vs Eager Loading

LAZY loading defers SQL until you access the relationship. This prevents loading unnecessary data but can cause N+1 problems.

```
@OneToMany(fetch = FetchType.LAZY)  // Default for collections
private List<Order> orders;

// When you need eager loading for a specific query:
@Query("SELECT u FROM User u JOIN FETCH u.orders")
List<User> findAllWithOrders();
```

Best Practice

Keep everything LAZY. Use JOIN FETCH or @EntityGraph when you need relations loaded.

Pattern 3: Dirty Checking

Hibernate automatically detects changes to managed entities. No save() call needed - just modify and commit.

```
@Transactional
public void updateName(Long id, String newName) {
    User user = repository.findById(id).orElseThrow();
    user.setName(newName);  // That's it! No save() needed
    // UPDATE generated automatically at commit
}
```

Pattern 4: Optimistic Locking

Prevents lost updates when two users edit the same record. Uses a version column checked at update time.

```
@Entity
public class Product {
    @Version
    private Integer version;  // Hibernate manages this
}

// Generated SQL:
// UPDATE products SET price=?, version=2 WHERE id=1 AND version=1
// If 0 rows updated -> OptimisticLockException
```

Pattern 5: Batch Operations

Batching groups multiple INSERTs into fewer database calls, dramatically improving performance.

```
# application.properties
spring.jpa.properties.hibernate.jdbc.batch_size=50
spring.jpa.properties.hibernate.order_inserts=true
```

```
// In code: flush periodically to avoid memory issues
for (int i = 0; i < 1000; i++) {
    em.persist(new User("User" + i));
    if (i % 50 == 0) {
        em.flush();
        em.clear();  // Clear persistence context
    }
}
```

JPA and Hibernate Best Practices

JDBC, JPA and Hibernate

15 common pitfalls and how to avoid them.

Why Best Practices Matter

JPA makes database access easy, but that simplicity hides complexity. A missing @Transactional silently reads stale data. An EAGER fetch multiplies your queries. A forgotten flush() causes OutOfMemoryError on batch inserts. These mistakes compile and often appear to work until production load exposes them.

1. Entity Design Best Practices

1. Never Use EAGER on Collections

EAGER loading on @OneToMany or @ManyToMany multiplies with other EAGER relations and loads data you may not need.

```
// BAD: EAGER on Collection
@OneToMany(fetch = FetchType.EAGER)
private List<Order> orders;
// Every time you load a User, ALL orders load too - even if you just need the name!
```

```
// GOOD: LAZY + Explicit Fetch
@OneToMany(fetch = FetchType.LAZY)
private List<Order> orders;

// When you need orders:
@Query("SELECT u FROM User u JOIN FETCH u.orders WHERE u.id = :id")
User findWithOrders(@Param("id") Long id);
```

2. Use @Version for Optimistic Locking

Without versioning, concurrent updates silently overwrite each other. Add @Version to detect conflicts.

```
// BAD: No Version Field
@Entity
public class Product {
    @Id
    private Long id;
    private String name;
    private BigDecimal price;
    // Last write wins - data loss!
}
```

```
// GOOD: Version Field Added
@Entity
public class Product {
    @Id
    private Long id;

    @Version
    private Integer version;

    private String name;
    private BigDecimal price;
}
```

3. Override equals() and hashCode() Correctly

Use business key or ID. Never use generated ID alone if entity might be in a Set before persisting.

```
// BAD: Uses Generated ID
@Override
public boolean equals(Object o) {
    if (this == o) return true;
    if (!(o instanceof User)) return false;
    User user = (User) o;
    return id != null && id.equals(user.id);
}
// Problem: id is null before persist()
// Two new Users in Set = duplicates!
```

```
// GOOD: Uses Business Key
@Override
public boolean equals(Object o) {
    if (this == o) return true;
    if (!(o instanceof User)) return false;
    User user = (User) o;
    return email != null &&
           email.equals(user.email);
}
// email is set before persist
// Works correctly in Sets
```

2. Query Best Practices

4. Fix N+1 with JOIN FETCH or @BatchSize

The N+1 problem is the most common JPA performance bug. Always check SQL logs in development.

```
// BAD: N+1 Queries
List<User> users = userRepo.findAll();
for (User u : users) {
    u.getOrders().size(); // N queries!
}
// 1 + N queries total; 100 users = 101 queries
```

```
// GOOD: JOIN FETCH
@Query("SELECT DISTINCT u FROM User u LEFT JOIN FETCH u.orders")
List<User> findAllWithOrders();
// 1 query total; all orders loaded with users
```

5. Use DTOs for Read-Only Queries

Loading full entities for display wastes memory and enables accidental modifications. Use projections.

```
// BAD: Full Entity for Display
// Loads entire User entity, tracks in persistence context, wastes memory for read-only use
List<User> users = userRepo.findAll();
return users.stream()
    .map(u -> u.getName())
    .collect(toList());
```

```
// GOOD: DTO Projection
// Only fetches what you need, not tracked in persistence context
@Query("SELECT new com.example.UserDTO(u.id, u.name, u.email) FROM User u")
List<UserDTO> findAllAsDTO();
```

6. Avoid SELECT * with Native Queries

Explicit column selection prevents loading unnecessary data and makes schema changes safer.

```
// BAD: SELECT *
@Query(value = "SELECT * FROM users WHERE status = 'ACTIVE'",
       nativeQuery = true)
List<User> findActive();
// Loads all columns; breaks if schema changes
```

```
// GOOD: Explicit Columns
@Query(value = "SELECT id, name, email FROM users WHERE status = 'ACTIVE'",
       nativeQuery = true)
List<Object[]> findActive();
// Or use JPQL for type safety
```

3. Transaction Best Practices

7. Always Use @Transactional for Write Operations

Without a transaction, each repository call is a separate transaction. Changes may not be atomic.

```
// BAD: No Transaction
public void transferMoney(Long from, Long to, BigDecimal amount) {
    accountRepo.debit(from, amount);
    // If this fails, debit already committed!
    accountRepo.credit(to, amount);
}
```

```
// GOOD: Explicit Transaction
@Transactional
public void transferMoney(Long from, Long to, BigDecimal amount) {
    accountRepo.debit(from, amount);
    accountRepo.credit(to, amount);
    // Both succeed or both rollback
}
```

8. Use Read-Only Transactions for Queries

readOnly = true hints to Hibernate to skip dirty checking, improving performance.

```
// BAD: Write Transaction for Read
@Transactional
public List<User> getAllUsers() {
    return userRepo.findAll();
}
// Hibernate tracks all entities, checks for changes at commit - wasted CPU cycles
```

```
// GOOD: Read-Only Transaction
@Transactional(readOnly = true)
public List<User> getAllUsers() {
    return userRepo.findAll();
}
// Hibernate skips dirty checking - better performance
```

9. Keep Transactions Short

Long transactions hold database locks, reducing concurrency and risking timeouts.

```
// BAD: External Call in Transaction
@Transactional
public void processOrder(Order order) {
    orderRepo.save(order);
    paymentService.charge(order); // HTTP call!
    // Holds DB connection during network I/O
    // Other requests wait for connection
}
```

```
// GOOD: External Call Outside
public void processOrder(Order order) {
    saveOrder(order);  // Transaction 1
    paymentService.charge(order); // No transaction
    markPaid(order);   // Transaction 2
}

@Transactional
void saveOrder(Order order) { ... }
```

4. Performance Best Practices

10. Use Batch Inserts with flush/clear

Without periodic flush/clear, all entities accumulate in memory causing OutOfMemoryError.

```
// BAD: No Flush/Clear
for (int i = 0; i < 100000; i++) {
    em.persist(new User("User" + i));
}
// 100,000 entities in memory - OutOfMemoryError likely
```

```
// GOOD: Periodic Flush/Clear
for (int i = 0; i < 100000; i++) {
    em.persist(new User("User" + i));
    if (i % 50 == 0) {
        em.flush();  // Send to DB
        em.clear();  // Free memory
    }
}
```

11. Configure Connection Pool Properly

Too few connections cause request queuing. Too many overwhelm the database.

```
# application.properties - HikariCP (Spring Boot default)
spring.datasource.hikari.maximum-pool-size=10
spring.datasource.hikari.minimum-idle=5
spring.datasource.hikari.idle-timeout=300000
spring.datasource.hikari.connection-timeout=20000

# Pool sizing formula:
# connections = (core_count * 2) + spindle_count
# For 4-core server with SSD: (4 * 2) + 1 = 9
```

12. Enable SQL Logging in Development

You cannot optimize what you cannot see. Always log SQL in development to catch N+1.

```
# application-dev.properties
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.format_sql=true

# Better: Use logging (can set level)
logging.level.org.hibernate.SQL=DEBUG
logging.level.org.hibernate.orm.jdbc.bind=TRACE  # Shows parameters
```

5. Common Pitfalls to Avoid

13. Handle LazyInitializationException Properly

This exception means you accessed a lazy relation after the persistence context closed.

```
// BAD: Access Outside Transaction
@GetMapping("/users/{id}")
public User getUser(@PathVariable Long id) {
    User user = userService.findById(id);
    user.getOrders().size(); // BOOM!
    // Transaction ended in service
    // LazyInitializationException
}
```

```
// GOOD: Fetch in Service Layer
// Service
@Transactional(readOnly = true)
public User findByIdWithOrders(Long id) {
    return userRepo.findByIdWithOrders(id);
}

// Repository
@Query("SELECT u FROM User u JOIN FETCH u.orders WHERE u.id = :id")
User findByIdWithOrders(@Param("id") Long id);
```

14. Avoid Modifying Detached Entities Without Merge

Changes to detached entities are not persisted. Use merge() to re-attach.

```
// BAD: Modify Detached Entity
User user = userService.findById(id);
// Transaction ended, user is detached
user.setName("New Name");
// Changes NOT saved! User is detached from persistence context
```

```
// GOOD: Merge or New Transaction
// Option 1: Merge
User detached = getDetachedUser();
detached.setName("New Name");
User managed = em.merge(detached);

// Option 2: Load and modify in transaction
@Transactional
public void updateName(Long id, String name) {
    User user = userRepo.findById(id).get();
    user.setName(name); // Auto-saved
}
```

15. Use Spring Data Repository Methods Correctly

save() on an existing entity does an unnecessary SELECT. Use direct updates for performance.

```
// BAD: save() for Update
User user = new User();
user.setId(existingId);  // Set existing ID
user.setName("New Name");
userRepo.save(user);
// Does SELECT first to check if exists, then UPDATE - 2 queries!
```

```
// GOOD: Direct Update Query
@Modifying
@Query("UPDATE User u SET u.name = :name WHERE u.id = :id")
int updateName(@Param("id") Long id, @Param("name") String name);
// Single UPDATE query, returns affected row count
```

Quick Reference Checklist

Before every code review:

- All @OneToMany and @ManyToMany use FetchType.LAZY?
- Entities that need concurrent editing have @Version?
- Write operations wrapped in @Transactional?
- Read-only operations use @Transactional(readOnly = true)?
- No HTTP/external calls inside transactions?
- N+1 checked with SQL logging?
- Batch operations use flush()/clear() pattern?
- DTOs used for read-only queries instead of full entities?
- No lazy access outside transaction boundaries?

Key Takeaways

- Keep everything LAZY, use JOIN FETCH when needed.
- Always enable SQL logging in development.
- Use @Transactional for writes, readOnly=true for reads.
- Keep transactions short - no external calls inside.
- Use DTOs for read-only queries to avoid tracking overhead.
- Add @Version to entities that might be concurrently edited.
- For batch operations, flush/clear every 50-100 entities.

What Actually Happens Inside map.put(key, value)?

HashMap is the most-used data structure in any Java codebase. It's also the most-asked interview question. Behind that one-line put call: a bucket array, a hash spread, a clever bitmask trick, a linked list, a red-black tree, and a resize that quietly doubles your map. Let's open it up.

HashMap Is a Fixed-Size Array Under the Hood

Forget anything fancy. At its core, HashMap is just an array of buckets. The default size is 16. Every key-value pair lives in one of those buckets, picked based on the key's hash. Two rules that explain everything else: the size is always a power of 2, and the array is lazily created on the first put().

Why power of 2?

A power-of-2 size lets HashMap replace the slow modulo operation with a single, fast bitwise AND. Every design decision flows from this.

HashMap Doesn't Trust Your hashCode

Your key's hashCode() returns an int, but HashMap only uses the lowest few bits to pick a bucket. If your hash codes only differ in the upper bits (which happens more than you'd think), every key would land in the same bucket. So HashMap spreads the hash by XORing it with its own upper half before using it.

The spreading trick

h = key.hashCode() ^ (h >>> 16) mixes upper and lower bits in a single, cheap operation.

(n - 1) & hash Instead of hash % n

To pick a bucket index, HashMap needs to map the spread hash to a value between 0 and (capacity - 1). The obvious choice is modulo: hash % n. But modulo is slow. When n is a power of 2, you can replace it with a bitwise AND against n - 1. Same result, much faster.

Why this is the reason for power-of-2 sizing

If n were 15, (n - 1) = 14 = 1110 in binary. The lowest bit can never be set. Hash distribution would be broken. n must be a power of 2 for the trick to work.

When Two Keys Want the Same Bucket, They Form a Chain

With only 16 buckets and potentially millions of keys, collisions are inevitable. HashMap doesn't crash - it builds a linked list inside the colliding bucket. Each node holds the key, value, hash, and a reference to the next node. Lookup walks this list calling equals() on each node's key.

The cost of collisions

An empty bucket lookup is O(1). A bucket with 5 entries is O(5) - the JVM walks each node calling equals(). Good hash distribution keeps chains short.

A Bucket of 8 Nodes Upgrades Itself to a Tree

Before Java 8, a bucket with thousands of colliding keys meant O(n) lookups. Java 8 introduced an automatic

upgrade: when a bucket reaches TREEIFY_THRESHOLD = 8 nodes, the linked list is converted into a red-black tree. Lookups in that bucket drop from O(n) to O(log n). If the bucket shrinks back to 6 nodes during resize, it untreeifies back to a list.

The capacity catch

Treeification only happens if the overall capacity is at least 64. Below that, HashMap just resizes instead. Trees only make sense when there's enough room for hashes to spread out.

At 75% Full, Capacity Doubles and Everything Moves

A HashMap with capacity 16 and load factor 0.75 will resize when it hits 12 entries. Capacity doubles to 32. A new array is allocated. Then every existing entry gets rehashed into its new bucket. This is O(n) and pauses your put. Pre-sizing the map saves you from many resizes.

Pre-sizing tip

If you know you'll insert N entries, create with new HashMap(N / 0.75 + 1). Java rounds up to a power of 2 and you skip all resizes.

One Full put() Walk, End to End

Now you can read the actual flow that runs on every put(). Hash. Spread. Index. Walk. Insert. Check thresholds. Maybe treeify. Maybe resize. All of it in microseconds.

This is what runs every time

That single line of Java is ~30 machine instructions in the common case, ~100 if it triggers resize or treeify. Still very fast.

HashMap Is NOT Thread-Safe - Use ConcurrentHashMap

HashMap is famously fast because it does zero locking. Two threads writing concurrently can corrupt the chain, lose entries, or even cause an infinite loop during resize. The fix is ConcurrentHashMap, which uses bucket-level locking - threads only block when they hit the same bucket.

The rule

Single-threaded or thread-confined: HashMap. Shared across threads: ConcurrentHashMap. Never wrap HashMap in synchronized blocks - it's neither correct nor fast.